

(12) **United States Patent**
Bernstein et al.

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(54) **SYSTEMS, DEVICES, AND METHODS FOR DYNAMICALLY PROVIDING USER INTERFACE CONTROLS AT A TOUCH-SENSITIVE SECONDARY DEVICE**

(58) **Field of Classification Search**
None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,956,745 A 5/1976 Ellis
6,111,527 A 8/2000 Susel
(Continued)

FOREIGN PATENT DOCUMENTS

CN 101197046 A 6/2008
CN 101232678 A 7/2008
(Continued)

OTHER PUBLICATIONS

Notice of Allowance, dated Aug. 13, 2024, received in Japanese Patent Application No. 2021-110169 (7205JP04), which corresponds with U.S. Appl. No. 15/113,779, 2 pages.
(Continued)

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(57) **ABSTRACT**

A computing system comprises one or more processors, memory, a first housing that includes a primary display, and a second housing at least partially containing a physical keyboard and a touch-sensitive secondary display that is distinct from the primary display. The computing system displays a first user interface on the primary display, the first user interface includes one or more user interface elements. The computing system identifies, from the one or more user interface elements, an active user interface element that is in focus on the primary display. The computer system, in accordance with a determination that the active user interface element that is in focus on the primary display is associated with the application executed by the computing system, displays, on the touch-sensitive secondary display, a
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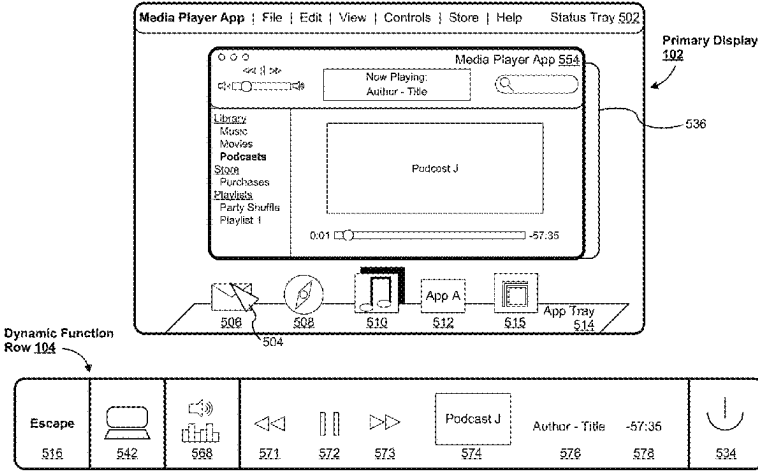
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second user interface that includes a first set of one or more affordances corresponding to the application.

33 Claims, 81 Drawing Sheets

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(56) References Cited

U.S. PATENT DOCUMENTS

6,396,483	B1	5/2002	Hiller	
7,239,728	B1	7/2007	Choi et al.	
9,256,719	B2 *	2/2016	Berini	G06V 40/197
9,369,635	B1	6/2016	Hilla	
9,521,375	B2	12/2016	Beaumier et al.	
10,551,995	B1 *	2/2020	Ho	G06F 3/0486
2001/0017934	A1	8/2001	Paloniemi et al.	
2002/0018585	A1	2/2002	Kim	
2002/0191029	A1	12/2002	Gillespie et al.	
2004/0004604	A1	1/2004	Numano	
2004/0108968	A1	6/2004	Finke-Anlauff	
2004/0239615	A1	12/2004	Firebaugh et al.	
2006/0036964	A1	2/2006	Satterfield et al.	
2006/0052885	A1	3/2006	Kong	
2007/0162574	A1	7/2007	Williamson et al.	
2008/0178117	A1	7/2008	Gelman et al.	
2008/0207273	A1	8/2008	Huo	
2008/0242343	A1	10/2008	Koh et al.	
2008/0252414	A1	10/2008	Crigger et al.	
2008/0320410	A1	12/2008	Whytock et al.	
2009/0100384	A1	4/2009	Louch	
2009/0183008	A1	7/2009	Jobmann	

2009/0199282	A1	8/2009	Tsitkova et al.	
2009/0201254	A1	8/2009	Rais	
2009/0265628	A1	10/2009	Bamford et al.	
2009/0271723	A1	10/2009	Matsushima et al.	
2009/0315867	A1	12/2009	Sakamoto et al.	
2010/0037064	A1	2/2010	Ku	
2010/0077346	A1	3/2010	Kawashima et al.	
2010/0088653	A1	4/2010	Yach et al.	
2010/0149101	A1	6/2010	Guo et al.	
2010/0153269	A1	6/2010	McCabe	
2010/0156595	A1	6/2010	Wong et al.	
2010/0182265	A1	7/2010	Kim et al.	
2010/0207727	A1	8/2010	Kanekiyo et al.	
2010/0265182	A1	10/2010	Ball et al.	
2010/0275163	A1	10/2010	Gillespie et al.	
2010/0295802	A1	11/2010	Lee et al.	
2011/0047459	A1 *	2/2011	Van Der Westhuizen	

G06F 3/04883
345/173

2011/0154342	A1	6/2011	Logan et al.	
2011/0187646	A1	8/2011	Mahmoud	
2011/0304644	A1	12/2011	Wada et al.	
2011/0314405	A1	12/2011	Turner et al.	
2011/0317886	A1	12/2011	Matsuoka	
2012/0068933	A1	3/2012	Larsen	
2012/0110515	A1	5/2012	Abramoff et al.	
2012/0235912	A1	9/2012	Laubach	
2013/0050135	A1	2/2013	Stewart et al.	
2013/0067545	A1	3/2013	Hanes	
2013/0080942	A1	3/2013	Reeves et al.	
2013/0086513	A1	4/2013	Rasmussen et al.	
2013/0132876	A1	5/2013	Kobayashi et al.	
2013/0160141	A1	6/2013	Tseng et al.	
2013/0167212	A1	6/2013	Azar et al.	
2013/0173925	A1	7/2013	Yen et al.	
2013/0207994	A1	8/2013	Rodeski et al.	
2013/0311931	A1	11/2013	Kocubo	
2013/0321340	A1	12/2013	Seo et al.	
2013/0322705	A1	12/2013	Wong	
2013/0332836	A1	12/2013	Cho	
2014/0068456	A1	3/2014	Chan et al.	
2014/0075373	A1	3/2014	Jitkoff et al.	
2014/0082548	A1	3/2014	Wu et al.	
2014/0092424	A1	4/2014	Grosz	
2014/0108929	A1	4/2014	Garmark et al.	
2014/0157209	A1	6/2014	Dalal et al.	
2014/0164997	A1	6/2014	Lee et al.	
2014/0184471	A1	7/2014	Martynov et al.	
2014/0232671	A1	8/2014	Chaudhri	
2014/0380239	A1	12/2014	Kang	
2015/0015512	A1	1/2015	Kwak et al.	
2015/0339031	A1 *	11/2015	Zeinstra	 B60K 35/10

715/747

FOREIGN PATENT DOCUMENTS

CN	101324821	A	12/2008
CN	101354791	A	1/2009
CN	101356493	A	1/2009
CN	101520720	A	2/2009
CN	101382842	A	3/2009
CN	101506867	A	8/2009
CN	101529368	A	9/2009
CN	101571785	A	11/2009
CN	101727287	A	6/2010

(56)

References Cited**FOREIGN PATENT DOCUMENTS**

CN	101876879	A	11/2010
CN	101893914	A	11/2010
CN	102096546	A	6/2011
CN	102279698	A	12/2011
CN	102385477	A	3/2012
CN	102405453	A	4/2012
CN	102414755	A	4/2012
CN	202562233	U	11/2012
CN	102934052	A	2/2013
CN	102968204	A	3/2013
CN	103124996	A	5/2013
CN	103155534	A	6/2013
CN	103164152	A	6/2013
CN	103282851	A	9/2013
CN	103324423	A	9/2013
CN	103380461	A	10/2013
CN	103425914	A	12/2013
CN	103477304	A	12/2013
CN	103502920	A	1/2014
CN	103729159	A	4/2014
CN	103729160	A	4/2014
CN	103778388	A	5/2014
CN	104008350	A	8/2014
CN	104715178	A	6/2015
CN	105138254	A	12/2015
CN	105260125	A	1/2016
CN	105574378	A	5/2016
CN	106445184	A	2/2017
EP	2 129 085	A1	2/2009
EP	2 500 807	A1	9/2012
EP	2 645 287	A1	10/2013
EP	2 660 692	A1	11/2013
GB	201204842		5/2012
JP	H 10124217	A	5/1998
JP	2000-122792	A	4/2000
JP	2001-344092	A	12/2001
JP	2006-155455	A	6/2006
JP	2007-289457	A	11/2007
JP	2009-169735	A	7/2009
JP	2009-258667	A	11/2009
JP	2010-009534	A	1/2010
JP	2010-066918	A	3/2010
JP	2010-108088	A	5/2010
JP	2010-129051	A	6/2010
JP	2011-018085	A	1/2011
JP	2011-141607	A	7/2011
JP	2012-003508	A	1/2012
JP	2013-532334	A	8/2013
JP	2015-201131	A	11/2015
JP	2017-504128	A	2/2017
JP	2001-103573	A	4/2021
KR	2009-0011323	A	2/2009
KR	2015/0050732	A	5/2015
KR	2015/0140302	A	12/2015
WO	WO 2015/112868	A1	7/2015
WO	WO 2016/104867	A1	6/2016

OTHER PUBLICATIONS

Office Action, dated Sep. 13, 2024, received in Chinese Patent Application No. 202110182280.2 (7394CN01), which corresponds with U.S. Appl. No. 15/655,707, 8 pages.

Notice of Allowance, dated Nov. 12, 2024, received in Chinese Patent Application No. 202110182280.2 (7394CN01), which corresponds with U.S. Appl. No. 15/655,707, 2 pages.

Grant Decision, dated Jul. 18, 2024, received in European Patent Application No. 20168044.4 (7394EP01), which corresponds with U.S. Appl. No. 15/655,707, 3 pages.

Office Action, dated Jul. 8, 2024, received in Chinese Patent Application No. 202110181406.4 (7577CN01), which corresponds with U.S. Appl. No. 16/361,122, 1 Page.

Office Action, dated Oct. 4, 2024, received in Japanese Patent Application No. 2023-184930 (7886JP), which corresponds with U.S. Appl. No. 17/863,189, 2 pages.

Office Action, dated Aug. 21, 2024, received in U.S. Appl. No. 18/408,528 (8011), 22 pages.

Extended European Search Report, dated Oct. 10, 2024, received in European Patent Application No. 24183896.0 (7886EP), 10 pages.

iOS 11 Updates, "Demo of iOS 9 Apps: Search, Siri, Photos App", <https://youtube.be.EBnXMM1X6xA>, Jun. 12, 2015, 2 pages.

MSAppliedSciences, "DisplayCover", <https://youtube/OSFIVmIOSso>, Aug. 20, 2015, 2 pages.

MSAppliedSciences, "UIST Student Innovation Contest 2010 Concept", <https://youtube/ojusRO38Tdc>, Aug. 9, 2010, 3 pages.

Response to Pending Examination Report for PA 201670555, "Systems, Devices, and Methods for Dynamically Providing User Interface Controls at A Touch-Sensitive Secondary Display", <https://onlineweb.dkpto.dk/Dokumenter2017/227/09194227.pdf>, Nov. 7, 2017, 161 pages.

Wagner et al., U.S. application filed on Apr. 15, 2014, titled "Device, Method, and Graphical User Interface for Navigating and Displaying Content in Context", 140 pages.

Sato, "Farewell on Google Android's Menu Button", Itmedia News, <https://www.itmedia.co.jp/news/articles/1201/27/news059.html>, Jan. 27, 2012, 4 pages.

Office Action, dated Aug. 13, 2018, received in U.S. Appl. No. 15/113,779 (7205), 23 pages.

Final Office Action, dated Jan. 22, 2019, received in U.S. Appl. No. 15/113,779 (7205), 25 pages.

Office Action, dated May 14, 2019, received in U.S. Appl. No. 15/113,779 (7205), 22 pages.

Notice of Allowance, dated Oct. 9, 2019, received in U.S. Appl. No. 15/113,779 (7205), 20 pages.

Notice of Allowance, dated Jan. 15, 2020, received in U.S. Appl. No. 15/113,779 (7205), 8 pages.

Office Action, dated Dec. 5, 2018, received in Chinese Patent Application No. 2015800055665.4 (7205CN), which corresponds with U.S. Appl. No. 15/113,779, 6 pages.

Office Action, dated Jul. 9, 2019, received in Chinese Patent Application No. 20150800055665.4 (7205CN), which corresponds with U.S. Appl. No. 15/113,779, 4 pages.

Office Action, dated Sep. 30, 2019, received in Chinese Patent Application No. 2015800055665.4 (7205CN), which corresponds with U.S. Appl. No. 15/113,779, 3 pages.

Notice of Allowance, dated Feb. 3, 2020, received in Chinese Patent Application No. 2015800055665.4 (7205CN), which corresponds with U.S. Appl. No. 15/113,779, 3 pages.

Patent, dated Mar. 31, 2020, received in Chinese Patent Application No. 2015800055665.4 (7205CN), which corresponds with U.S. Appl. No. 15/113,779, 6 pages.

Office Action, dated Jun. 15, 2018 received in Chinese Patent Application No. 201610840116.5 (7205CN01), which corresponds with U.S. Appl. No. 15/113,779, 5 pages.

Notice of Allowance, dated Apr. 2, 2019, received in Chinese Patent Application No. 201610840116.5 (7205CN01), which corresponds with U.S. Appl. No. 15/113,779, 3 pages.

Patent, dated May 17, 2019, received in Chinese Patent Application No. 201610840116.5 (7205CN01), which corresponds with U.S. Appl. No. 15/113,779, 6 pages.

Office Action, dated Feb. 13, 2023, received in Chinese Patent Application No. 202010307026.6 (7205CN02), which corresponds with U.S. Appl. No. 15/113,779, 10 pages.

Office Action, dated Sep. 20, 2023, received in Chinese Patent Application No. 202010307026.6 (7205CN02), which corresponds with U.S. Appl. No. 15/113,779, 2 pages.

Office Action, dated Jan. 18, 2024, received in Chinese Patent Application No. 202010307026.6 (7205CN02), which corresponds with U.S. Appl. No. 15/113,779, 2 pages.

Patent, dated Mar. 8, 2024, received in Chinese Patent Application No. 202010307026.6 (7205CN02), which corresponds with U.S. Appl. No. 15/113,779, 6 pages.

Office Action, dated Dec. 30, 2022, received in Chinese Patent Application No. 202010307686.4 (7205CN03), which corresponds with U.S. Appl. No. 15/113,779, 5 pages.

Notice of Allowance, dated Oct. 27, 2023, received in Chinese Patent Application No. 202010307686.4 (7205CN03), which corresponds with U.S. Appl. No. 15/113,779, 3 pages.

(56)

References Cited**OTHER PUBLICATIONS**

Patent, dated Nov. 28, 2023, received in Chinese Patent Application No. 202010307686.4 (7205CN03), which corresponds with U.S. Appl. No. 15/113,779, 6 pages.

Notice of Allowance, dated Jan. 11, 2023, received in Chinese Patent Application No. 202010305200.3 (7205CN04), which corresponds with U.S. Appl. No. 15/113,779, 3 pages.

Patent, dated Mar. 24, 2023, received in Chinese Patent Application No. 202010305200.3 (7205CN04), which corresponds with U.S. Appl. No. 15/113,779, 6 pages.

Office Action, dated Jun. 17, 2024, received in Japanese Patent Application No. 2023-095728 (7205JP05), which corresponds with U.S. Appl. No. 15/113,779, 6 pages.

Office Action, dated Oct. 26, 2016, received in Danish Patent Application No. 201670555 (7205DK), which corresponds with U.S. Appl. No. 15/113,779, 10 pages.

Office Action, dated May 8, 2017, received in Danish Patent Application No. 201670555 (7205DK), which corresponds with U.S. Appl. No. 15/113,779, 3 pages.

Office Action, dated Nov. 16, 2017, received in Danish Patent Application No. 201670555 (7205DK), which corresponds with U.S. Appl. No. 15/113,779, 2 pages.

Notice of Allowance, dated Feb. 12, 2018, received in Danish Patent Application No. 201670555 (7205DK), which corresponds with U.S. Appl. No. 15/113,779, 2 pages.

Patent, dated Oct. 11, 2018, received in Danish Patent Application No. 201670555 (7205DK), which corresponds with U.S. Appl. No. 15/113,779, 3 pages.

Notice of Allowance, dated Feb. 12, 2018, received in Danish Patent Application No. 201670559 (7205DK01), which corresponds with U.S. Appl. No. 15/113,779, 2 pages.

Office Action, dated Oct. 18, 2016, received in Danish Patent Application No. 201670559 (7205DK01), which corresponds with U.S. Appl. No. 15/113,779, 9 pages.

Office Action, dated May 8, 2017, received in Danish Patent Application No. 201670559 (7205DK01), which corresponds with U.S. Appl. No. 15/113,779, 4 pages.

Patent, dated Apr. 23, 2018, received in Danish Patent Application No. 201670559 (7205DK01), which corresponds with U.S. Appl. No. 15/113,779, 3 pages.

Office Action, dated Nov. 16, 2017, received in Danish Patent Application No. 201670559 (7205DK01), which corresponds with U.S. Appl. No. 15/113,779, 2 pages.

Office Action, dated May 29, 2017, received in European Patent Application No. 1570293.5.6 (7205EP), which corresponds with U.S. Appl. No. 14/113,779, 9 pages.

Summons to Attend Oral Proceedings, dated Jan. 8, 2018, received in European Patent Application No. 15702935.6 (7205EP), which corresponds with U.S. Appl. No. 14/113,779, 18 pages.

Intention to Grant, dated Jun. 18, 2021, received in European Patent Application No. 15702935.6 (7205EP), which corresponds with U.S. Appl. No. 14/113,779, 7 pages.

Patent, dated Sep. 29, 2021, received in European Patent Application No. 15702935.6 (7205EP), which corresponds with U.S. Appl. No. 14/113,779, 3 pages.

Office Action, dated Jan. 29, 2018, received in European Patent Application No. 16181866.1 (7205EP01), which corresponds with U.S. Appl. No. 15/113,779, 11 pages.

Intention to Grant, dated Aug. 16, 2018, received in European Patent Application No. 16181866.1 (7205EP01), which corresponds with U.S. Appl. No. 15/113,779, 5 pages.

Patent, dated Jan. 23, 2019, received in European Patent Application No. 16181866.1 (7205EP01), which corresponds with U.S. Appl. No. 15/113,779, 3 pages.

Office Action, dated Apr. 5, 2024, received in European Patent Application No. 21164867.0 (7205EP02), which corresponds with U.S. Appl. No. 15/113,779, 5 pages.

Notice of Allowance, dated Nov. 13, 2017, received in Japanese Patent Application No. 2016-548096 (7205JP), which corresponds with U.S. Appl. No. 14/113,779, 5 pages.

Patent, dated Dec. 8, 2017, received in Japanese Patent Application No. 2016-548096 (7205JP), which corresponds with U.S. Appl. No. 14/113,779, 3 pages.

Office Action, dated Jun. 2, 2017, received in Japanese Patent Application No. 2016-158867 (7205JP01), which corresponds with U.S. Appl. No. 15/113,779, 6 pages.

Office Action, dated Feb. 5, 2018, received in Japanese Patent Application No. 2016-158867 (7205JP01), which corresponds with U.S. Appl. No. 15/113,779, 5 pages.

Patent, dated Jun. 29, 2018, received in Japanese Patent Application No. 2016-158867 (7205JP01), which corresponds with U.S. Appl. No. 15/113,779, 3 pages.

Office Action, dated Apr. 9, 2019, received in Japanese Patent Application No. 2018-116341 (7205JP02), which corresponds with U.S. Appl. No. 15/113,779, 7 pages.

Office Action, dated Dec. 13, 2019, received in Japanese Patent Application No. 2018-116341 (7205JP02), which corresponds with U.S. Appl. No. 15/113,779, 5 pages.

Notice of Allowance, dated Jun. 5, 2020, received in Japanese Patent Application No. 2018-116341 (7205JP02), which corresponds with U.S. Appl. No. 15/113,779, 5 pages.

Patent, dated Jul. 3, 2020, received in Japanese Patent Application No. 2018-116341 (7205JP02), which corresponds with U.S. Appl. No. 15/113,779, 4 pages.

Notice of Allowance, dated Jun. 7, 2021, received in Japanese Patent Application No. 2020-071844 (7205JP03), which corresponds with U.S. Appl. No. 15/113,779, 2 pages.

Patent, dated Jul. 5, 2021, received in Japanese Patent Application No. 2020-071844 (7205JP03), which corresponds with U.S. Appl. No. 15/113,779, 4 pages.

Office Action, dated Jul. 4, 2022, received in Japanese Patent Application No. 2021-110169 (7205JP04), which corresponds with U.S. Appl. No. 15/113,779, 2 pages.

Final Office Action, dated Feb. 14, 2023, received in Japanese Patent Application No. 2021-110169 (7205JP04), which corresponds with U.S. Appl. No. 15/113,779, 4 pages.

Office Action, dated Apr. 7, 2024, received in Japanese Patent Application No. 2021-110169 (7205JP04), which corresponds with U.S. Appl. No. 15/113,779, 3 pages.

Office Action, dated Aug. 10, 2018, received in U.S. Appl. No. 15/655,707 (7394), 15 pages.

Innovation Patent, dated Sep. 19, 2017, received in Australia Patent Application No. 2017100879 (7394AU), which corresponds with U.S. Appl. No. 15/655,707, 1 page.

Office Action, dated Feb. 1, 2019, received in Australia Patent Application No. 2017302420 (7394AU01), which corresponds with U.S. Appl. No. 15/655,707, 6 pages.

Notice of Acceptance, dated Jul. 15, 2019, received in Australia Patent Application No. 2017302420 (7394AU01), which corresponds with U.S. Appl. No. 15/655,707, 3 pages.

Certificate of Grant, dated Nov. 14, 2019, received in Australia Patent Application No. 2017302420 (7394AU01), which corresponds with U.S. Appl. No. 15/655,707, 4 pages.

Office Action, dated Aug. 18, 2020, received in Australian Patent Application No. 2019219760 (7394AU02), which corresponds with U.S. Appl. No. 15/655,707, 6 pages.

Notice of Acceptance, dated Jul. 12, 2021, received in Australian Patent Application No. 2019219760 (7394AU02), which corresponds with U.S. Appl. No. 15/655,707, 3 pages.

Certificate of Grant, dated Nov. 18, 2021, received in Australian Patent Application No. 2019219760 (7394AU02), which corresponds with U.S. Appl. No. 15/655,707, 4 pages.

Office Action, dated Aug. 25, 2017, received in Chinese Patent Application No. 201710632979.8 (7394CN), which corresponds with U.S. Appl. No. 15/655,707, 3 pages.

Office Action, dated Jun. 3, 2020, received in Chinese Patent Application No. 201710632979.8 (7394CN), which corresponds with U.S. Appl. No. 15/655,707, 8 pages.

Notice of Allowance, dated Nov. 30, 2020, received in Chinese Patent Application No. 201710632979.8 (7394CN), which corresponds with U.S. Appl. No. 15/655,707, 5 pages.

(56)

References Cited**OTHER PUBLICATIONS**

Patent, dated Feb. 23, 2021, received in Chinese Patent Application No. 201710632979.8 (7394CN), which corresponds with U.S. Appl. No. 15/655,707, 7 pages.

Office Action, dated Jan. 4, 2024, received in Chinese Patent Application No. 202110182280.2 (7394CN01), which corresponds with U.S. Appl. No. 15/655,707, 5 pages.

Office Action, dated May 16, 2024, received in Chinese Patent Application No. 202110182280.2 (7394CN01), which corresponds with U.S. Appl. No. 15/655,707, 3 pages.

Office Action, dated Oct. 10, 2017, received in Danish Patent Application No. 2017-70561 (7394DK), which corresponds with U.S. Appl. No. 15/655,707, 9 pages.

Office Action, dated Mar. 14, 2018, received in Danish Patent Application No. 2017-70561 (7394DK), which corresponds with U.S. Appl. No. 15/655,707, 3 pages.

Office Action, dated Oct. 31, 2018, received in Danish Patent Application No. 2017-70561 (7394DK), which corresponds with U.S. Appl. No. 15/655,707, 5 pages.

Intention to Grant, dated Jan. 2, 2019, received in Danish Patent Application No. 2017-70561 (7394DK), which corresponds with U.S. Appl. No. 15/655,707, 2 pages.

Notice of Allowance, dated Feb. 7, 2019, received in Danish Patent Application No. 2017-70561 (7394DK), which corresponds with U.S. Appl. No. 15/655,707, 2 pages.

Patent, dated Apr. 3, 2019, received in Danish Patent Application No. 2017-70561 (7394DK), which corresponds with U.S. Appl. No. 15/655,707, 5 pages.

Office Action, dated Jun. 27, 2019, received in European Patent Application No. 17745225.7 (7394EP), which corresponds with U.S. Appl. No. 15/655,707, 6 pages.

Intention to Grant, dated Jan. 16, 2020, received in European Patent Application No. 17745225.7 (7394EP), which corresponds with U.S. Appl. No. 15/655,707, 8 pages.

Decision to Grant, dated Jul. 23, 2020, received in European Patent Application No. 17745225.7 (7394EP), which corresponds with U.S. Appl. No. 15/655,707, 2 pages.

Patent, dated Aug. 19, 2020, received in European Patent Application No. 17745225.7 (7394EP), which corresponds with U.S. Appl. No. 15/655,707, 3 pages.

Office Action, dated Jan. 13, 2023, received in European Patent Application No. 20168044.4 (7394EP01), which corresponds with U.S. Appl. No. 15/655,707, 6 pages.

Intention to Grant, dated Nov. 10, 2023, received in European Patent Application No. 20168044.4 (7394EP01), which corresponds with U.S. Appl. No. 15/655,707, 12 pages.

Intention to Grant, dated Mar. 19, 2024, received in European Patent Application No. 20168044.4 (7394EP01), which corresponds with U.S. Appl. No. 15/655,707, 10 pages.

Office Action, dated Sep. 2, 2019, received in Japanese Patent Application No. 2019-503292 (7394JP), which corresponds with U.S. Appl. No. 15/655,707, 7 pages.

Notice of Allowance, dated Apr. 6, 2020, received in Japanese Patent Application No. 2019-503292 (7394JP), which corresponds with U.S. Appl. No. 15/655,707, 5 pages.

Patent, dated May 7, 2020, received in Japanese Patent Application No. 2019-503292 (7394JP), which corresponds with U.S. Appl. No. 15/655,707, 5 pages.

Office Action, dated Aug. 2, 2021, received in Japanese Patent Application No. 2020-060636 (7394JP01), which corresponds with U.S. Appl. No. 15/655,707, 2 pages.

Notice of Allowance, dated Oct. 14, 2022, received in Japanese Patent Application No. 2020-060636 (7394JP01), which corresponds with U.S. Appl. No. 15/655,707, 2 pages.

Patent, dated Nov. 9, 2022, received in Japanese Patent Application No. 2020-060636 (7394JP01), which corresponds with U.S. Appl. No. 15/655,707, 3 pages.

Office Action, dated Jun. 6, 2019, received in Korean Patent Application No. 2019-7004584 (7394KR), which corresponds with U.S. Appl. No. 15/655,707, 6 pages.

Notice of Allowance, dated Oct. 24, 2019, received in Korean Patent Application No. 2019-7004584 (7394KR), which corresponds with U.S. Appl. No. 15/655,707, 5 pages.

Patent, dated Jan. 7, 2020, received in Korean Patent Application No. 2019-7004584 (7394KR), which corresponds with U.S. Appl. No. 15/655,707, 4 pages.

Office Action, dated Jan. 6, 2021, received in Korean Patent Application No. 2020-7028796 (7394KR01), which corresponds with U.S. Appl. No. 15/655,707, 5 pages.

Notice of Allowance, dated Sep. 28, 2021, received in Korean Patent Application No. 2020-7028796 (7394KR01), which corresponds with U.S. Appl. No. 15/655,707, 2 pages.

Patent, dated Dec. 28, 2021, received in Korean Patent Application No. 2020-7028796 (7394KR01), which corresponds with U.S. Appl. No. 15/655,707, 6 pages.

Office Action, dated Mar. 23, 2022, received in Korean Patent Application No. 2021-7042875 (7394KR02), which corresponds with U.S. Appl. No. 15/655,707, 8 pages.

Notice of Allowance, dated Nov. 1, 2022, received in Korean Patent Application No. 2021-7042875 (7394KR02), which corresponds with U.S. Appl. No. 15/655,707, 2 pages.

Office Action, dated Jun. 18, 2018, received in U.S. Appl. No. 15/273,627 (7422), 20 pages.

Office Action, dated Jan. 17, 2019, received in U.S. Appl. No. 15/273,627 (7422), 24 pages.

Notice of Allowance, dated Nov. 21, 2019, received in U.S. Appl. No. 15/273,627 (7422), 5 pages.

Notice of Allowance, dated Jan. 27, 2020, received in U.S. Appl. No. 15/273,627 (7422), 7 pages.

Office Action, dated Feb. 8, 2019, received in U.S. Appl. No. 15/275,298 (7445), 16 pages.

Final Office Action, dated Jul. 3, 2019, received in U.S. Appl. No. 15/275,298 (7445), 16 pages.

Notice of Allowance, dated Apr. 17, 2020, received in U.S. Appl. No. 15/275,298 (7445), 5 pages.

Notice of Allowance, dated Jun. 18, 2020, received in U.S. Appl. No. 15/275,298 (7445), 5 pages.

Office Action, dated Jul. 31, 2019, received in U.S. Appl. No. 16/361,109 (7576), 13 pages.

Notice of Allowance, dated Dec. 18, 2019, received in U.S. Appl. No. 16/361,109 (7576), 5 pages.

Notice of Allowance, dated Apr. 21, 2020, received in U.S. Appl. No. 16/361,109 (7576), 5 pages.

Office Action, dated Sep. 9, 2022, received in Australian Patent Application No. 2021254570 (7576 AU), which corresponds with U.S. Appl. No. 16/361,109, 2 pages.

Notice of Allowance, dated Dec. 14, 2022, received in Australian Patent Application No. 2021254570 (7576 AU), which corresponds with U.S. Appl. No. 16/361,109, 3 pages.

Patent, dated Apr. 27, 2023, received in Australian Patent Application No. 2021254570 (7576 AU), which corresponds with U.S. Appl. No. 16/361,109, 4 pages.

Office Action, dated Jan. 9, 2024, received in Chinese Patent Application No. 202110181990.3 (7576CN01), which corresponds with U.S. Appl. No. 16/361,109, 5 pages.

Office Action, dated May 31, 2024, received in Chinese Patent Application No. 202110181990.3 (7576CN01), which corresponds with U.S. Appl. No. 16/361,109, 1 page.

Patent, dated May 20, 2020, received in Korean Patent Application No. 2115714 (7576KR), which corresponds with U.S. Appl. No. 16/361,109, 4 pages.

Notice of Allowance, dated Aug. 4, 2020, received in Korean Patent Application No. 2020-7014317 (7576KR01), which corresponds with U.S. Appl. No. 16/361,109, 5 pages.

Patent, dated Oct. 8, 2020, received in Korean Patent Application No. 2020-7014317 (7576KR01), which corresponds with U.S. Appl. No. 16/361,109, 7 pages.

Office Action, dated Aug. 13, 2019, received in U.S. Appl. No. 16/361,122 (7577), 13 pages.

Notice of Allowance, dated Dec. 5, 2019, received in U.S. Appl. No. 16/361,122 (7577), 9 pages.

(56)

References Cited**OTHER PUBLICATIONS**

Office Action, dated Jun. 14, 2024, received in Australian Patent Application No. 2023251543 (7577AU), which corresponds with U.S. Appl. No. 16/361,122, 2 pages.

Office Action, dated Oct. 1, 2021, received in U.S. Appl. No. 16/862,427 (7578), 14 pages.

Notice of Allowance, dated Apr. 26, 2022, received in U.S. Appl. No. 16/862,427 (7578), 9 pages.

Office Action, dated Jan. 18, 2024, received in Chinese Patent Application No. 202110181406.4 (7577CN), which corresponds with U.S. Appl. No. 16/361,122, 5 pages.

Notice of Allowance, dated Oct. 6, 2023, received in Japanese Patent Application No. 2022-121971 (7578JP), 1 page.

Patent, dated Oct. 31, 2023, received in Japanese Patent Application No. 2022-121971 (7578JP), 3 pages.

Office Action, dated Sep. 8, 2023, received in Korean Patent Application No. 2023-7001774 (7577KR), which corresponds with U.S. Appl. No. 16/361,122, 8 pages.

Notice of Allowance, dated Oct. 1, 2021, received in U.S. Appl. No. 16/835,096 (7687), 9 pages.

Notice of Allowance, dated Dec. 27, 2021, received in U.S. Appl. No. 17/086,248 (7751), 9 pages.

Office Action, dated Oct. 31, 2022, received in U.S. Appl. No. 17/705,186 (7866), 14 pages.

Final Office Action, dated Jan. 31, 2023, received in U.S. Appl. No. 17/705,186 (7866), 15 pages.

Office Action, dated Aug. 9, 2023, received in U.S. Appl. No. 17/705,186 (7866), 14 pages.

Final Office Action, dated Apr. 1, 2024, received in U.S. Appl. No. 17/705,186 (7866), 17 pages.

Office Action, dated Feb. 17, 2023, received in U.S. Appl. No. 17/863,189 (7886), 6 pages.

Notice of Allowance, dated Jun. 28, 2023, received in U.S. Appl. No. 17/863,189 (7886), 5 pages.

Notice of Allowance, dated Jan. 4, 2024, received in U.S. Appl. No. 17/863,189 (7886), 5 pages.

Notice of Allowance, dated Oct. 25, 2023, received in U.S. Appl. No. 17/863,189 (7886AU), 3 pages.

Certificate of Grant, dated Feb. 22, 2024, received in U.S. Appl. No. 17/863,189 (7886AU), 4 pages.

Extended European Search Report, dated Jul. 9, 2021, received in European Patent Application No. 211634867.0 (7205CN02), which corresponds with U.S. Appl. No. 15/113,779, 26 pages.

International Search Report and Written Opinion, dated Apr. 8, 2015, received in International Patent Application No. PCT/US2015/012694, which corresponds with U.S. Appl. No. 15/113,779, 14 pages.

International Preliminary Report on Patentability, dated Jul. 26, 2016, received in International Patent Application No. PCT/US2015/012694, which corresponds with U.S. Appl. No. 15/113,779, 9 pages.

Extended European Search Report, dated Dec. 7, 2016, received in European Patent Application No. 1618866.1 (7295EP), which corresponds with U.S. Appl. No. 15/113,779, 10 pages.

Extended European Search Report, dated Jul. 24, 2020, received in European Patent Application No. 20168044.4 (7394EP01), which corresponds with U.S. Appl. No. 15/655,707, 9 pages.

International Search Report and Written Opinion, dated Jan. 2, 2018, received in International Patent Application No. PCT/US2017/041959 (7394WO), which corresponds with U.S. Appl. No. 15/655,707, 15 pages.

International Preliminary Report on Patentability, dated Jan. 29, 2019, received in International Application No. PCT/US2017/041959 (7394WO), which corresponds with U.S. Appl. No. 15/655,707, 9 pages.

Patent, dated Jan. 21, 2025, received in Chinese Patent Application No. 202110182280.2 (7394CN01), which corresponds with U.S. Appl. No. 15/655,707, 5 pages.

Grant Certificate, dated Mar. 13, 2025, received in Australian Patent Application No. 2023251543 (7577AU), which corresponds with U.S. Appl. No. 16/361,122, 3 pages.

* cited by examiner

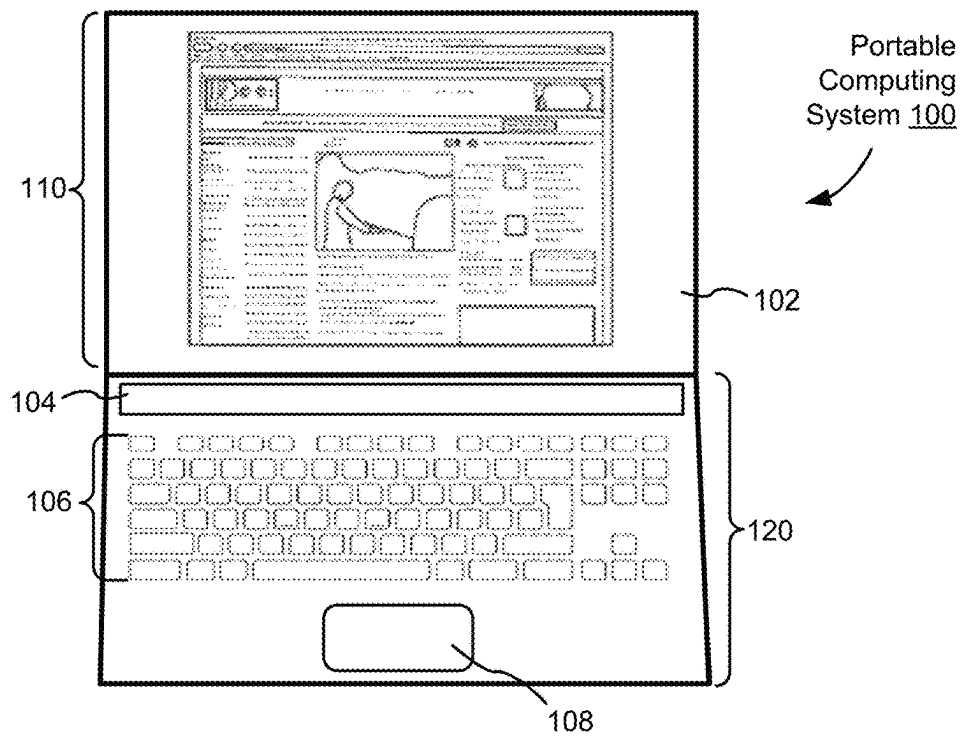


Figure 1A

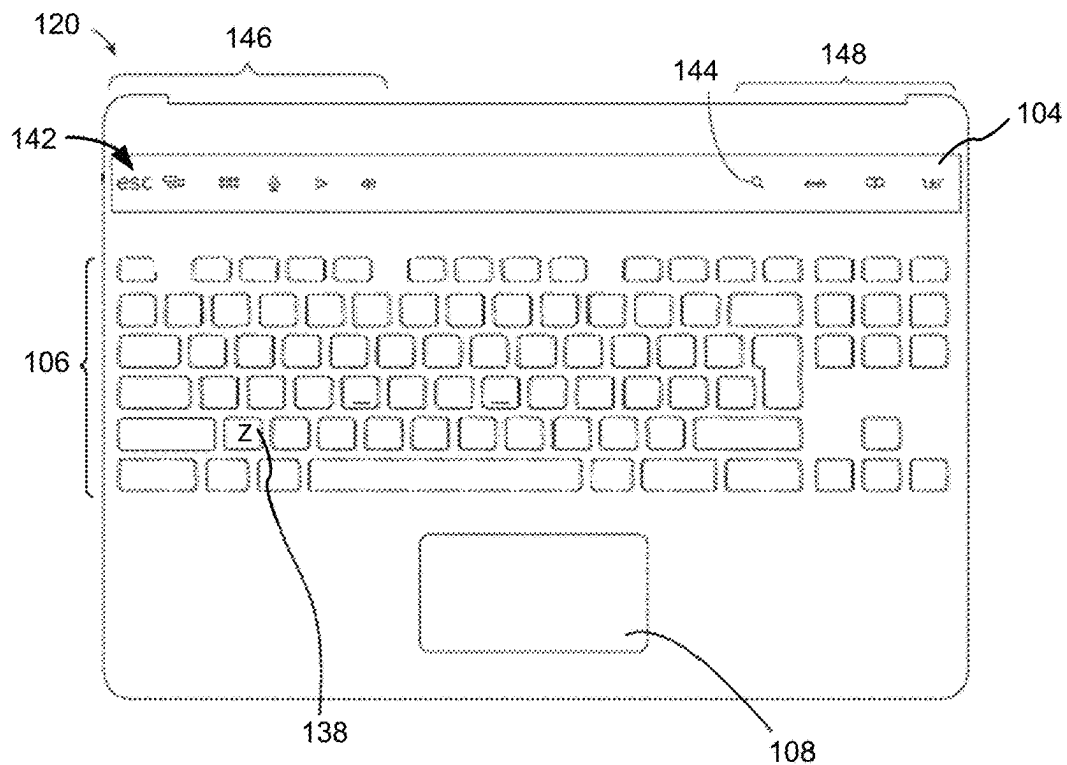


Figure 1B

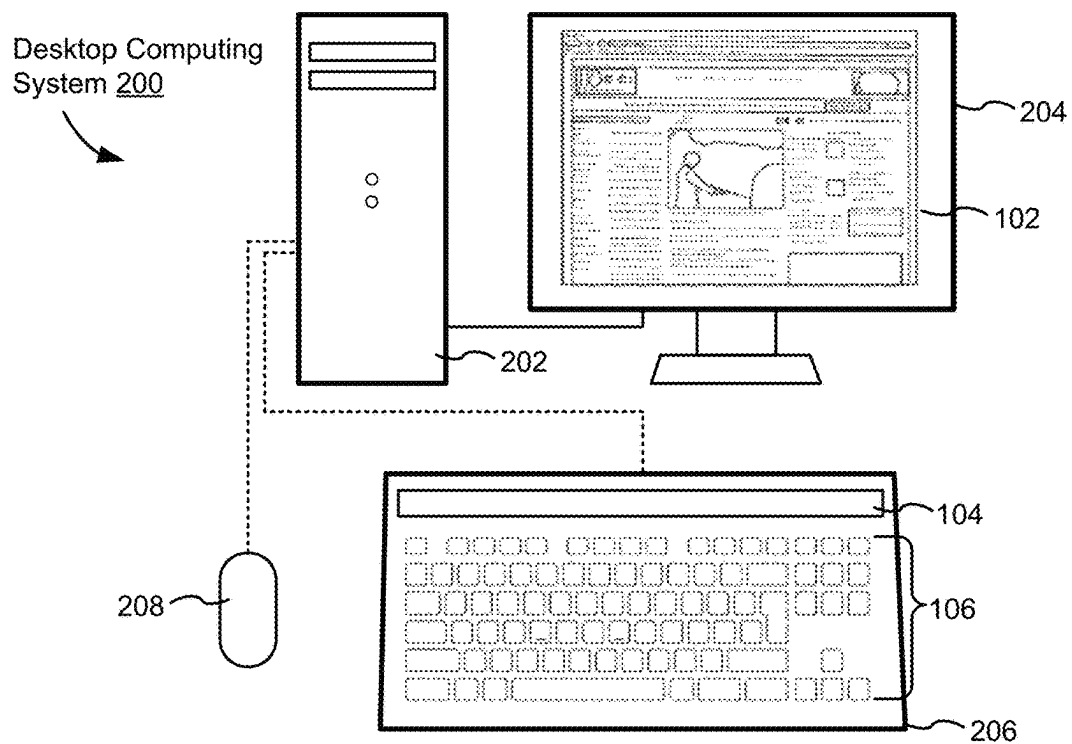


Figure 2A

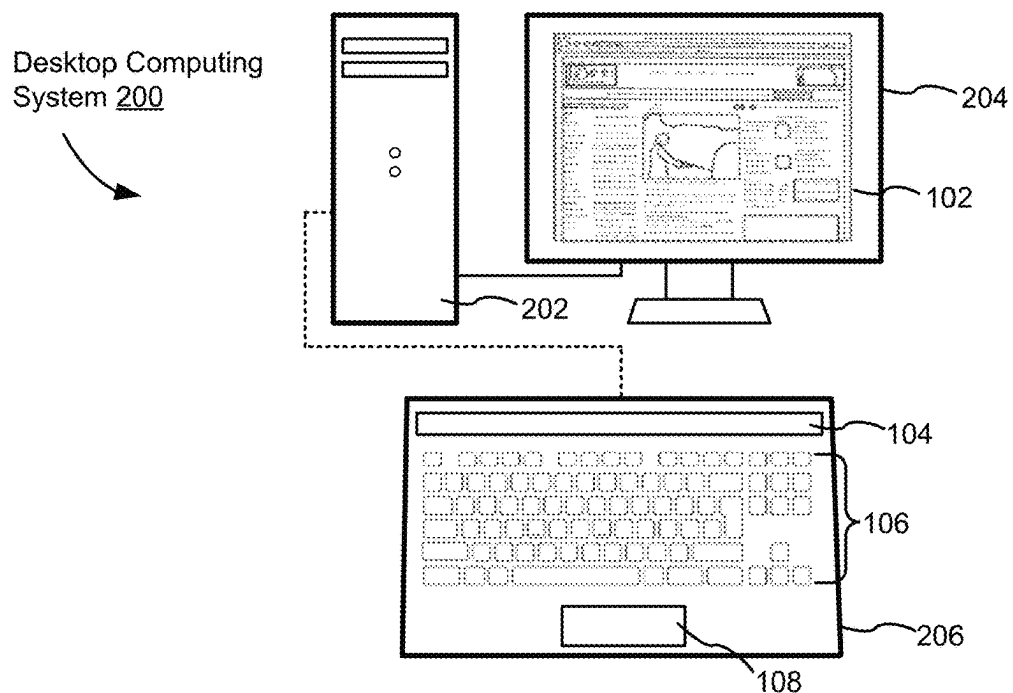


Figure 2B

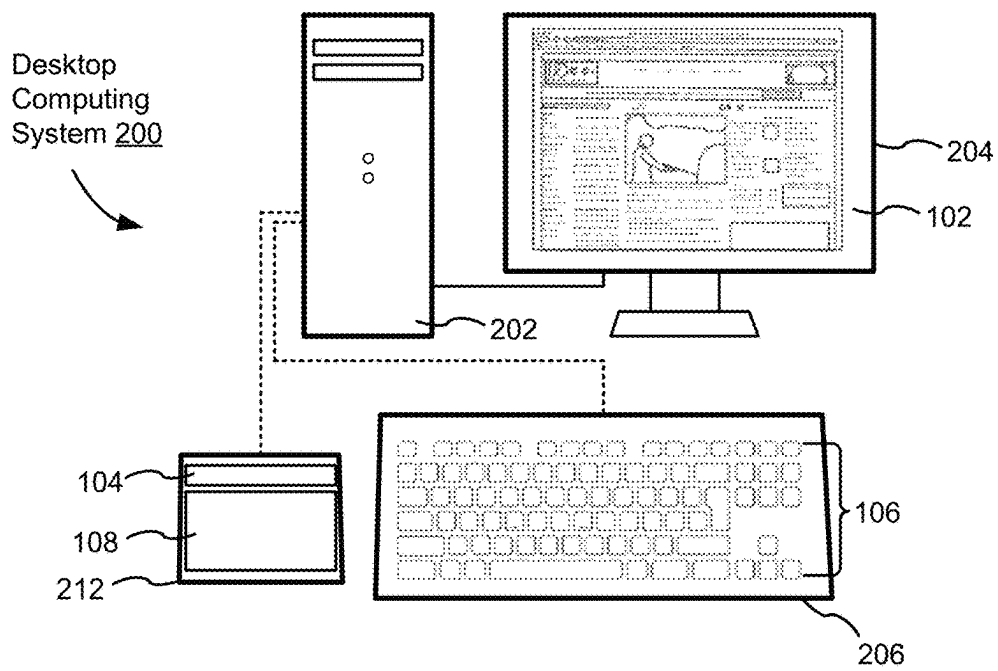


Figure 2C

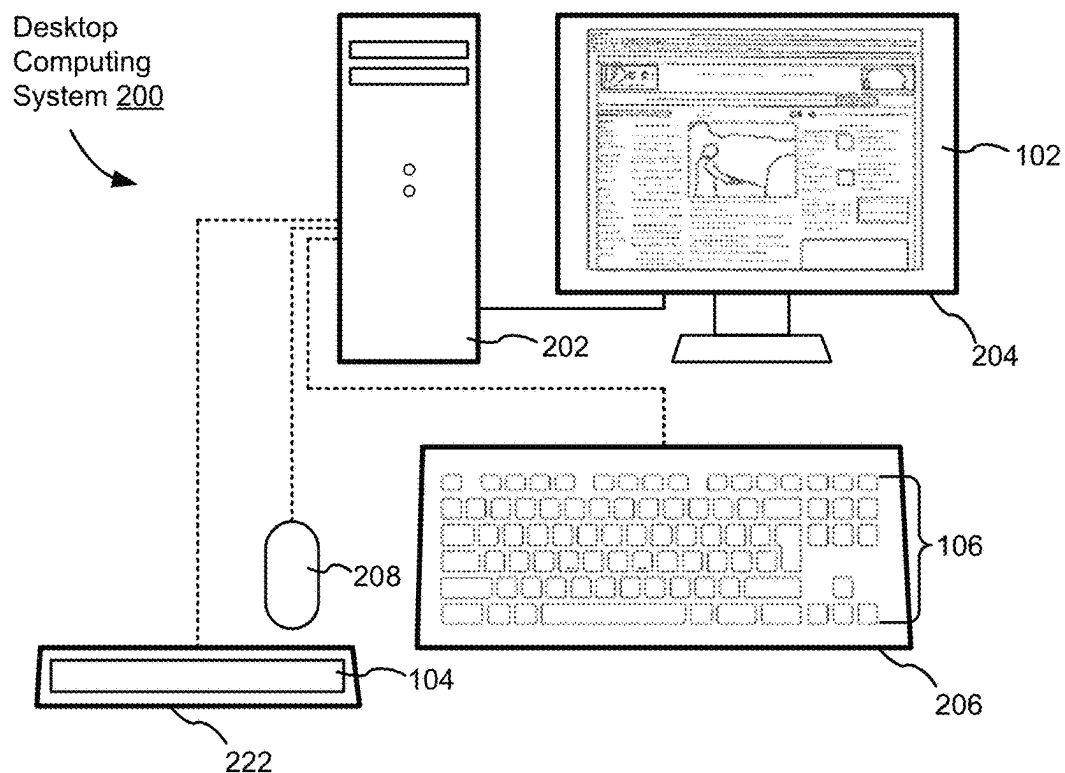


Figure 2D

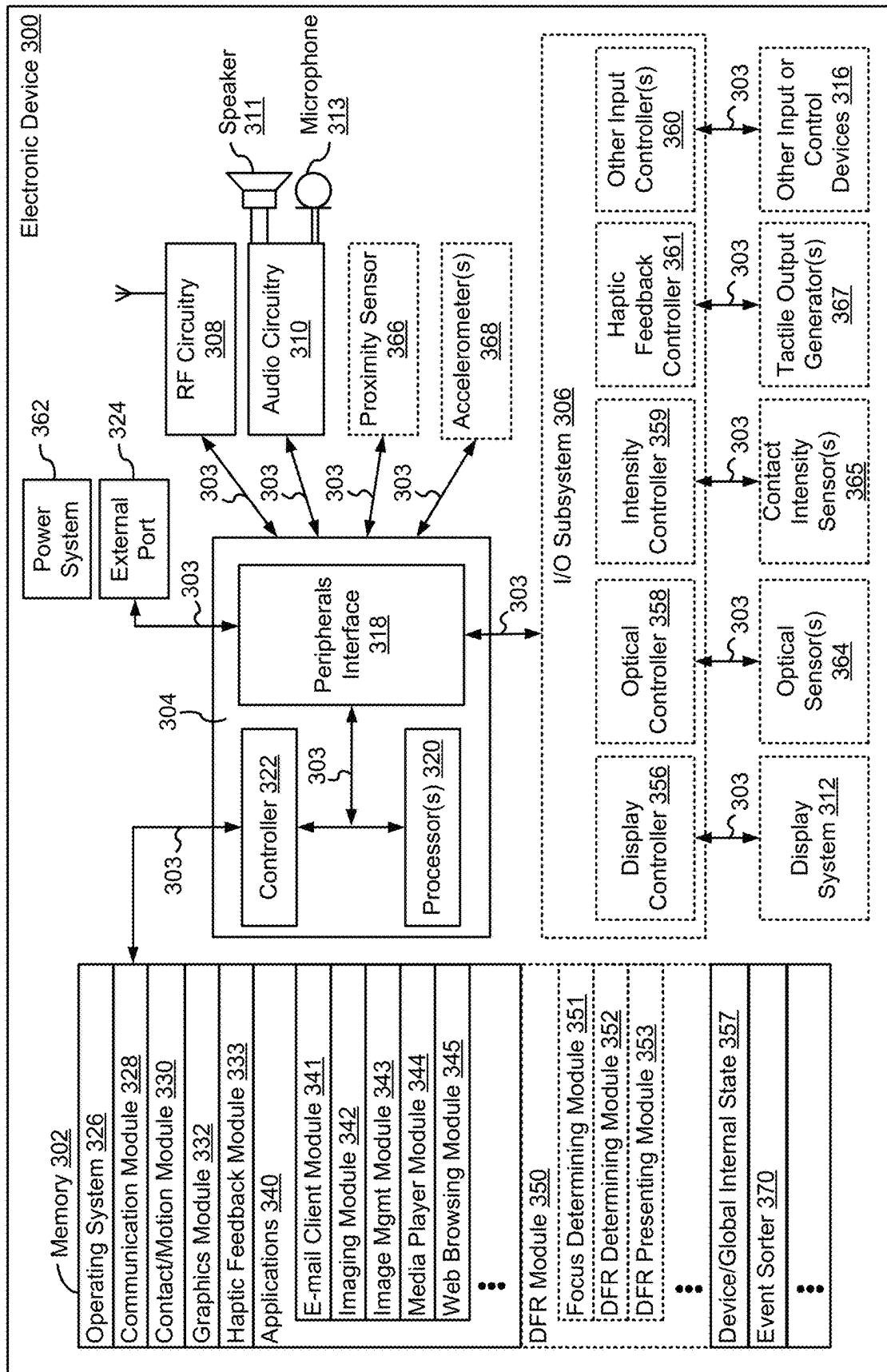


Figure 3A

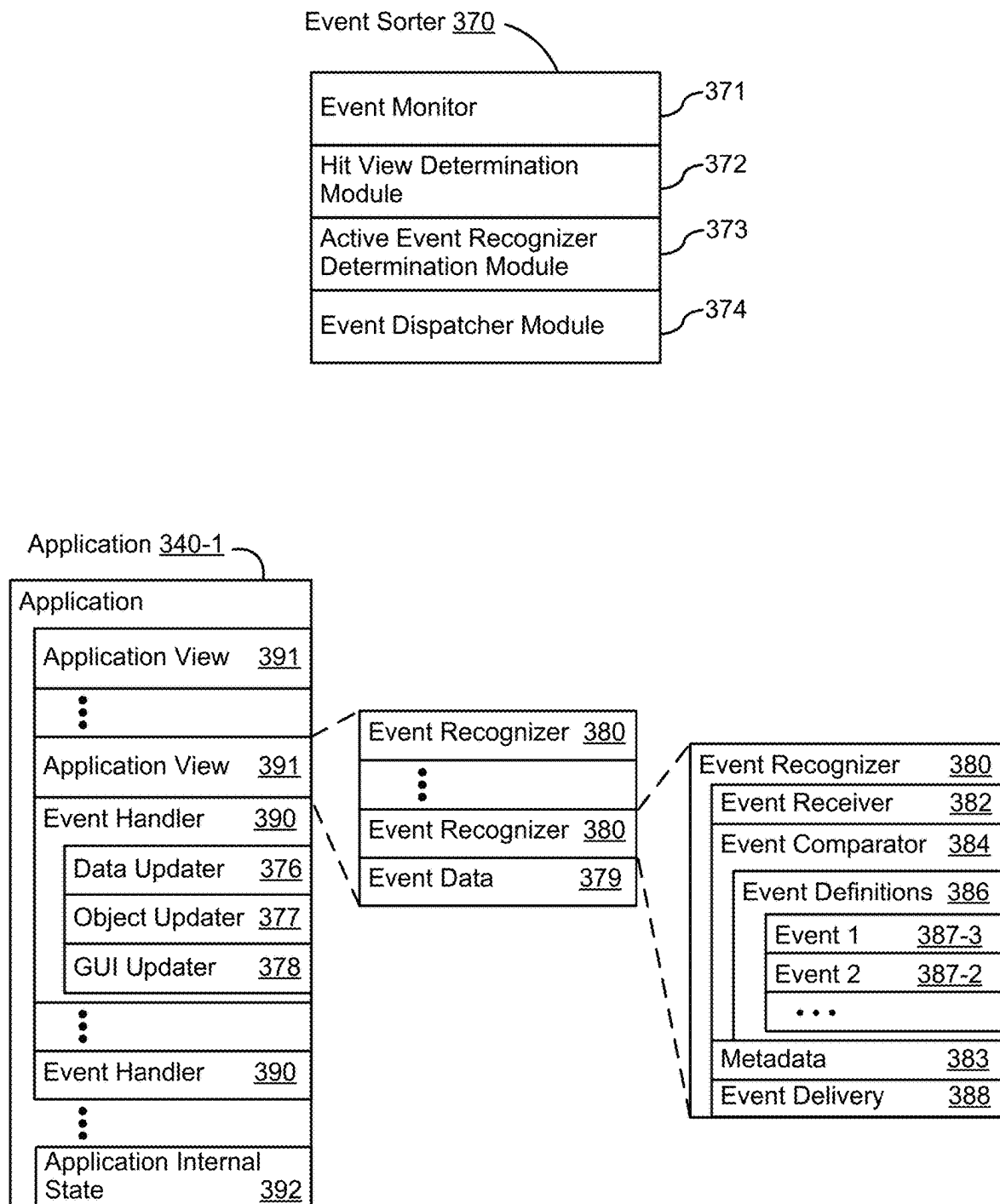


Figure 3B

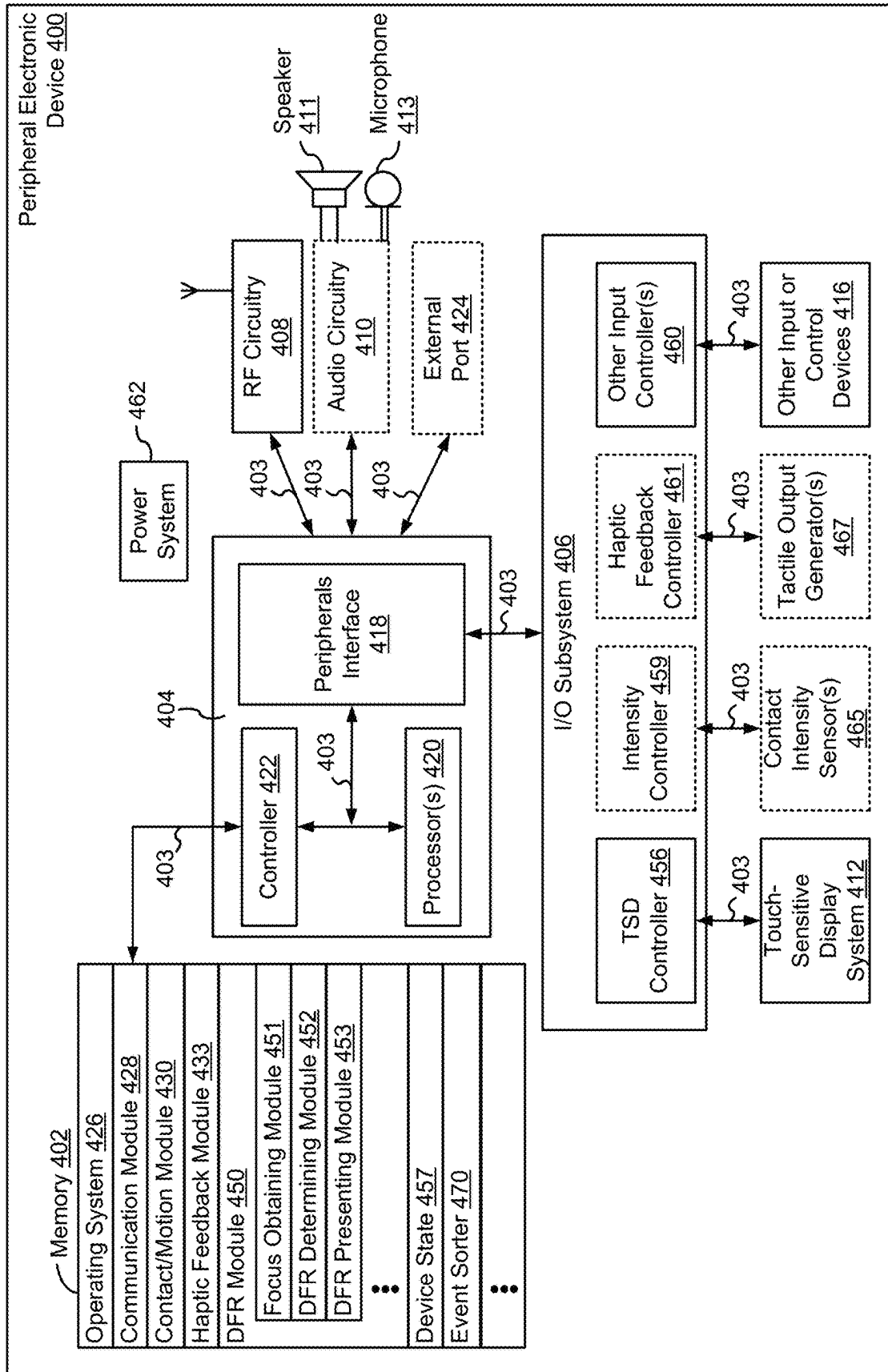


Figure 4

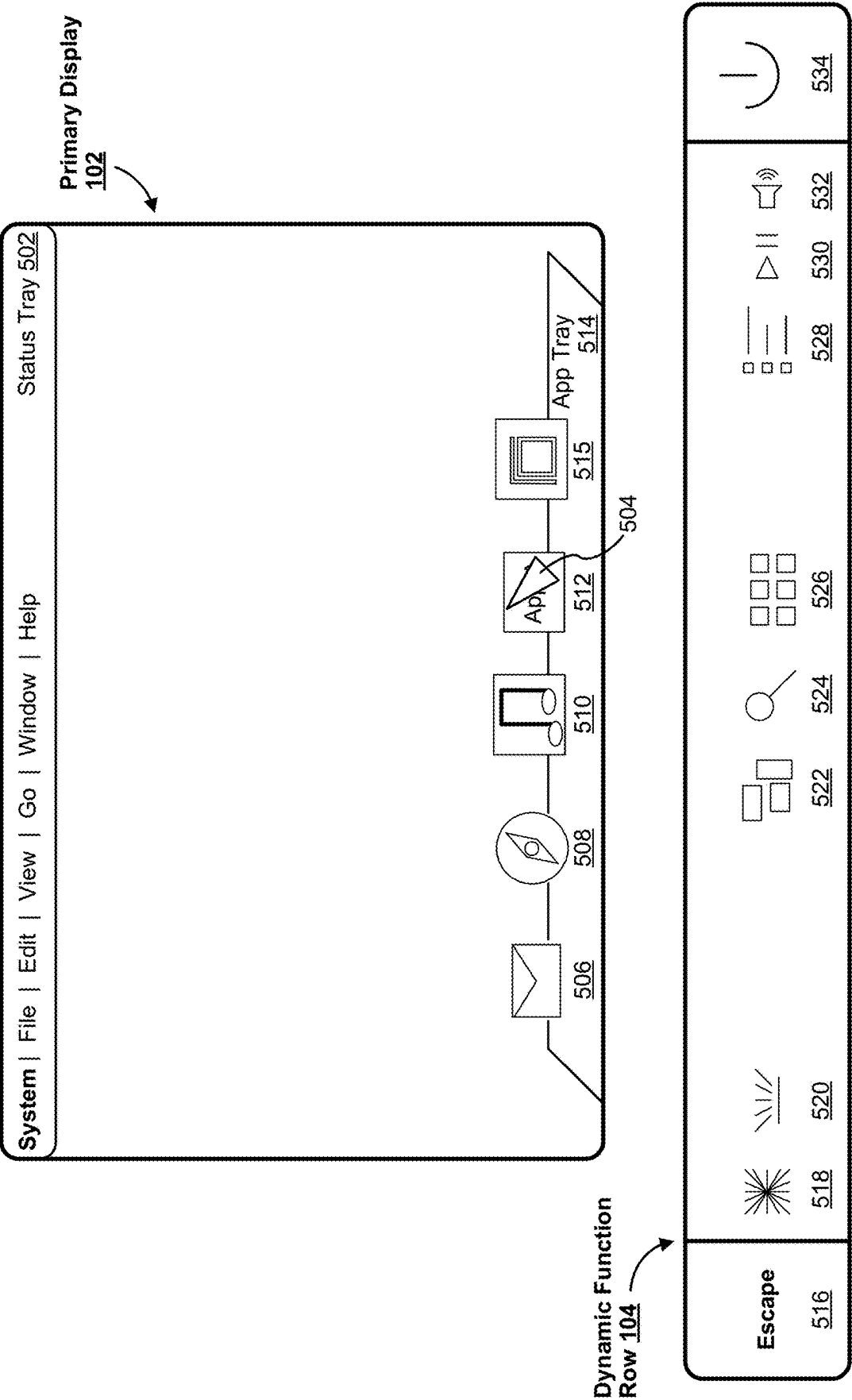


Figure 5A

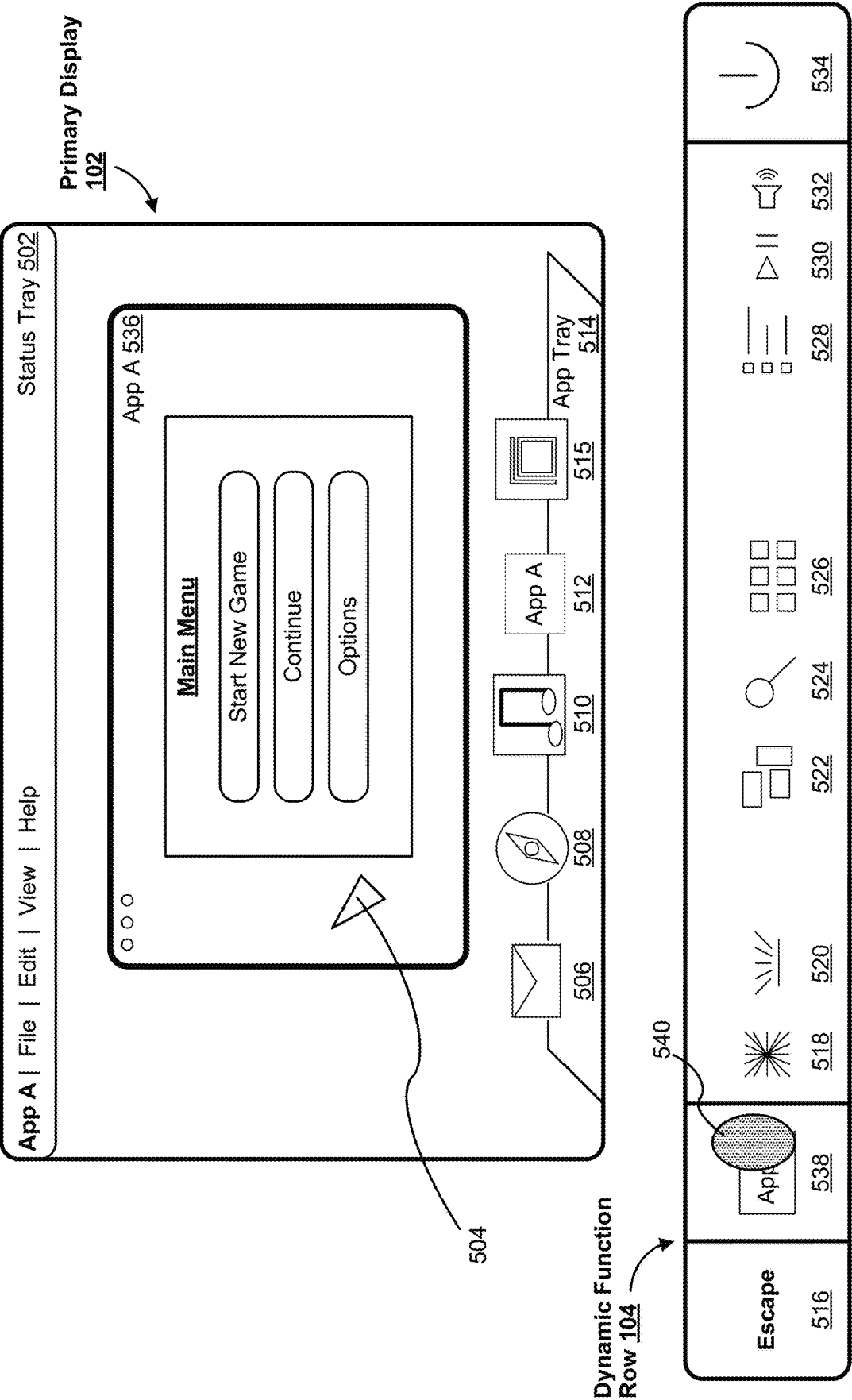


Figure 5B

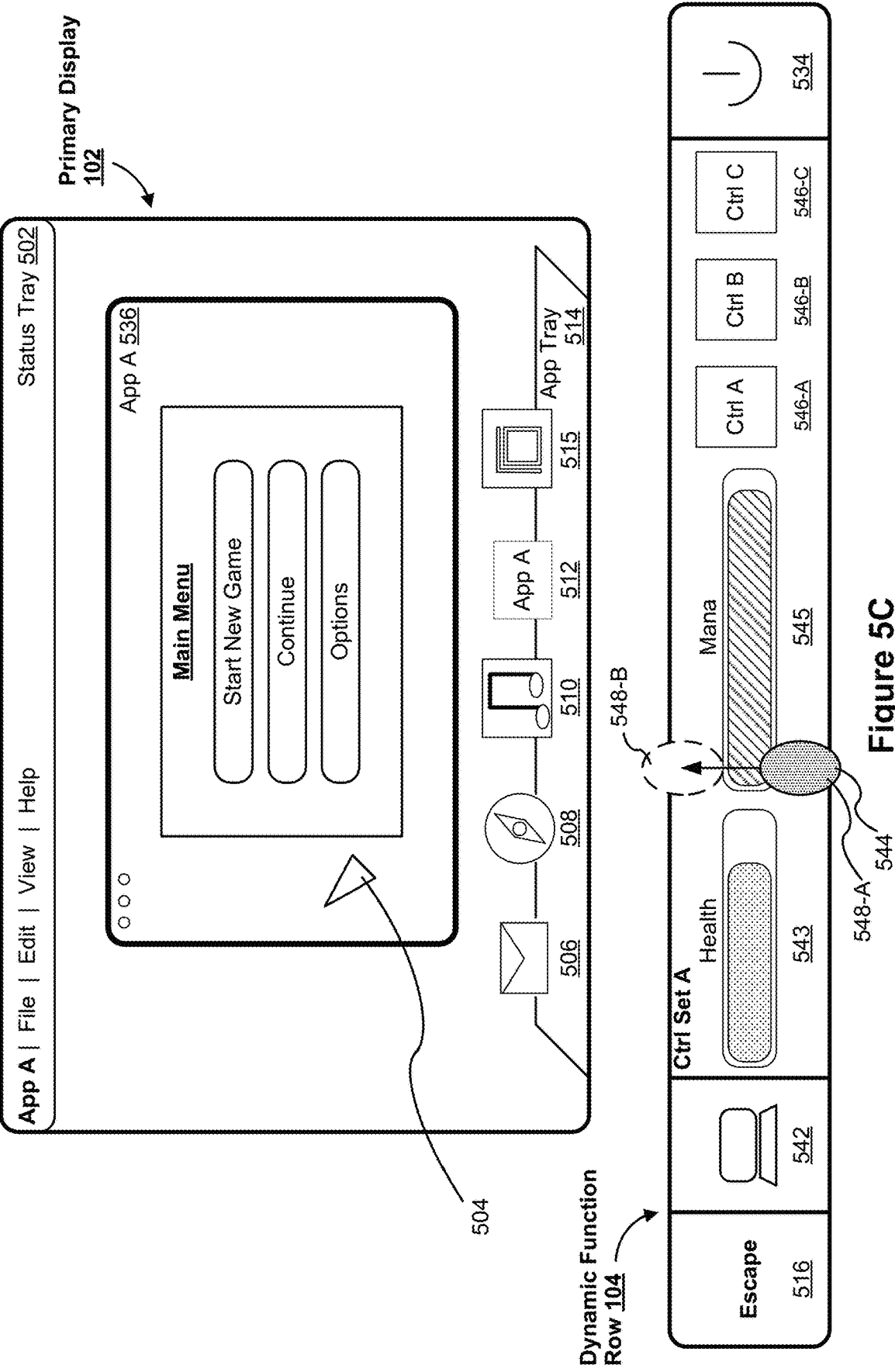


Figure 5C

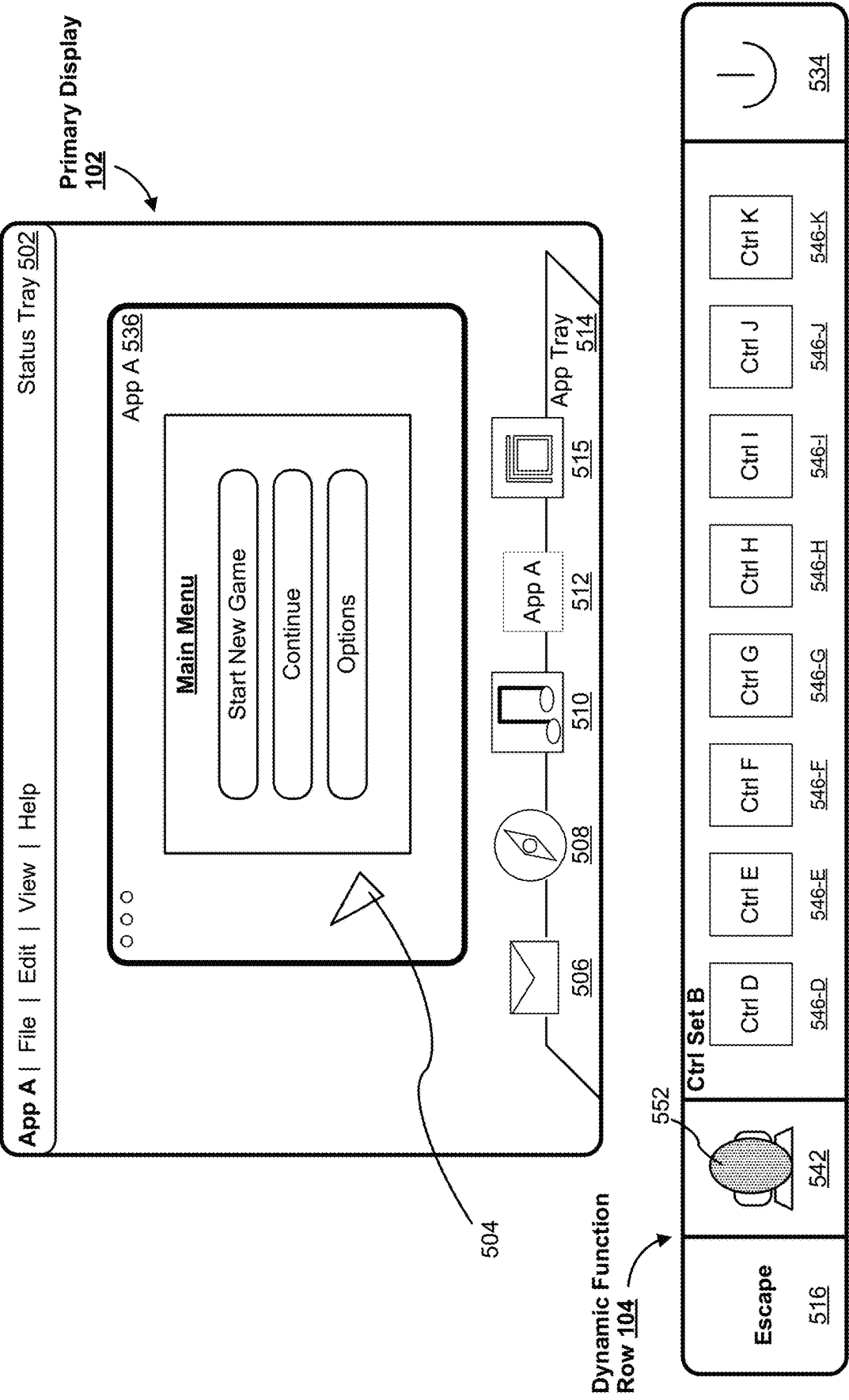


Figure 5D

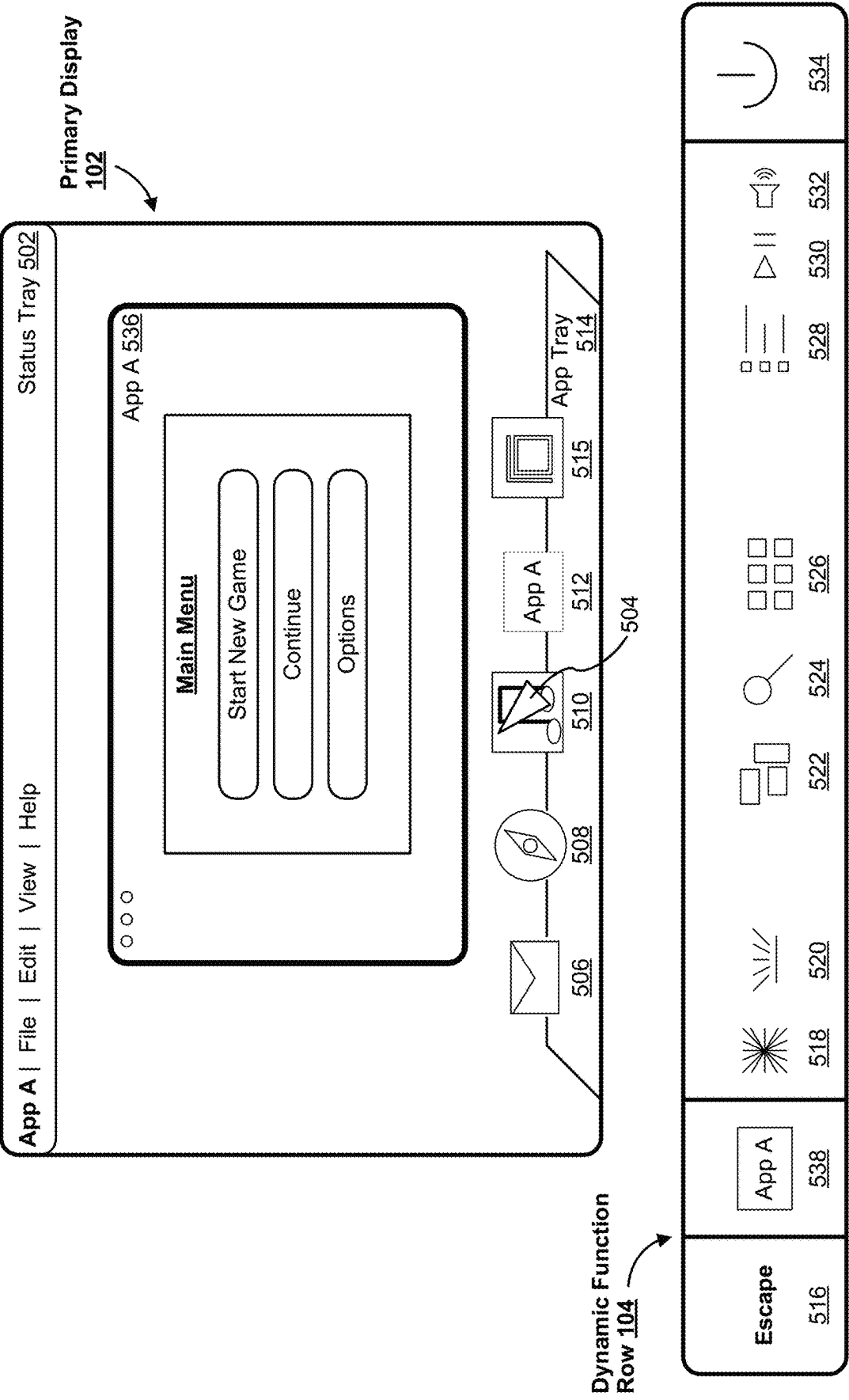


Figure 5E

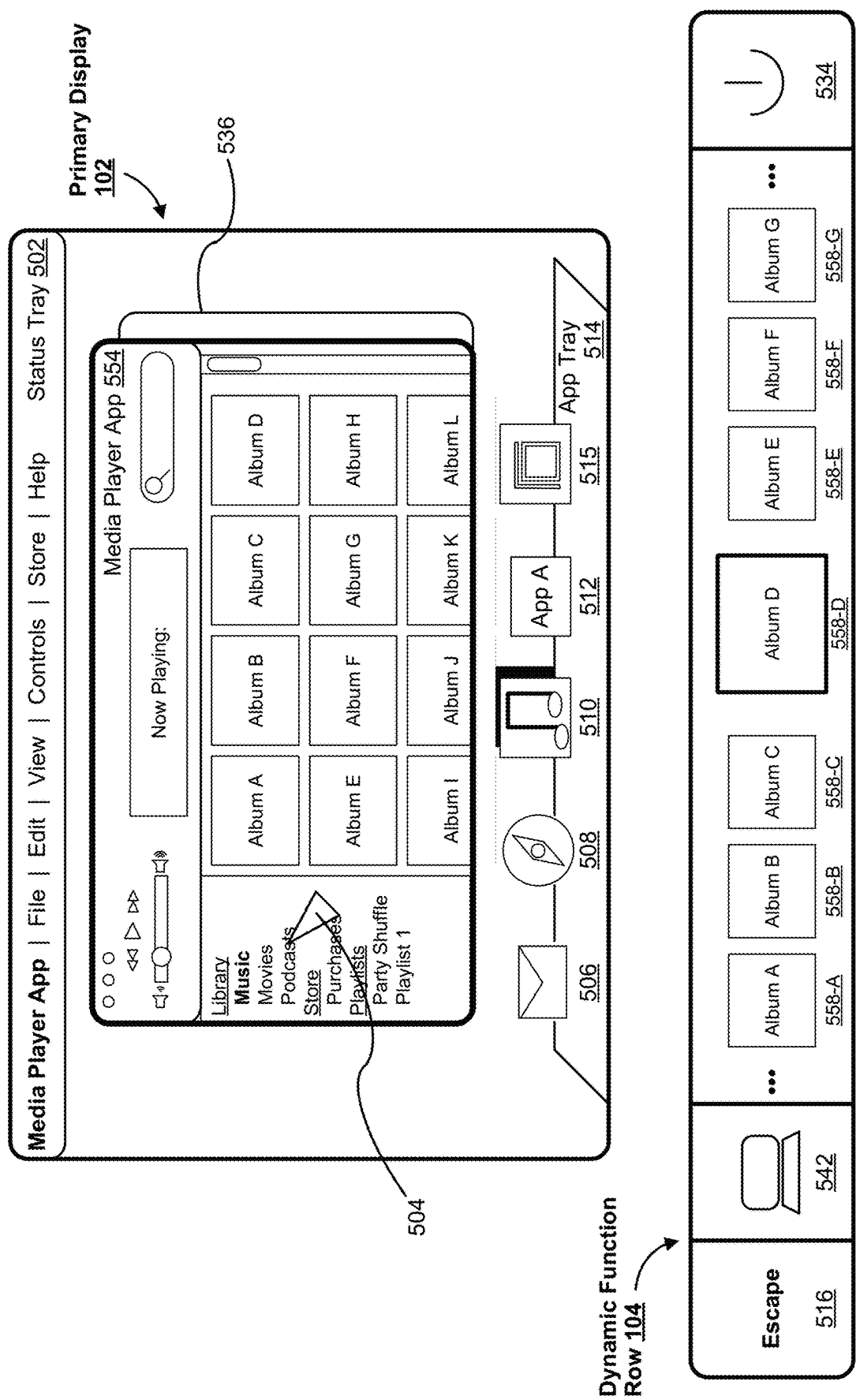


Figure 5F

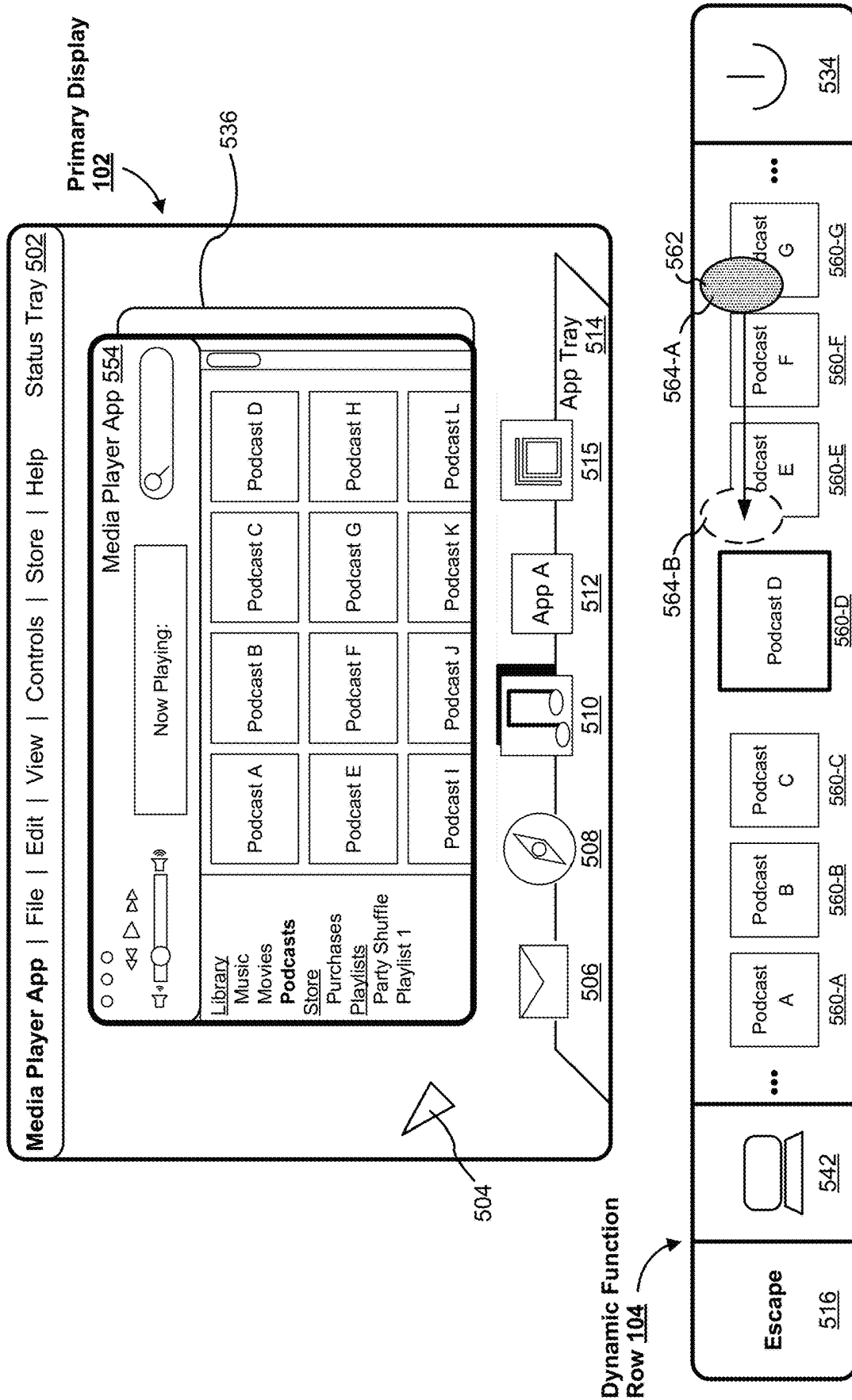


Figure 5G

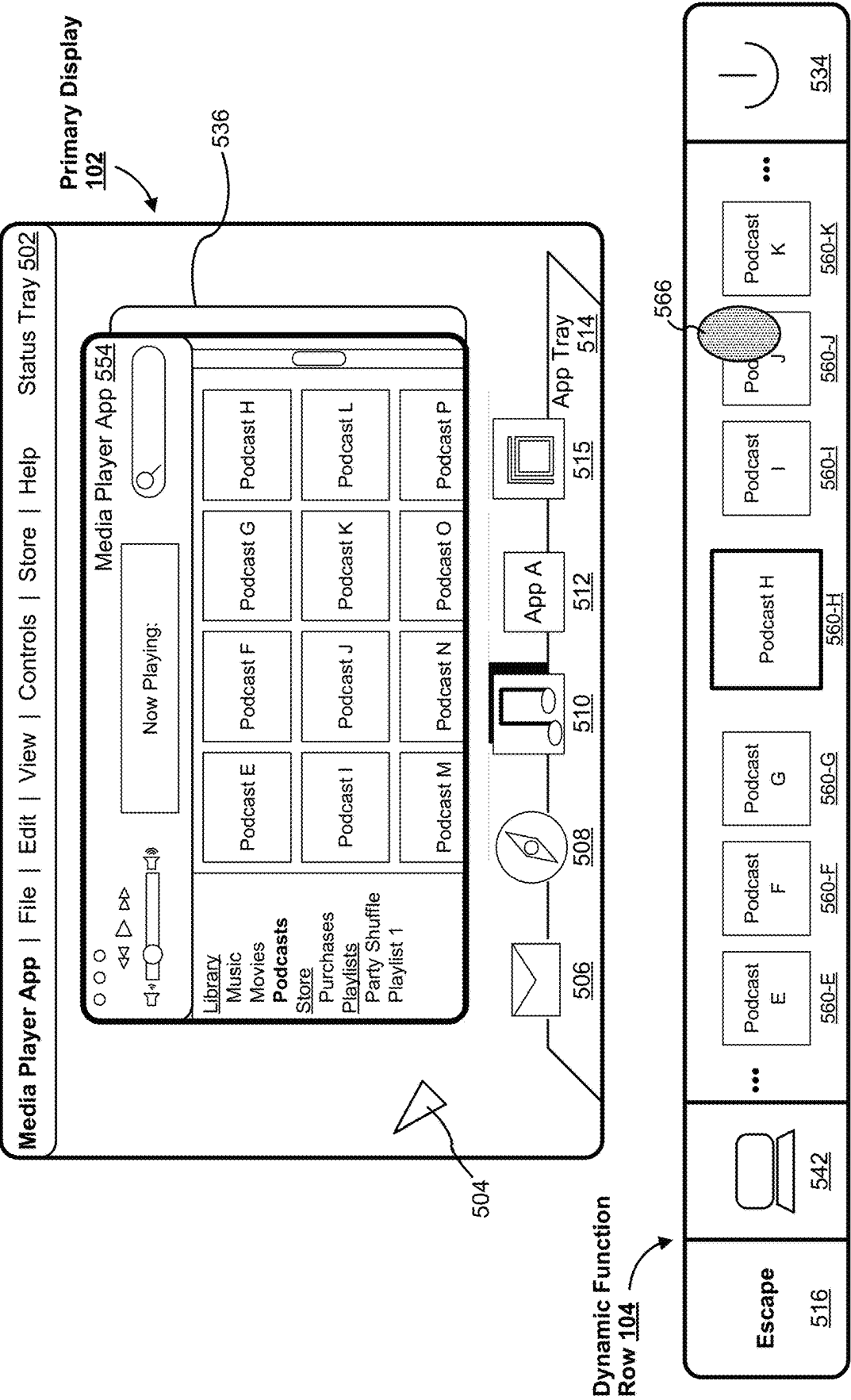


Figure 5H

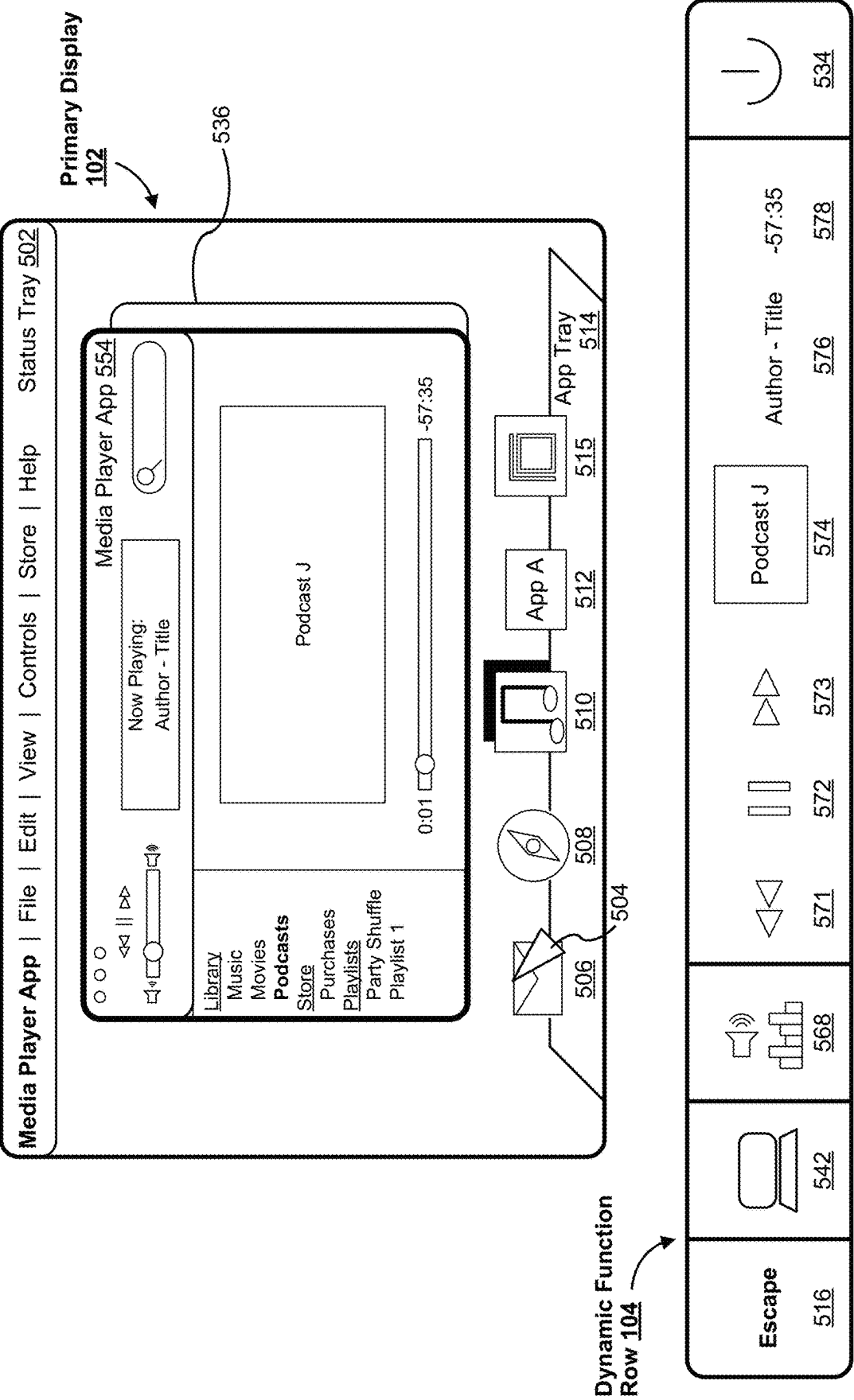


Figure 5I

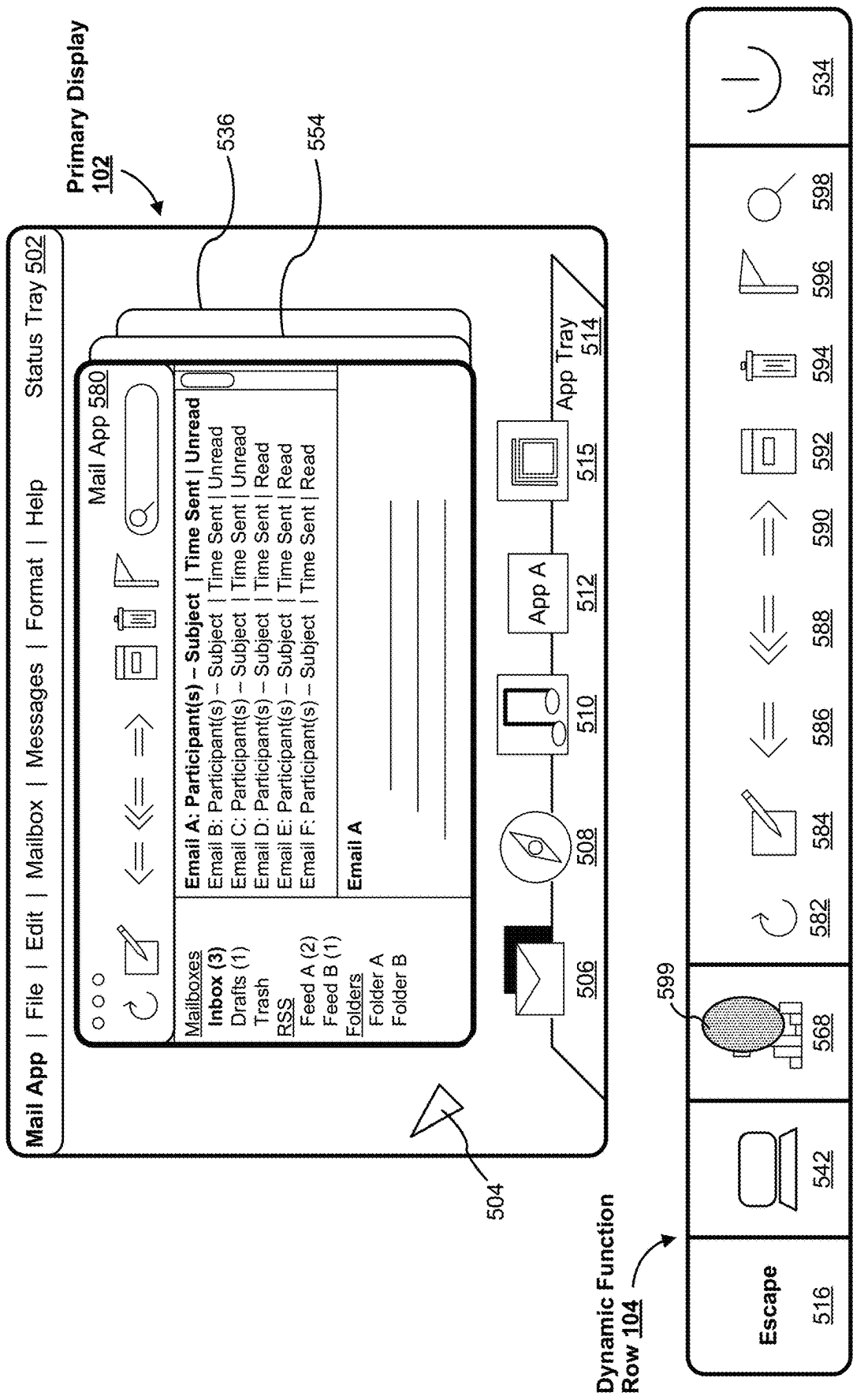


Figure 5J

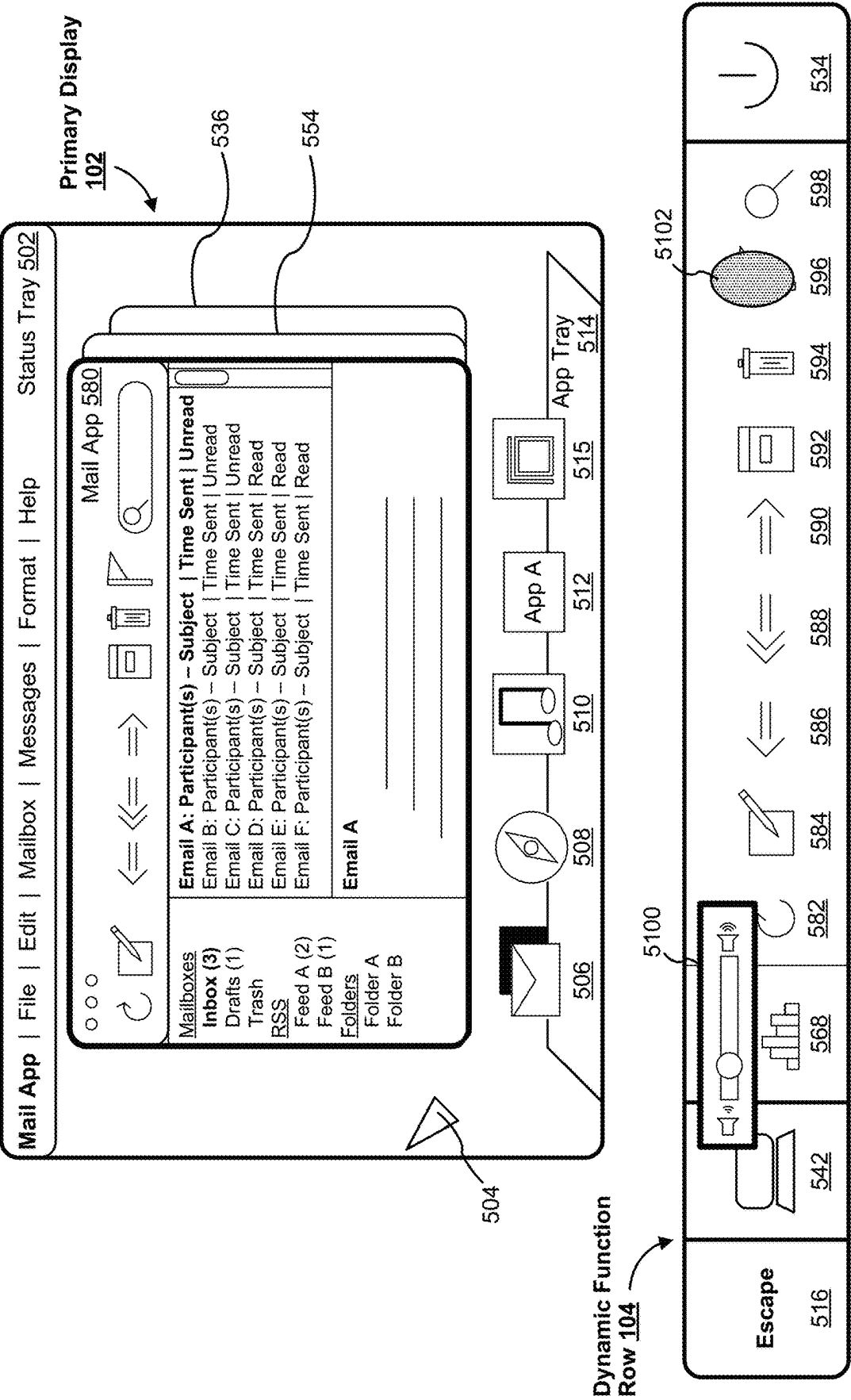


Figure 5K

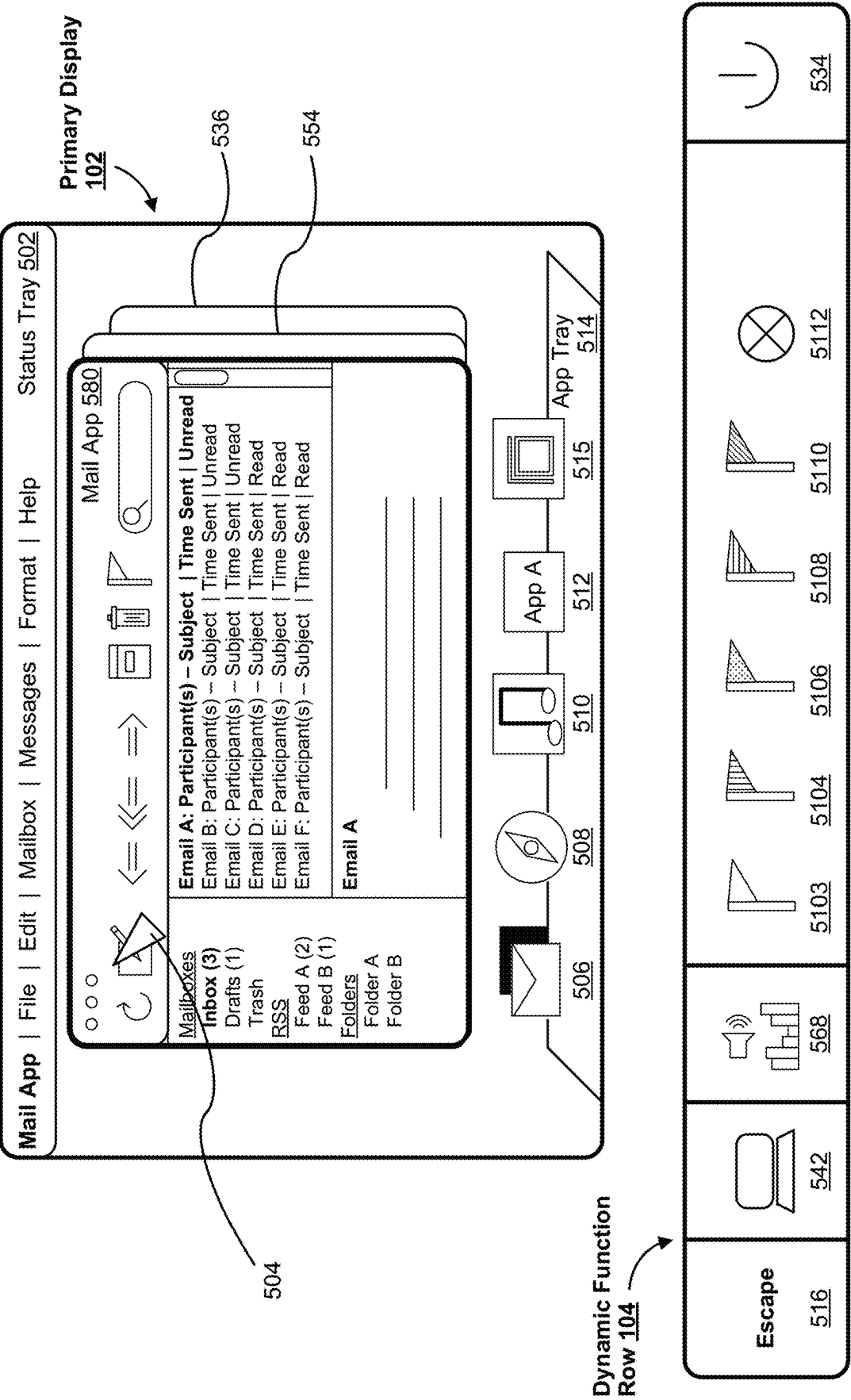


Figure 5L

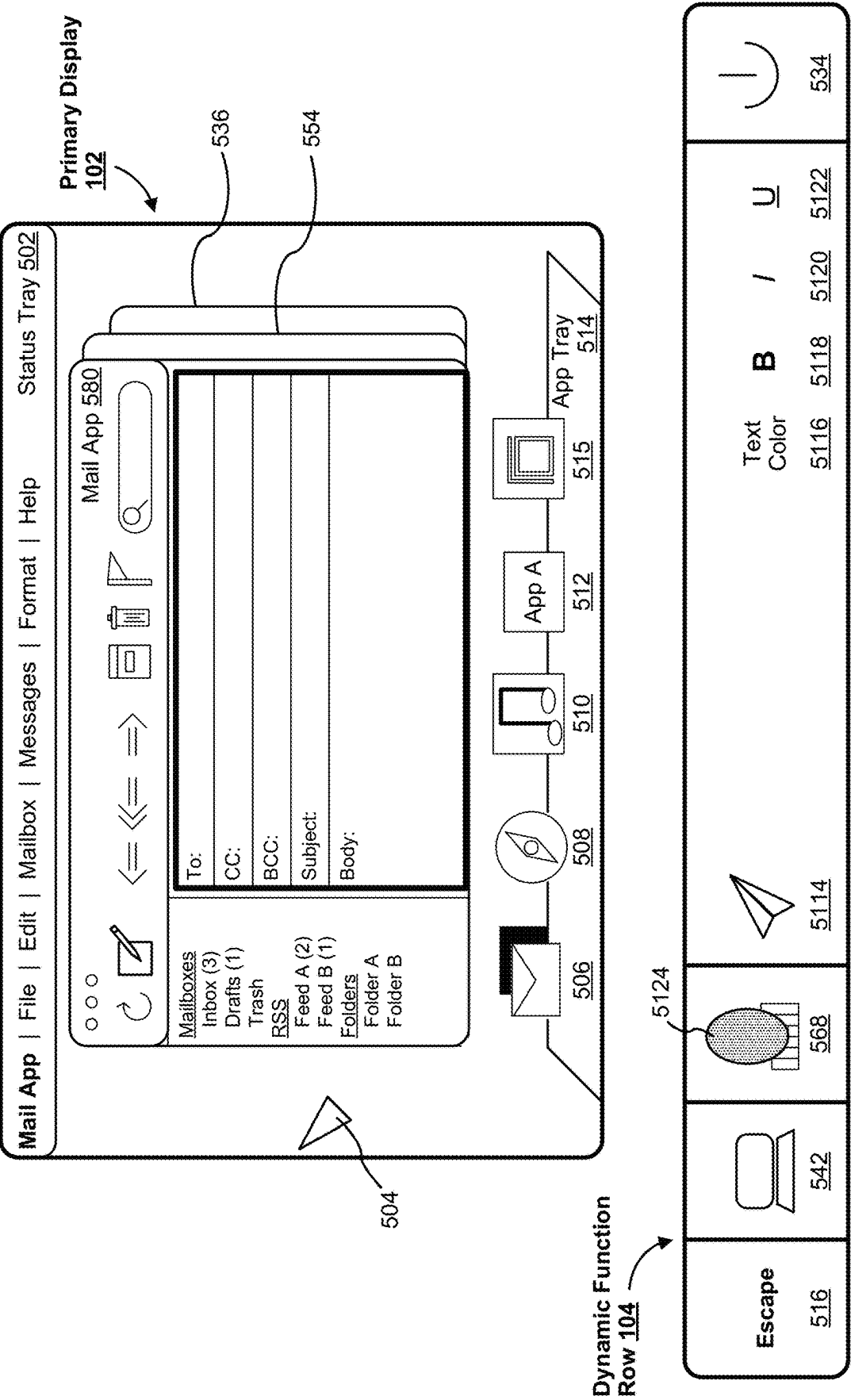
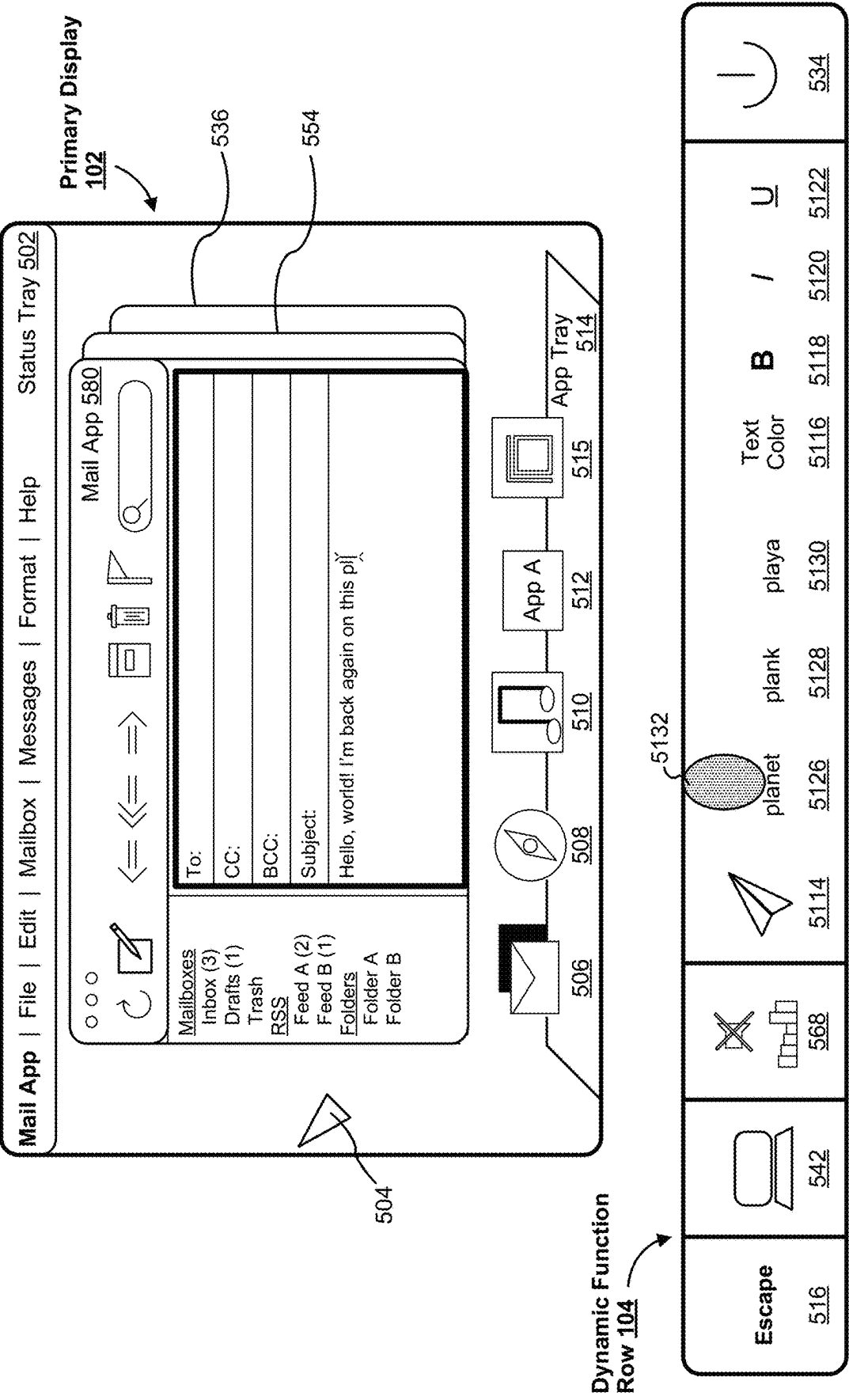


Figure 5M



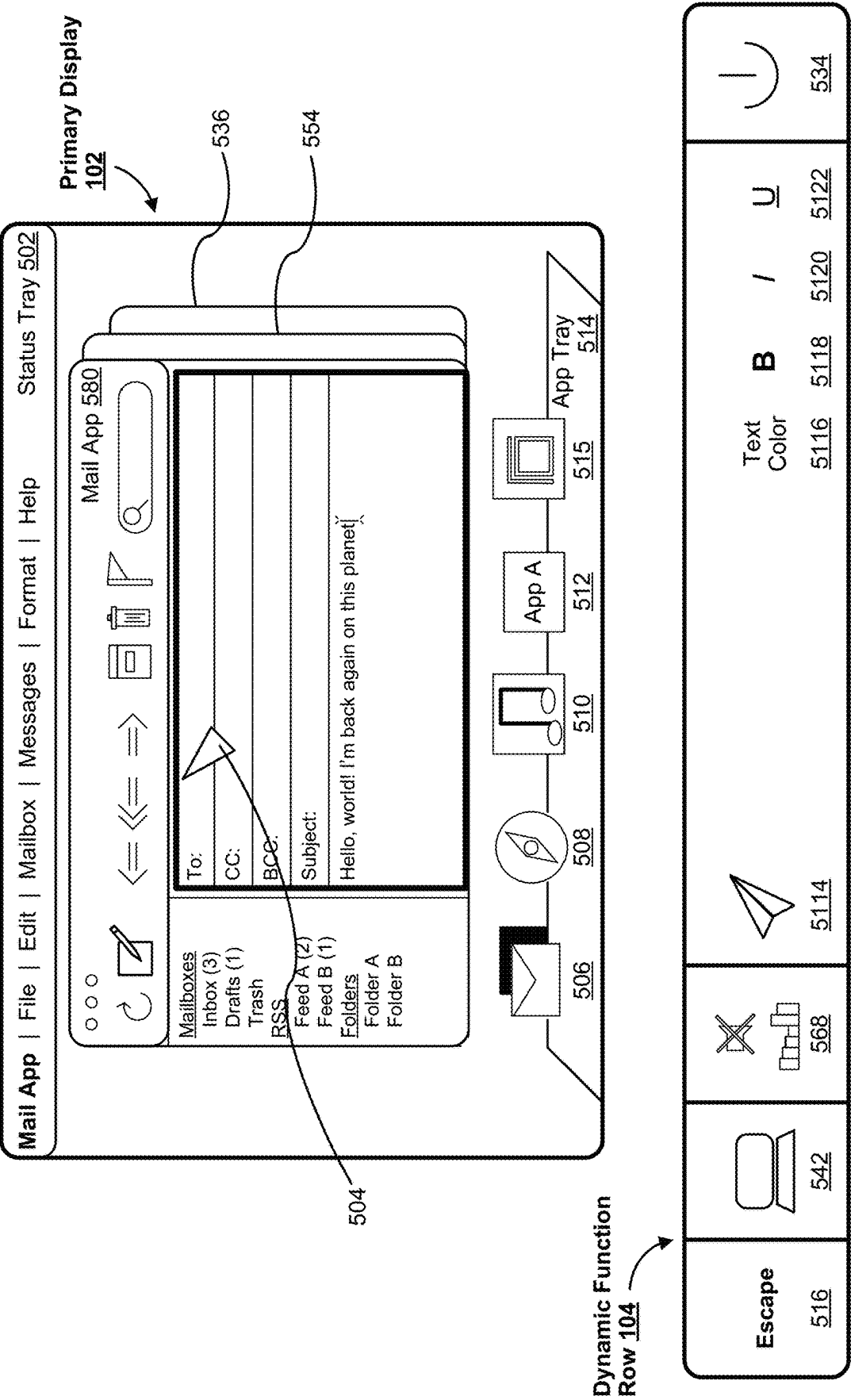
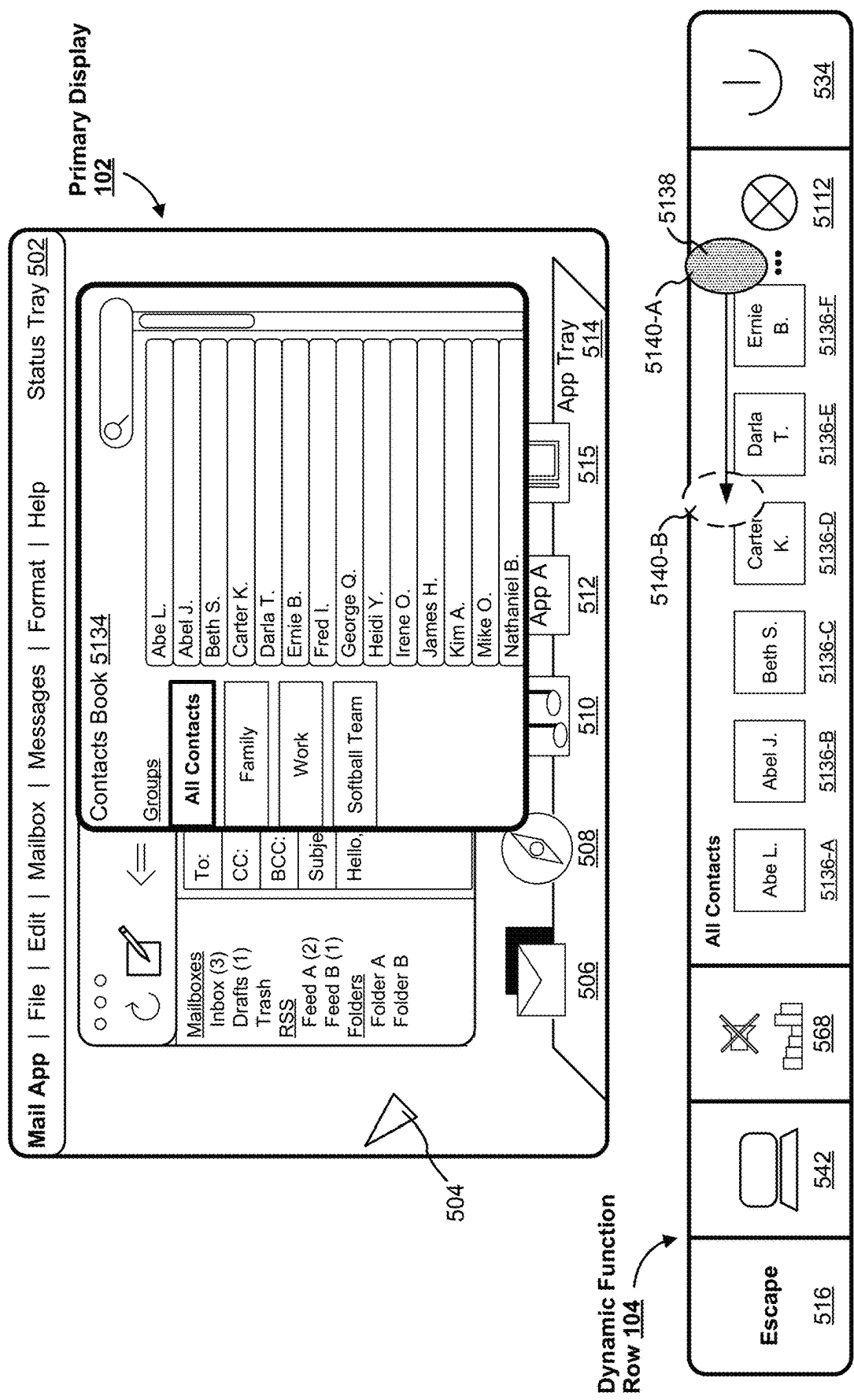
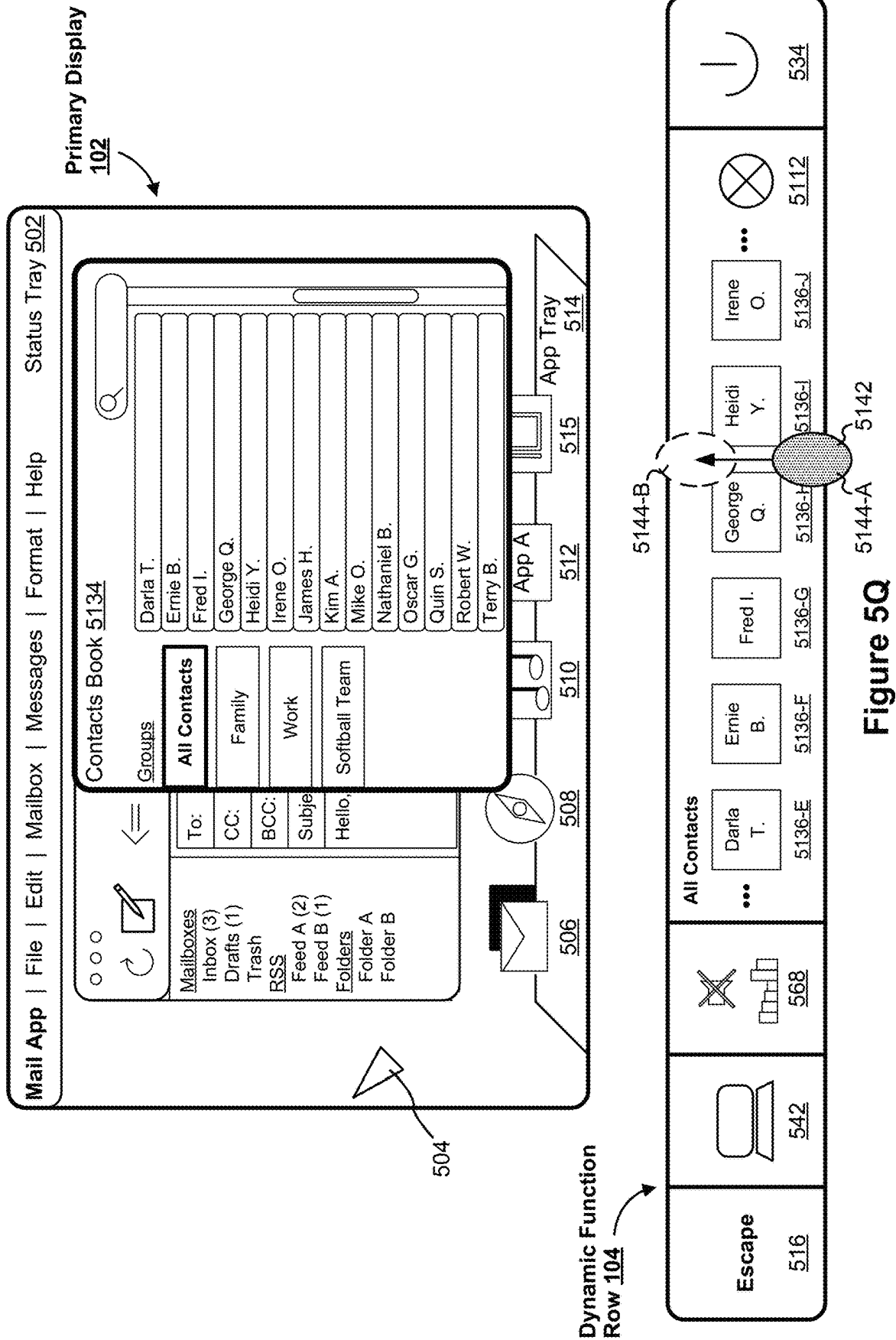
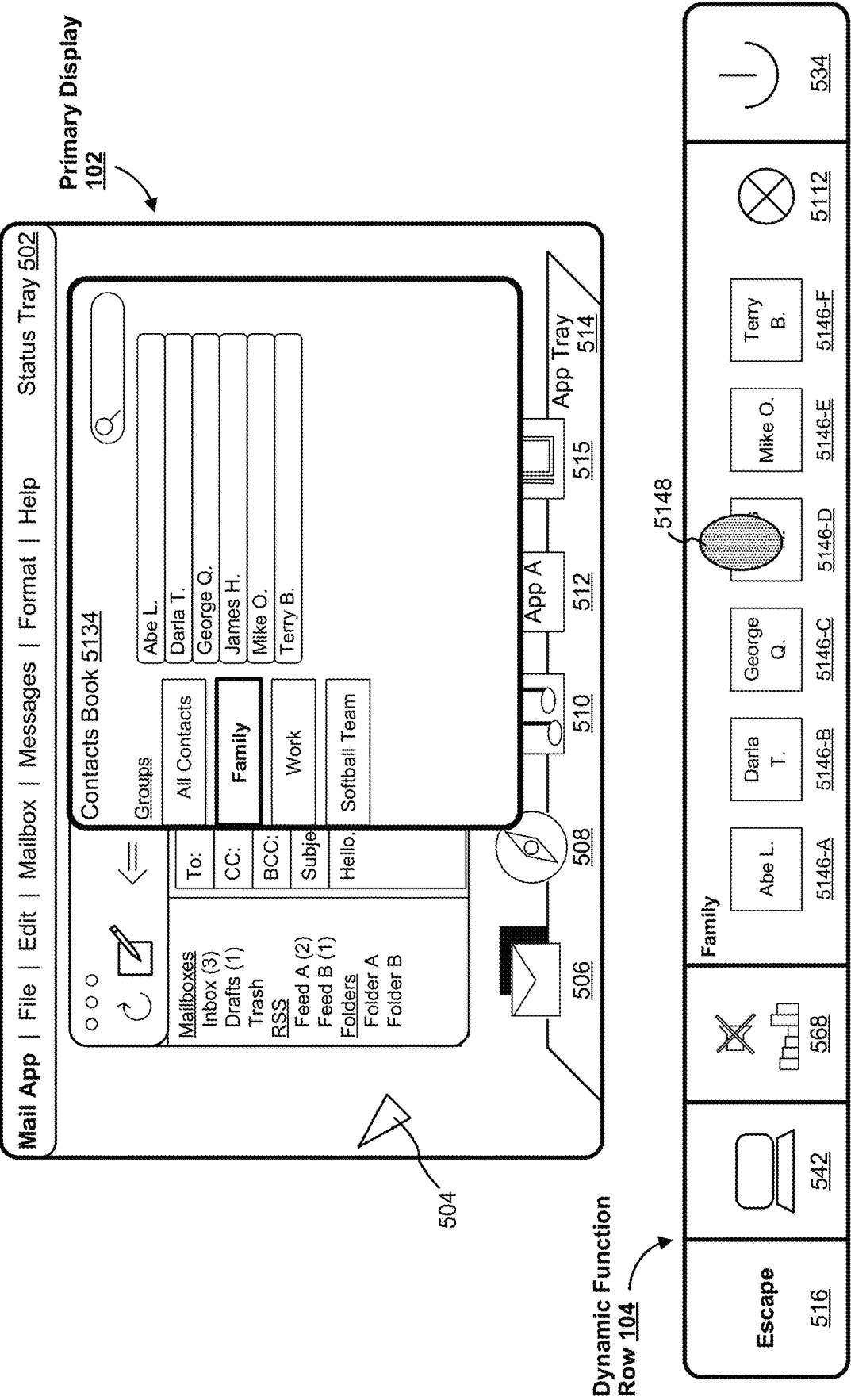


Figure 50







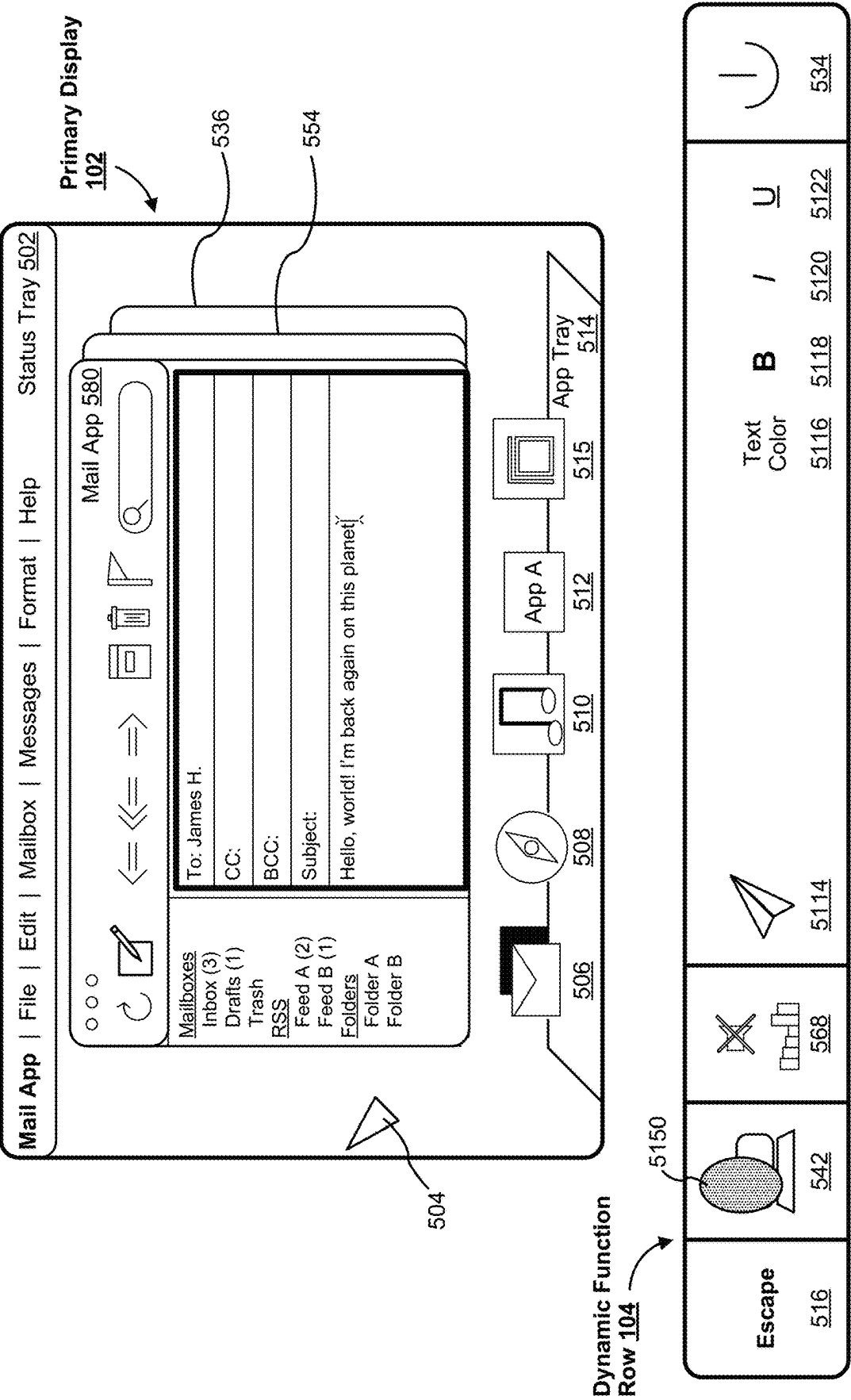


Figure 5S

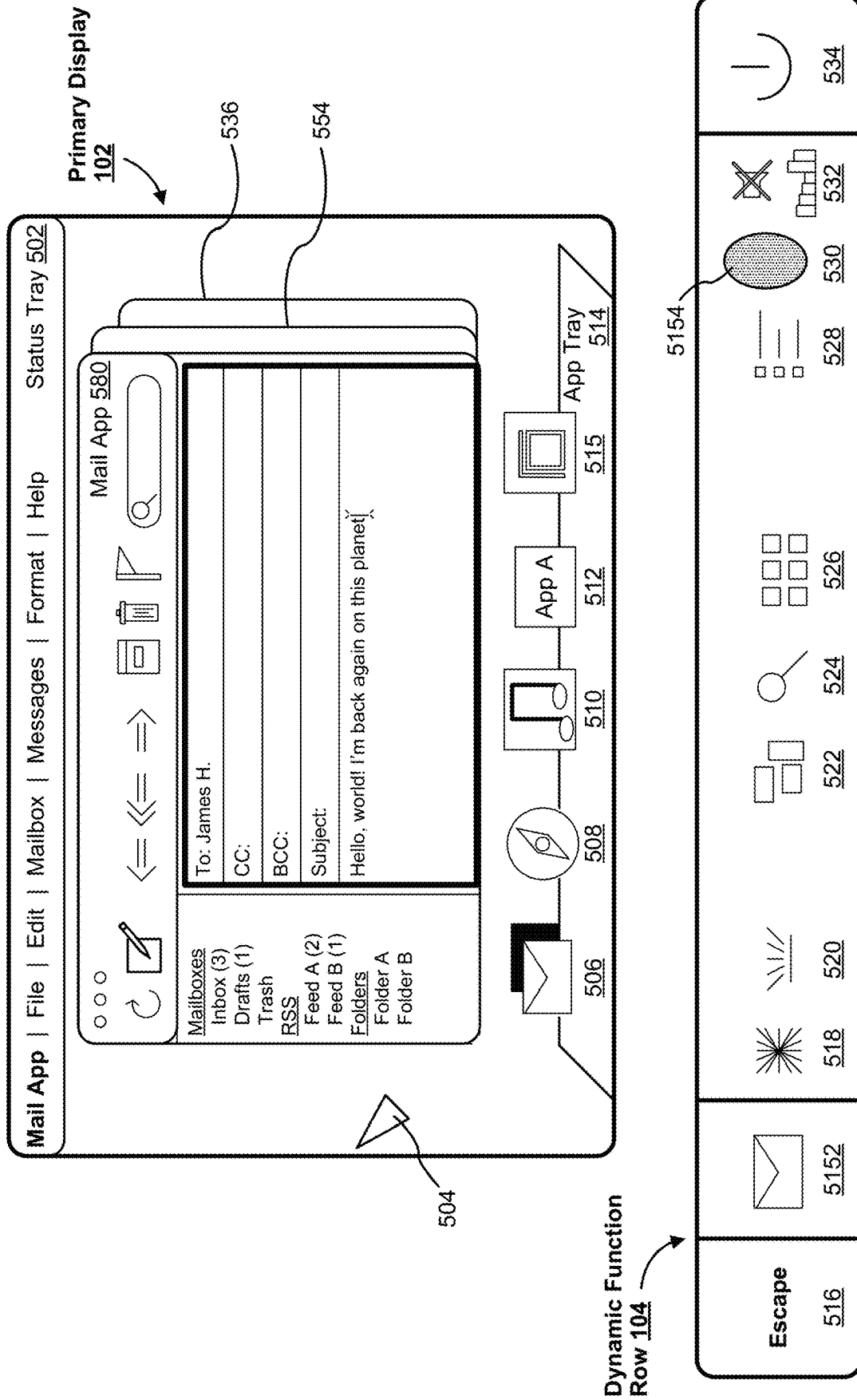


Figure 5T

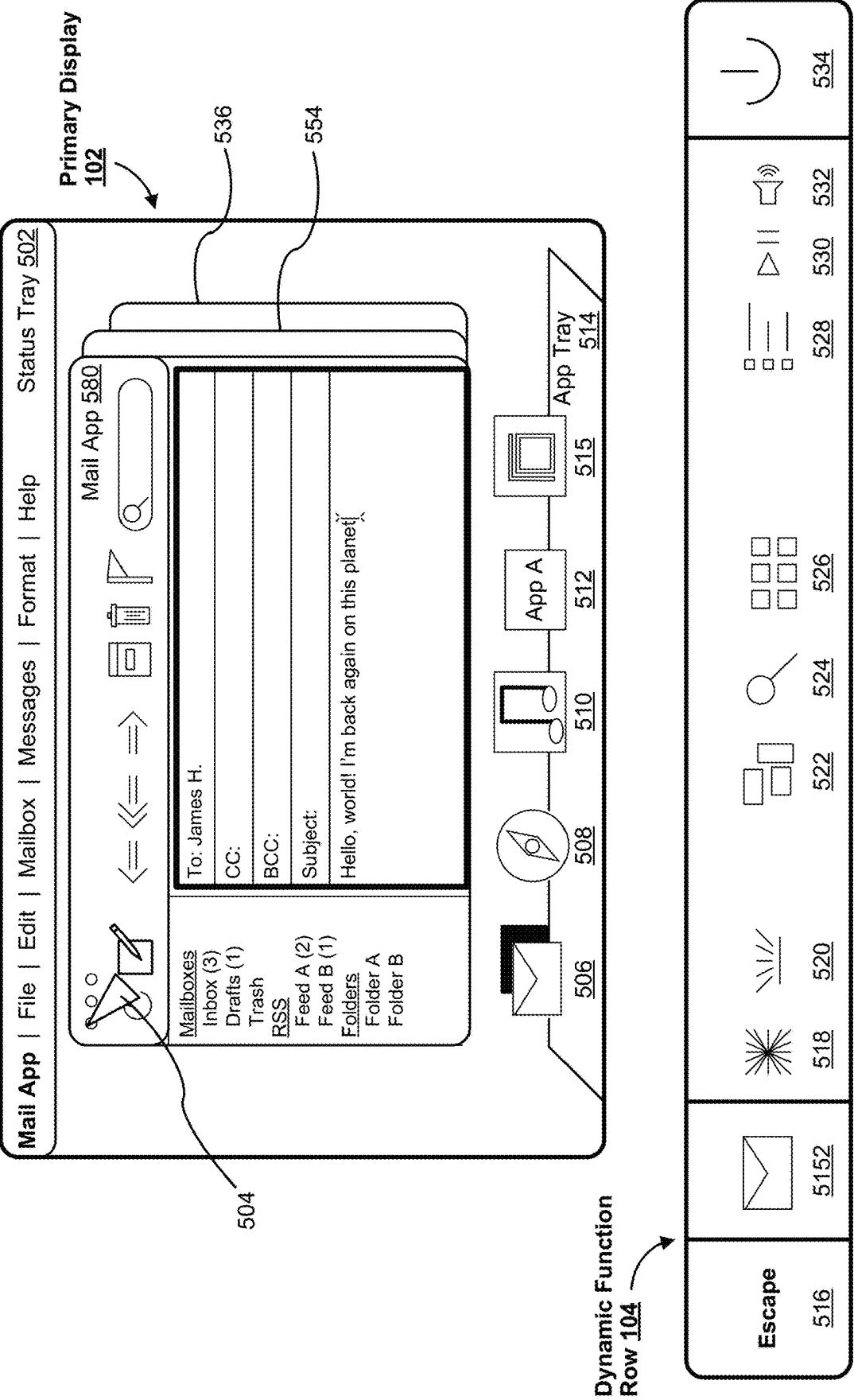


Figure 5U

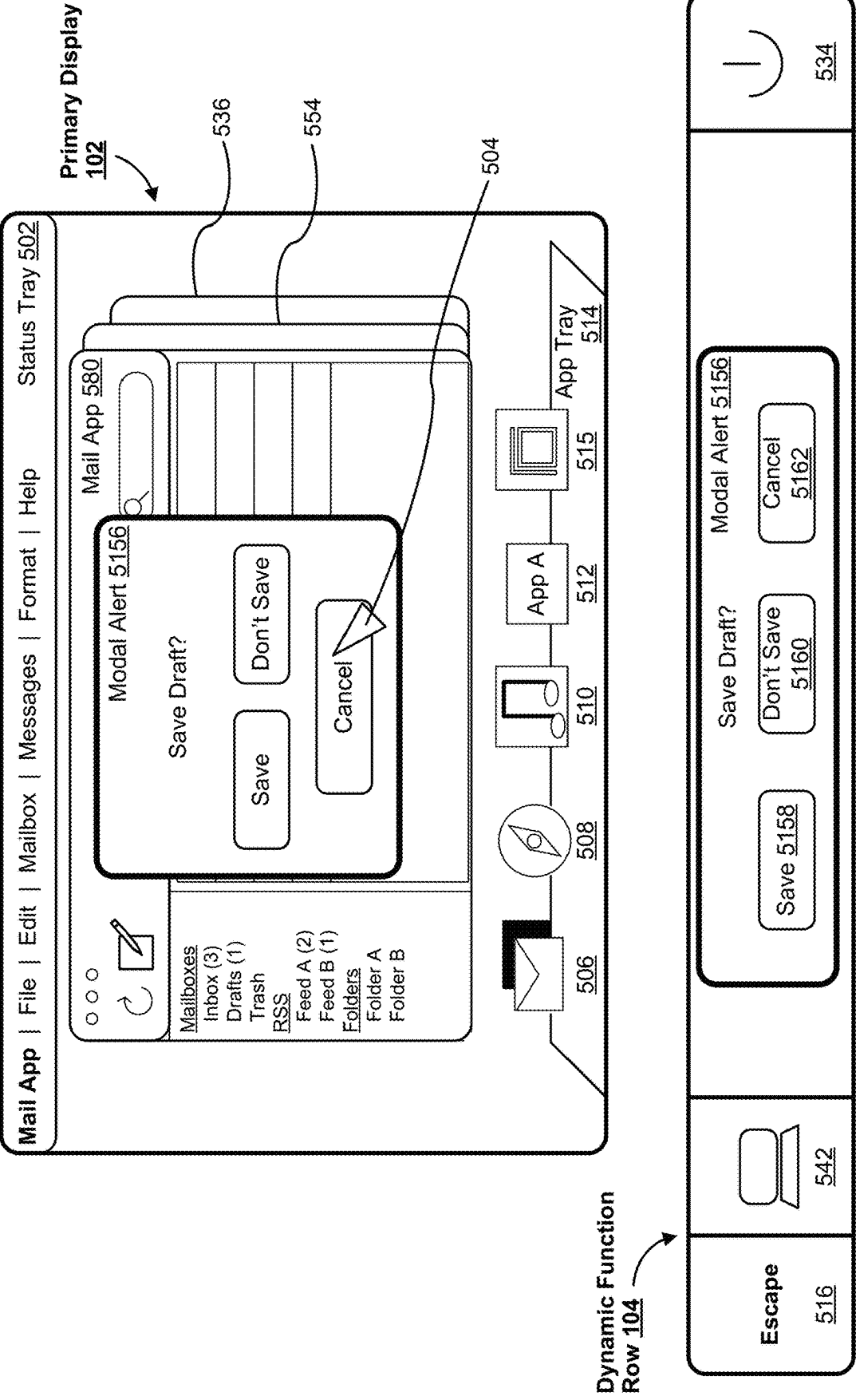


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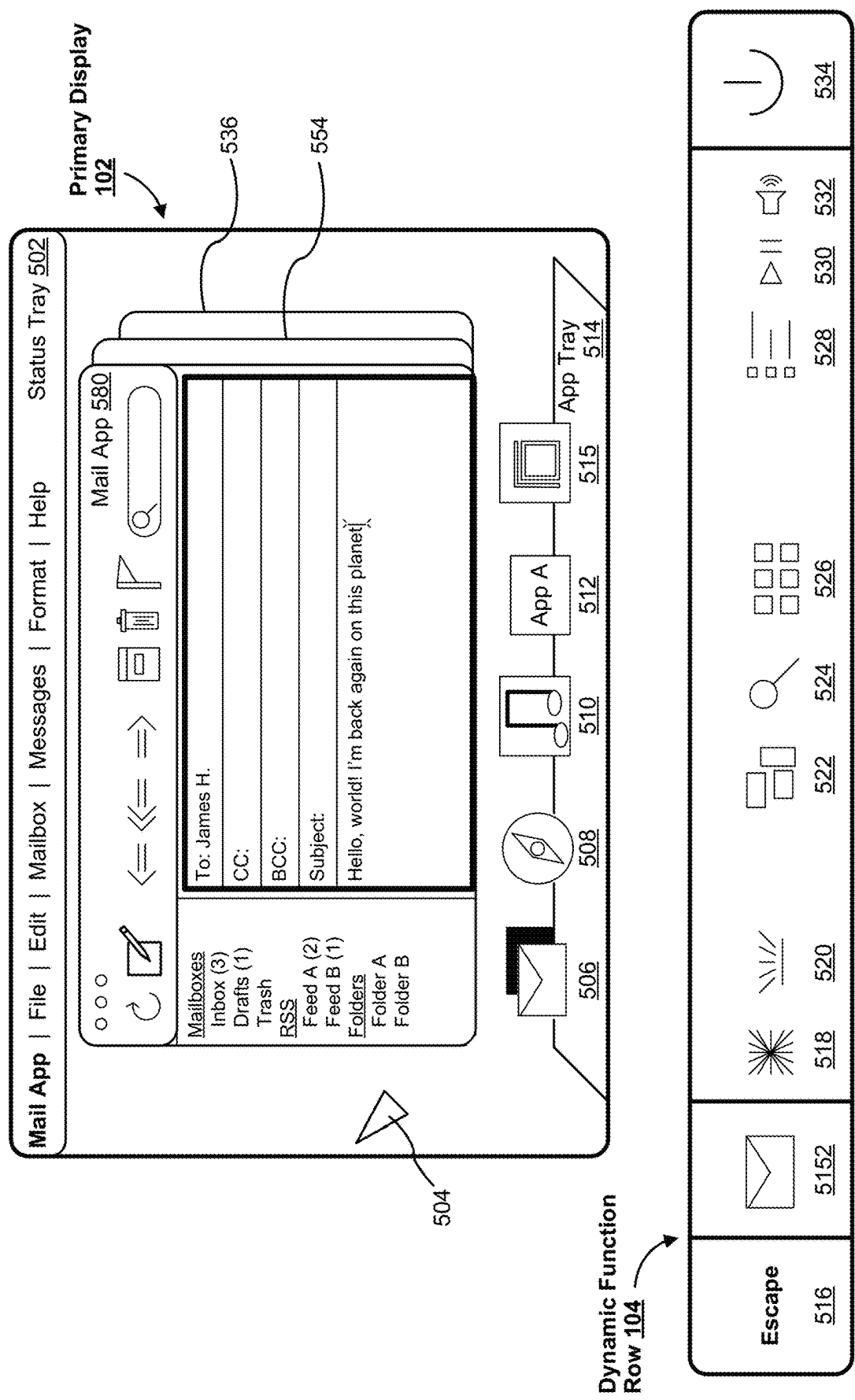


Figure 5W

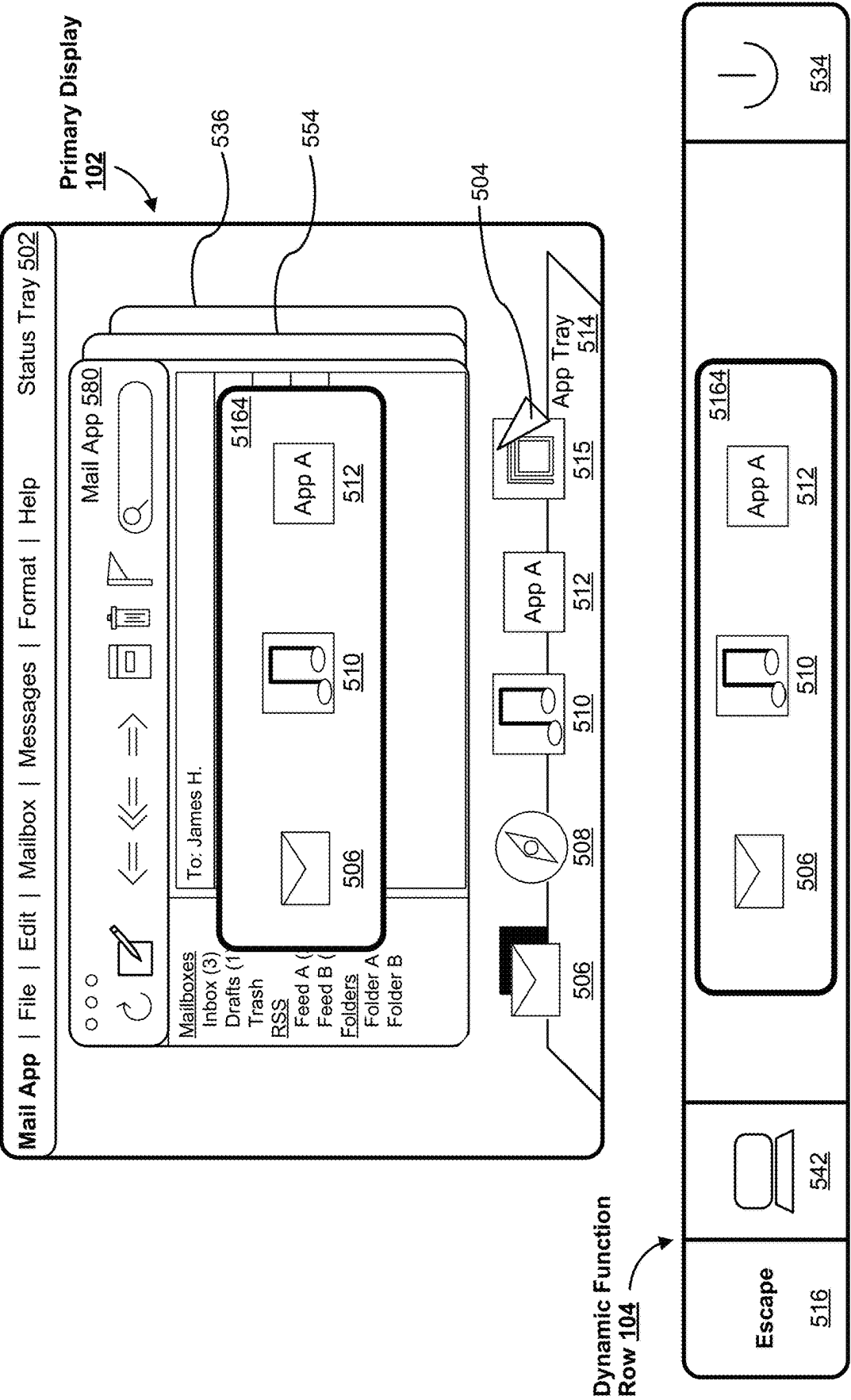


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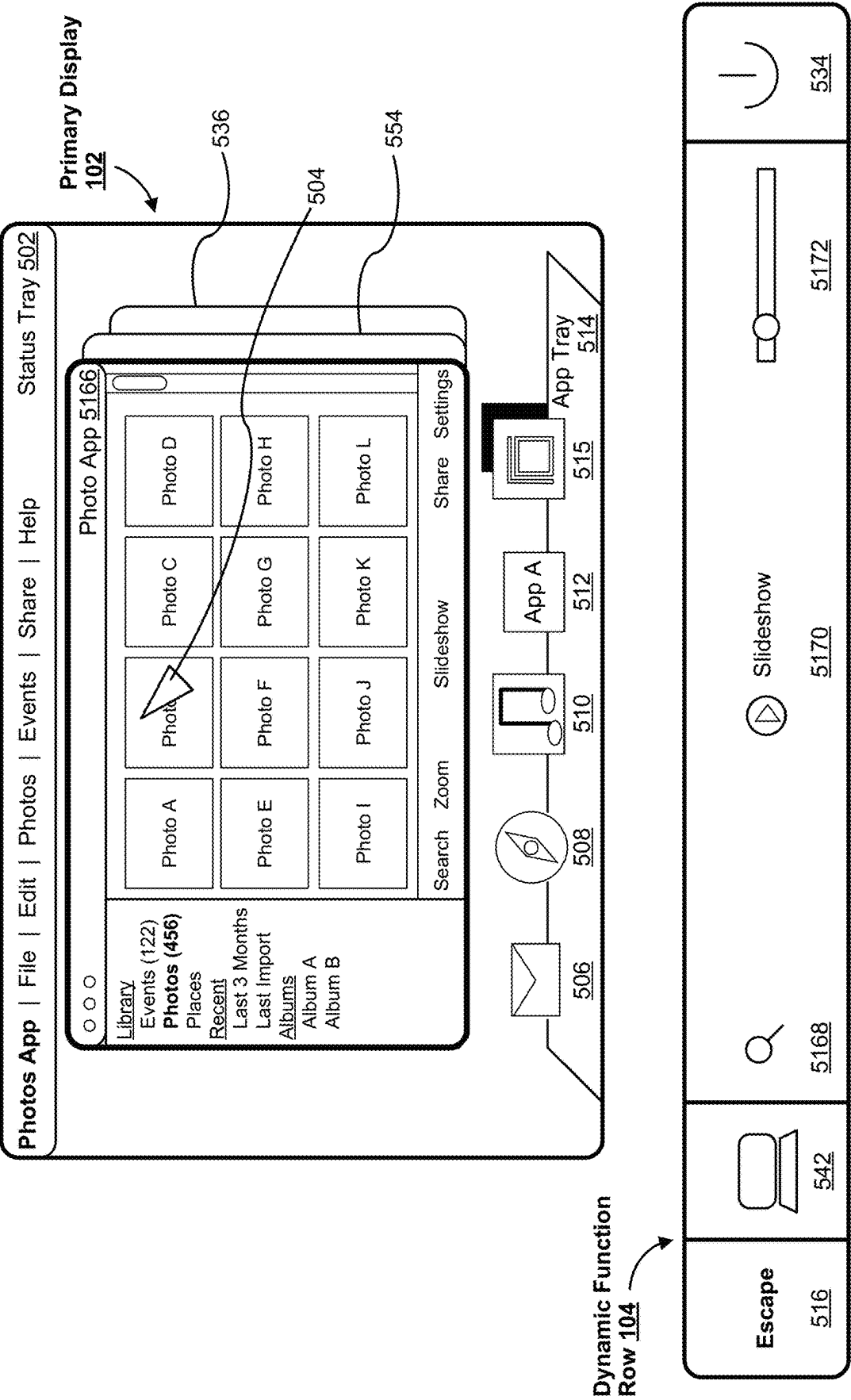


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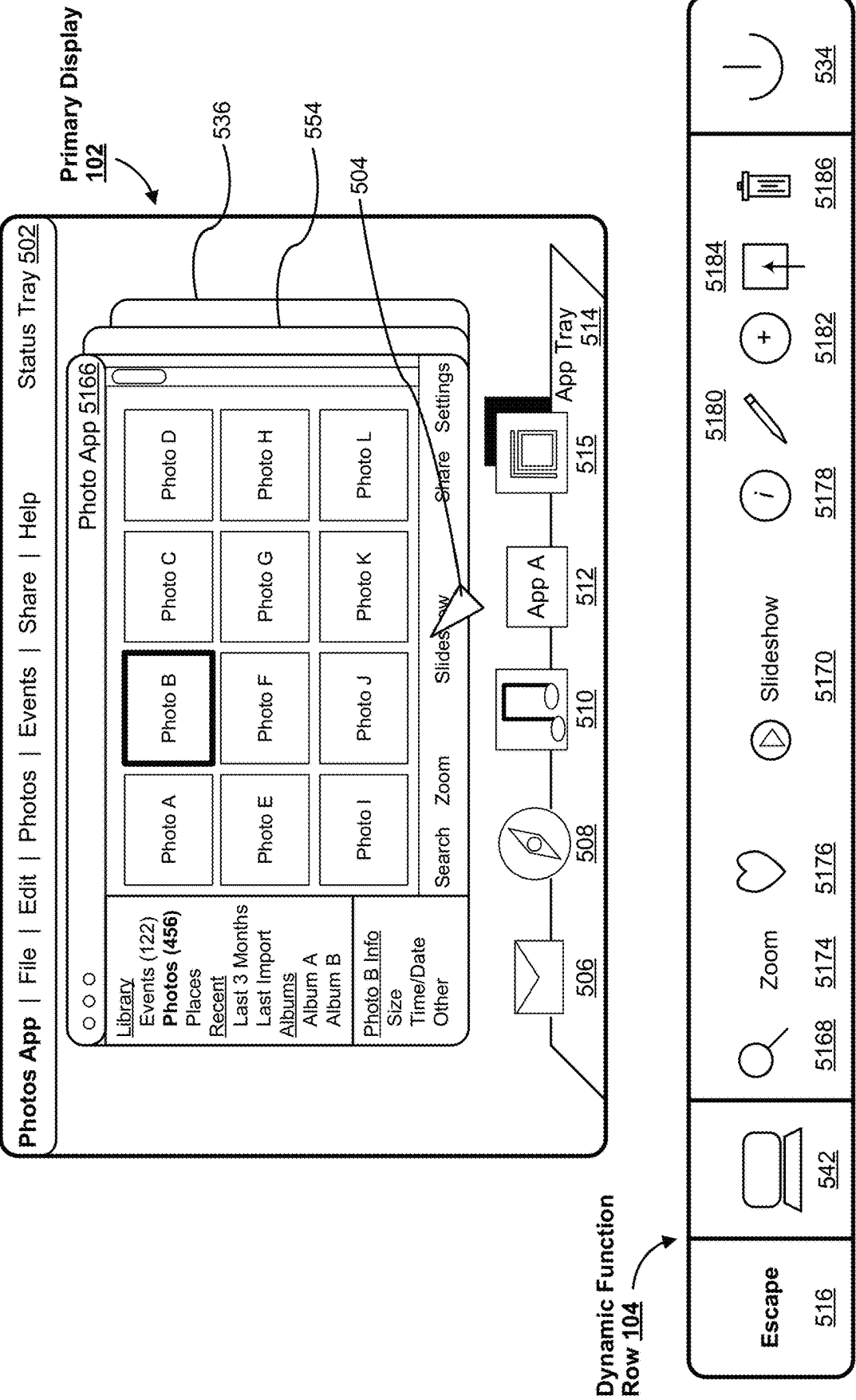


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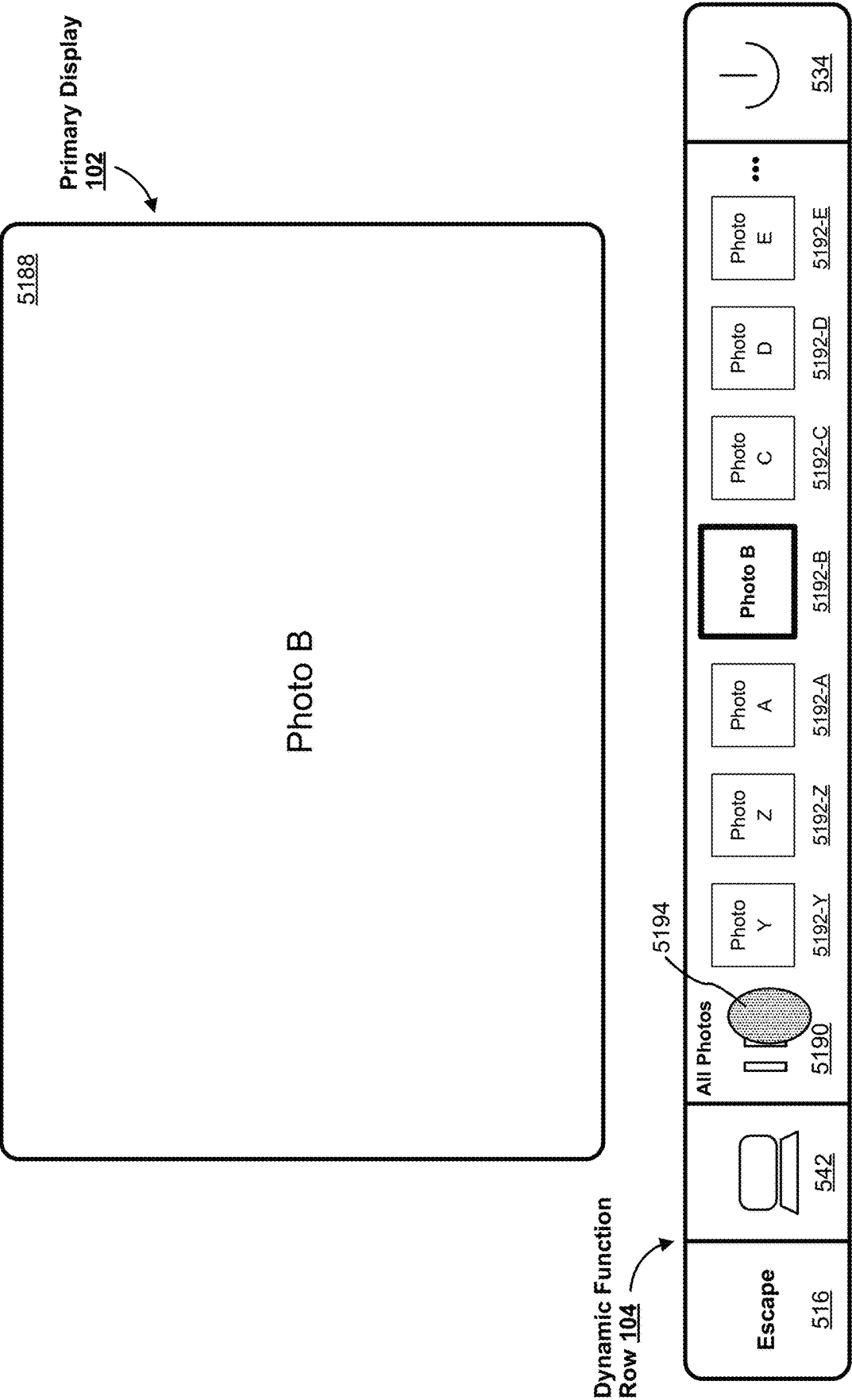


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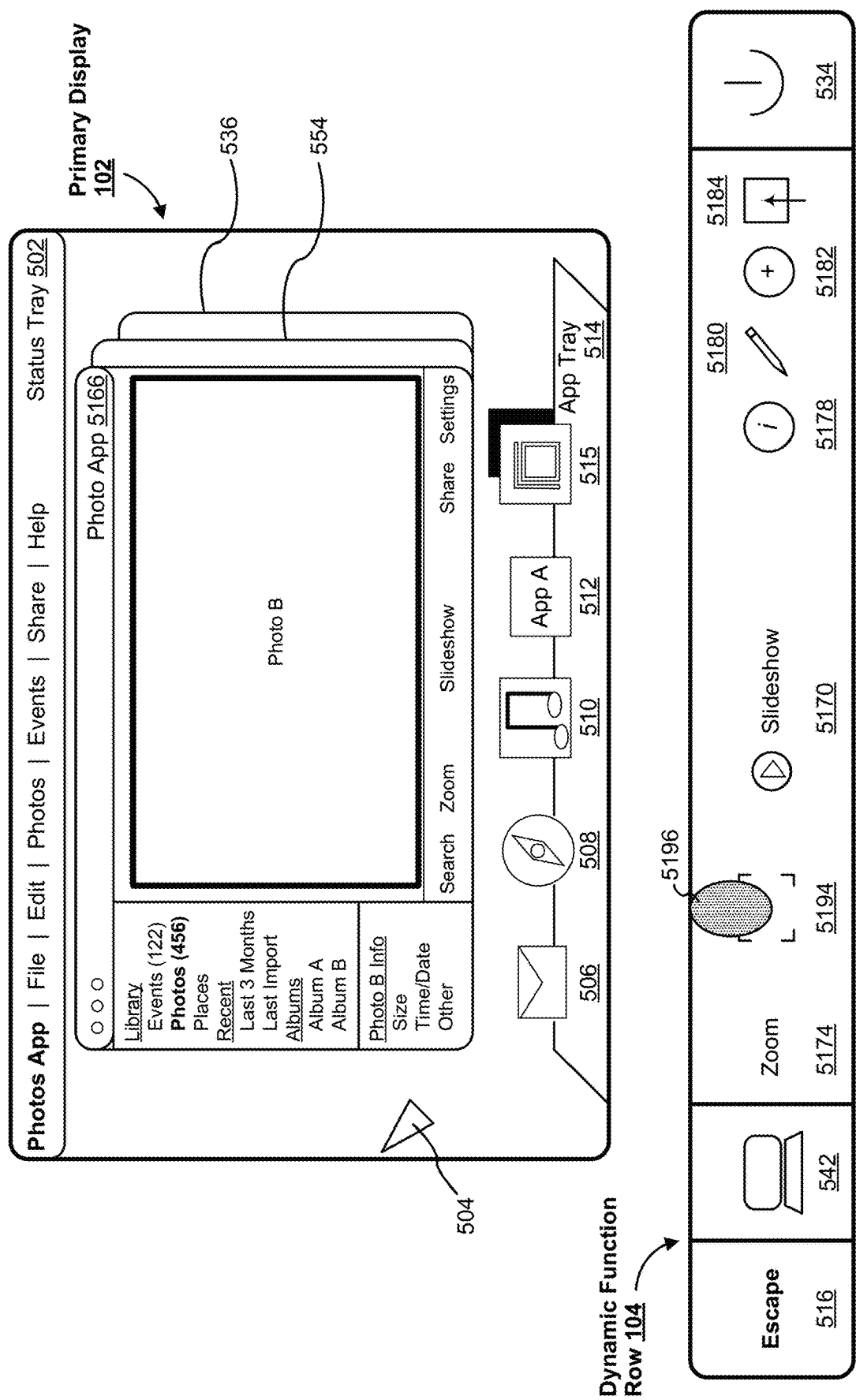


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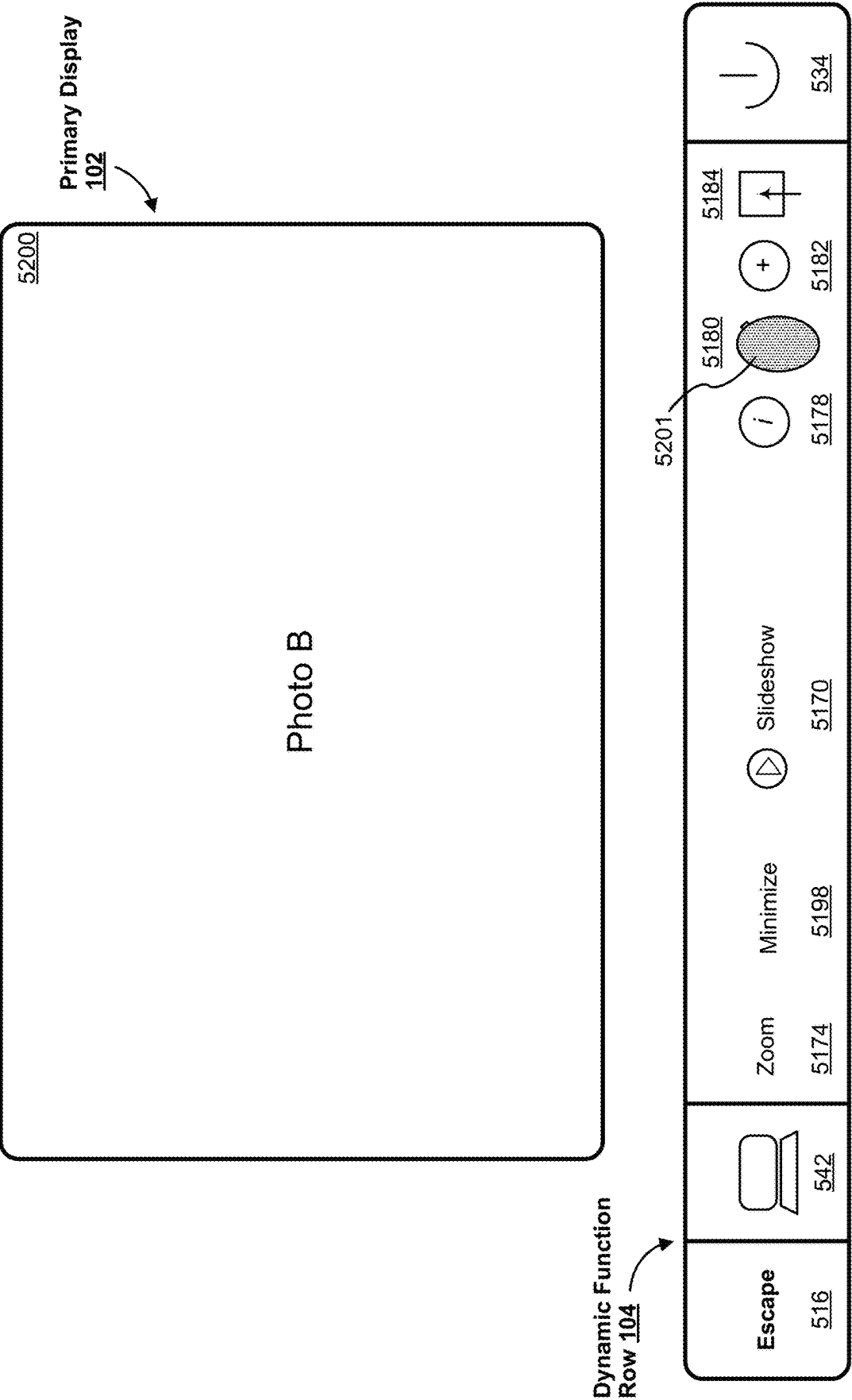


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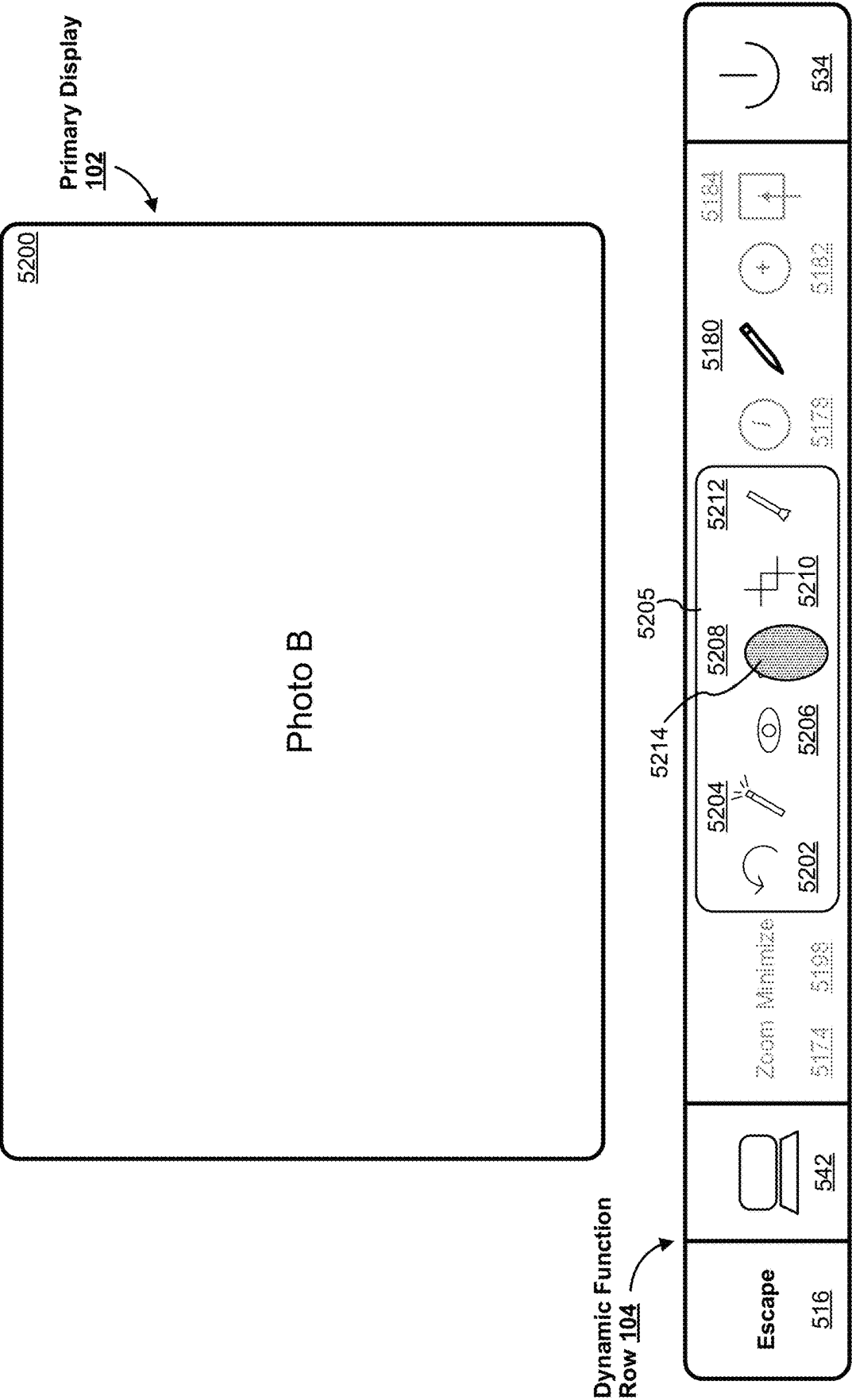


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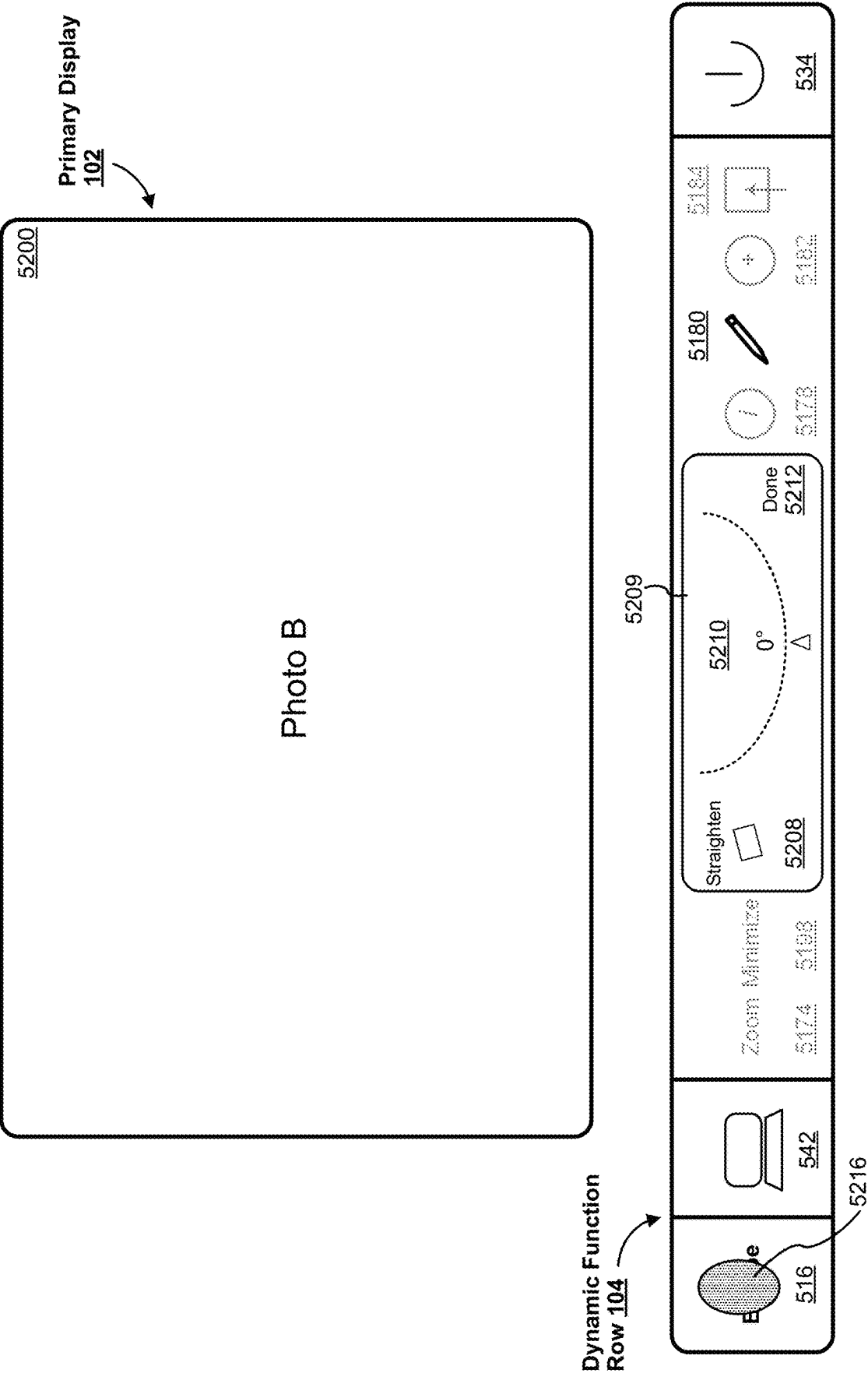


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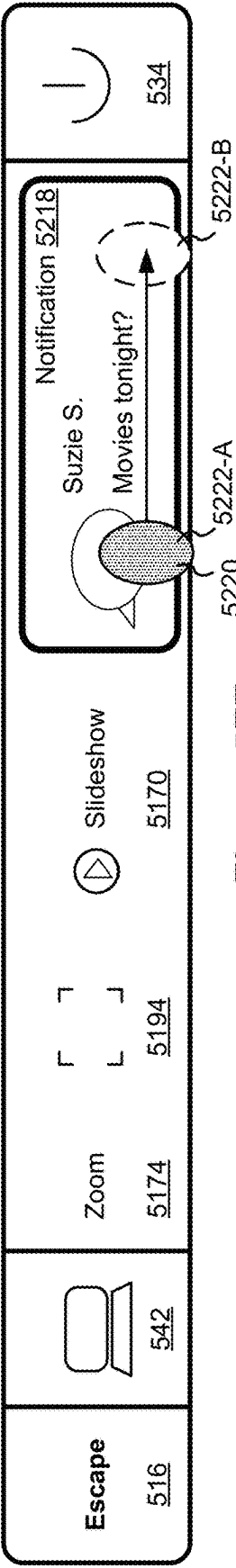
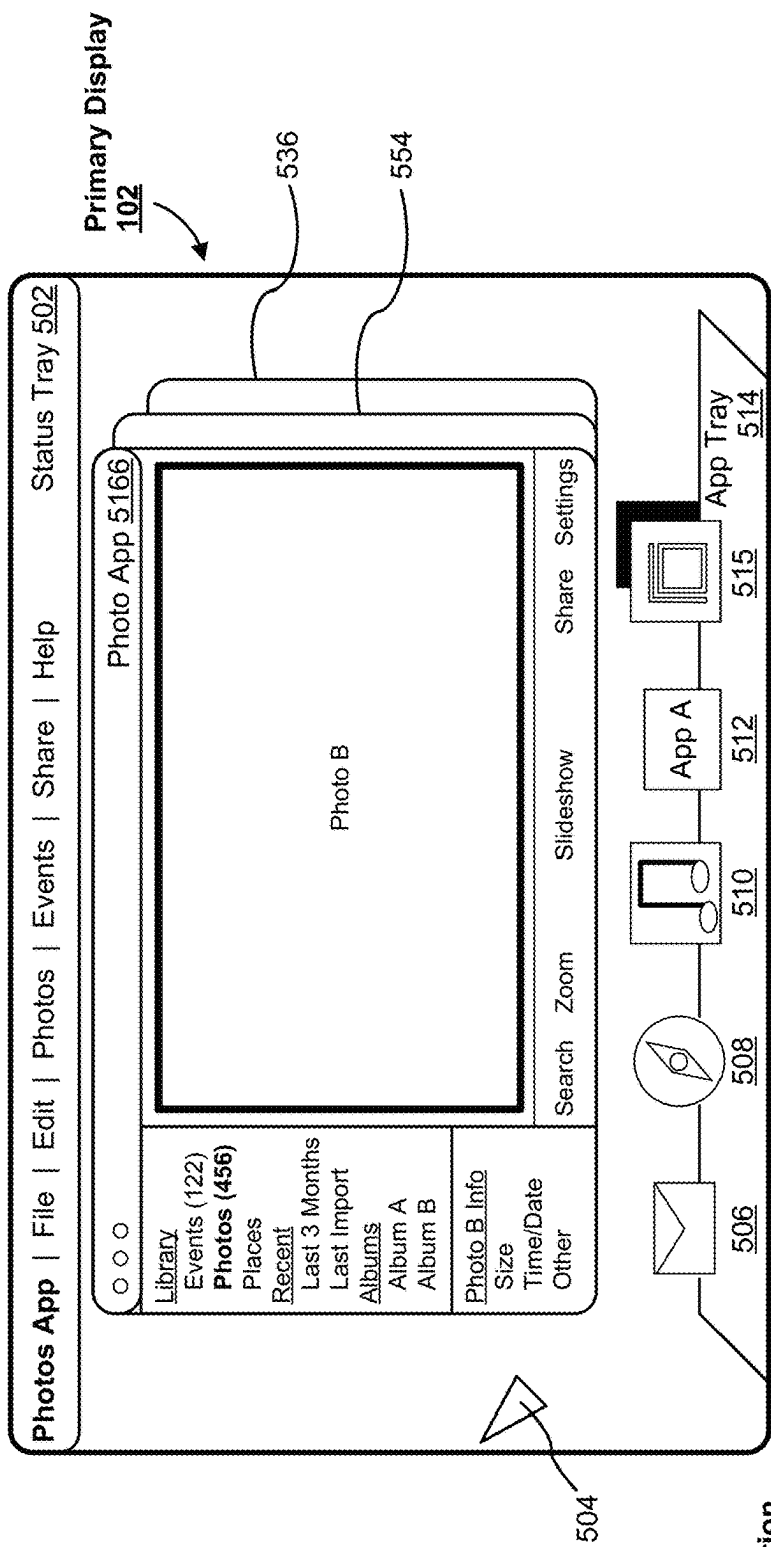


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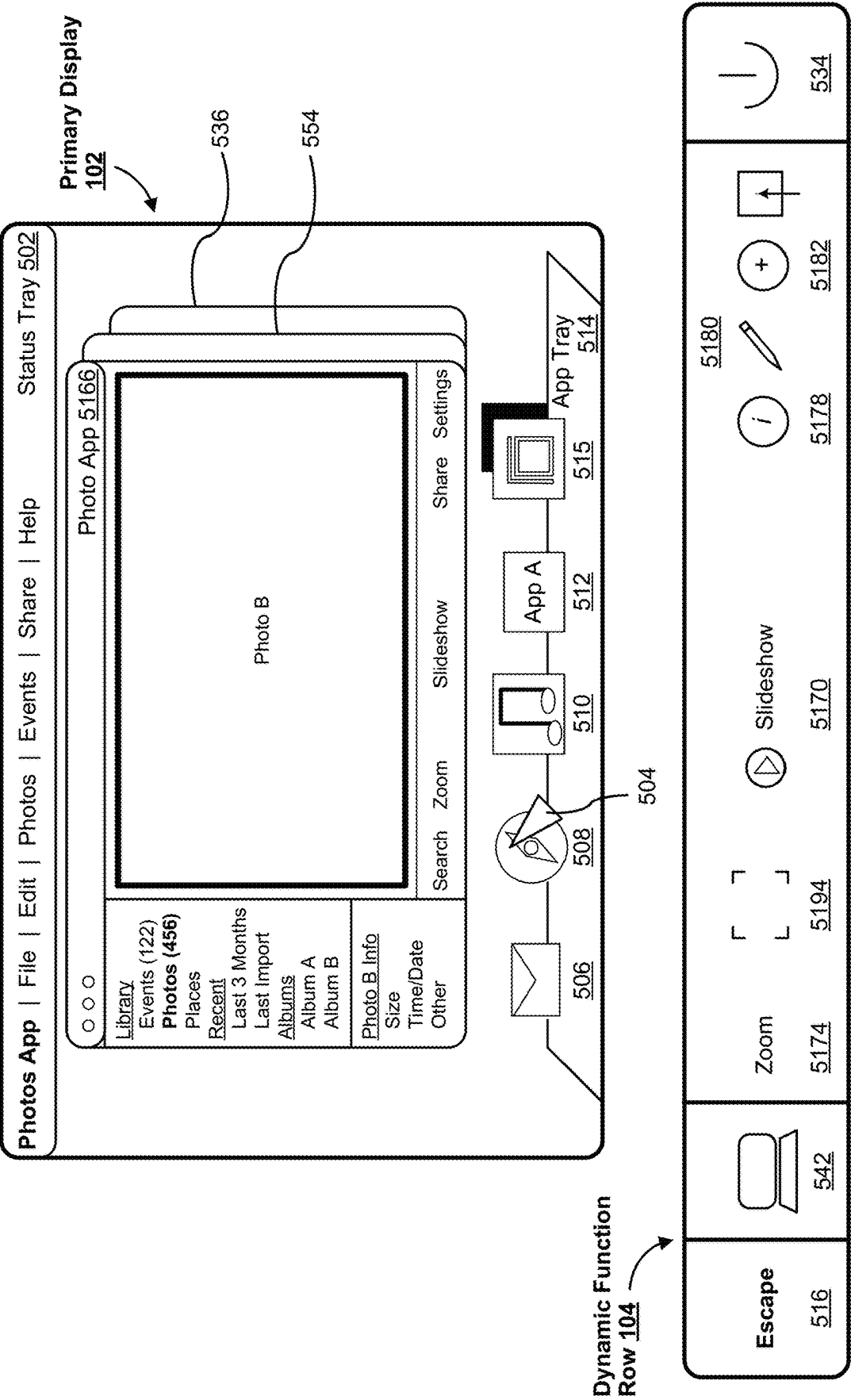


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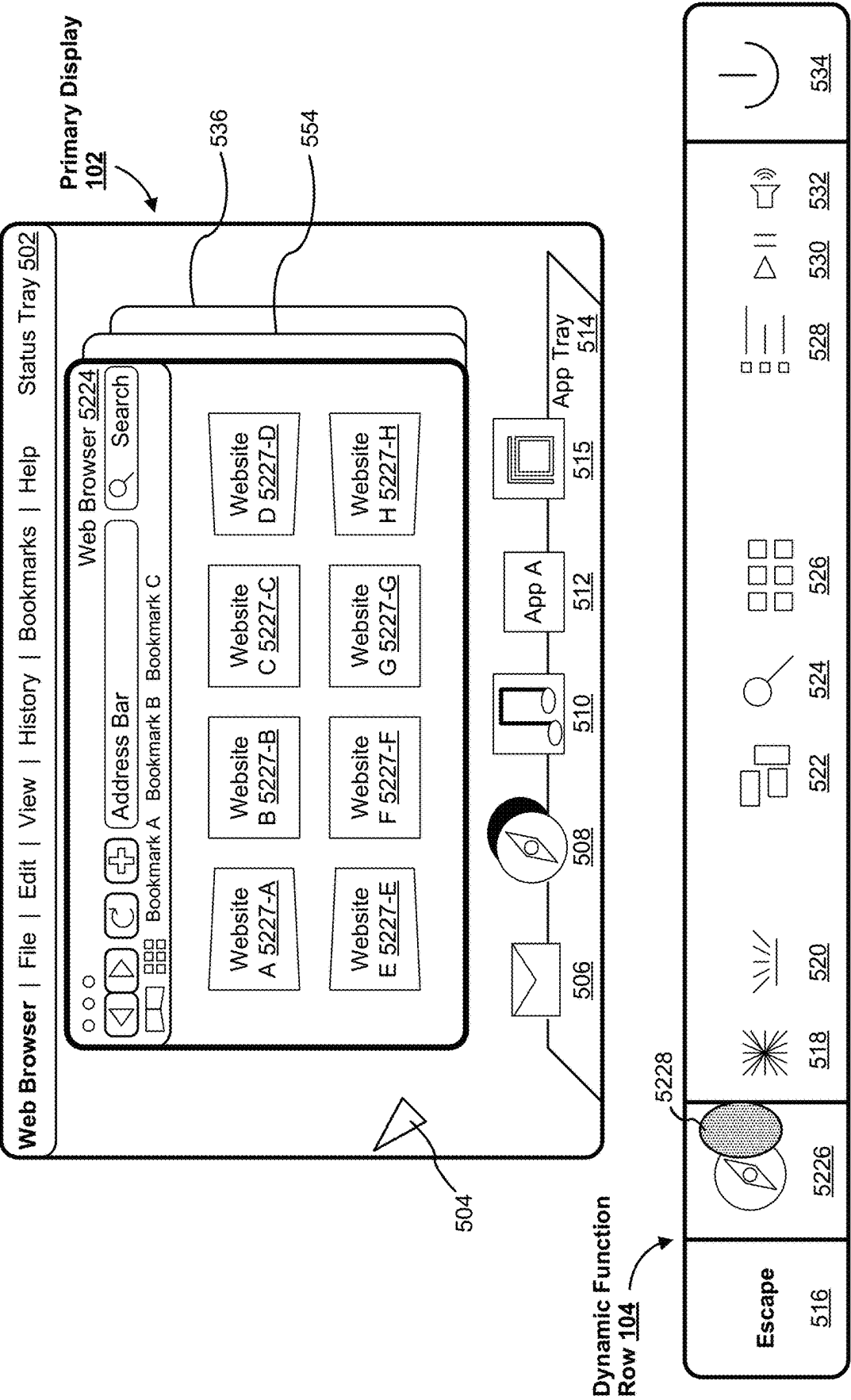


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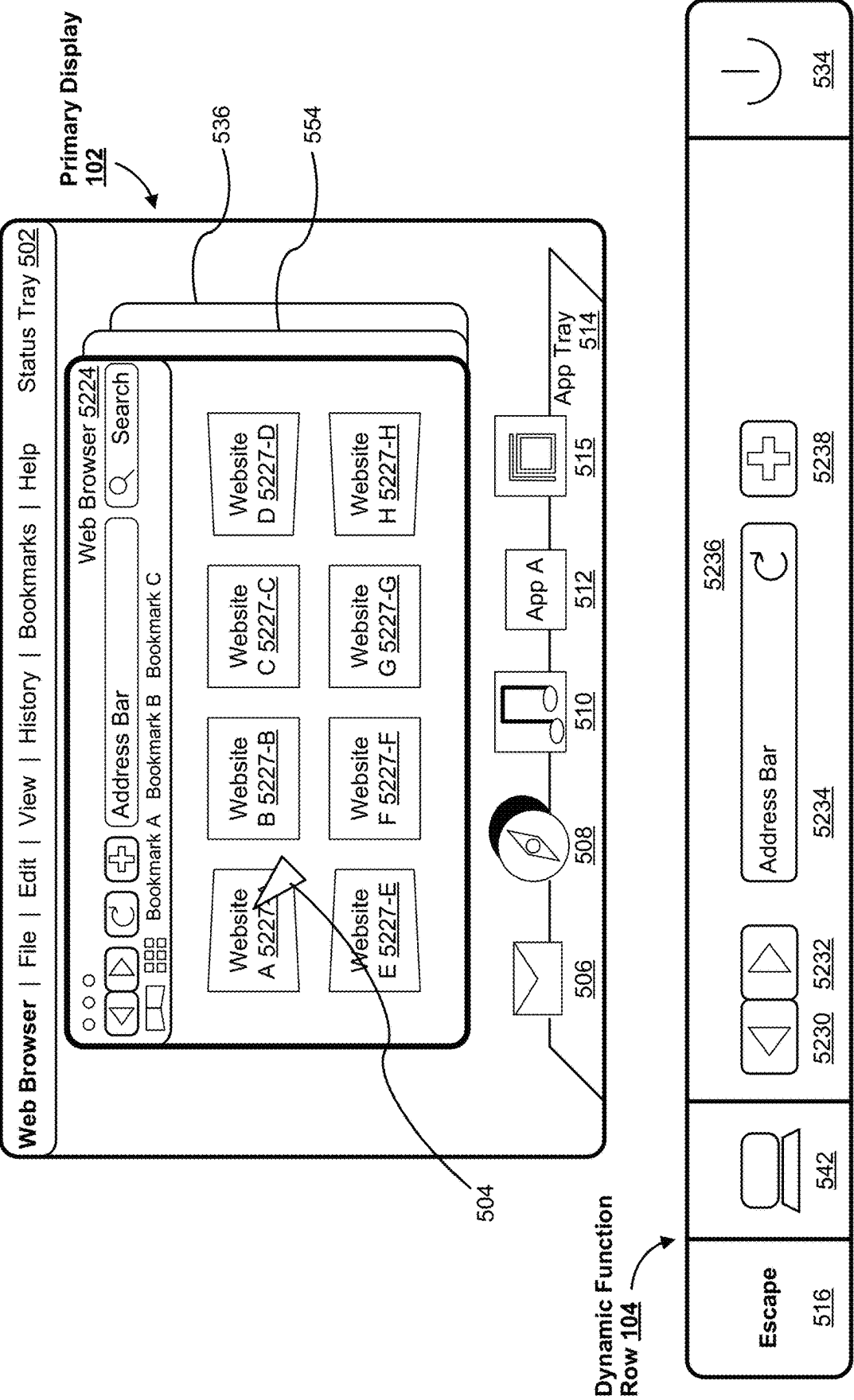


Figure 5II

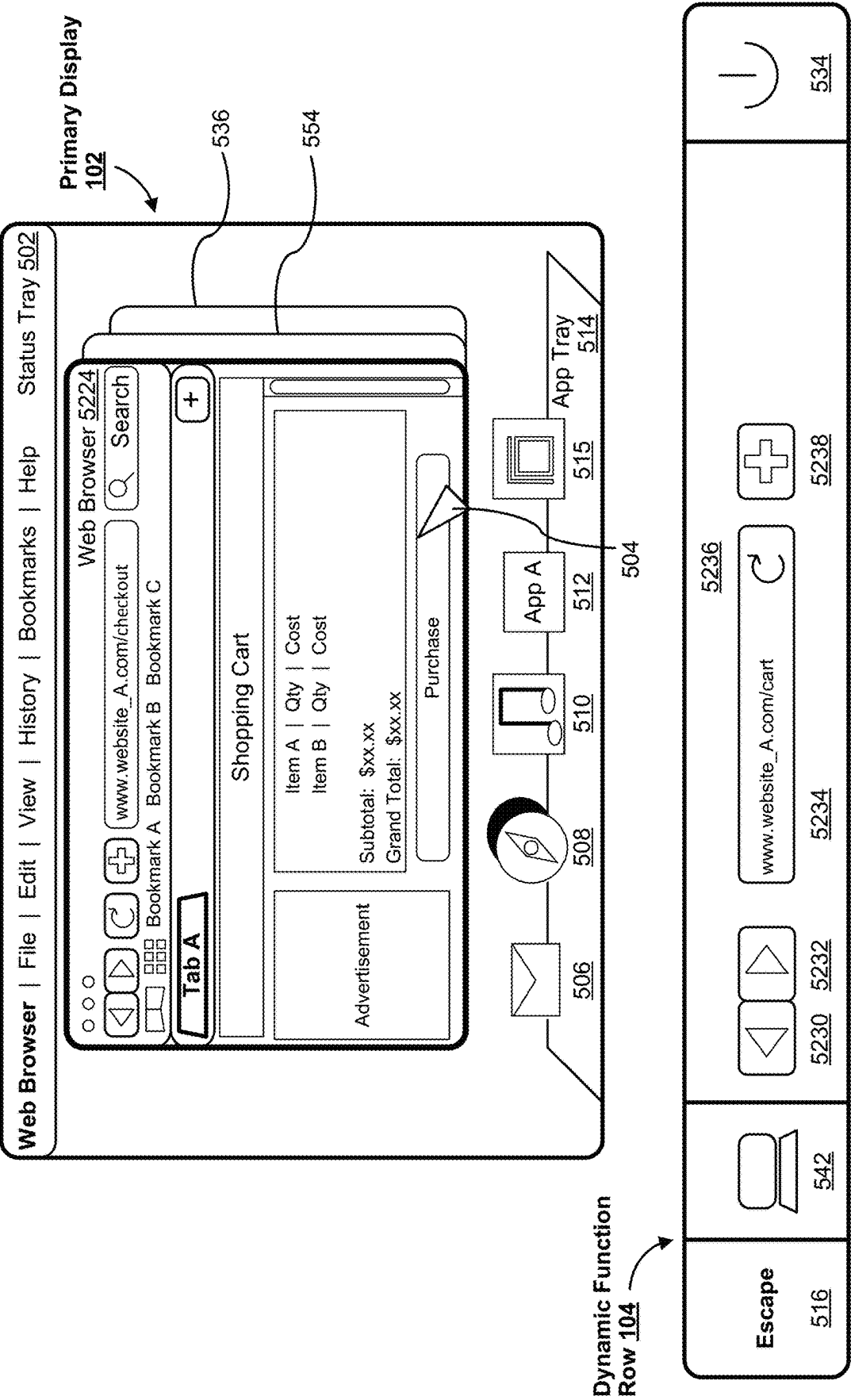


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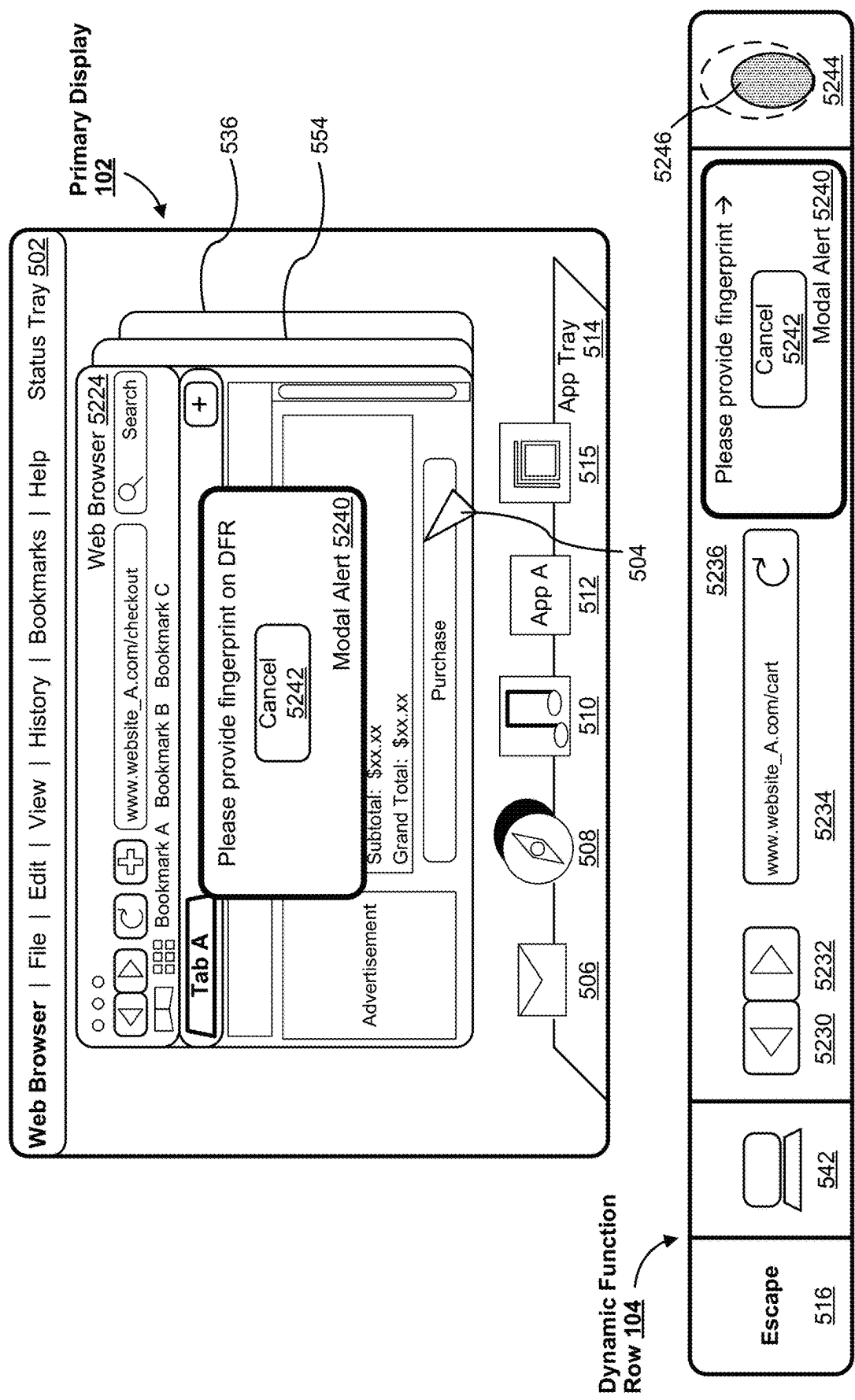
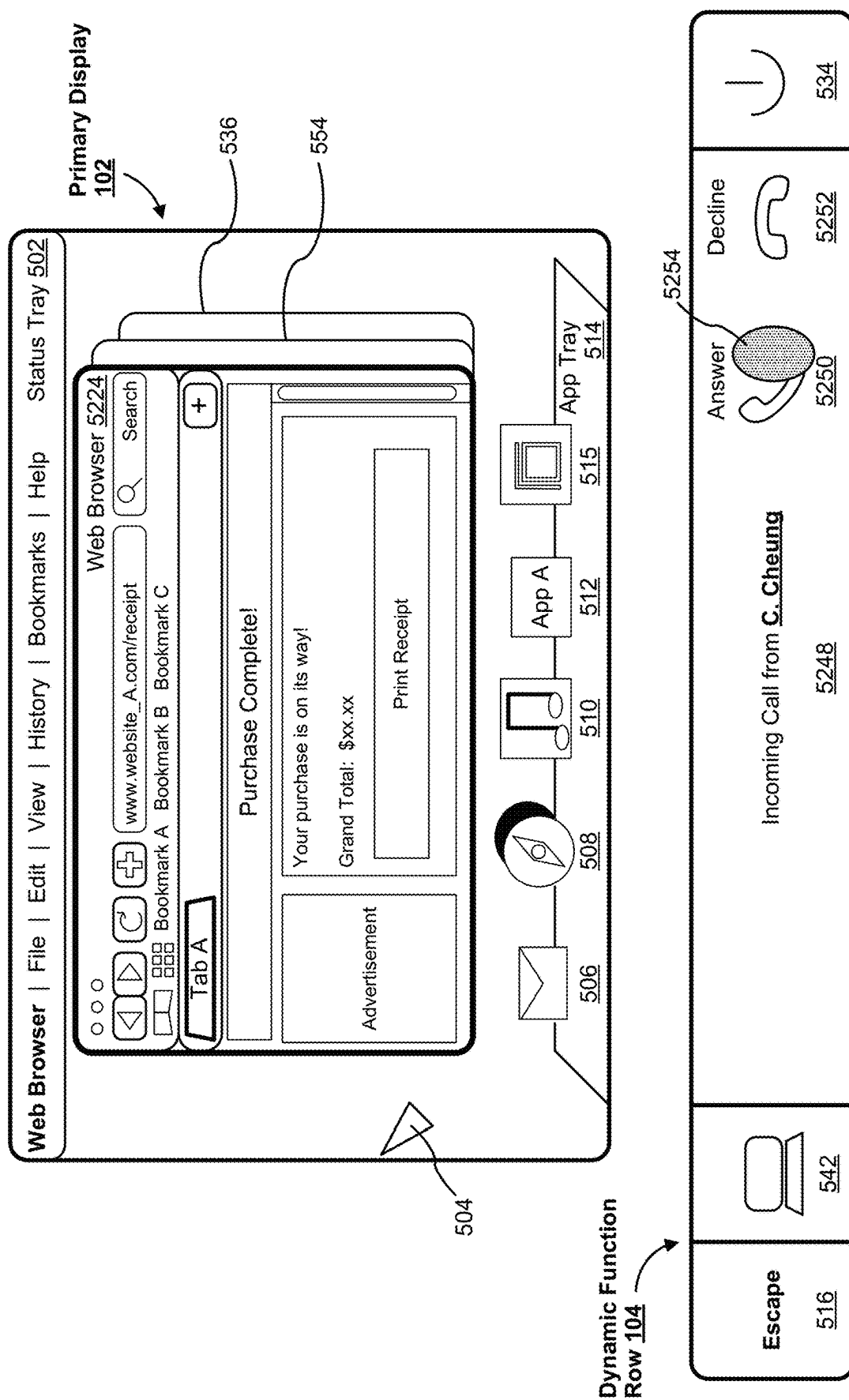


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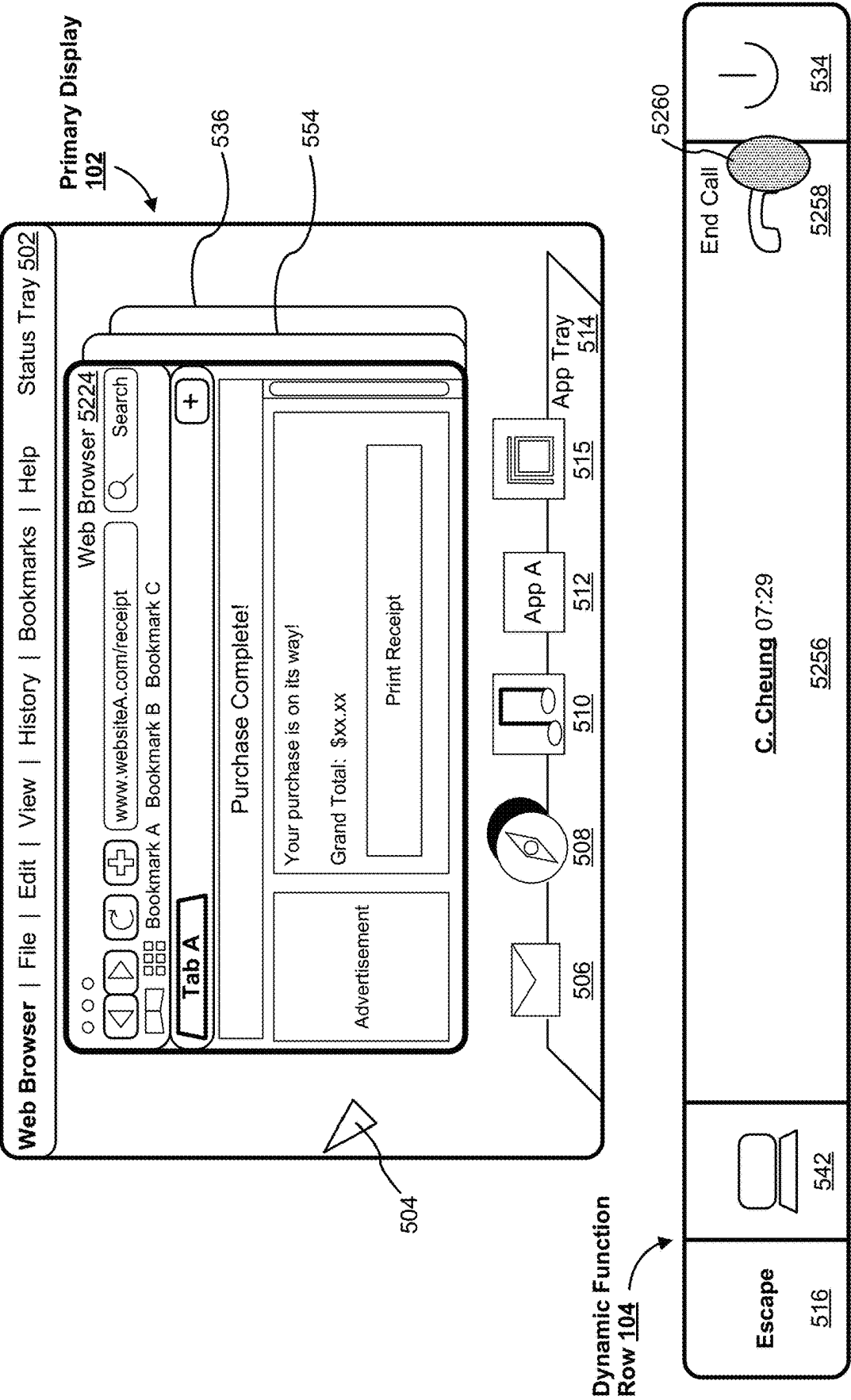


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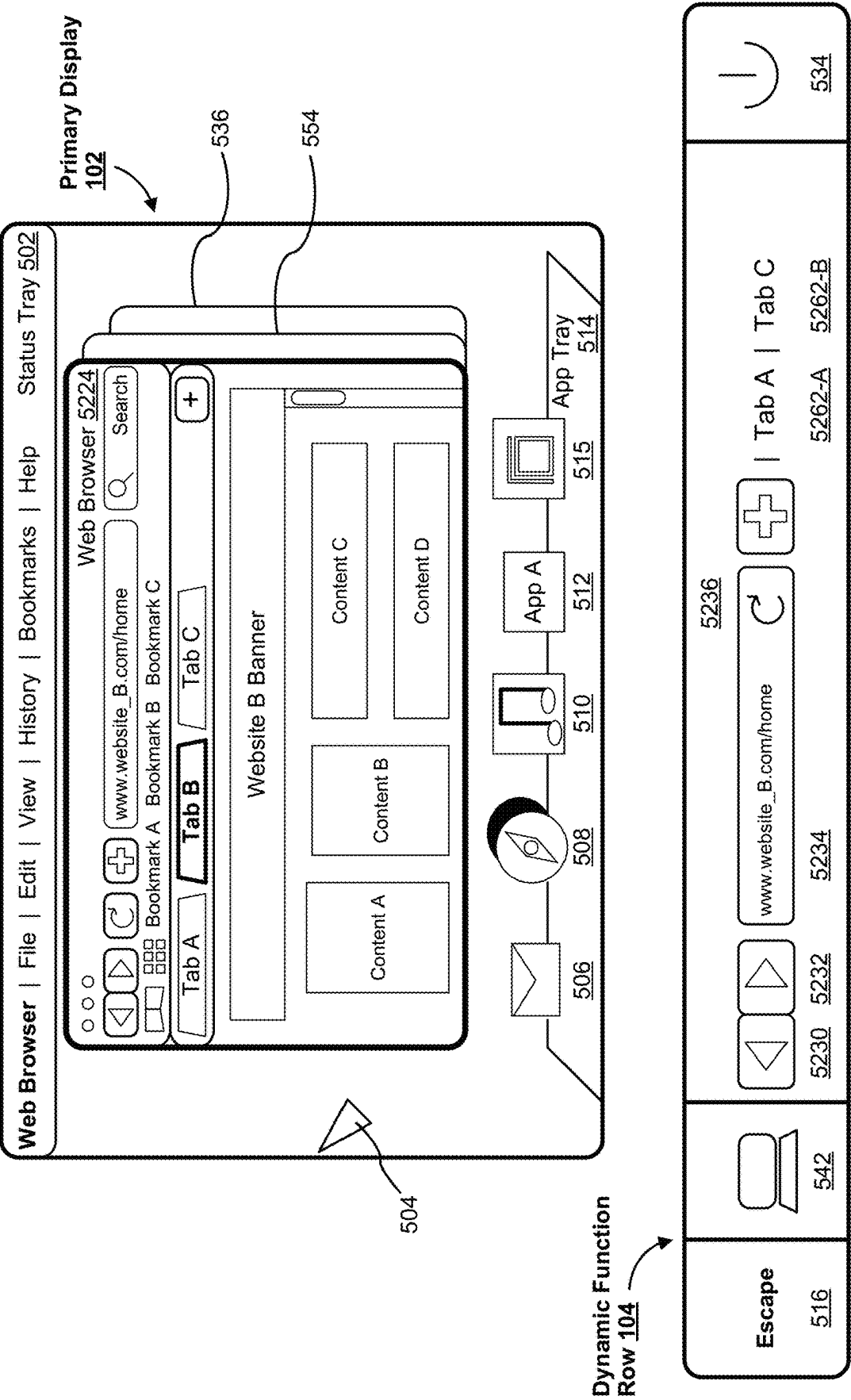
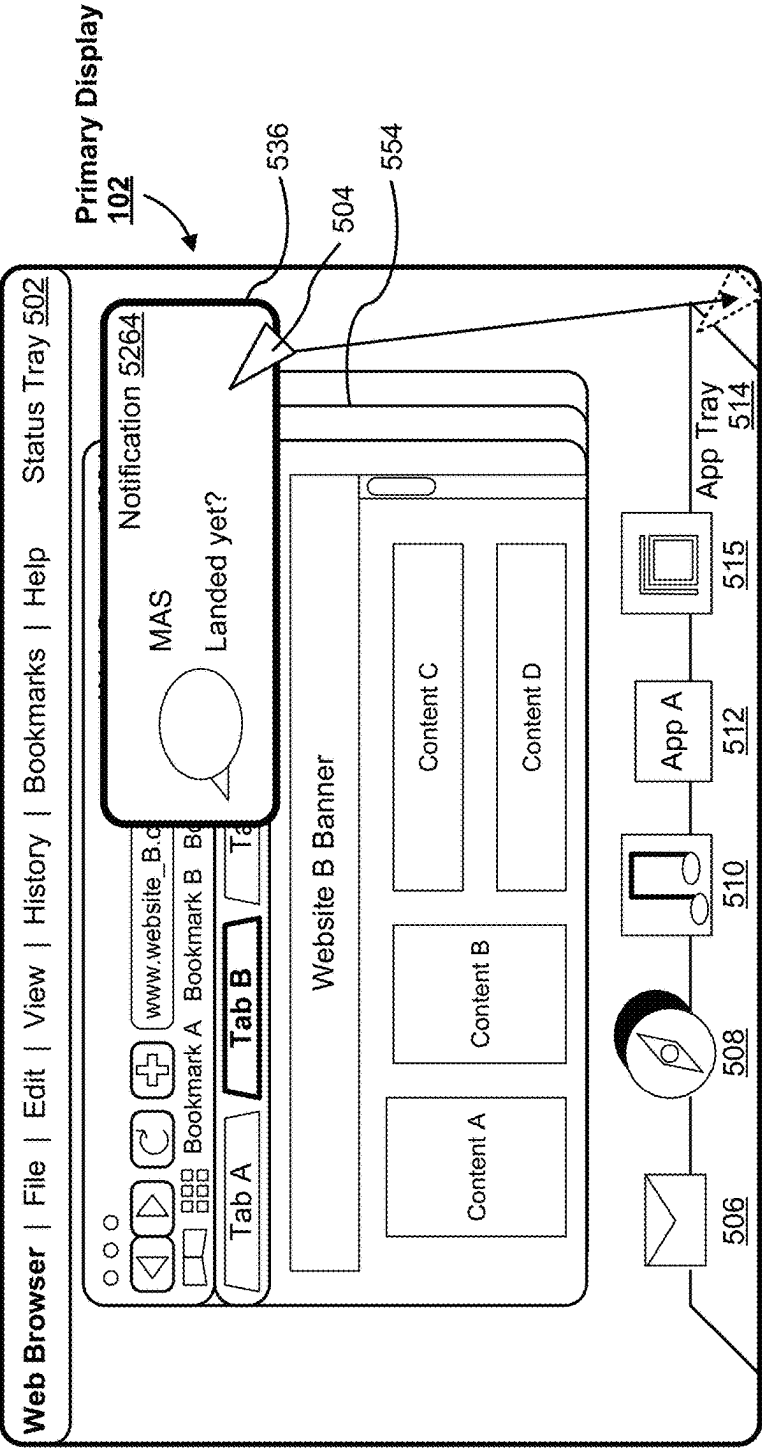


Figure 5NN



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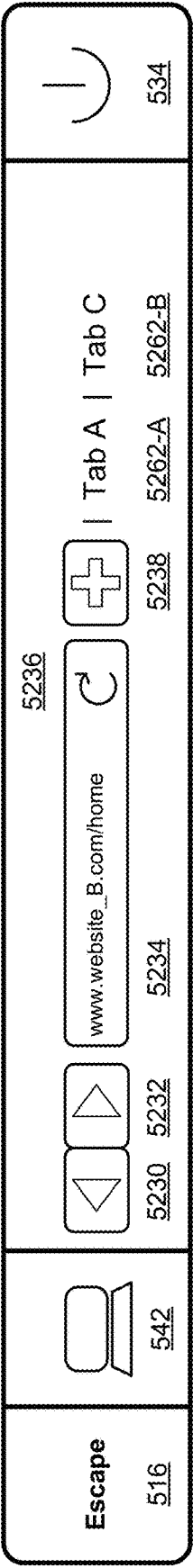


Figure 500

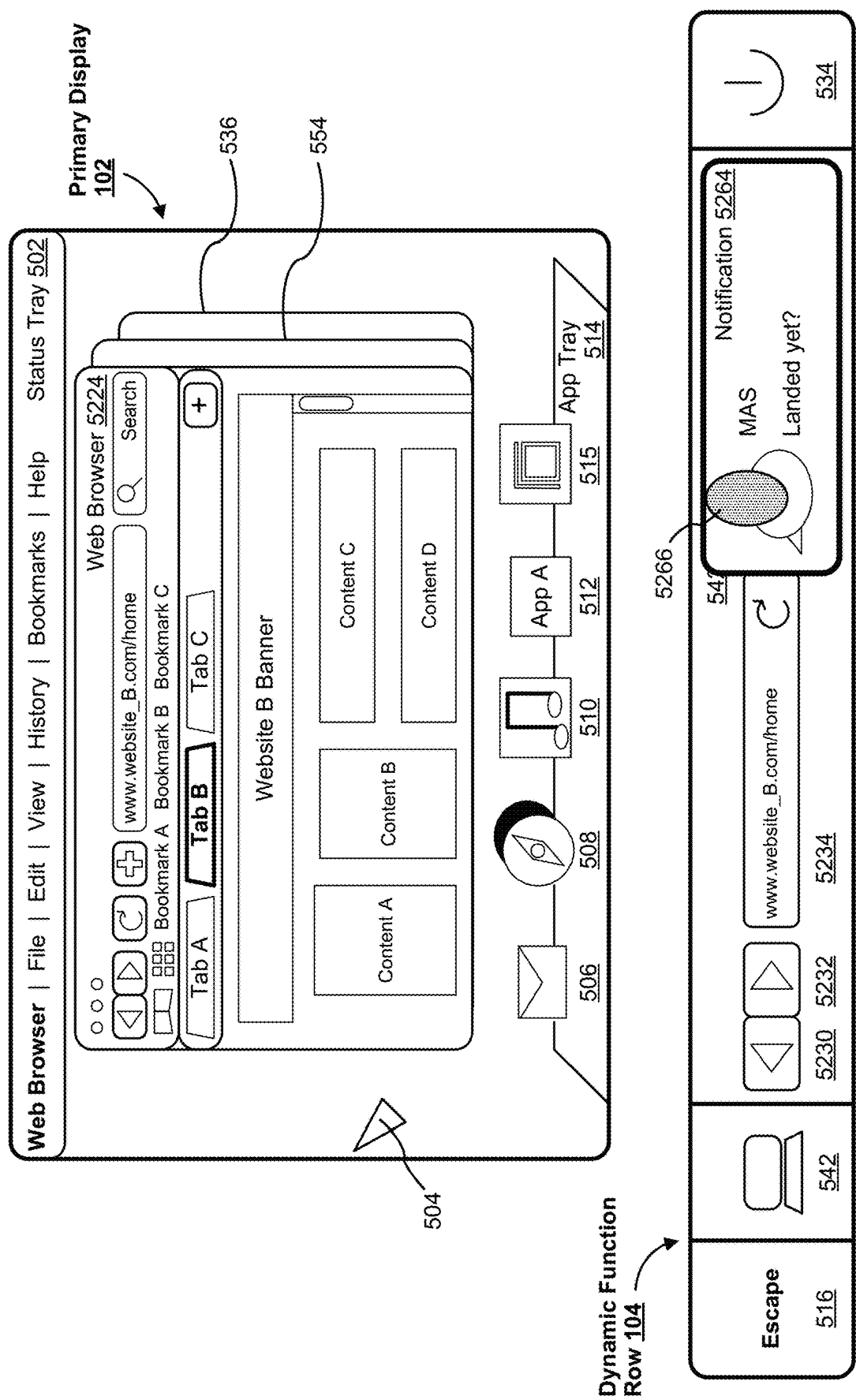


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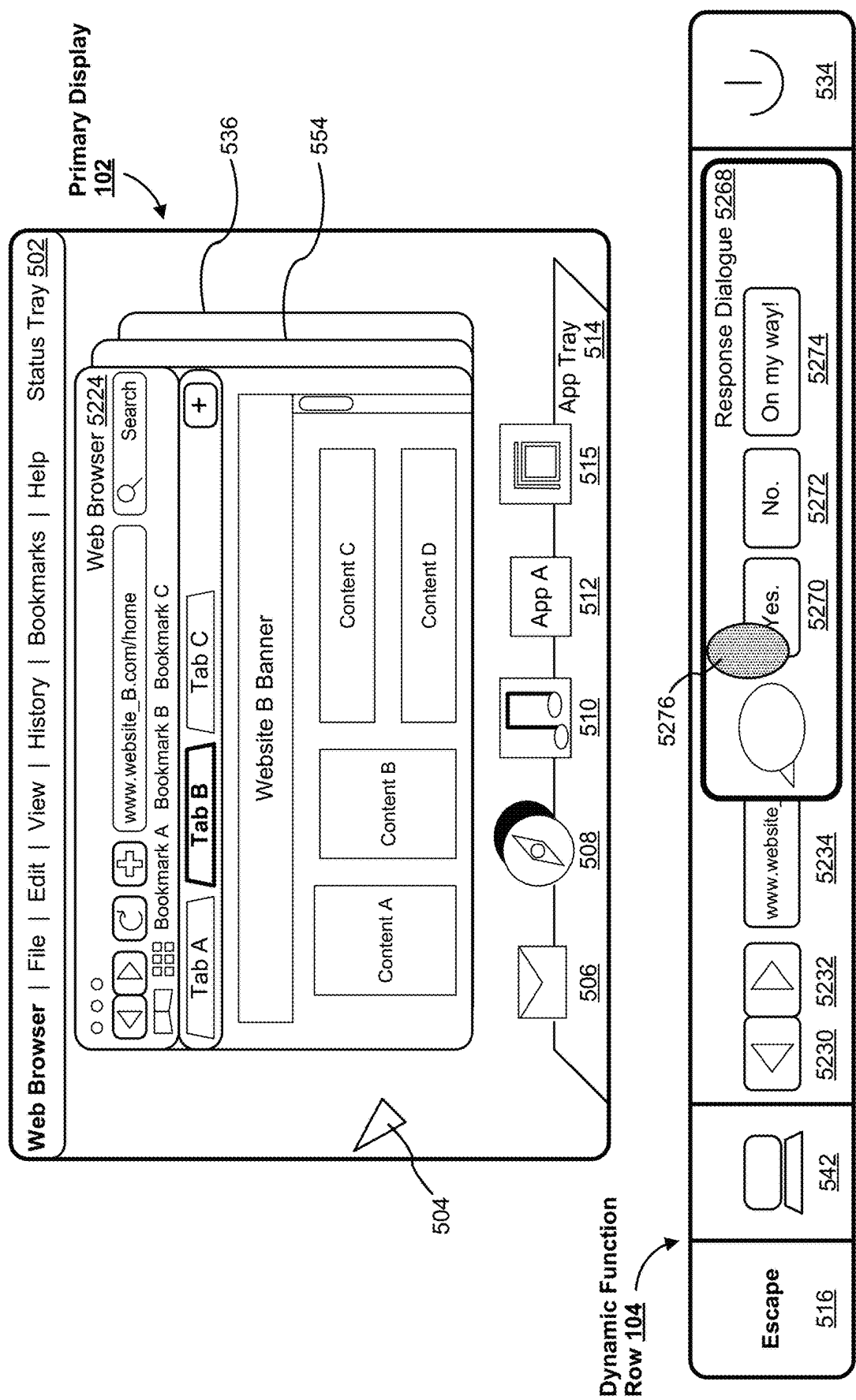
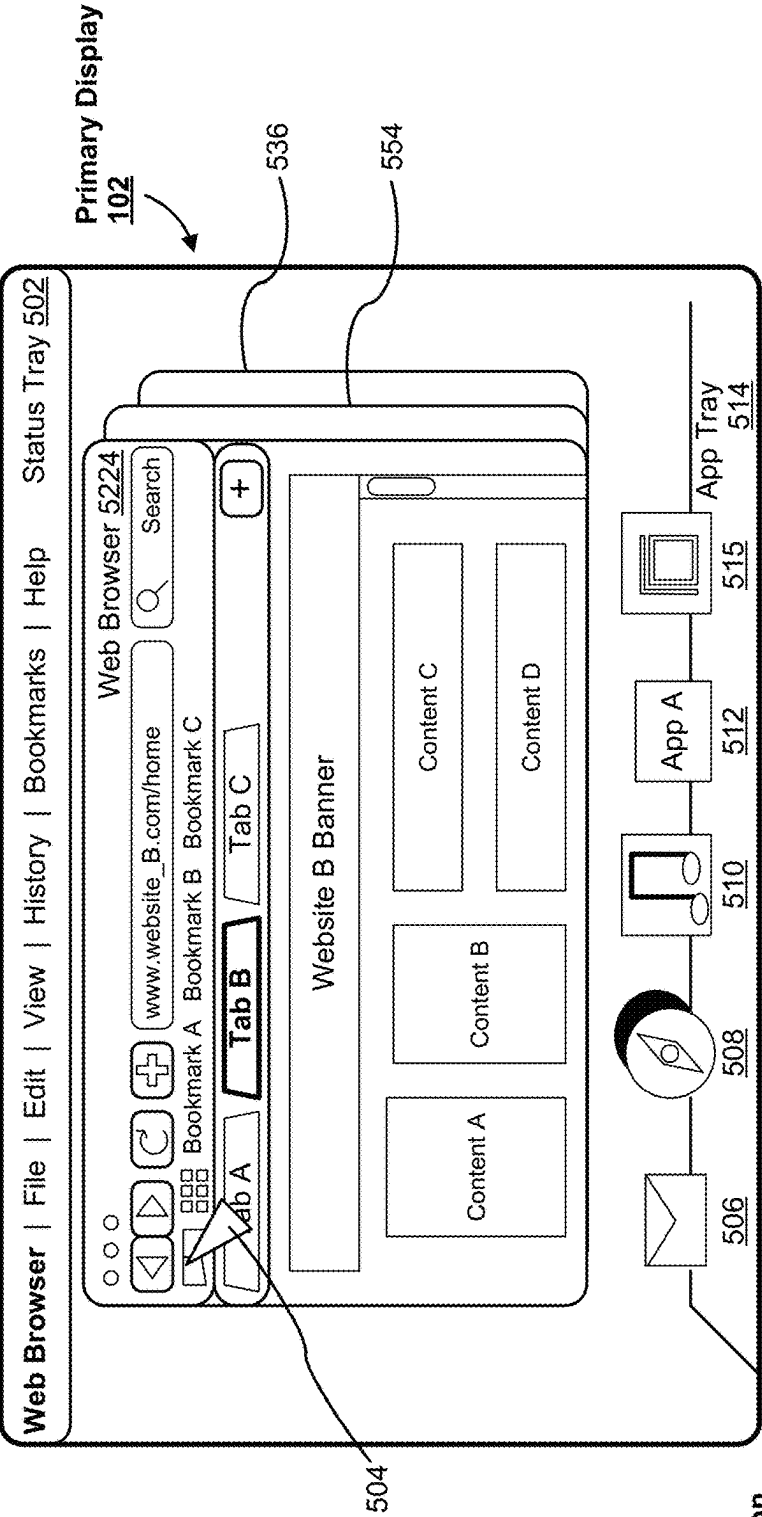


Figure 50Q



Dynamic Function Row 104

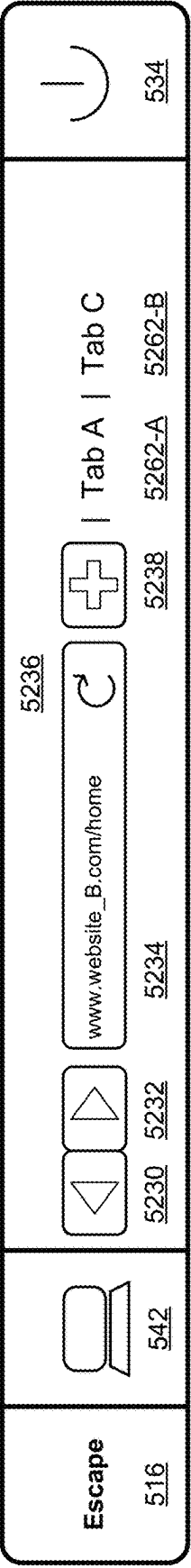
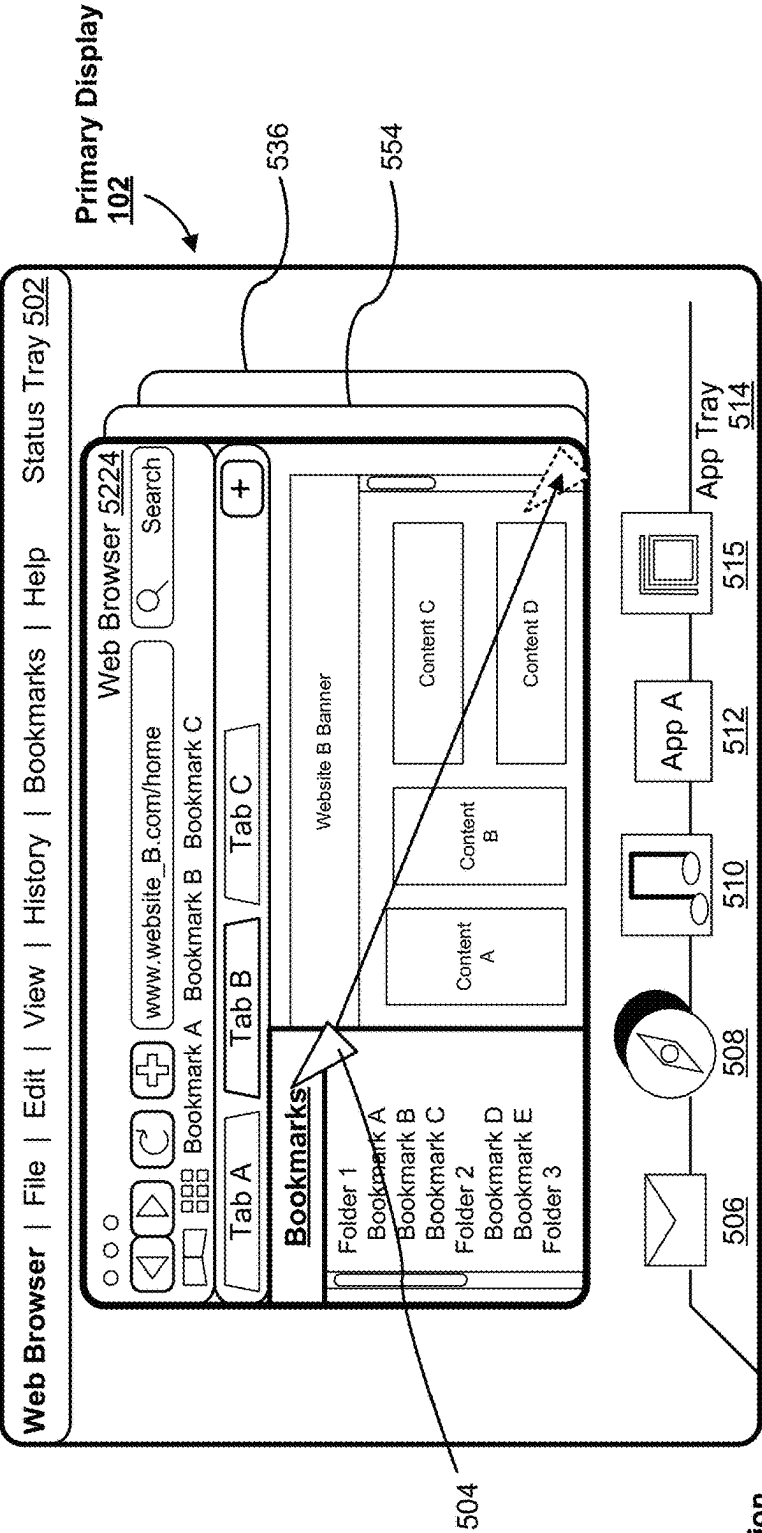


Figure 5RR



Dynamic Function Row 104

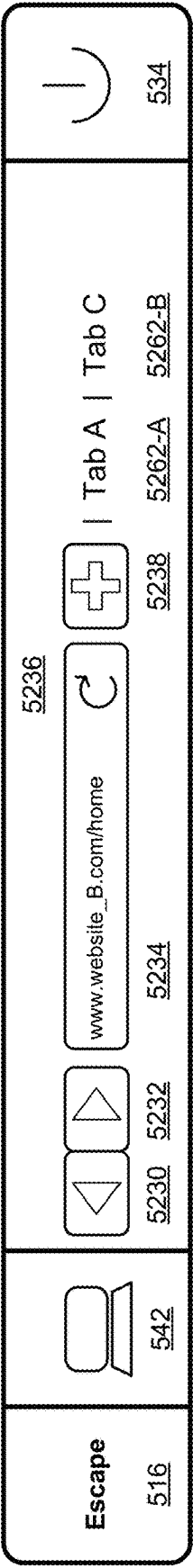


Figure 5SS

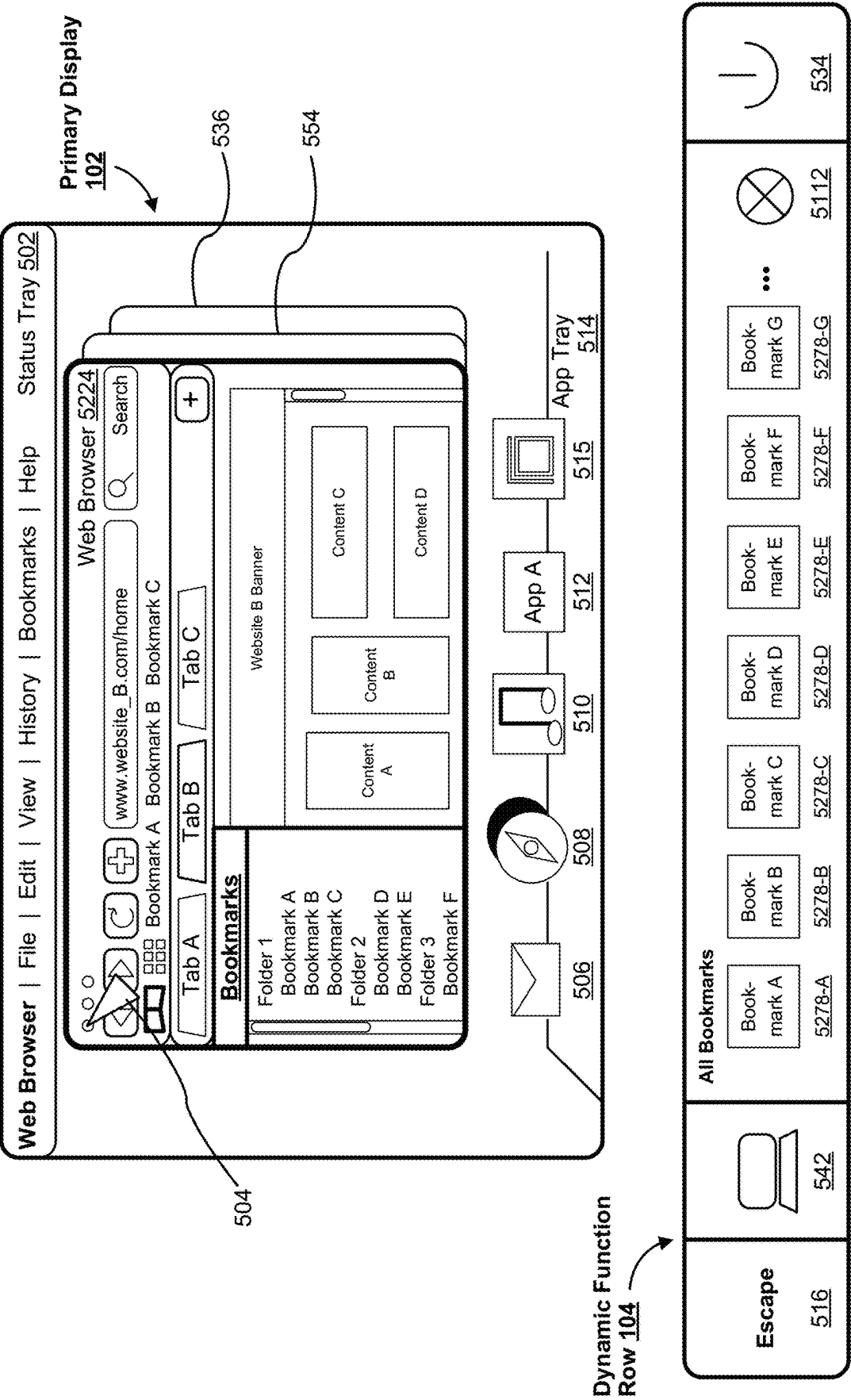


Figure 5TT

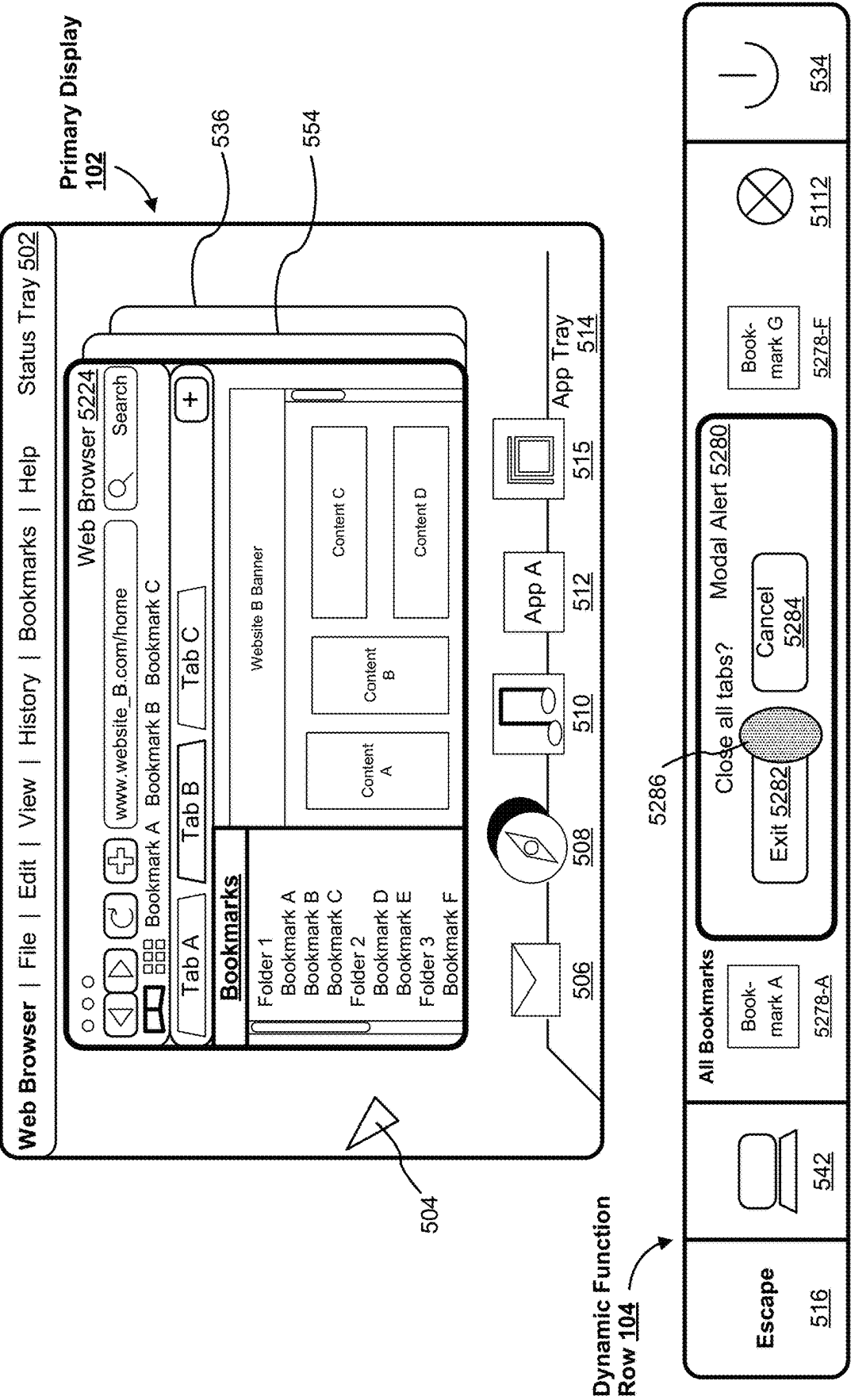
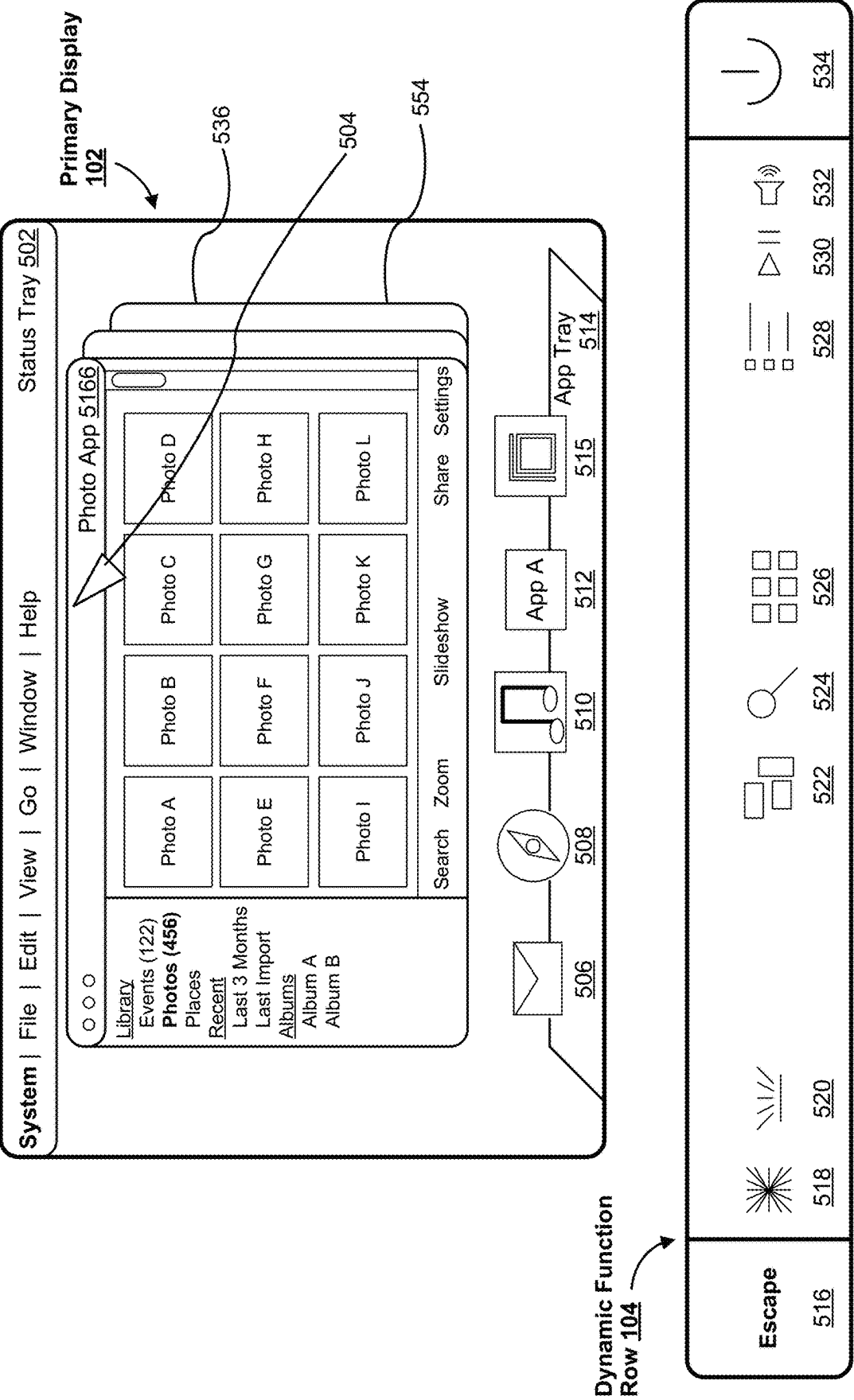


Figure 5UU



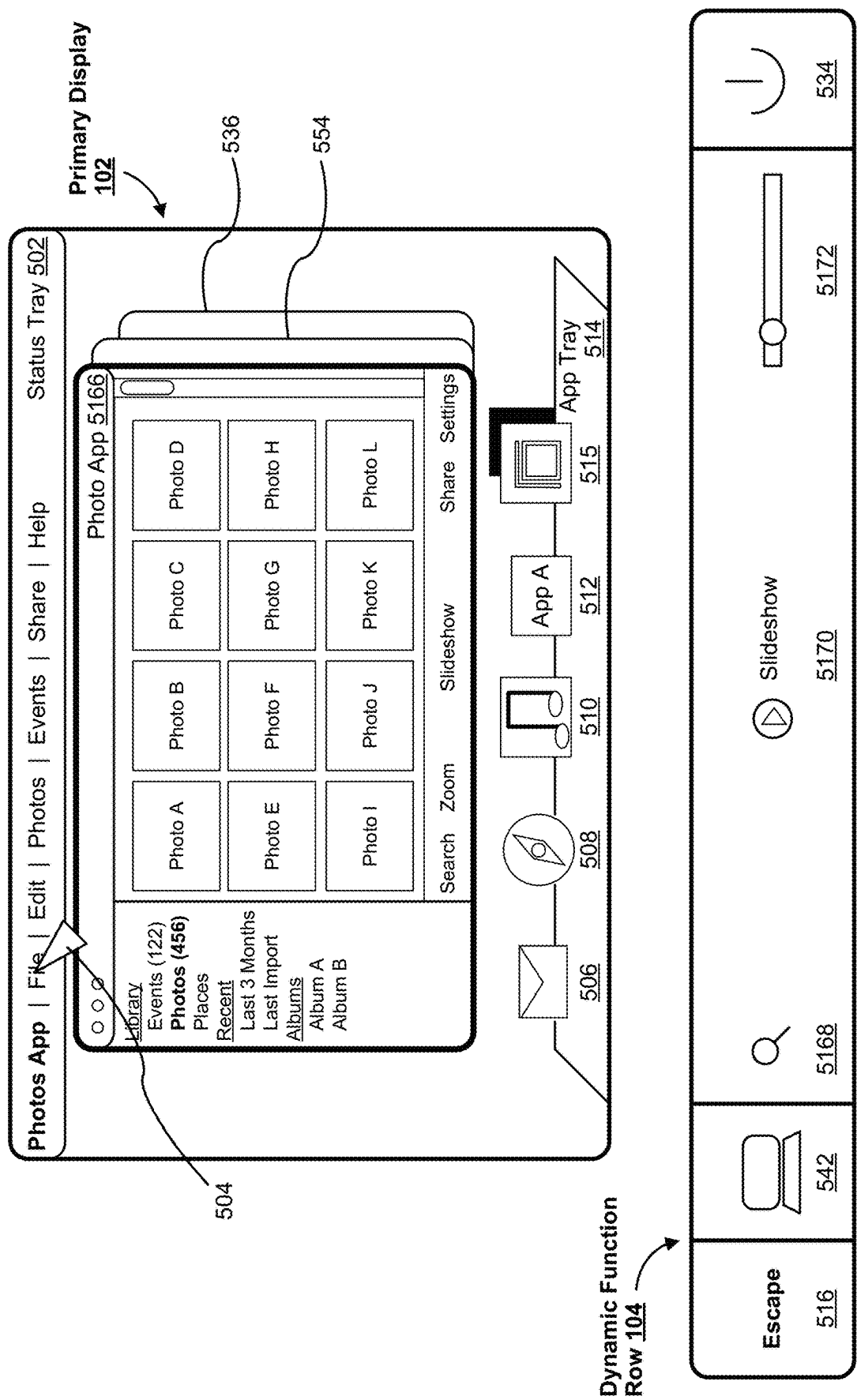


Figure 5WW

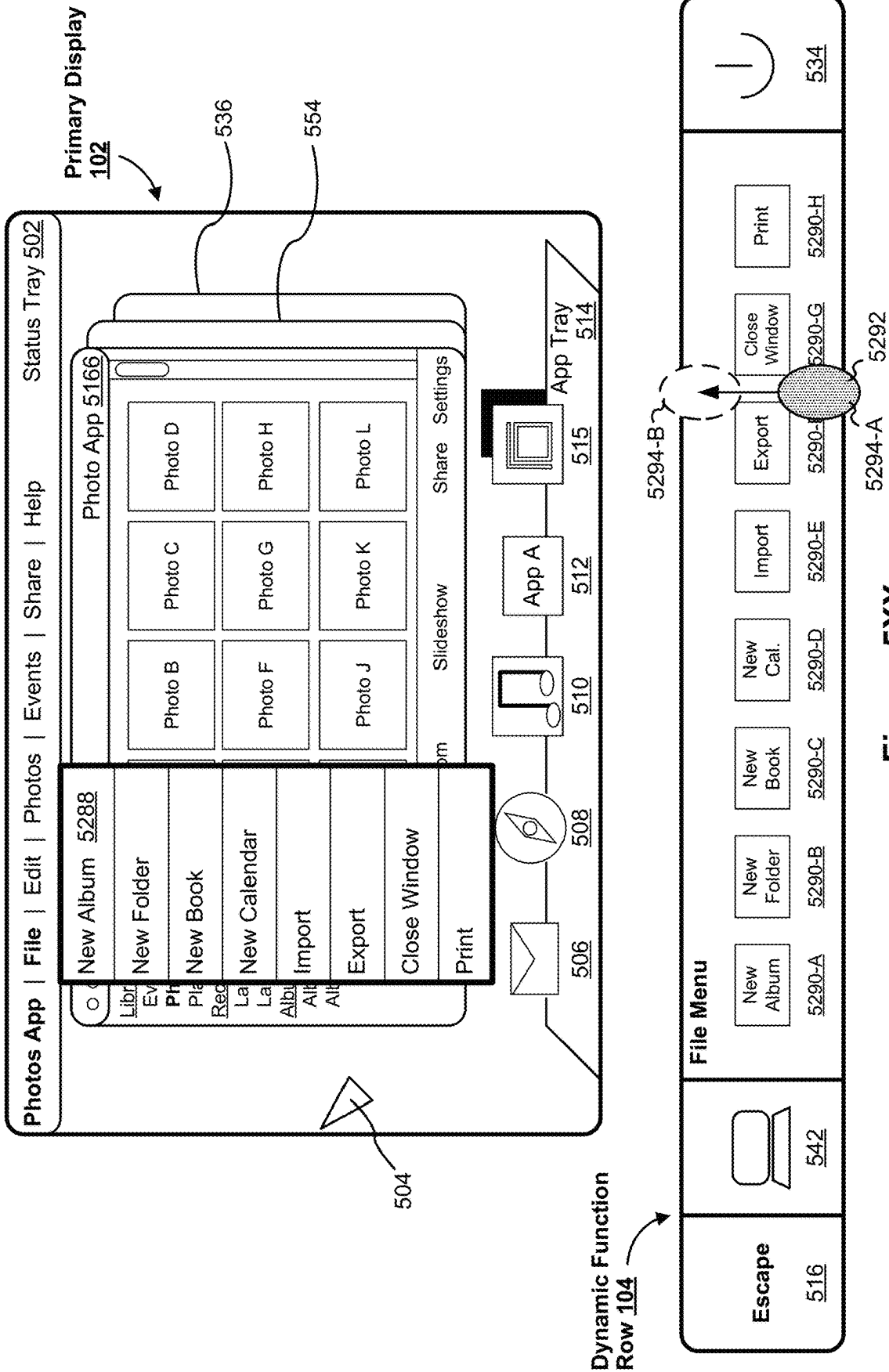


Figure 5XX

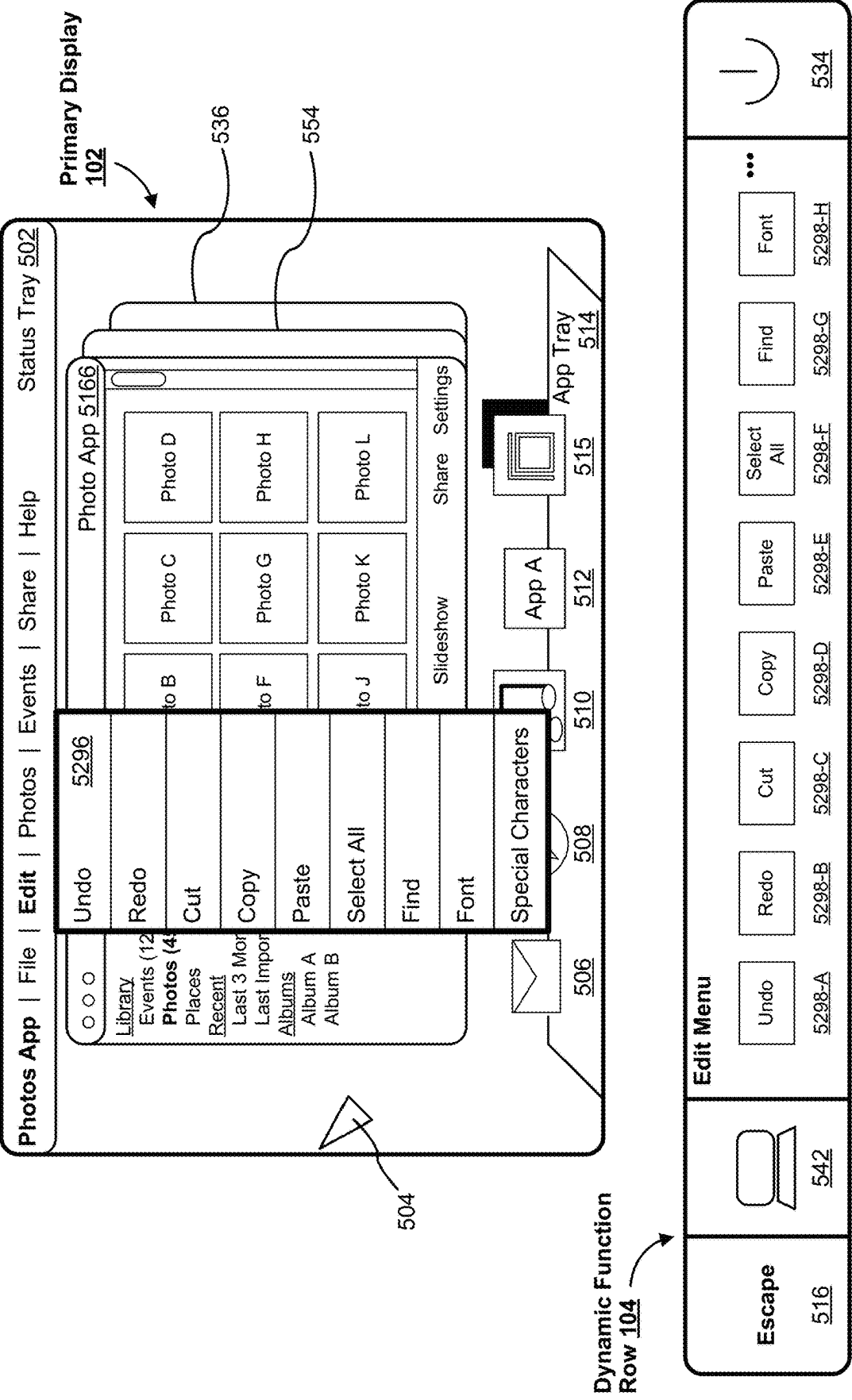


Figure 5YY

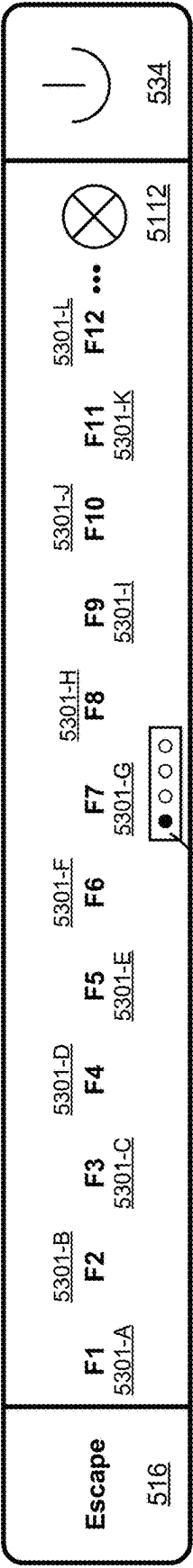
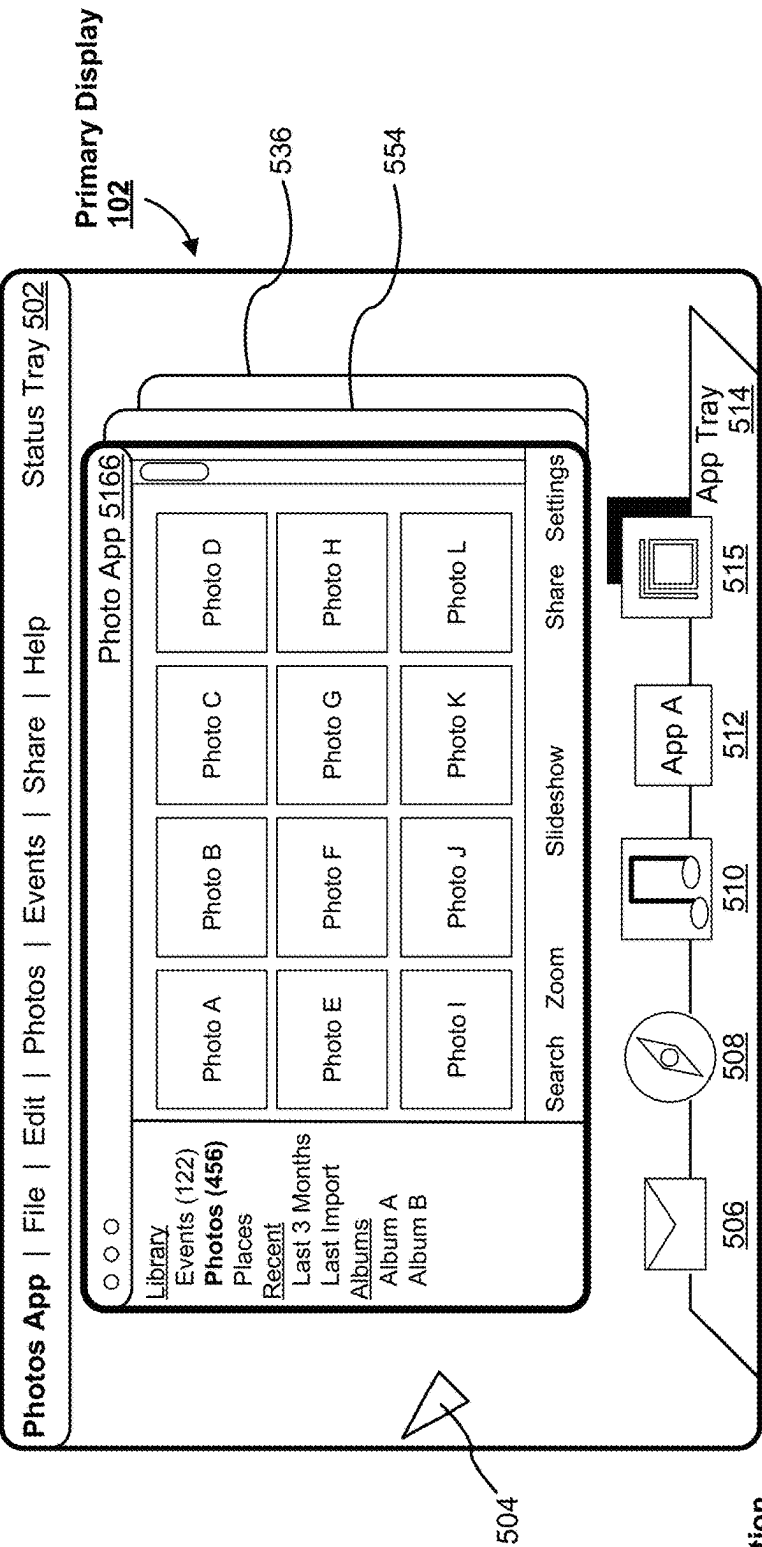


Figure 5ZZ

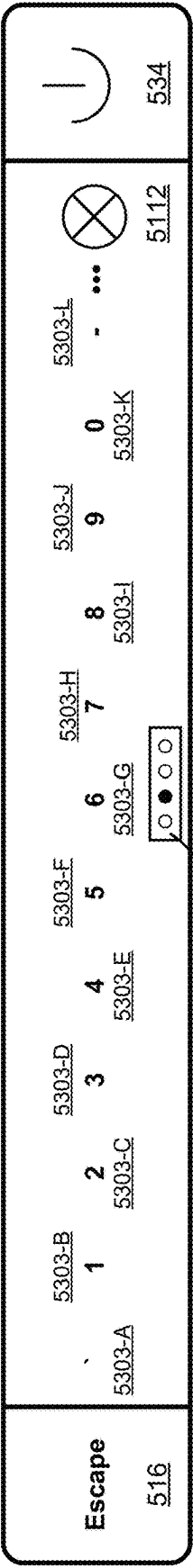
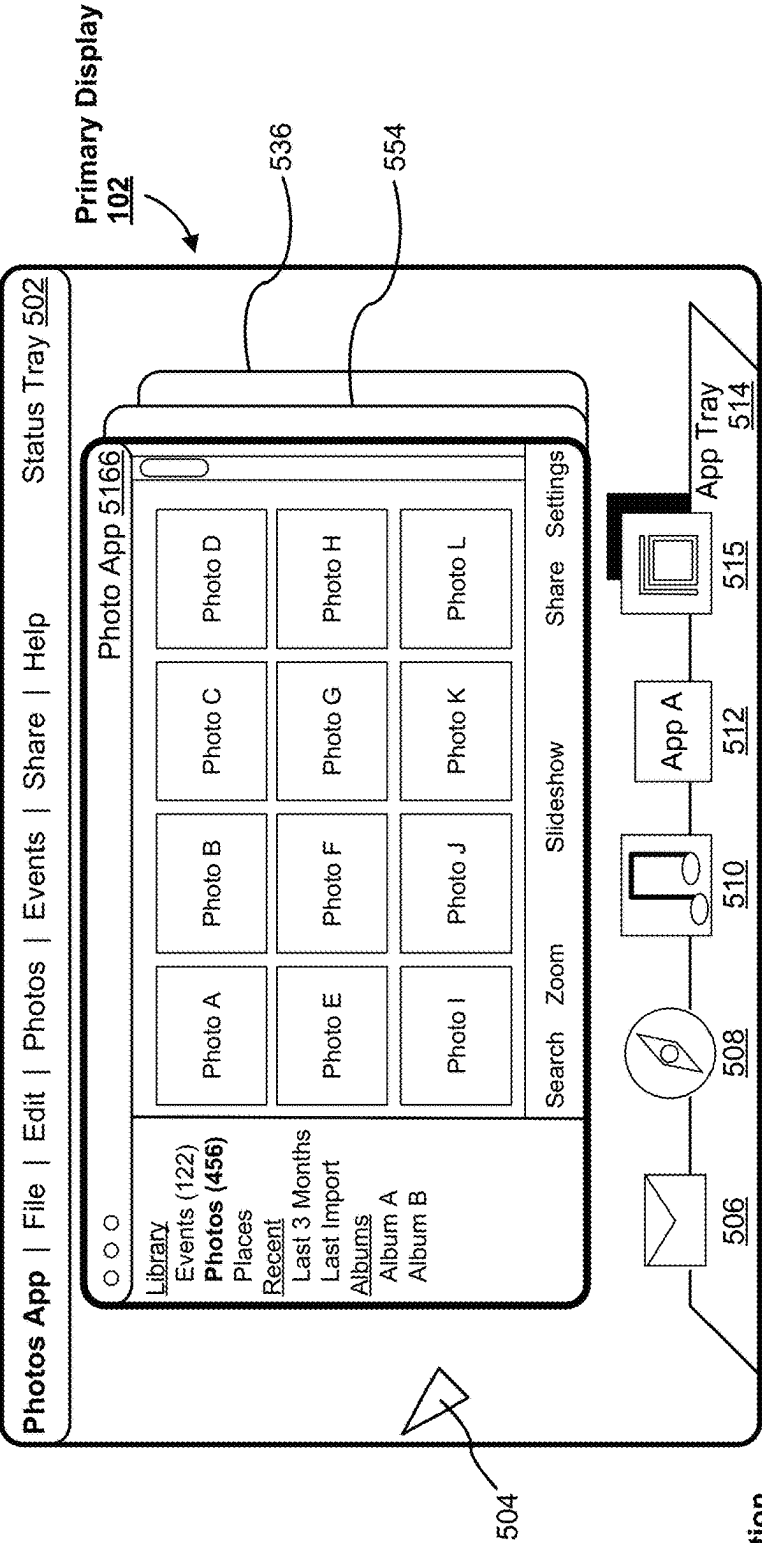
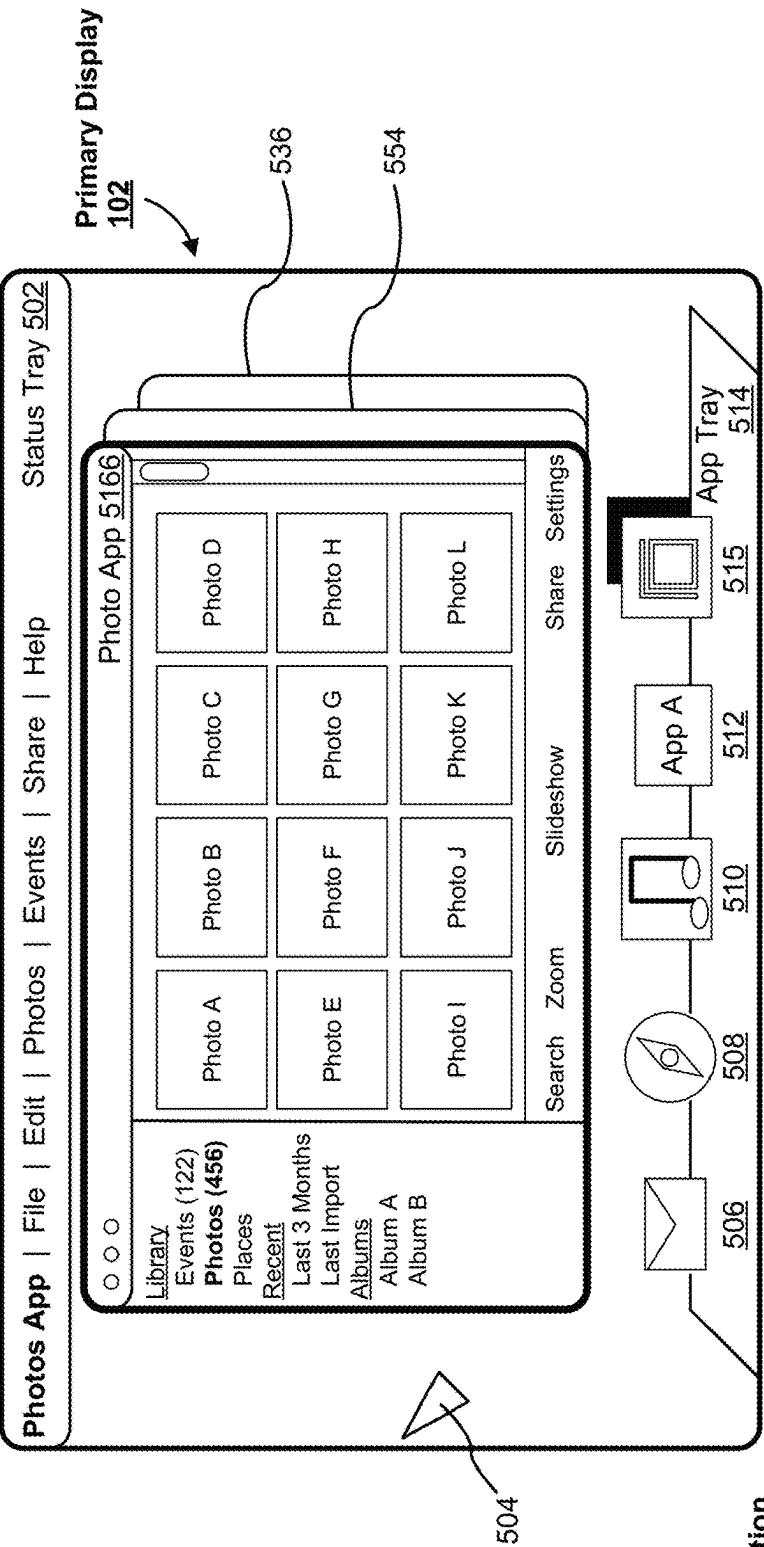
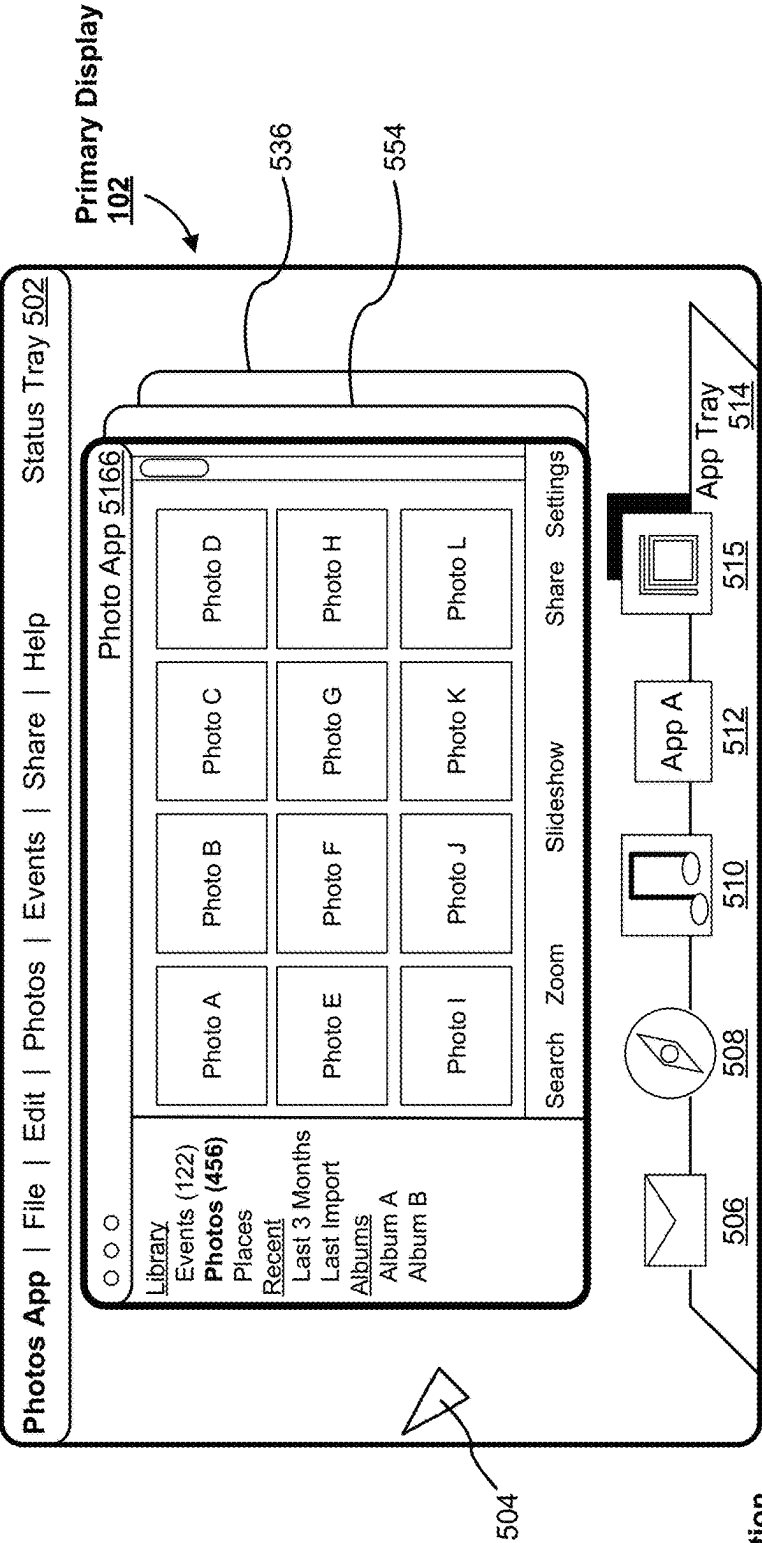


Figure 5AAA



Escape 516	5304-A ~	5304-B !	5304-C @	5304-D #	5304-E \$	5304-F %	5304-G ^	5304-H &	5304-I *	5304-J (	5304-K)	5304-L -	5304-M ...	5112 ⊗	534 ⌵
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Figure 5BBB



Dynamic Function Row 104

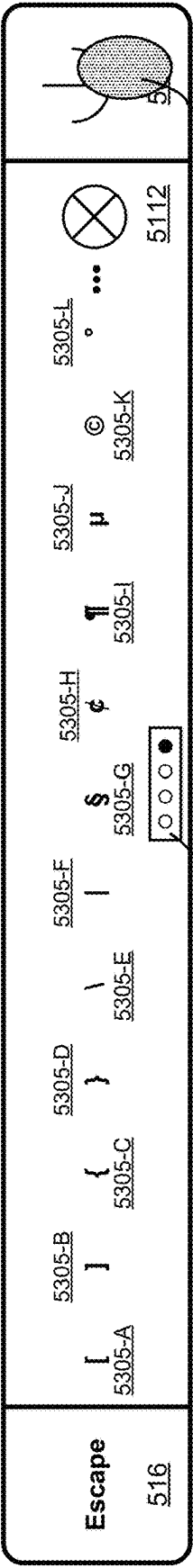


Figure 502

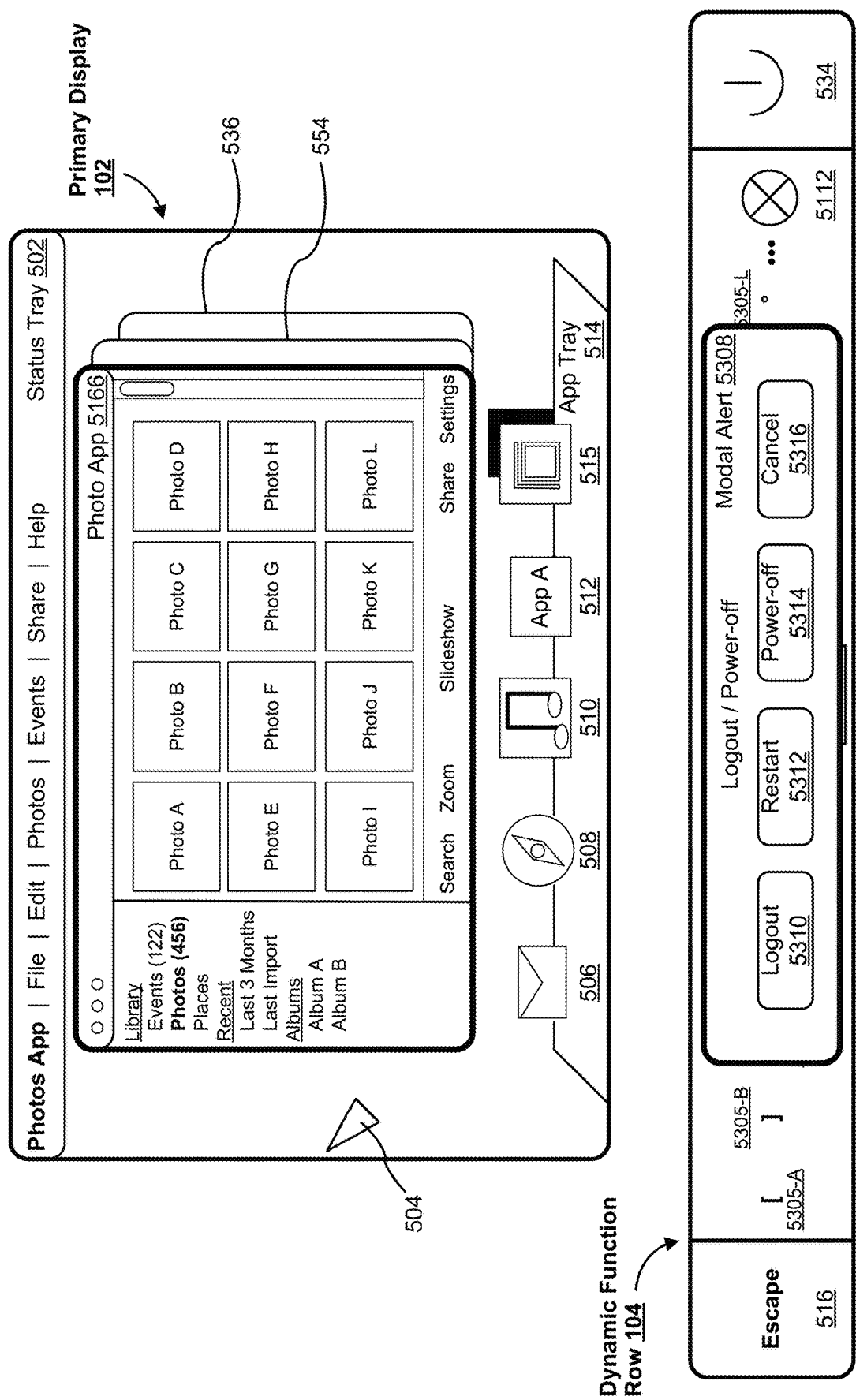


Figure 5DDD

600

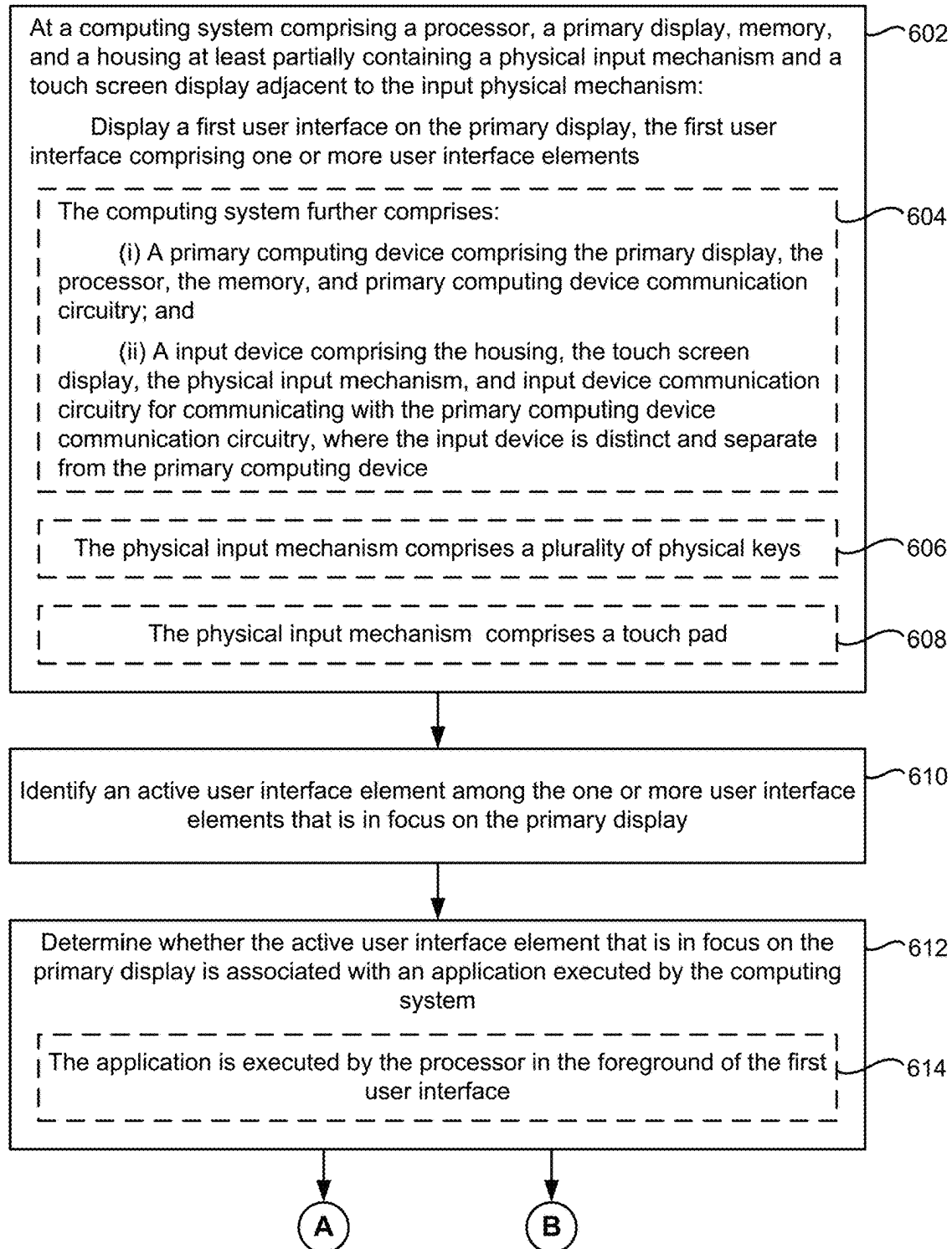


Figure 6A

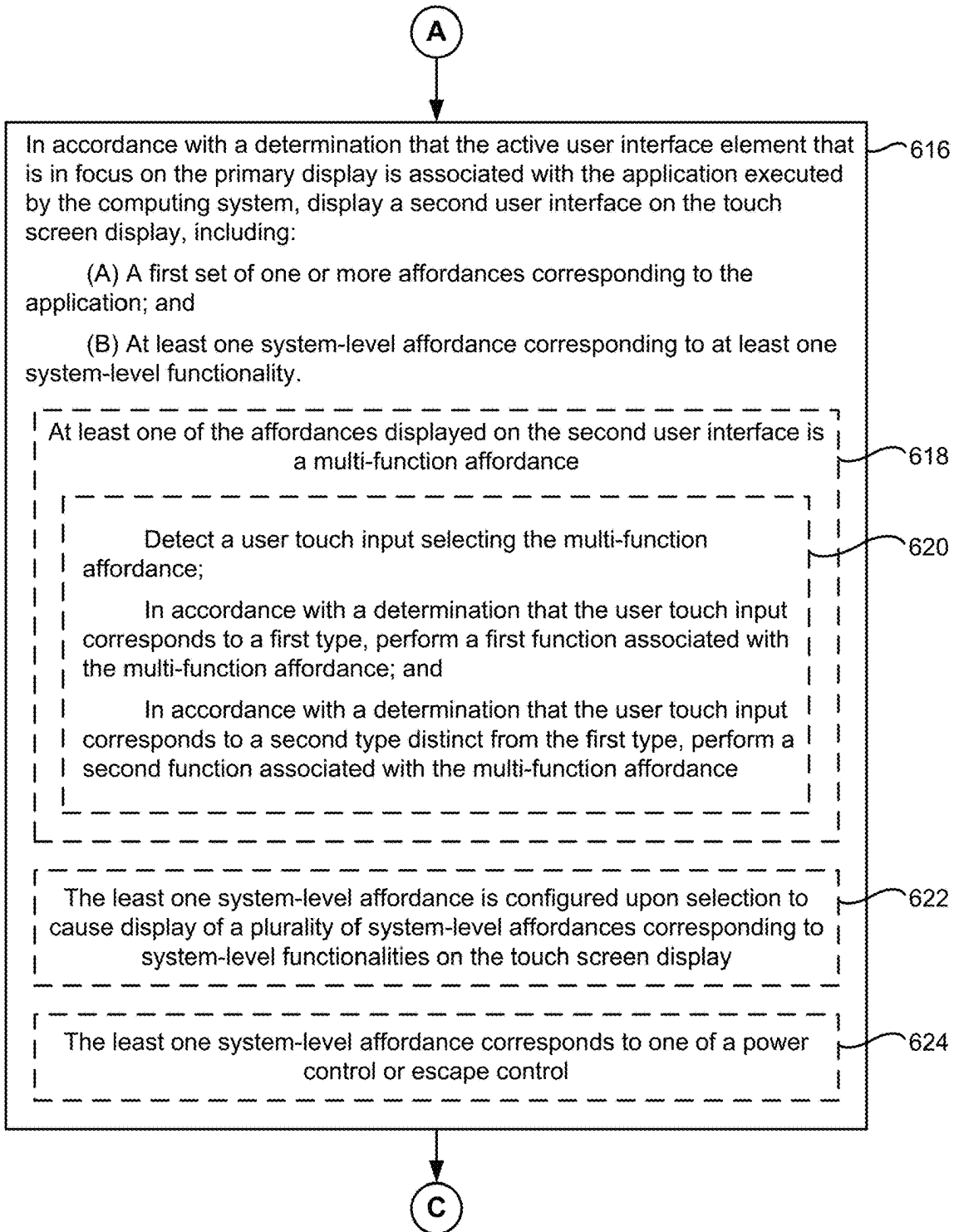
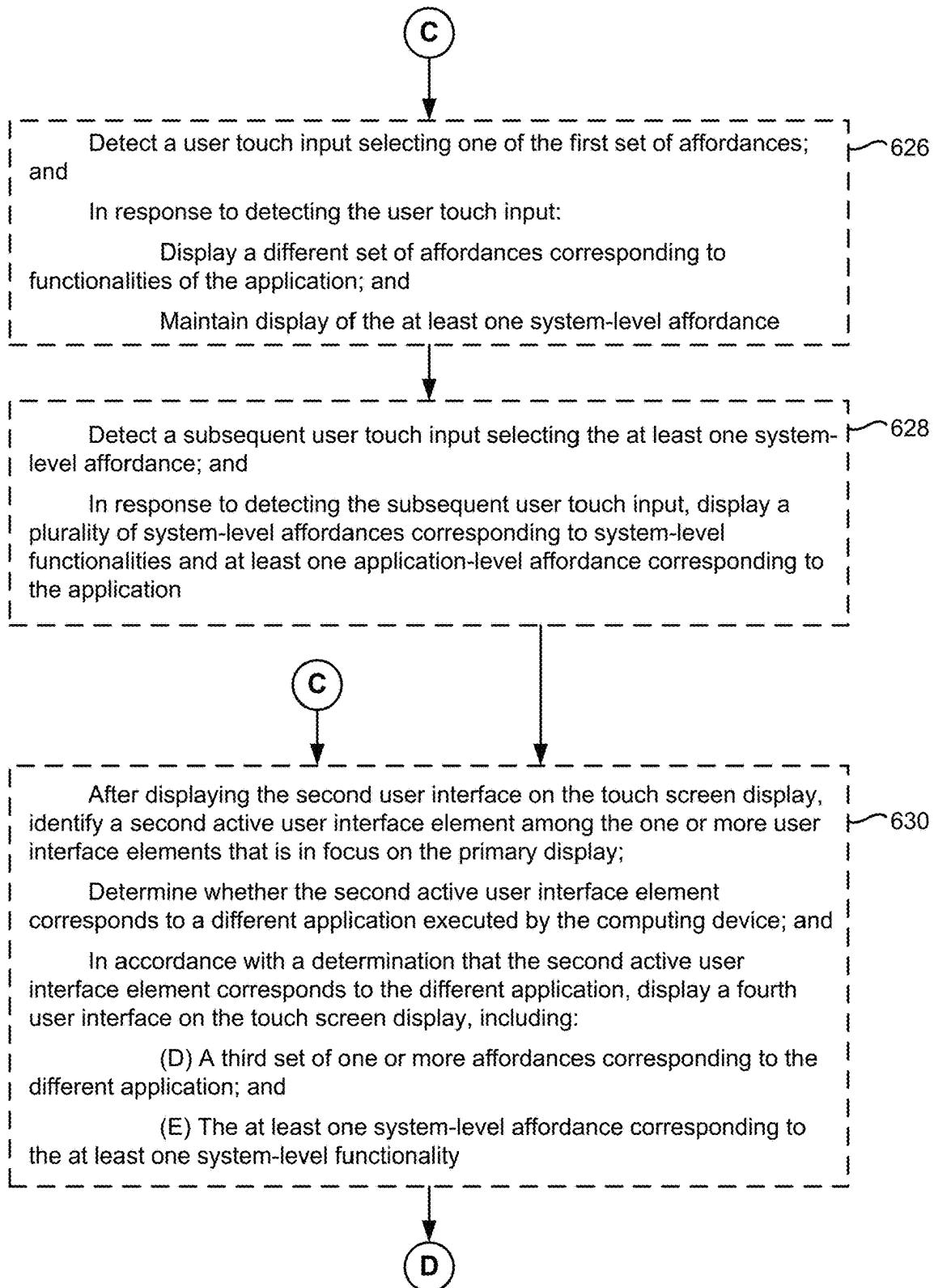
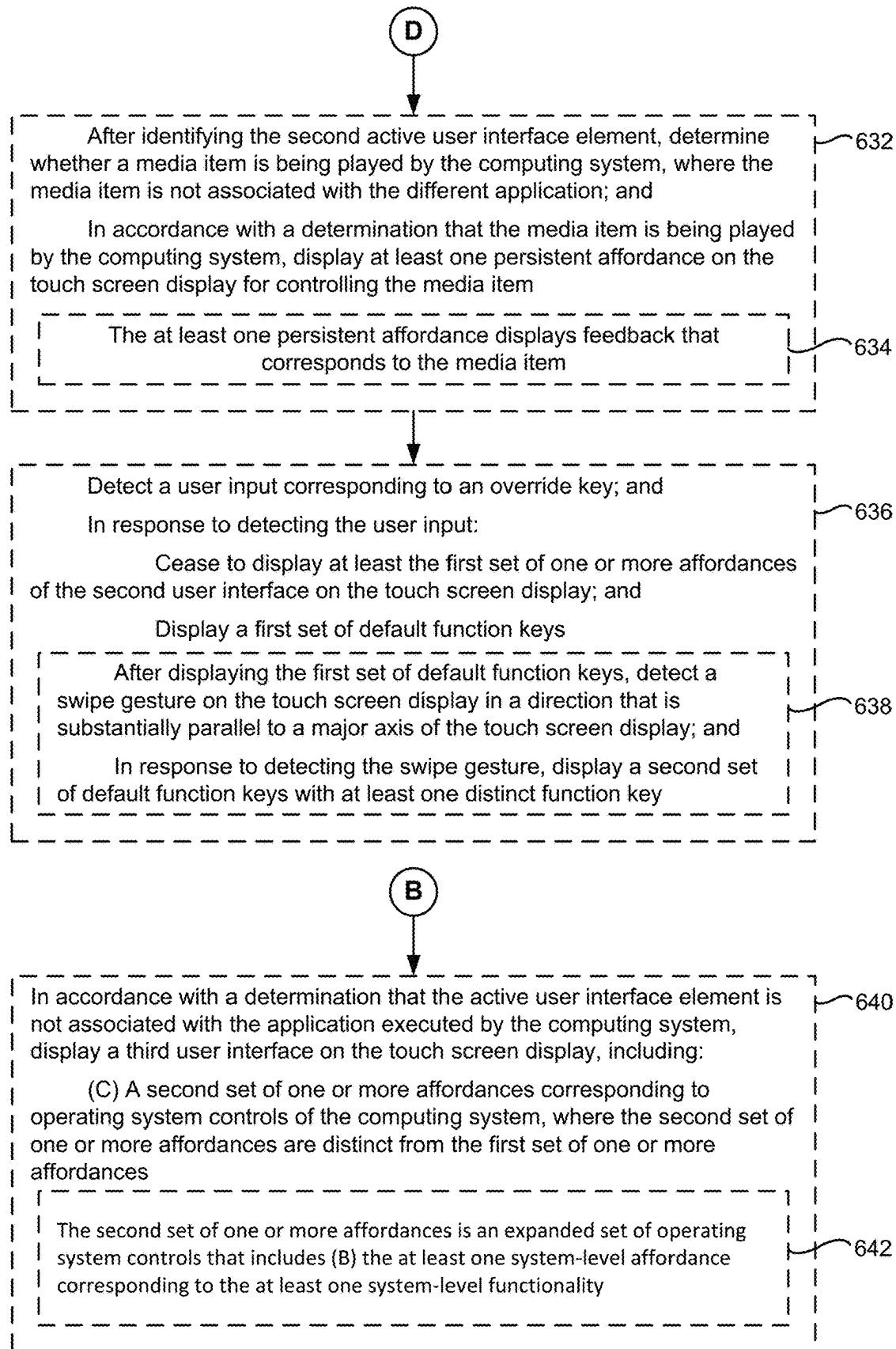
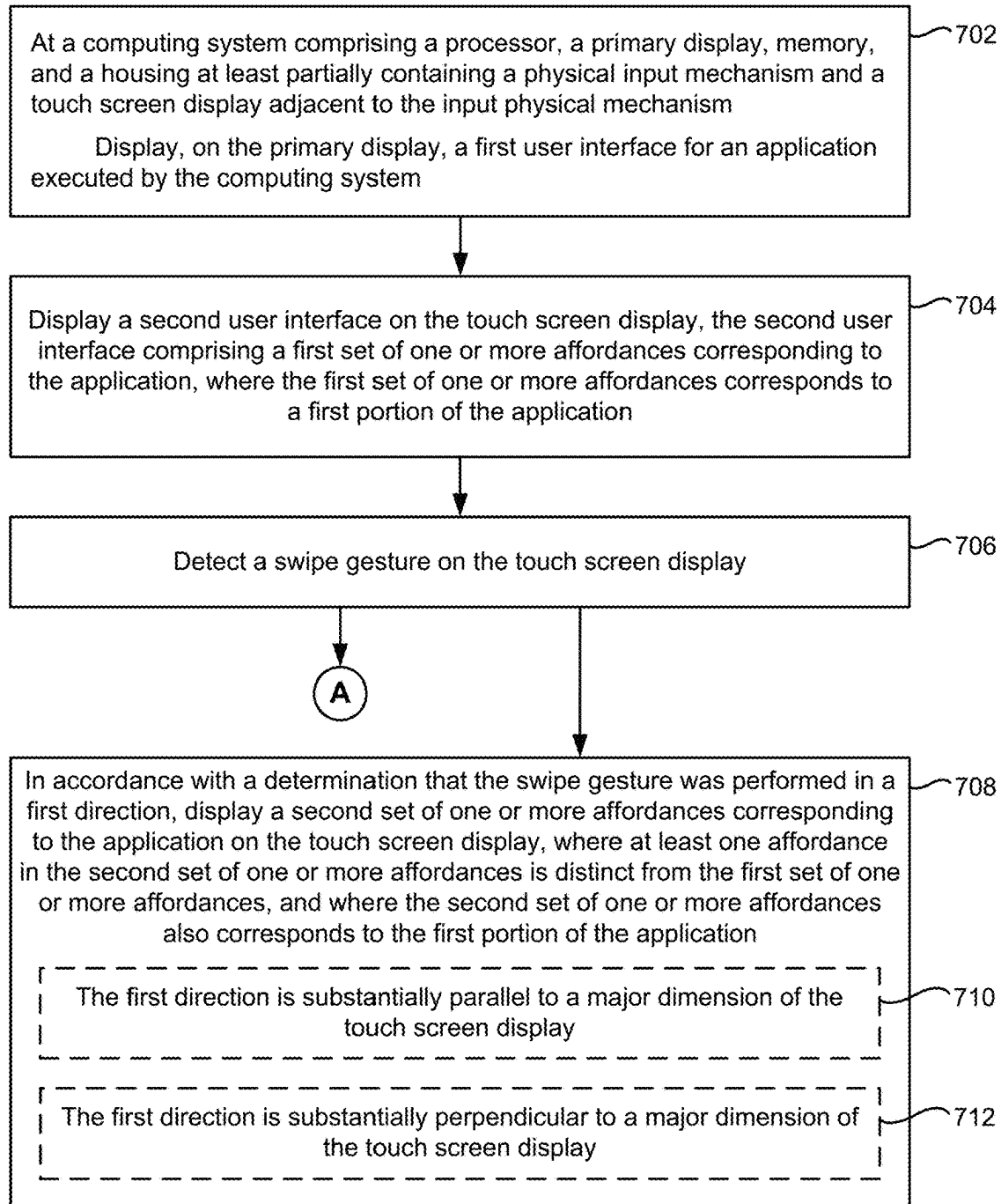


Figure 6B

**Figure 6C**

**Figure 6D**

700**Figure 7A**

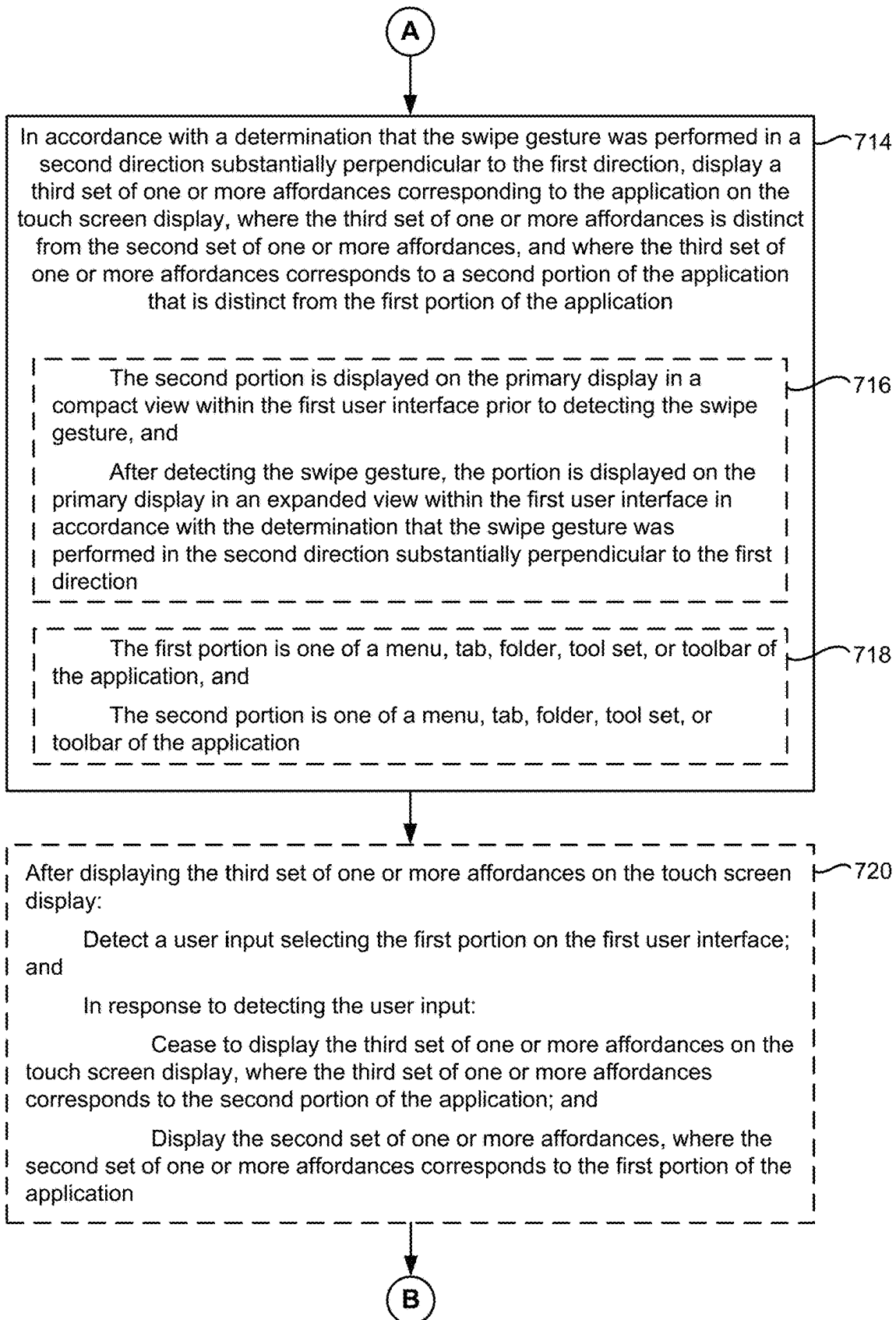
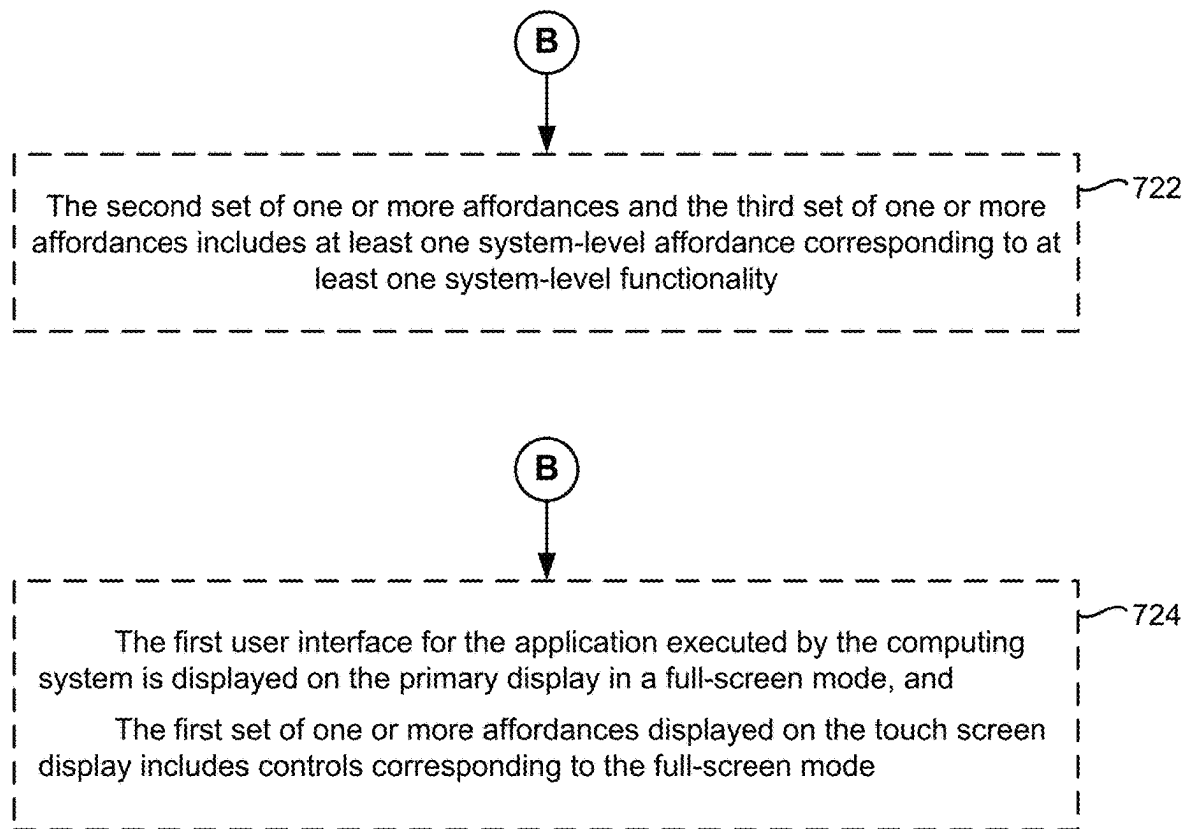


Figure 7B

**Figure 7C**

800

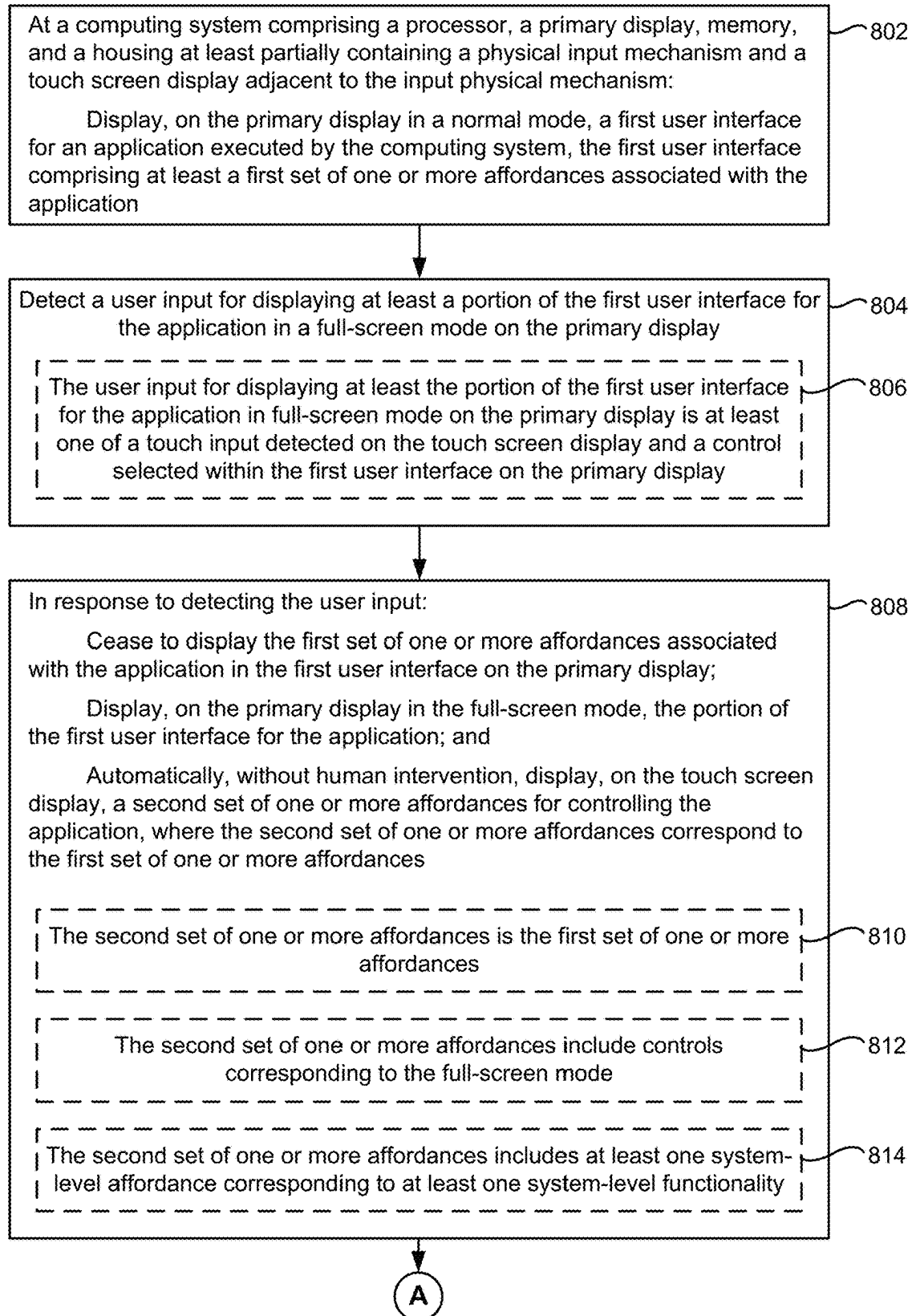


Figure 8A

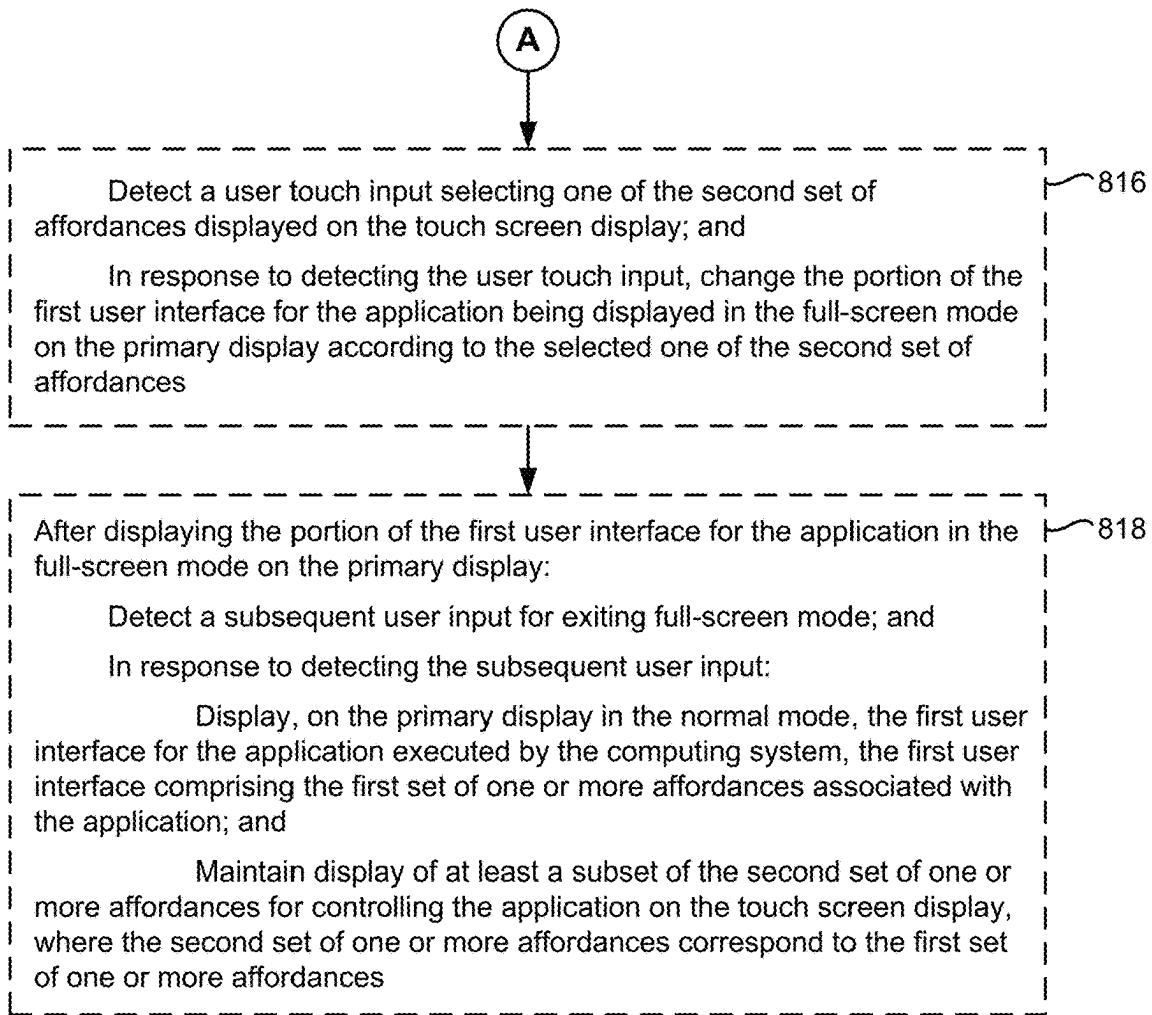
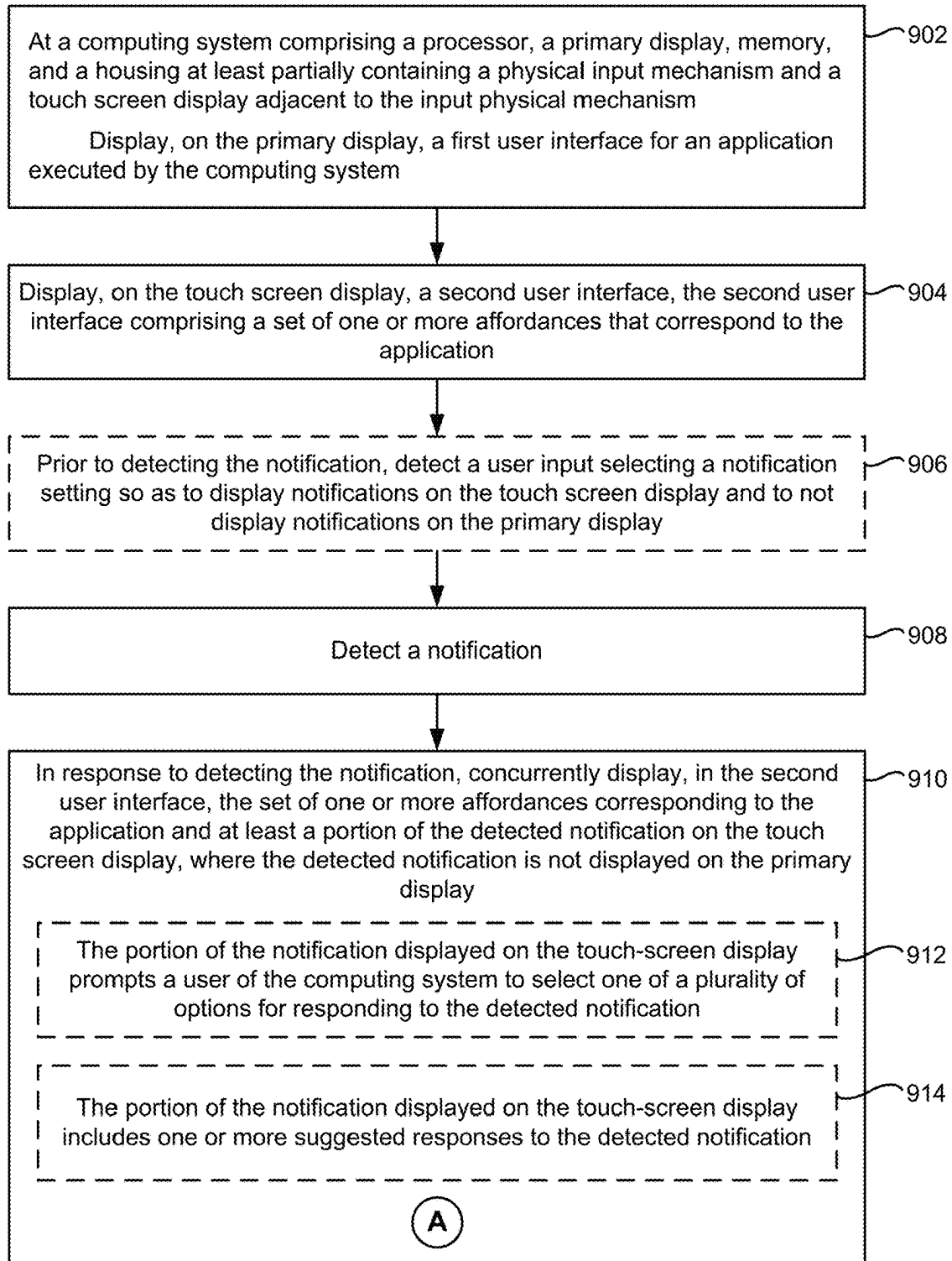


Figure 8B

900**Figure 9A**

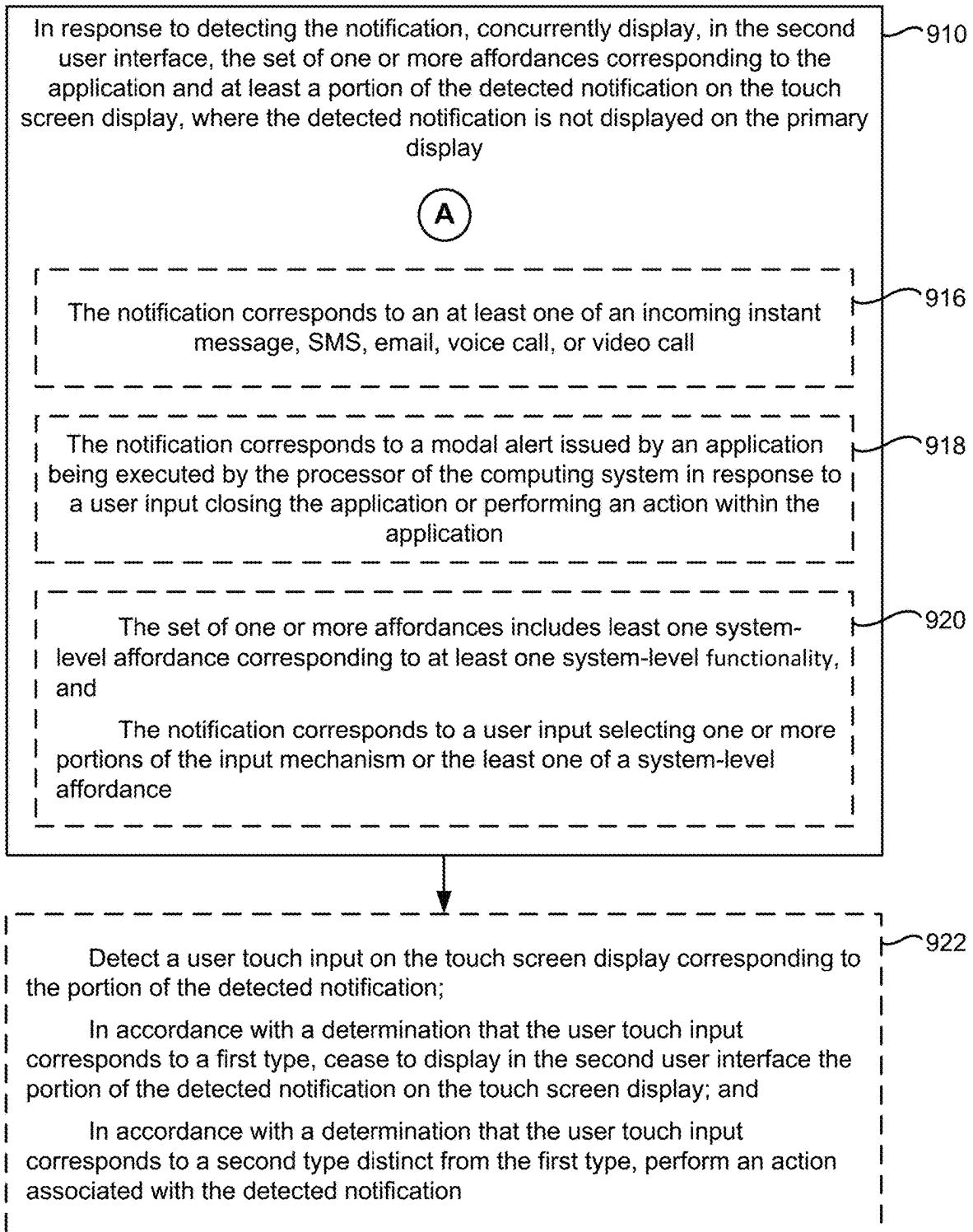
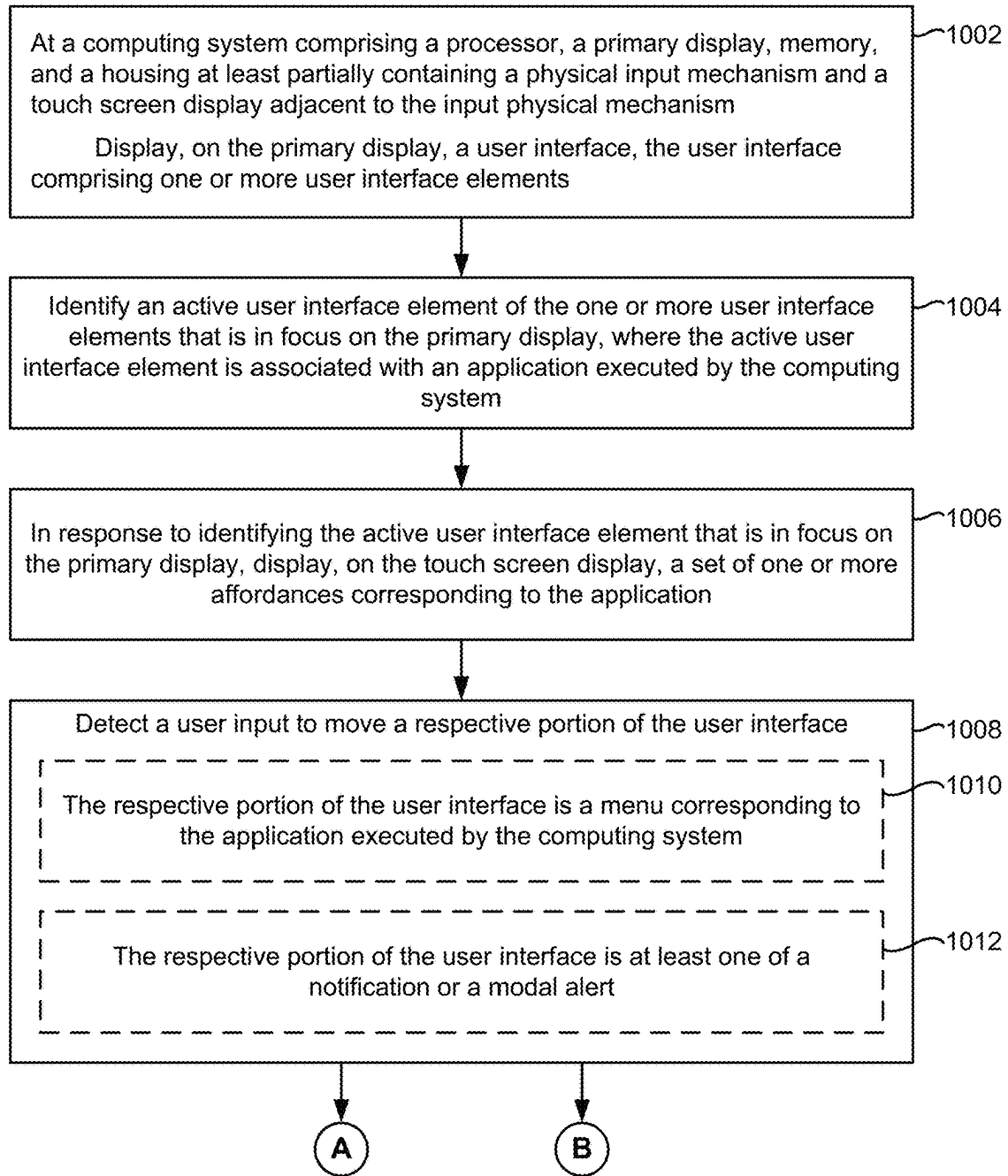


Figure 9B

1000**Figure 10A**

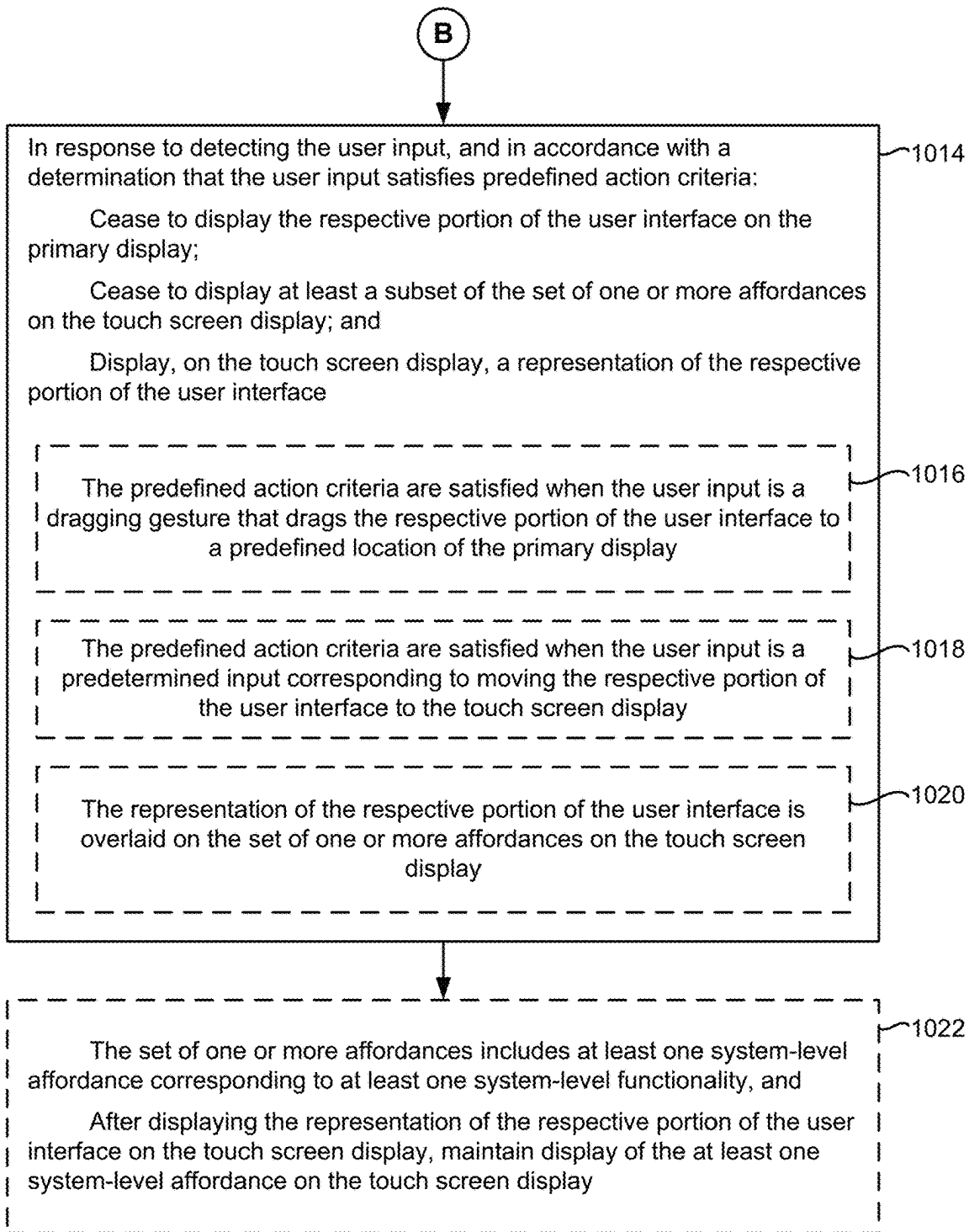
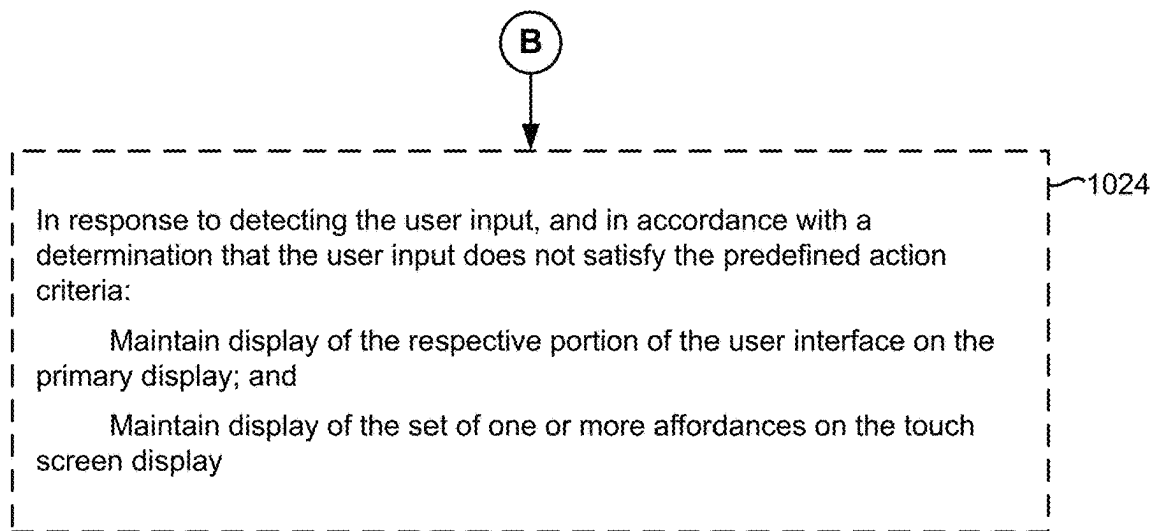


Figure 10B

**Figure 10C**

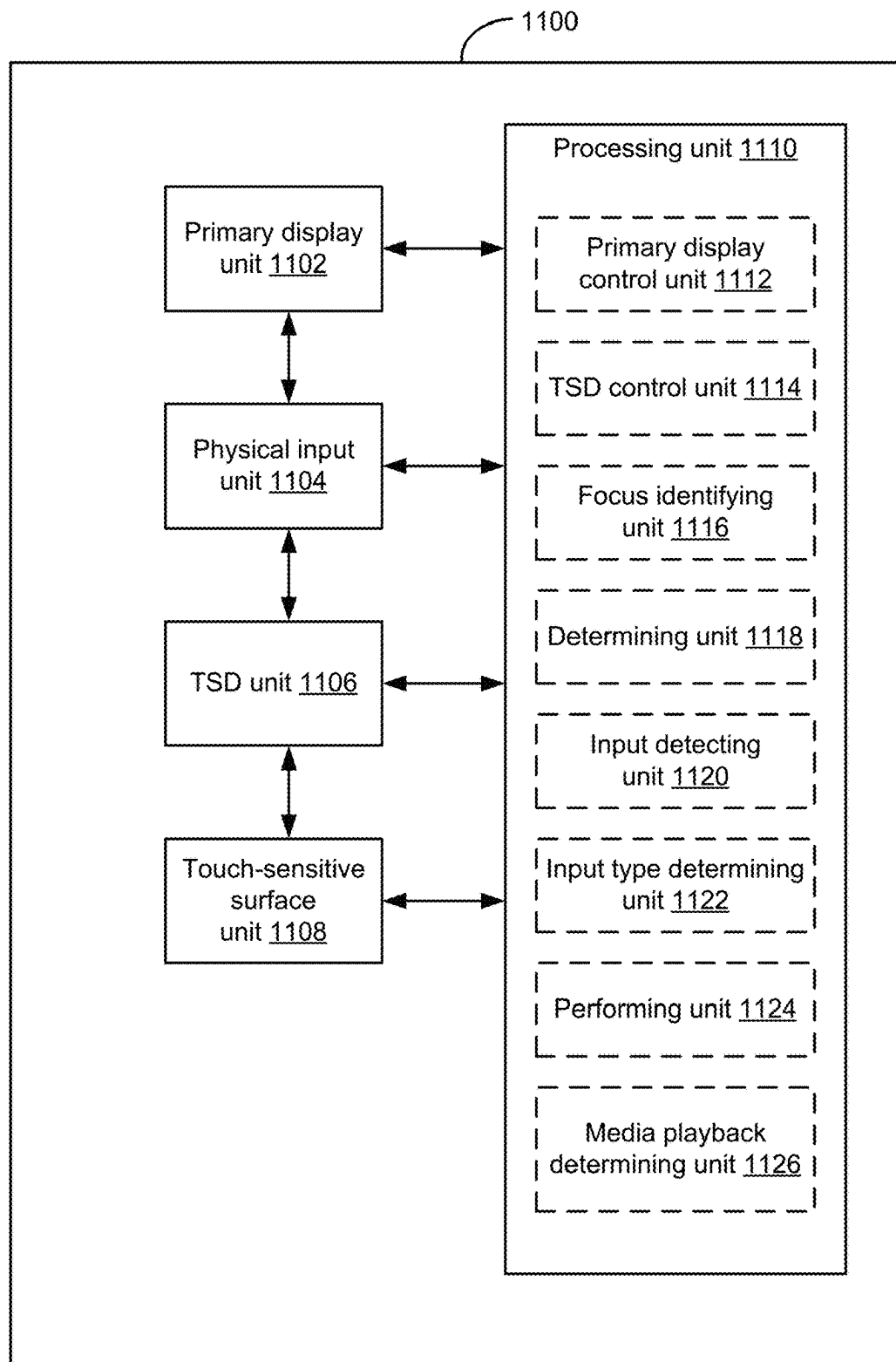


Figure 11

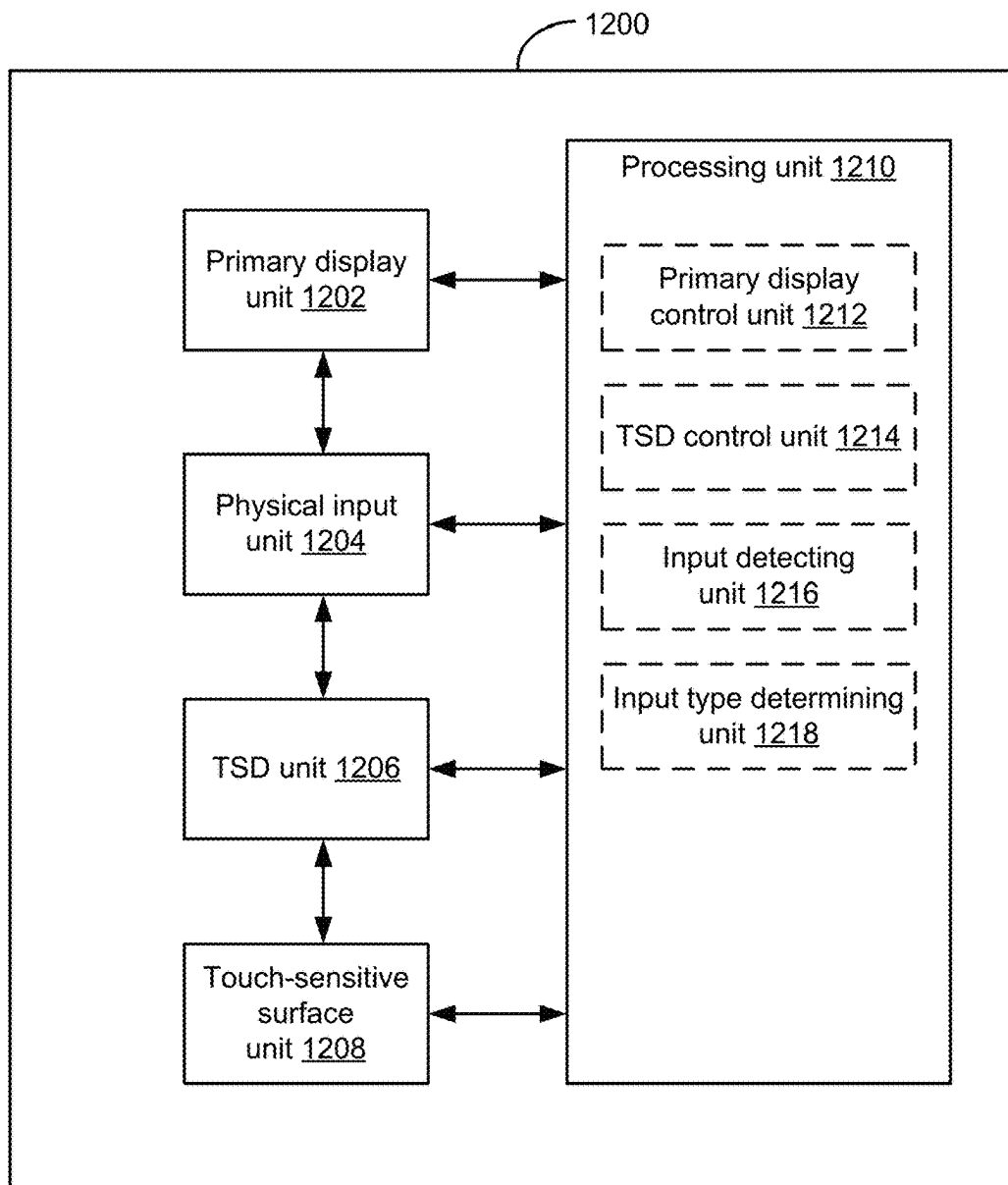


Figure 12

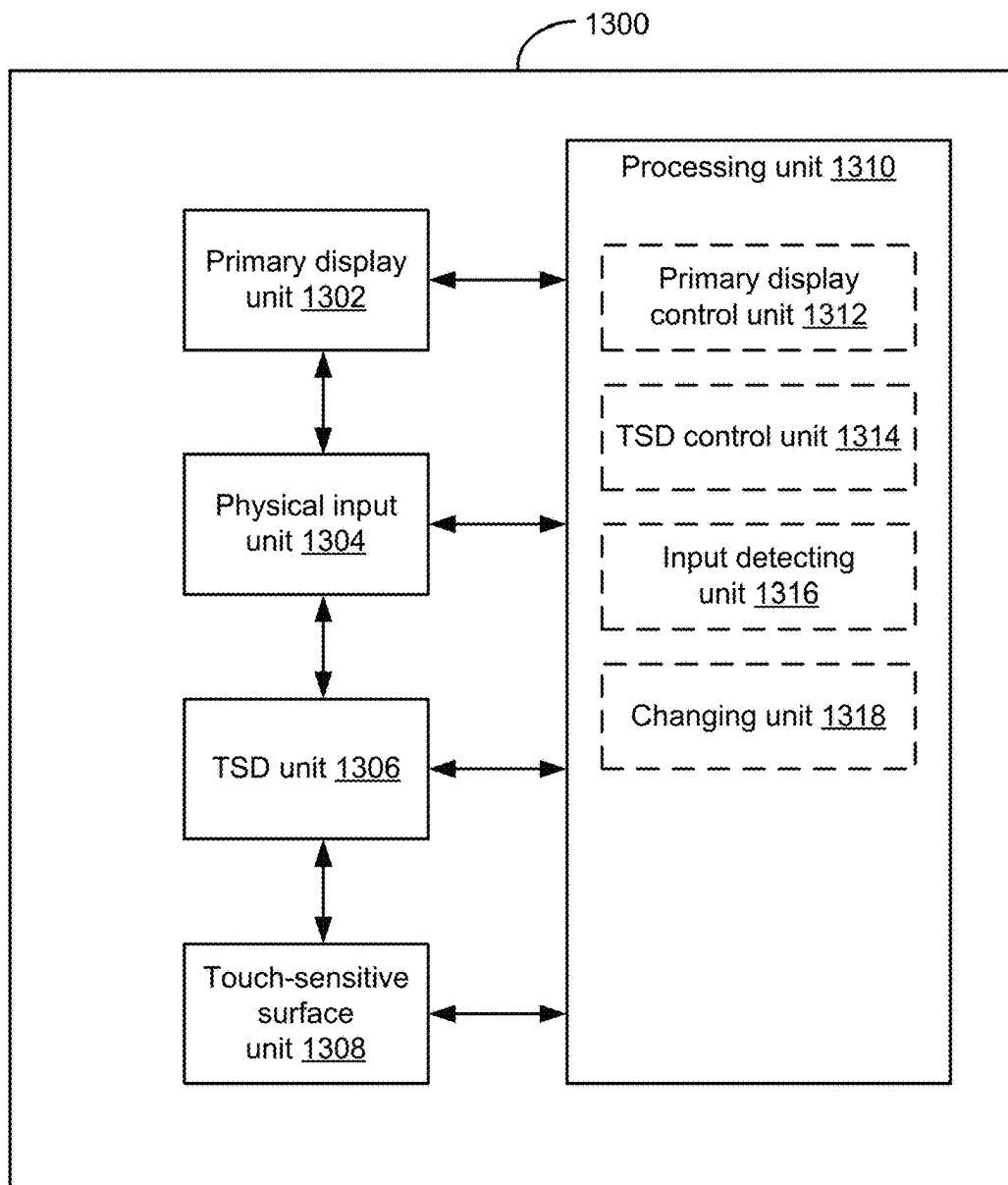


Figure 13

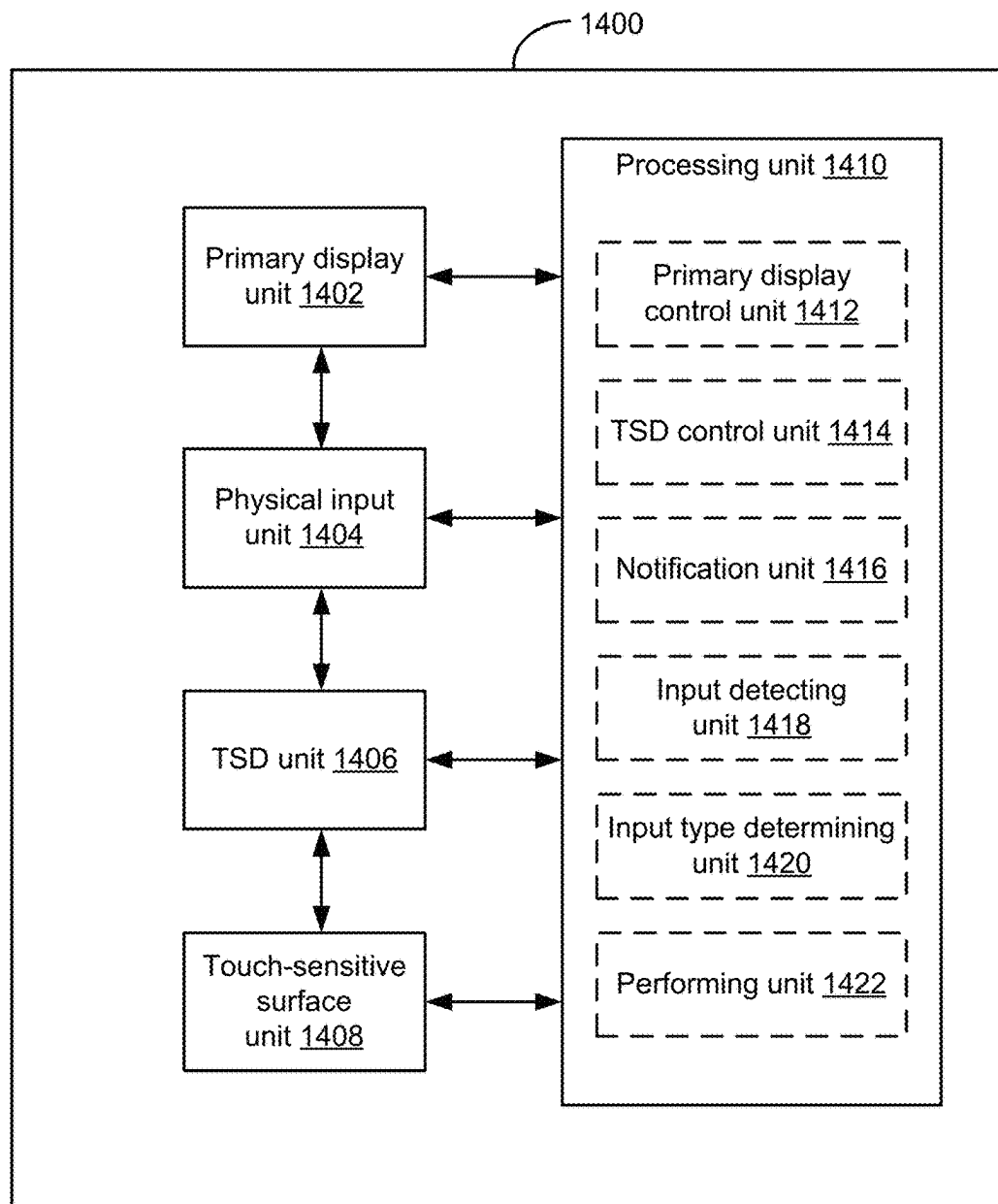


Figure 14

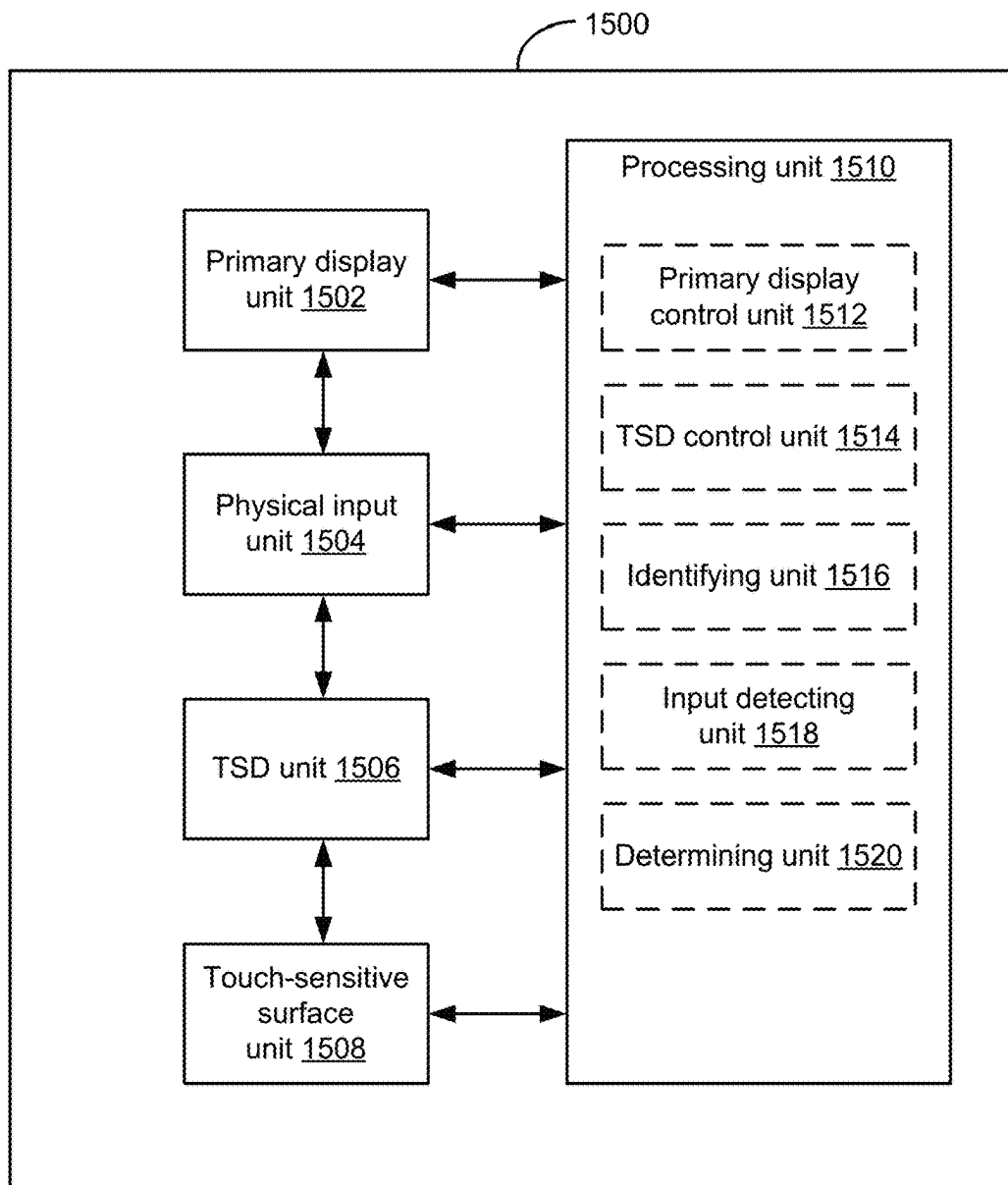


Figure 15

SYSTEMS, DEVICES, AND METHODS FOR DYNAMICALLY PROVIDING USER INTERFACE CONTROLS AT A TOUCH-SENSITIVE SECONDARY DEVICE

RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/705,186, filed Mar. 25, 2022, which is a continuation of U.S. patent application Ser. No. 17/086,248, filed Oct. 30, 2020, now U.S. Pat. No. 11,321,041, which is a continuation of U.S. patent application Ser. No. 16/835,096, filed Mar. 30, 2020, now U.S. Pat. No. 10,908,864, which is a continuation of U.S. patent application Ser. No. 15/113,779, filed Jul. 22, 2016, now U.S. Pat. No. 10,606,539, which is a national phase entry of International Application No. PCT/US2015/012694, filed Jan. 23, 2015, which claims priority to U.S. Provisional Application Ser. No. 62/104,023, filed Jan. 15, 2015 and U.S. Provisional Application Ser. No. 61/930,663, filed Jan. 23, 2014, each of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

The disclosed embodiments relate to keyboards and, more specifically, to improved techniques for receiving input via a dynamic input and output (I/O) device.

BACKGROUND

Conventional keyboards include any number of physical keys for inputting information (e.g., characters) into the computing device. Typically, the user presses or otherwise movably actuates a key to provide input corresponding to the key. In addition to providing inputs for characters, a keyboard may include movably actuated keys related to function inputs. For example, a keyboard may include an “escape” or “esc” key to allow a user to activate an escape or exit function. In many keyboards, a set of functions keys for function inputs are located in a “function row.” Typically, a set of keys for alphanumeric characters is located in a part of the keyboard that is closest to the user and a function row is located in a part of the keyboard that is further away from the user but adjacent to the alphanumeric characters. A keyboard may also include function keys that are not part of the aforementioned function row.

With the advent and popularity of portable computing devices, such as laptop computers, the area consumed by the dedicated keyboard may be limited by the corresponding size of a display. Compared with a peripheral keyboard for a desktop computer, a dedicated keyboard that is a component of a portable computing device may have fewer keys, smaller keys, or keys that are closer together to allow for a smaller overall size of the portable computing device.

Conventional dedicated keyboards are static and fixed in time regardless of the changes on a display. Furthermore, the functions of software application displayed on a screen are typically accessed via toolbars and menus that a user interacts with by using a mouse. This periodically requires the user to switch modes and move the location of his/her hands between keyboard and mouse. Alternatively, the application’s functions are accessed via complicated key combinations that require memory and practice. As such, it is desirable to provide an I/O device and method that addresses the shortcomings of conventional systems.

SUMMARY

The embodiments described herein address the above shortcomings by providing dynamic and space efficient I/O

devices and methods. Such devices and methods optionally complement or replace conventional input devices and methods. Such devices and methods also reduce the amount of mode switching (e.g., moving one’s hands between keyboard and mouse, and also moving one’s eyes from keyboard to display) required of a user and produce a more efficient human-machine interface.

In accordance with some embodiments, a method of updating a dynamic input and output device is performed at a computing system comprising a processor, a primary display, memory, and a housing at least partially containing a physical input mechanism and a touch screen display adjacent to the physical input mechanism (e.g., portable computing system **100**, FIG. **1A-1B**). In some embodiments, the housing is separate and distinct from the primary display (e.g., desktop computing system **200**, FIGS. **2A-2C**). In other embodiments, the housing is separate and distinct from the processor, the primary display, and the memory (e.g., desktop computing system **200**, FIGS. **2A-2C**). In some embodiments, the method includes: displaying a first user interface on the primary display, the first user interface comprising one or more user interface elements; identifying an active user interface element among the one or more user interface elements that is in focus on the primary display; and determining whether the active user interface element that is in focus on the primary display is associated with an application executed by the computing system. In accordance with a determination that the active user interface element that is in focus on the primary display is associated with the application executed by the computing system, the method includes displaying a second user interface on the touch screen display, including: (A) a first set of one or more affordances corresponding to the application; and (B) at least one system-level affordance corresponding to at least one system-level functionality.

Some embodiments provide a different method including: displaying, on the primary display, a first user interface for an application executed by the computing system; displaying, on the touch screen display, a second user interface, the second user interface comprising a first set of one or more affordances corresponding to the application, where the first set of one or more affordances corresponds to a first portion of the application; and detecting a swipe gesture on the touch screen display. In accordance with a determination that the swipe gesture was performed in a first direction, the method includes displaying a second set of one or more affordances corresponding to the application on the touch screen display, where at least one affordance in the second set of one or more affordances is distinct from the first set of one or more affordances, and where the second set of one or more affordances also corresponds to the first portion of the application. In accordance with a determination that the swipe gesture was performed in a second direction substantially perpendicular to the first direction, the method includes displaying a third set of one or more affordances corresponding to the application on the touch screen display, where the third set of one or more affordances is distinct from the second set of one or more affordances, and where the third set of one or more affordances corresponds to a second portion of the application that is distinct from the first portion of the application. An examples of a different portions of the first user interface include the menu of file controls **5288** associated with the photo application in FIG. **5XX** and the menu of edit controls **5296** associated with the photo application in FIG. **5YY**.

Other embodiments provide a different method including: displaying, on the primary display in a normal mode (i.e.,

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non-full-screen mode), a first user interface for the application executed by the computing system, the first user interface comprising a first set of one or more affordances associated with the application; and detecting a user input for displaying at least a portion of the first user interface for the application in a full-screen mode on the primary display. In response to detecting the user input, the method includes: ceasing to display the first set of one or more affordances associated with the application in the first user interface on the primary display; displaying, on the primary display in the full-screen mode, the portion of the first user interface for the application; and automatically, without human intervention, displaying, on the touch screen display, a second set of one or more affordances for controlling the application, where the second set of one or more affordances correspond to the first set of one or more affordances.

In some embodiments, the method includes: displaying, on the primary display, a first user interface for an application executed by the computing system; displaying, on the touch screen display, a second user interface, the second user interface comprising a set of one or more affordances corresponding to the application; and detecting a notification. In response to detecting the notification, the method includes concurrently displaying, in the second user interface, the set of one or more affordances corresponding to the application and at least a portion of the detected notification on the touch screen display, where the detected notification is not displayed on the primary display.

In other embodiments, the method includes: displaying, on the primary display, a user interface, the user interface comprising one or more user interface elements; identifying an active user interface element of the one or more user interface elements that is in focus on the primary display, where the active user interface element is associated with an application executed by the computing system; in response to identifying the active user interface element that is in focus on the primary display, displaying, on the touch screen display, a set of one or more affordances corresponding to the application; and detecting a user input to move a respective portion of the user interface. In response to detecting the user input, and in accordance with a determination that the user input satisfies predefined action criteria, the method includes: ceasing to display the respective portion of the user interface on the primary display; ceasing to display at least a subset of the set of one or more affordances on the touch screen display; and displaying, on the touch screen display, a representation of the respective portion of the user interface.

In accordance with some embodiments, an electronic device is provided that includes a primary display unit (e.g., primary display **102**, FIGS. **1A** and **2A-2D**) configured to display information, a physical input unit configured to receive user inputs, a touch screen display unit (e.g., dynamic function row **104**, FIGS. **1A-1B** and **2A-2D**) configured to display information, a touch-sensitive surface unit configured to receive user touch inputs on the touch screen display unit, and a processing unit coupled to the primary display unit, the physical input unit, the touch screen display unit, and the touch-sensitive surface unit. The processing unit is configured to: cause display of a first user interface on the primary display unit, the first user interface comprising one or more user interface elements; identify an active user interface element among the one or more user interface elements that is in focus on the primary display unit; and determine whether the active user interface element that is in focus on the primary display unit is associated with an application executed by the processing unit. In accordance

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with a determination that the active user interface element that is in focus on the primary display is associated with the application executed by the computing system, the processing unit is configured to cause display of a second user interface on the touch screen display unit, including: (A) a first set of one or more affordances corresponding to the application; and (B) at least one system-level affordance corresponding to at least one system-level functionality.

Some embodiments provide a processing unit configured to: cause display of a first user interface on the primary display unit for an application executed by the processing unit; cause display of a second user interface on the touch screen display unit, the second user interface comprising a first set of one or more affordances corresponding to the application, where the first set of one or more affordances corresponds to a first portion of the application; and detecting a swipe gesture on the touch-sensitive surface unit. In accordance with a determination that the swipe gesture was performed in a first direction, the processing unit is configured to cause display of a second set of one or more affordances corresponding to the application on the touch screen display unit, where at least one affordance in the second set of one or more affordances is distinct from the first set of one or more affordances, and where the second set of one or more affordances also corresponds to the first portion of the application. In accordance with a determination that the swipe gesture was performed in a second direction substantially perpendicular to the first direction, the processing unit is configured to cause display of a third set of one or more affordances corresponding to the application on the touch screen display unit, where the third set of one or more affordances is distinct from the second set of one or more affordances, and where the third set of one or more affordances corresponds to a second portion of the application that is distinct from the first portion of the application.

Other embodiments provide a processing unit configured to: cause display of a first user interface for the application executed by the processing unit on the primary display unit in a normal mode, the first user interface comprising a first set of one or more affordances associated with the application; and detect a user input for displaying at least a portion of the first user interface for the application in a full-screen mode on the primary display unit. In response to detecting the user input, the processing unit is configured to: cease to display the first set of one or more of affordances associated with the application in the first user interface on the primary display unit; cause display of the portion of the first user interface for the application in the full-screen mode on the primary display unit; and automatically, without human intervention, cause display of a second set of one or more affordances for controlling the application on the touch screen display unit, where the second set of one or more affordances correspond to the first set of one or more affordances.

In some embodiments, the processing unit is configured to: cause display of a first user interface, on the primary display unit, for an application executed by the processing unit; cause display of a second user interface, on the touch screen display unit, the second user interface comprising a set of one or more affordances corresponding to the application; and detect a notification. In response to detecting the notification, the processing unit is configured to cause concurrent display of, in the second user interface on the touch screen display unit, the set of one or more affordances corresponding to the application and at least a portion of the

detected notification, where the detected notification is not displayed on the primary display unit.

In other embodiments, the processing unit is configured to: cause display of a user interface, on the primary display unit, the user interface comprising one or more user interface elements; identify an active user interface element of the one or more user interface elements that is in focus on the primary display unit, where the active user interface element is associated with an application executed by the computing system; in response to identifying the active user interface element that is in focus on the primary display, cause display of a set of one or more affordances corresponding to the application on the touch screen display unit; and detect a user input to move a respective portion of the user interface. In response to detecting the user input, and in accordance with a determination that the user input satisfies predefined action criteria, the processing unit is configured to: cease to display the respective portion of the user interface on the primary display unit; cease to display at least a subset of the set of one or more affordances on the touch screen display unit; and cause display of a representation of the respective portion of the user interface on the touch screen display unit.

In accordance with some embodiments, a computing system includes a processor, a primary display, memory storing one or more programs, and a housing at least partially containing a physical input mechanism and a touch screen display adjacent to the physical input mechanism; the one or more programs are configured to be executed by the processor and include instructions for performing or causing performance of the operations of any of the methods described herein. In accordance with some embodiments, a non-transitory computer readable storage medium has stored therein instructions which when executed by a processor of a computing system with a primary display and a housing at least partially containing a physical input mechanism and a touch screen display adjacent to the physical input mechanism; cause the computing system to perform or cause performance of the operations of any of the methods referred described herein. In accordance with some embodiments, a graphical user interface on a computing system that includes a processor, a primary display, memory storing one or more programs, and a housing at least partially containing a physical input mechanism and a touch screen display adjacent to the physical input mechanism; the one or more programs are configured to be executed by the processor and include instructions for displaying or causing display of one or more of the elements displayed in any of the methods described above, which are updated in response to user inputs, as described in any of the methods described herein. In accordance with some embodiments, a computing system includes: a primary display; a housing at least partially containing a physical input mechanism and a touch screen display adjacent to the physical input mechanism; and means for performing or causing performance of the operations of any of the methods described herein. In accordance with some embodiments, an information processing apparatus, for use in a computing system with a primary display and a housing at least partially containing a physical input mechanism and a touch screen display adjacent to the physical input mechanism; includes means for performing or causing performance of the operations of any of the methods described herein.

BRIEF DESCRIPTION OF DRAWINGS

For a better understanding of the various described embodiments, reference should be made to the Description

of Embodiments below, in conjunction with the following drawings in which like reference numerals refer to corresponding parts throughout the figures.

FIG. 1A is an illustrative diagram of a portable computing system (e.g., a laptop computer), in accordance with some embodiments.

FIG. 1B is an illustrative diagram of a body portion of the portable computing system in FIG. 1A, in accordance with some embodiments.

FIG. 2A is an illustrative diagram of a first implementation of a desktop computing system, in accordance with some embodiments.

FIG. 2B is an illustrative diagram of a second implementation of a desktop computing system, in accordance with some embodiments.

FIG. 2C is an illustrative diagram of a third implementation of a desktop computing system, in accordance with some embodiments.

FIG. 2D is an illustrative diagram of a fourth implementation of a desktop computing system, in accordance with some embodiments.

FIG. 3A is a block diagram of an electronic device, in accordance with some embodiments.

FIG. 3B is a block diagram of components for event handling of FIG. 3A, in accordance with some embodiments.

FIG. 4 is a block diagram of a peripheral electronic device, in accordance with some embodiments.

FIGS. 5A-5DDD illustrate example user interfaces for updating a dynamic input and output device, in accordance with some embodiments.

FIGS. 6A-6D are a flowchart of a method of updating a dynamic input and output device, in accordance with some embodiments.

FIGS. 7A-7C are a flowchart of a method of updating a dynamic input and output device, in accordance with some embodiments.

FIGS. 8A-8B are a flowchart of a method of maintaining functionality of an application while in full-screen mode, in accordance with some embodiments.

FIGS. 9A-9B are a flowchart of a method of displaying notifications on a touch screen display, in accordance with some embodiments.

FIGS. 10A-10C are a flowchart of a method of moving user interface portions, in accordance with some embodiments.

FIGS. 11-15 illustrate functional block diagrams of an electronic device, in accordance with some embodiments.

DESCRIPTION OF EMBODIMENTS

FIGS. 1A-1B, 2A-2D, 3A-3B, and 4 provide a description of example devices. FIGS. 5A-5DDD illustrate example user interfaces for updating a dynamic input and output device. FIGS. 6A-6D are a flowchart of a method 600 of updating a dynamic input and output device. FIGS. 7A-7C are a flowchart of a method 700 of updating a dynamic input and output device. FIGS. 8A-8B are a flowchart of a method 800 of maintaining functionality of an application while in full-screen mode. FIGS. 9A-9B are a flowchart of a method 900 of displaying notifications on a touch screen display. FIGS. 10A-10C are a flowchart of a method 1000 of moving user interface portions. The user interfaces in FIGS. 5A-5DDD are used to illustrate the methods and/or processes in FIGS. 6A-6D, 7A-7C, 8A-8B, 9A-9B, and 10A-10C.

Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the various described embodiments. However, it will be apparent to one of ordinary skill in the art that the various described embodiments may be practiced without these specific details. In other instances, well-known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.

It will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first contact could be termed a second contact, and, similarly, a second contact could be termed a first contact, without departing from the scope of the various described embodiments. The first contact and the second contact are both contacts, but they are not the same contact.

The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

As used herein, the term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

FIG. 1A is an illustrative diagram of a portable computing system 100, in accordance with some embodiments. Portable computing system 100 may be, for example, a laptop computer, such as a MACBOOK® device, or any other portable computing device. Portable computing system 100 includes: (A) a display portion 110 with a primary display 102; and (B) a body portion 120 with a dynamic function row 104, a set of physical (i.e., movably actuated) keys 106, and a touch pad 108 partially contained within a same housing. Display portion 110 is typically mechanically, electrically, and communicatively coupled with body portion 120 of portable computing system 100. For example, portable computing system 100 may include a hinge, allowing display portion 110 to be rotated relative to body portion 120. Portable computing system 100 includes one or more processors and memory storing one or more programs for execution by the one or more processors to perform any of the embodiments described herein. In some embodiments,

dynamic function row 104, which is described in more detail with reference to FIG. 1B, is a touch screen display using resistive sensing, acoustic sensing, capacitive sensing, optical sensing, infrared sensing, or the like to detect user touch inputs and selections. In some embodiments, primary display 102 of display portion 110 is also a touch screen display.

FIG. 1B is an illustrative diagram of body portion 120 of portable computing system 100 in accordance with some embodiments. Body portion 120 include a set of physical keys 106, a dynamic function row 104, and a touch pad 108 partially contained within a same housing. In some embodiments, dynamic function row 104, which is a touch screen, replaces a function row of the set of physical keys 106 allowing the space consumed by the set of physical keys 106 to be reduced, allowing for a smaller overall body portion 120 or allowing other portions, such as touch pad 108, to be larger. In some embodiments, dynamic function row 104 is approximately 18 inches in length relative to a major dimension of the set of physical keys 106. Although called a “row” for ease of explanation, in some other embodiments, the touch screen comprising dynamic function row 104 in FIG. 1A may take any other form such as a square, circle, a plurality of rows, column, a plurality of columns, a plurality of separate sectors, or the like. Although FIGS. 1A-1B show dynamic function row 104 replacing the function row of the set of physical keys 106, in some other embodiments, dynamic function row 104 may additionally and/or alternatively replace a numpad section, editing/function section, or the like of the set of physical keys 106.

Each physical key of the set of physical keys 106 has at least one associated input. The input may be a printable character, non-printable character, function, or other input. The input associated with a physical key may be shown by a letter, word, symbol, or other indicia shown (e.g., printed) on the surface of the key in Latin script, Arabic characters, Chinese characters, or any other script. For example, the particular physical key indicated at 138 is associated with alphabetic character “z” as indicated by the letter z shown on the key. In another example, a physical key labeled with the word “command” may be associated with a command function. For example, the set of physical keys 106 is associated with a QWERTY, Dvorak, or other keyboard layouts with alphanumeric, numeric, and/or editing/function sections (e.g., standard, extended, or compact) according to ISO/IEC 9995, ANSI-INCITS 154-1988, JIS X 6002-1980, or other similar standards.

A signal corresponding to an input associated with a physical key may be received by the processor of portable computing system 100 (or computing device 202 in FIGS. 2A-2D or peripheral keyboard 206 in FIGS. 2A-2B) when a key has been activated by a user. In an illustrative example, each key of the set of physical keys 106 includes two plates and a spring. A user may activate a key by pressing down on the key, which compresses the spring. When the spring is compressed, the two plates may come into contact, allowing electric current to flow through the connected plates. An input corresponding to the key may be provided to a processor in response to the flow of the current through the connected plates. For example, in response to activation of one of the set of keys 106 of peripheral keyboard 206 in FIG. 2C, an input corresponding to the activated key is provided to computing device 202. It will be recognized that other systems for movably actuated keys could be used.

In some embodiments, dynamic function row 104 is a touch screen display that displays one or more user-selectable symbols 142 (sometimes also herein called “user inter-

face elements,” “user interface components,” “affordances,” “buttons,” or “soft keys”). For example, dynamic function row 104 replaces the function row keys on a typical keyboard. A user may select a particular one of the one or more user-selectable symbols 142 by touching a location on the touch screen display that corresponds to the particular one of the one or more user-selectable symbols 142. For example, a user may select the user-selectable symbol indicated by magnifying glass symbol 144 by tapping dynamic function row 104 such that the user’s finger contacts dynamic function row 104 at the position of the magnifying glass indicator 214. In some embodiments, a tap contact or a tap gesture includes touch-down of a contact and lift-off of the contact within a predetermined amount of time (e.g., 250 ms or the like). In some embodiments, the touch screen display of dynamic function row 104 is implemented using resistive sensing, acoustic sensing, capacitive sensing, optical sensing, infrared sensing, or the like to detect user inputs and selections.

When a user selects a particular one of the one or more user-selectable symbols 142, a signal corresponding to the particular one of the one or more user-selectable symbols 142 is generated by dynamic function row 104. For example, when a user taps “esc” on dynamic function row 104, dynamic function row 104 transmits a signal indicating a user input corresponding to an escape function to the processor of portable computing system 100 (or computing device 202 in FIGS. 2A-2D, or the processor of peripheral keyboard 206 in FIGS. 2A-2B, or the processor of first input mechanism 212, FIG. 2C or the processor of second input mechanism 222, FIG. 2D).

In some embodiments, when a particular one of the one or more user-selectable symbols 142 is selected, dynamic function row 104 transmits a signal corresponding to a position on the touch screen display where the particular one of the one or more user-selectable symbols 142 is displayed, to the processor of portable computing system 100 (or computing device 202 in FIGS. 2A-2D, or the processor of peripheral keyboard 206 in FIGS. 2A-2B, or the processor of first input mechanism 212, FIG. 2C or the processor of second input mechanism 222, FIG. 2D). For example, dynamic function row 104 may transmit a signal including a position value (0 to 20) depending on the position on the touch screen display of the particular one of the one or more user-selectable symbols 142 that was selected. In the illustrative example of FIG. 1B, the “esc” symbol may have a position value of 0, magnifying glass symbol 144 may have a position value of 16, and so on. A processor of portable computing system 100 (or computing device 202 in FIGS. 2A-2D, or the processor of peripheral keyboard 206 in FIGS. 2A-2B, or the processor of first input mechanism 212, FIGS. 2C, or the processor of second input mechanism 222, FIG. 2D) may receive the signal indicating the position value of the selected user-selectable symbol and interpret the position value using contextual information, such as an element of a graphical user interface displayed on primary display 102 of display portion 110 (or peripheral display device 204, FIGS. 2A-2D) that is currently active or that has focus.

Each of the one or more user-selectable symbols 142 may include an indicator, such as a symbol (e.g., a magnifying glass symbol as shown at 144), an abbreviated word (e.g., “esc”), an unabbreviated word, a character, an image, an animated image, a video, or the like. In some embodiments, a respective one of the one or more user-selectable symbols 142 is capable of receiving user input(s).

An input may be associated with each of the one or more user-selectable symbols 142. The input may be a function,

character, numerical value, and the like. A respective one of the one or more user-selectable symbols 142 may include an indicator that corresponds to the input for the respective one of the one or more user-selectable symbols 142. For example, in FIG. 1B, the user-selectable symbol with the abbreviated word “esc” indicates to the user that an escape function is associated with the user-selectable symbol. A function associated with the one or more user-selectable symbols 142 may be activated when the user selects a user-selectable symbol. For example, an escape function may be activated when a user selects the user-selectable symbol with the indicator “esc.” Activation of the function may have different effects depending on the current state of portable computing system 100 (or computing device 202 in FIGS. 2A-2D). For example, when a dialog box is open on primary display 102 of display portion 110 (or peripheral display device 204, FIGS. 2A-2D), activating an escape function on dynamic function row 104 may close the dialog box. In another example, when a game application is being executed by a processor of portable computing system 100 (or computing device 202 in FIGS. 2A-2D), activating an escape function on dynamic function row 104 may pause the game.

In some embodiments, functions may be associated with combinations of movably actuated keys and/or user-selectable symbols. For example, simultaneous actuation of a command key and “c” key (i.e., command+c) may be associated with a “copy” function. In another example, simultaneous actuation of the command key and selection of the user-selectable symbol with the indicator “esc” (i.e., command+esc) may activate a function to open a particular application such as a media player application. In yet another example, simultaneous selection of two user-selectable symbols (e.g., the user-selectable symbol with the indicator “esc” and the user-selectable symbol 144 with the magnifying glass indicator) may result in activation of a function, such as a specialized search function.

In some embodiments, a first subset 146 of the one or more user-selectable symbols 142 of dynamic function row 104 may be associated with one group of functions and a second subset 148 of the one or more user-selectable symbols 142 of function row 104 may be associated with a second group of functions. For example, the user-selectable symbols in first subset 146 may be global functions (e.g., system-level functions or affordances), and the user-selectable symbols in second subset 148 may be application-specific functions. As such, the user-selectable symbols in first subset 148 change when the focus shifts from a first element of a graphical user interface displayed on primary display 102 (e.g., a first window corresponding to an Internet browser application) to a second element of the graphical user interface (e.g., a second window corresponding to an e-mail application). In contrast, the user-selectable symbols in first subset 146 are maintained when the focus shifts from the first element of the graphical user interface to the second element of the graphical user interface.

In some embodiments, the user-selectable symbols in second subset 148 are determined based on an active user interface element display on primary display 102 that is in focus. In some embodiments, the term “in focus” can refer to the active element of the user interface (e.g., a window associated with an application, a particular toolbar or menu associated with an application, or the operating system) that is currently in the foreground and actively running or is controllable by input received from a user of the computing system such as a key press, mouse click, voice command, gestural motion, or the like.

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In some embodiments, the first subset **146** of the one or more user-selectable symbols **142** corresponding to global user-selectable symbols occupies a first area of dynamic function row **104** (e.g., the left half of dynamic function row **104**), and the second subset **148** of the one or more user-selectable symbols **142** occupies a second area of dynamic function row **104** (e.g., the right half of dynamic function row **104**). It will be realized that other proportions of function row **104** may be allocated to the first subset **146** and the second subset **148**. In some embodiments, when no application has focus, the second area of dynamic function row **104** may not include any user-selectable symbols. In some embodiments, dynamic function row **104** includes three or more subsets of user-selectable symbols. In some embodiments, dynamic function row **104** includes a single set of user-selectable symbols that are not divided into subsets. While a single row of user-selectable symbols are shown in dynamic function row **104** in FIG. 1B, it will be recognized that dynamic function row **104** may include multiple rows of user-selectable symbols.

In some embodiments, the change in focus changes which element of the graphical user interface displayed on primary display **102** of display portion **110** (or peripheral display device **204**, FIGS. 2A-2D) is active and which element will receive user input. The user input may be received from a keyboard, mouse, touch pad, or other user input device. Additionally and/or alternatively, in some embodiments, the change in focus changes an element that is shown in the foreground of a graphical user interface displayed on primary display **102** of display portion **110** (or peripheral display device **204**, FIGS. 2A-2D).

In some embodiments, the change in focus occurs in response to user input, for example, in response to user selection of an element of a graphical user interface (e.g., a different window) displayed on primary display **102** of display portion **110** (or peripheral display device **204**, FIGS. 2A-2D) or in response to user selection of a user-selectable symbol (e.g., one of the affordances/symbols displayed on dynamic function row **104**). The user selection may be a key stroke, a mouse click, a mouse over, a command+tab input, or the like. In some embodiments, the change in focus occurs in response to a determination by an operating system of portable system **100** (or computing device **202** in FIGS. 2A-2D). For example, when a user closes an application window that has focus, the operating system may give focus to a different application, such as an application that had focus prior to the closed application window. In another example, when a user closes an application window that has focus, the operating system may give focus to a dialog box prompting the user to save changes made to a document via the application.

In some embodiments, the change in focus may be a change from one element associated with an application to another element associated with the same application (e.g., from an e-mail composition window of an e-mail application to an inbox list window of an e-mail application or from one tab of an Internet browser application to another tab of an Internet browser application). In some embodiments, the change in focus may be a change from an element associated with one application to an element associated with another application (e.g., from an Internet browser window to an e-mail application window). Further, in some embodiments, the change in focus may be a change from an element associated with an application to an element associated with an operating system, such as a system dialog box, a system setting control (e.g., volume control), a window associated with a file/folder navigation application (e.g., Apple Inc.'s

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FINDER application), etc. Additionally, focus may also be directed to a dialog box, file directory, setting control (e.g., volume control), or any other element of a graphical user interface for which information can be presented to a user and/or user input can be received.

FIG. 2A is an illustrative diagram of a first implementation of desktop computing system **200** in accordance with some embodiments. Desktop computing system **200** includes a computing device **202**, a peripheral display device **204** with primary display **102**, a peripheral keyboard **206**, and a peripheral mouse **208**. Computing device **202** includes one or more processors and memory storing one or more programs for execution by the one or more processors. In some embodiments, peripheral display device **204** may be integrated with computing device **202** such as an iMAC® device. In some embodiments, primary display **102** of peripheral display device **204** is a touch screen display. In FIG. 2A, peripheral display device **204**, peripheral keyboard **206**, and peripheral mouse **208** are communicatively coupled to computing device **202** via a wired connection, such as USB or PS/2, or via a wireless communication link, using a communication protocol such as Bluetooth, Wi-Fi, or the like. For example, peripheral keyboard **206** is not more than fifteen feet from computing device **202** (e.g., approximately three feet away). In FIG. 2A, peripheral keyboard **206** includes dynamic function row **104** and a set of physical keys **106** at least partially contained within a same housing. In some embodiments, dynamic function row **104**, which is described in more detail with reference to FIG. 1B, is a touch screen display. In some embodiments, peripheral keyboard **206** includes one or more processors and memory storing one or more programs that may be executed by the one or more processors of peripheral keyboard **206** to perform any of the embodiments described herein. In some embodiments, peripheral keyboard **206** relays signals indicating user inputs (e.g., key strokes and selections of user-selectable symbols/affordances displayed by dynamic function row **104**) to computing device **202**.

FIG. 2B is an illustrative diagram of a second implementation of desktop computing system **200** in accordance with some embodiments. In FIG. 2B, desktop computing system **200** includes a computing device **202**, a peripheral display device **204** with primary display **102**, and a peripheral keyboard **206**. In FIG. 2B, peripheral display device **204** and peripheral keyboard **206** are communicatively coupled to computing device **202** via a wired connection, such as USB or PS/2, or via a wireless communication link, using a communication protocol such as Bluetooth, Wi-Fi, or the like. In FIG. 2B, peripheral keyboard **206** includes dynamic function row **104**, a set of physical keys **106**, and touch pad **108** at least partially contained within a same housing. In some embodiments, dynamic function row **104**, which is described in more detail with reference to FIG. 1B, is a touch screen display. In some embodiments, peripheral keyboard **206** includes one or more processors and memory storing one or more programs that may be executed by the one or more processors of peripheral keyboard **206** to perform any of the embodiments described herein. In some embodiments, peripheral keyboard **206** relays signals indicating user inputs (e.g., key strokes, user interactions with touch pad **108**, and selections of user-selectable symbols/affordances displayed by dynamic function row **104**) to computing device **202**.

FIG. 2C is an illustrative diagram of a third implementation of desktop computing system **200** in accordance with some embodiments. In FIG. 2C, desktop computing system **200** includes a computing device **202**, a peripheral display device **204** with primary display **102**, a peripheral keyboard

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206, and a first peripheral input mechanism 212. In FIG. 2C, peripheral display device 204, peripheral keyboard 206, and the first peripheral input mechanism 212 are communicatively coupled to computing device 202 via a wired connection, such as USB or PS/2, or via a wireless communication link, using a communication protocol such as Bluetooth, Wi-Fi, or the like. In FIG. 2C, peripheral keyboard 206 includes a set of physical keys 106, and the first peripheral input mechanism 212 includes dynamic function row 104 and touch pad 108 at least partially contained within a same housing. In some embodiments, dynamic function row 104, which is described in more detail with reference to FIG. 1B, is a touch screen display. In some embodiments, the first peripheral input mechanism 212 includes one or more processors and memory storing one or more programs that may be executed by the one or more processors of the first peripheral input mechanism 212 to perform any of the embodiments described herein. In some embodiments, the first peripheral input mechanism 212 relays signals indicating user inputs (e.g., user interactions with touch pad 108 and user selections of user-selectable symbols/affordances displayed by dynamic function row 104) to computing device 202.

FIG. 2D is an illustrative diagram of a fourth implementation of desktop computing system 200 in accordance with some embodiments. In FIG. 2D, desktop computing system 200 includes a computing device 202, a peripheral display device 204 with primary display 102, a peripheral keyboard 206, a peripheral mouse 208, and a second peripheral input mechanism 222. In FIG. 2D, peripheral display device 204, peripheral keyboard 206, peripheral mouse 208, and the second peripheral input mechanism 222 are communicatively coupled to computing device 202 via a wired connection, such as USB or PS/2, or via a wireless communication link, using a communication protocol such as Bluetooth, Wi-Fi, or the like. In FIG. 2A, peripheral keyboard 206 includes dynamic function row 104 and a set of physical keys 106. In FIG. 2D, peripheral keyboard 206 includes a set of physical keys 106, and the second peripheral input mechanism 222 includes dynamic function row 104 at least partially contained within the housing of the second peripheral input mechanism 222. In some embodiments, dynamic function row 104, which is described in more detail with reference to FIG. 1B, is a touch screen display. In some embodiments, the second peripheral input mechanism 222 includes one or more processors and memory storing one or more programs that may be executed by the one or more processors of the second peripheral input mechanism 222 to perform any of the embodiments described herein. In some embodiments, the second peripheral input mechanism 222 relays signals indicating user inputs (e.g., user selections of user-selectable symbols/affordances displayed by dynamic function row 104) to computing device 202.

FIG. 3A is a block diagram of an electronic device 300, in accordance with some embodiments. In some embodiments, electronic device 300 is a portable electronic device, such as a laptop (e.g., portable computing system 100, FIG. 1A). In some embodiments, electronic device 300 is not a portable device, but is a desktop computer (e.g., computing device 202 of desktop computing system 200, FIGS. 2A-2D), which is communicatively coupled with a peripheral display system (e.g., peripheral display device 204, FIGS. 2A-2D) and optionally a peripheral touch-sensitive surface (e.g., a touch pad 108, FIGS. 2B-2C and/or a

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touch-sensitive display, such as peripheral display device 204, FIGS. 2A-2D and/or dynamic function row 104, FIGS. 2A-2D).

Electronic device 300 typically supports a variety of applications, such as one or more of the following: a drawing application, a presentation application, a word processing application, a website creation application, a disk authoring application, a spreadsheet application, a gaming application, a video conferencing application, an e-mail application, an instant messaging application, an image management application, a digital camera application, a digital video camera application, a web browser application, and/or a media player application.

The various applications that are executed on electronic device 300 optionally use at least one common physical user-interface device, such as the touch-sensitive surface. One or more functions of the touch-sensitive surface as well as corresponding information displayed by electronic device 300 are, optionally, adjusted and/or varied from one application to the next and/or within an application. In this way, a common physical architecture (such as the touch-sensitive surface) of electronic device 300 optionally supports the variety of applications with user interfaces that are intuitive and transparent to the user.

Electronic device 300 includes memory 302 (which optionally includes one or more computer readable storage mediums), memory controller 322, one or more processing units (CPU(s)) 320, peripherals interface 318, RF circuitry 308, audio circuitry 310, speaker 311, microphone 313, input/output (I/O) subsystem 306, other input or control devices 316, and external port 324. Electronic device 300 optionally includes a display system 312 (e.g., primary display 102 of display portion 110, FIG. 1A and/or dynamic function row 104, FIGS. 1A-1B), which may be a touch-sensitive display (sometimes also herein called a “touch screen” or a “touch screen display”). Electronic device 300 optionally includes one or more optical sensors 364. Electronic device 300 optionally includes one or more intensity sensors 365 for detecting intensity of contacts on a touch-sensitive surface such as touch-sensitive display or a touch pad. Electronic device 300 optionally includes one or more tactile output generators 367 for generating tactile outputs on a touch-sensitive surface such as touch-sensitive display or a touch pad (e.g., touch pad 108, FIGS. 1A-1B). These components optionally communicate over one or more communication buses or signal lines 303.

As used in the specification, the term “intensity” of a contact on a touch-sensitive surface refers to the force or pressure (force per unit area) of a contact (e.g., a finger contact) on the touch sensitive surface, or to a substitute (proxy) for the force or pressure of a contact on the touch sensitive surface. The intensity of a contact has a range of values that includes at least four distinct values and more typically includes hundreds of distinct values (e.g., at least 256). Intensity of a contact is, optionally, determined (or measured) using various approaches and various sensors or combinations of sensors. For example, one or more force sensors underneath or adjacent to the touch-sensitive surface are, optionally, used to measure force at various points on the touch-sensitive surface. In some implementations, force measurements from multiple force sensors are combined (e.g., a weighted average) to determine an estimated force of a contact. Similarly, a pressure-sensitive tip of a stylus is, optionally, used to determine a pressure of the stylus on the touch-sensitive surface. Alternatively, the size of the contact area detected on the touch-sensitive surface and/or changes thereto, the capacitance of the touch-sensitive surface proxi-

mate to the contact and/or changes thereto, and/or the resistance of the touch-sensitive surface proximate to the contact and/or changes thereto are, optionally, used as a substitute for the force or pressure of the contact on the touch-sensitive surface. In some implementations, the substitute measurements for contact force or pressure are used directly to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is described in units corresponding to the substitute measurements). In some implementations, the substitute measurements for contact force or pressure are converted to an estimated force or pressure and the estimated force or pressure is used to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is a pressure threshold measured in units of pressure).

As used in the specification and claims, the term “tactile output” refers to physical displacement of a device relative to a previous position of the device, physical displacement of a component (e.g., a touch-sensitive surface) of a device relative to another component (e.g., housing) of the device, or displacement of the component relative to a center of mass of the device that will be detected by a user with the user’s sense of touch. For example, in situations where the device or the component of the device is in contact with a surface of a user that is sensitive to touch (e.g., a finger, palm, or other part of a user’s hand), the tactile output generated by the physical displacement will be interpreted by the user as a tactile sensation corresponding to a perceived change in physical characteristics of the device or the component of the device. For example, movement of a touch-sensitive surface (e.g., a touch-sensitive display or touch/track pad) is, optionally, interpreted by the user as a “down click” or “up click” of a physical actuator button. In some cases, a user will feel a tactile sensation such as an “down click” or “up click” even when there is no movement of a physical actuator button associated with the touch-sensitive surface that is physically pressed (e.g., displaced) by the user’s movements. As another example, movement of the touch-sensitive surface is, optionally, interpreted or sensed by the user as “roughness” of the touch-sensitive surface, even when there is no change in smoothness of the touch-sensitive surface. While such interpretations of touch by a user will be subject to the individualized sensory perceptions of the user, there are many sensory perceptions of touch that are common to a large majority of users. Thus, when a tactile output is described as corresponding to a particular sensory perception of a user (e.g., an “up click,” a “down click,” “roughness”), unless otherwise stated, the generated tactile output corresponds to physical displacement of the device or a component thereof that will generate the described sensory perception for a typical (or average) user.

It should be appreciated that electronic device 300 is only an example and that electronic device 300 optionally has more or fewer components than shown, optionally combines two or more components, or optionally has a different configuration or arrangement of the components. The various components shown in FIG. 3A are implemented in hardware, software, firmware, or a combination thereof, including one or more signal processing and/or application specific integrated circuits.

Memory 302 optionally includes high-speed random access memory and optionally also includes non-volatile memory, such as one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices. Access to memory 302 by other components of electronic device 300, such as CPU(s) 320 and

peripherals interface 318, is, optionally, controlled by memory controller 322. Peripherals interface 318 can be used to couple input and output peripherals to CPU(s) 320 and memory 302. The one or more processing units 320 run or execute various software programs and/or sets of instructions stored in memory 302 to perform various functions for electronic device 300 and to process data. In some embodiments, peripherals interface 318, CPU(s) 320, and memory controller 322 are, optionally, implemented on a single chip, such as chip 304. In some other embodiments, they are, optionally, implemented on separate chips.

RF (radio frequency) circuitry 308 receives and sends RF signals, also called electromagnetic signals. RF circuitry 308 converts electrical signals to/from electromagnetic signals and communicates with communications networks and other communications devices via the electromagnetic signals. RF circuitry 308 optionally includes well-known circuitry for performing these functions, including but not limited to an antenna system, an RF transceiver, one or more amplifiers, a tuner, one or more oscillators, a digital signal processor, a CODEC chipset, a subscriber identity module (SIM) card, memory, and so forth. RF circuitry 308 optionally communicates with networks, such as the Internet, also referred to as the World Wide Web (WWW), an intranet and/or a wireless network, such as a cellular telephone network, a wireless local area network (LAN) and/or a metropolitan area network (MAN), and other devices by wireless communication. The wireless communication optionally uses any of a plurality of communications standards, protocols and technologies, including but not limited to Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), high-speed downlink packet access (HSDPA), high-speed uplink packet access (HSUPA), Evolution, Data-Only (EV-DO), HSPA, HSPA+, Dual-Cell HSPA (DC-HSPDA), long term evolution (LTE), near field communication (NFC), wideband code division multiple access (W-CDMA), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Wireless Fidelity (Wi-Fi) (e.g., IEEE 802.11a, IEEE 802.11b, IEEE 802.11g, and/or IEEE 802.11n), voice over Internet Protocol (VOIP), Wi-MAX, a protocol for e-mail (e.g., Internet message access protocol (IMAP) and/or post office protocol (POP)), instant messaging (e.g., extensible messaging and presence protocol (XMPP), Session Initiation Protocol for Instant Messaging and Presence Leveraging Extensions (SIMPLE), Instant Messaging and Presence Service (IMPS)), and/or Short Message Service (SMS), or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document.

Audio circuitry 310, speaker 311, and microphone 313 provide an audio interface between a user and electronic device 300. Audio circuitry 310 receives audio data from peripherals interface 318, converts the audio data to an electrical signal, and transmits the electrical signal to speaker 311. Speaker 311 converts the electrical signal to human-audible sound waves. Audio circuitry 310 also receives electrical signals converted by microphone 313 from sound waves. Audio circuitry 310 converts the electrical signals to audio data and transmits the audio data to peripherals interface 318 for processing. Audio data is, optionally, retrieved from and/or transmitted to memory 302 and/or RF circuitry 308 by peripherals interface 318. In some embodiments, audio circuitry 310 also includes a headset jack. The headset jack provides an interface between audio circuitry 310 and removable audio input/output

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peripherals, such as output-only headphones or a headset with both output (e.g., a headphone for one or both ears) and input (e.g., a microphone).

I/O subsystem 306 couples the input/output peripherals of electronic device 300, such as display system 312 and other input or control devices 316, to peripherals interface 318. I/O subsystem 306 optionally includes display controller 356, optical sensor controller 358, intensity sensor controller 359, haptic feedback controller 361, and one or more other input controllers 360 for other input or control devices. The one or more other input controllers 360 receive/send electrical signals from/to other input or control devices 316. The other input or control devices 316 optionally include physical buttons (e.g., push buttons, rocker buttons, etc.), dials, slider switches, joysticks, click wheels, and so forth. In some alternate embodiments, other input controller(s) 360 are, optionally, coupled with any (or none) of the following: a keyboard, infrared port, USB port, and a pointer device such as a mouse. The one or more physical buttons optionally include an up/down button for volume control of speaker 311 and/or microphone 313.

Display system 312 (e.g., primary display 102 of display portion 110, FIG. 1A and/or dynamic function row 104, FIGS. 1A-1B) provides an output interface (and, optionally, an input interface when it is a touch-sensitive display) between electronic device 300 and a user. Display controller 356 receives and/or sends electrical signals from/to display system 312. Display system 312 displays visual output to the user. The visual output optionally includes graphics, text, icons, video, and any combination thereof (collectively termed "graphics"). In some embodiments, some or all of the visual output corresponds to user-interface objects/elements.

In some embodiments, display system 312 (e.g., primary display 102 of display portion 110, FIG. 1A and/or dynamic function row 104, FIGS. 1A-1B) is a touch-sensitive display with a touch-sensitive surface, sensor, or set of sensors that accepts input from the user based on haptic and/or tactile contact. As such, display system 312 and display controller 356 (along with any associated modules and/or sets of instructions in memory 302) detect contact (and any movement or breaking of the contact) on display system 312 and convert the detected contact into interaction with user-interface objects (e.g., one or more soft keys, icons, web pages, or images) that are displayed on display system 312. In one example embodiment, a point of contact between display system 312 and the user corresponds to an area under a finger of the user.

Display system 312 (e.g., primary display 102 of display portion 110, FIG. 1A and/or dynamic function row 104, FIGS. 1A-1B) optionally uses LCD (liquid crystal display) technology, LPD (light emitting polymer display) technology, LED (light emitting diode) technology, or OLED (organic light emitting diode) technology, although other display technologies are used in other embodiments. In some embodiments, when display system 312 is a touch-sensitive display, display system 312 and display controller 356 optionally detect contact and any movement or breaking thereof using any of a plurality of touch sensing technologies now known or later developed, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with display system 312. In one example embodiment, projected mutual capacitance sensing technology is used, such as that found in the iPHONER, IPOD TOUCH®, and iPAD® from Apple Inc. of Cupertino, California.

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Display system 312 (e.g., primary display 102 of display portion 110, FIG. 1A and/or dynamic function row 104, FIGS. 1A-1B) optionally has a video resolution in excess of 400 dpi (e.g., 500 dpi, 800 dpi, or greater). In some embodiments, display system 312 is a touch-sensitive display with which the user optionally makes contact using a stylus, a finger, and so forth. In some embodiments, the user interface is designed to work primarily with finger-based contacts and gestures. In some embodiments, electronic device 300 translates the rough finger-based input into a precise pointer/cursor position or command for performing the actions desired by the user.

In some embodiments, in addition to display system 312, electronic device 300 optionally includes a touch pad (e.g., touch pad 108, FIGS. 1A-1B) for activating or deactivating particular functions. In some embodiments, the touch pad is a touch-sensitive area of electronic device 300 that, unlike display system 312, does not display visual output. In some embodiments, when display system 312 is a touch-sensitive display, the touch pad is, optionally, a touch-sensitive surface that is separate from display system 312, or an extension of the touch-sensitive surface formed by display system 312.

Electronic device 300 also includes power system 362 for powering the various components. Power system 362 optionally includes a power management system, one or more power sources (e.g., battery, alternating current (AC), etc.), a recharging system, a power failure detection circuit, a power converter or inverter, a power status indicator (e.g., a light-emitting diode (LED)) and any other components associated with the generation, management and distribution of power in portable devices.

Electronic device 300 optionally also includes one or more optical sensors 364 coupled with optical sensor controller 358 in I/O subsystem 306. Optical sensor(s) 364 optionally includes charge-coupled device (CCD) or complementary metal-oxide semiconductor (CMOS) phototransistors. Optical sensor(s) 364 receive light from the environment, projected through one or more lens, and converts the light to data representing an image. In conjunction with imaging module 343, optical sensor(s) 364 optionally capture still images or video. In some embodiments, an optical sensor is located on the front of electronic device 300 so that the user's image is, optionally, obtained for video-conferencing while the user views the other video conference participants on display system 312.

Electronic device 300 optionally also includes one or more contact intensity sensor(s) 365 coupled with intensity sensor controller 359 in I/O subsystem 306. Contact intensity sensor(s) 365 optionally includes one or more piezoresistive strain gauges, capacitive force sensors, electric force sensors, piezoelectric force sensors, optical force sensors, capacitive touch-sensitive surfaces, or other intensity sensors (e.g., sensors used to measure the force (or pressure) of a contact on a touch-sensitive surface). Contact intensity sensor(s) 365 receives contact intensity information (e.g., pressure information or a proxy for pressure information) from the environment. In some embodiments, at least one contact intensity sensor is collocated with, or proximate to, a touch-sensitive surface (e.g., touch pad 108, FIGS. 1A-1B or display system 312 when it is a touch-sensitive display).

Electronic device 300 optionally also includes one or more tactile output generators 367 coupled with haptic feedback controller 361 in I/O subsystem 306. Tactile output generator(s) 367 optionally includes one or more electroacoustic devices such as speakers or other audio components and/or electromechanical devices that convert energy into

linear motion such as a motor, solenoid, electroactive polymer, piezoelectric actuator, electrostatic actuator, or other tactile output generating component (e.g., a component that converts electrical signals into tactile outputs on the device). Contact intensity sensor(s) **365** receives tactile feedback generation instructions from haptic feedback module **333** and generates tactile outputs that are capable of being sensed by a user of electronic device **300**. In some embodiments, at least one tactile output generator is collocated with, or proximate to, a touch-sensitive surface (e.g., touch pad **108**, FIGS. 1A-1B or display system **312** when it is a touch-sensitive display) and, optionally, generates a tactile output by moving the touch-sensitive surface vertically (e.g., in/out of a surface of electronic device **300**) or laterally (e.g., back and forth in the same plane as a surface of electronic device **300**).

Electronic device **300** optionally also includes one or more proximity sensors **366** coupled with peripherals interface **318**. Alternately, proximity sensor(s) **366** are coupled with other input controller(s) **360** in I/O subsystem **306**. Electronic device **300** optionally also includes one or more accelerometers **368** coupled with peripherals interface **318**. Alternately, accelerometer(s) **368** are coupled with other input controller(s) **360** in I/O subsystem **306**.

In some embodiments, the software components stored in memory **302** include operating system **326**, communication module **328** (or set of instructions), contact/motion module **330** (or set of instructions), graphics module **332** (or set of instructions), applications **340** (or sets of instructions), and dynamic function row module **350** (or sets of instructions). Furthermore, in some embodiments, memory **302** stores device/global internal state **357** (or sets of instructions), as shown in FIG. 3A. Device/global internal state **357** includes one or more of: active application state, indicating which applications, if any, are currently active and/or in focus; display state, indicating what applications, views or other information occupy various regions of display system **312** (e.g., primary display **102** of display portion **110**, FIG. 1A and/or dynamic function row **104**, FIGS. 1A-1B) and/or a peripheral display system (e.g., primary display **102** of peripheral display device **204**, FIGS. 2A-2D and/or dynamic function row **104**, FIGS. 2A-2D); sensor state, including information obtained from various sensors and input or control devices **316** of electronic device **300**; and location information concerning the location and/or attitude of electronic device **300**.

Operating system **326** (e.g., DARWIN, RTXC, LINUX, UNIX, OS X, WINDOWS, or an embedded operating system such as VXWorks) includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device control, power management, etc.) and facilitates communication between various hardware and software components.

Communication module **328** facilitates communication with other devices (e.g., peripheral display device **202**, FIGS. 2A-2D; peripheral mouse **208**, FIGS. 2A and 2D; peripheral keyboard **206**, FIGS. 2A-2B; first peripheral input mechanism **212**, FIG. 2C; and/or second peripheral input mechanism **222**, FIG. 2D) over one or more external ports **324** and/or RF circuitry **308** and also includes various software components for sending/receiving data via RF circuitry **308** and/or external port **324**. External port **324** (e.g., Universal Serial Bus (USB), FIREWIRE, etc.) is adapted for coupling directly to other devices or indirectly over a network (e.g., the Internet, wireless LAN, etc.). In some embodiments, external port **324** is a multi-pin (e.g.,

30-pin) connector that is the same as, or similar to and/or compatible with the 30-pin connector used on iPod® devices.

Contact/motion module **330** optionally detects contact with display system **312** when it is a touch-sensitive display (in conjunction with display controller **356**) and other touch sensitive devices (e.g., a touch pad or physical click wheel). Contact/motion module **330** includes various software components for performing various operations related to detection of contact, such as determining if contact has occurred (e.g., detecting a finger-down event), determining an intensity of the contact (e.g., the force or pressure of the contact or a substitute for the force or pressure of the contact), determining if there is movement of the contact and tracking the movement across the touch-sensitive surface (e.g., detecting one or more finger-dragging events), and determining if the contact has ceased (e.g., detecting a finger-up event or a break in contact). Contact/motion module **330** receives contact data from the touch-sensitive surface. Determining movement of the point of contact, which is represented by a series of contact data, optionally includes determining speed (magnitude), velocity (magnitude and direction), and/or an acceleration (a change in magnitude and/or direction) of the point of contact. These operations are, optionally, applied to single contacts (e.g., one finger contacts) or to multiple simultaneous contacts (e.g., "multitouch"/multiple finger contacts). In some embodiments, contact/motion module **330** also detects contact on a touch pad (e.g., touch pad **108**, FIGS. 1A-1B).

In some embodiments, contact/motion module **330** uses a set of one or more intensity thresholds to determine whether an operation has been performed by a user (e.g., to determine whether a user has selected or "clicked" on an affordance). In some embodiments at least a subset of the intensity thresholds are determined in accordance with software parameters (e.g., the intensity thresholds are not determined by the activation thresholds of particular physical actuators and can be adjusted without changing the physical hardware of electronic device **300**). For example, a mouse "click" threshold of a trackpad or touch screen display can be set to any of a large range of predefined thresholds values without changing the trackpad or touch screen display hardware. Additionally, in some implementations a user of the device is provided with software settings for adjusting one or more of the set of intensity thresholds (e.g., by adjusting individual intensity thresholds and/or by adjusting a plurality of intensity thresholds at once with a system-level click "intensity" parameter).

Contact/motion module **330** optionally detects a gesture input by a user. Different gestures on the touch-sensitive surface have different contact patterns (e.g., different motions, timings, and/or intensities of detected contacts). Thus, a gesture is, optionally, detected by detecting a particular contact pattern. For example, detecting a finger tap contact includes detecting a finger-down event followed by detecting a finger-up (lift off) event at the same position (or substantially the same position) as the finger-down event (e.g., at the position of an icon). As another example, detecting a finger swipe gesture on the touch-sensitive surface includes detecting a finger-down event followed by detecting one or more finger-dragging events, and in some embodiments also followed by detecting a finger-up (lift off) event.

Graphics module **332** includes various known software components for rendering and causing display of graphics on primary display **102** (e.g., primary display **102** of display portion **110**, FIG. 1A or primary display **102** of peripheral

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display device **204**, FIGS. 2A-2D) or other display, including components for changing the visual impact (e.g., brightness, transparency, saturation, contrast or other visual property) of graphics that are displayed. As used herein, the term “graphics” includes any object that can be displayed to a user, including without limitation text, web pages, icons (such as user-interface objects including soft keys), digital images, videos, animations and the like. In some embodiments, graphics module **332** stores data representing graphics to be used. Each graphic is, optionally, assigned a corresponding code. Graphics module **332** receives, from applications etc., one or more codes specifying graphics to be displayed along with, if necessary, coordinate data and other graphic property data, and then generates screen image data to output to display controller **356**.

Haptic feedback module **333** includes various software components for generating instructions used by tactile output generator(s) **367** to produce tactile outputs at one or more locations on electronic device **300** in response to user interactions with electronic device **300**.

Applications **340** optionally include the following modules (or sets of instructions), or a subset or superset thereof: e-mail client module **341** (sometimes also herein called “mail app” or “e-mail app”) for receiving, sending, composing, and viewing e-mails; imaging module **342** for capturing still and/or video images; image management module **343** (sometimes also herein called “photo app”) for editing and viewing still and/or video images; media player module **344** (sometimes also herein called “media player app”) for playback of audio and/or video; and web browsing module **345** (sometimes also herein called “web browser”) for connecting to and browsing the Internet.

Examples of other applications **340** that are, optionally, stored in memory **302** include messaging and communications applications, word processing applications, other image editing applications, drawing applications, presentation applications, JAVA-enabled applications, encryption applications, digital rights management applications, voice recognition applications, and voice replication applications.

In conjunction with one or more of RF circuitry **308**, display system **312** (e.g., primary display **102** of display portion **110**, FIG. 1A and/or dynamic function row **104**, FIGS. 1A-1B), display controller **356**, and contact module **330**, graphics module **332**, e-mail client module **341** includes executable instructions to create, send, receive, and manage e-mail in response to user instructions. In conjunction with image management module **343**, e-mail client module **341** makes it very easy to create and send e-mails with still or video images taken with imaging module **342**.

In conjunction with one or more of display system **312** (e.g., primary display **102** of display portion **110**, FIG. 1A and/or dynamic function row **104**, FIGS. 1A-1B), display controller **356**, optical sensor(s) **364**, optical sensor controller **358**, contact module **330**, graphics module **332**, and image management module **343**, imaging module **342** includes executable instructions to capture still images or video (including a video stream) and store them into memory **302**, modify characteristics of a still image or video, or delete a still image or video from memory **302**.

In conjunction with one or more of display system **312** (e.g., primary display **102** of display portion **110**, FIG. 1A and/or dynamic function row **104**, FIGS. 1A-1B), display controller **356**, contact module **330**, graphics module **332**,

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and imaging module **342**, image management module **343** includes executable instructions to arrange, modify (e.g., edit), or otherwise manipulate, label, delete, present (e.g., in a digital slide show or album), and store still and/or video images.

In conjunction with one or more of display system **312** (e.g., primary display **102** of display portion **110**, FIG. 1A and/or dynamic function row **104**, FIGS. 1A-1B), display controller **356**, contact module **330**, graphics module **332**, audio circuitry **310**, speaker **311**, RF circuitry **308**, and web browsing module **347**, media player module **345** includes executable instructions that allow the user to download and play back recorded music and other sound files stored in one or more file formats, such as MP3 or AAC files, and executable instructions to display, present or otherwise play back videos (e.g., on primary display **102** of display portion **110**, FIG. 1A or primary display **102** of peripheral display device **2014**, FIGS. 2A-2B connected via external port **324**).

In conjunction with one or more of RF circuitry **308**, display system **312** (e.g., primary display **102** of display portion **110**, FIG. 1A and/or dynamic function row **104**, FIGS. 1A-1B), display controller **356**, contact module **330**, and graphics module **332**, web browsing module **347** includes executable instructions to browse the Internet in accordance with user instructions, including searching, linking to, receiving, and displaying web pages or portions thereof, as well as attachments and other files linked to web pages.

Dynamic function row (DFR) module **350** includes: focus determining module **351**, DFR determining module **352**, and DFR presenting module **353**. In some embodiments, focus determining module **351** is configured to determine an active user interface element that is in focus on the graphical user interface displayed by display system **312** (e.g., primary display **102** of display portion **110**, FIG. 1A) or a peripheral display system (e.g., peripheral display device **204**, FIGS. 2A-2D). In some embodiments, DFR determining module **352** is configured to determine graphics (e.g., a set of one or more affordances) based on the active user interface element that is in focus. In some embodiments, DFR presenting module **353** is configured to render the graphics determined by DFR determining module **352** on display system **312** (e.g., dynamic function row **104**, FIGS. 1A-1B). DFR presenting module **353** includes various known software components for rendering and causing display of graphics on display system **312** (e.g., dynamic function row **104**, FIGS. 1A-1B), including components for changing the visual impact (e.g., brightness, transparency, saturation, contrast or other visual property) of graphics that are displayed. As used herein, the term “graphics” includes any object that can be displayed to a user, including without limitation text, web pages, icons (such as user-interface objects including soft keys), digital images, videos, animations, and the like. In some embodiments, DFR module **350** includes other modules for: adjusting the sensitivity of dynamic function row **104**; adjusting the audible and/or haptic feedback provided by dynamic function row **104**; adjusting the settings of affordances and information displayed by dynamic function row **104** (e.g., size, brightness, font, language, and the like); adjusting the current power mode of dynamic function row **104** (e.g., normal and low-power modes); and the like.

Each of the above identified modules and applications correspond to a set of executable instructions for performing one or more functions described above and the methods described in this application (e.g., the computer-implemented methods and other information processing methods described herein). These modules (i.e., sets of instructions)

need not be implemented as separate software programs, procedures or modules, and thus various subsets of these modules are, optionally, combined or otherwise re-arranged in various embodiments. In some embodiments, memory 302 optionally stores a subset of the modules and data structures identified above. Furthermore, memory 302 optionally stores additional modules and data structures not described above.

FIG. 3B is a block diagram of components for event handling of FIG. 3A, in accordance with some embodiments. In some embodiments, memory 302 (FIG. 3A) includes event sorter 370 (e.g., in operating system 326) and an application 340-1 (e.g., any of the aforementioned applications 341, 342, 343, 344, or 345).

Event sorter 370 receives event information and determines the application 340-1 and application view 391 of application 340-1 to which to deliver the event information. Event sorter 370 includes event monitor 371 and event dispatcher module 374. In some embodiments, application 340-1 includes application internal state 392, which indicates the current application view(s) displayed on display system 312 (e.g., primary display 102 of display portion 110, FIG. 1A and/or dynamic function row 104, FIGS. 1A-1B) when the application is active or executing. In some embodiments, device/global internal state 357 is used by event sorter 370 to determine which application(s) is (are) currently active or in focus, and application internal state 392 is used by event sorter 370 to determine application views 391 to which to deliver event information.

In some embodiments, application internal state 392 includes additional information, such as one or more of: resume information to be used when application 340-1 resumes execution, user interface state information that indicates information being displayed or that is ready for display by application 340-1, a state queue for enabling the user to go back to a prior state or view of application 340-1, and a redo/undo queue of previous actions taken by the user.

Event monitor 371 receives event information from peripherals interface 318. Event information includes information about a sub-event (e.g., a user touch on display system 312 when it is a touch-sensitive display, as part of a multi-touch gesture). Peripherals interface 318 transmits information it receives from I/O subsystem 306 or a sensor, such as proximity sensor(s) 366, accelerometer(s) 368, and/or microphone 313 (through audio circuitry 310). Information that peripherals interface 318 receives from I/O subsystem 306 includes information from display system 312 when it is a touch-sensitive display or another touch-sensitive surface (e.g., touch pad 108, FIGS. 1A-1B).

In some embodiments, event monitor 371 sends requests to the peripherals interface 318 at predetermined intervals. In response, peripherals interface 318 transmits event information. In other embodiments, peripheral interface 318 transmits event information only when there is a significant event (e.g., receiving an input above a predetermined noise threshold and/or for more than a predetermined duration).

In some embodiments, event sorter 370 also includes a hit view determination module 372 and/or an active event recognizer determination module 373.

Hit view determination module 372 provides software procedures for determining where a sub-event has taken place within one or more views, when display system 312 displays more than one view, where views are made up of controls and other elements that a user can see on the display.

Another aspect of the user interface associated with an application is a set of views, sometimes herein called

application views or user interface windows, in which information is displayed and touch-based gestures occur. The application views (of an application) in which a touch is detected optionally correspond to programmatic levels within a programmatic or view hierarchy of the application. For example, the lowest level view in which a touch is detected is, optionally, called the hit view, and the set of events that are recognized as proper inputs are, optionally, determined based, at least in part, on the hit view of the initial touch that begins a touch-based gesture.

Hit view determination module 372 receives information related to sub-events of a touch-based gesture. When an application has multiple views organized in a hierarchy, hit view determination module 372 identifies a hit view as the lowest view in the hierarchy which should handle the sub-event. In most circumstances, the hit view is the lowest level view in which an initiating sub-event occurs (i.e., the first sub-event in the sequence of sub-events that form an event or potential event). Once the hit view is identified by the hit view determination module, the hit view typically receives all sub-events related to the same touch or input source for which it was identified as the hit view.

Active event recognizer determination module 373 determines which view or views within a view hierarchy should receive a particular sequence of sub-events. In some embodiments, active event recognizer determination module 373 determines that only the hit view should receive a particular sequence of sub-events. In other embodiments, active event recognizer determination module 373 determines that all views that include the physical location of a sub-event are actively involved views, and therefore determines that all actively involved views should receive a particular sequence of sub-events. In other embodiments, even if touch sub-events were entirely confined to the area associated with one particular view, views higher in the hierarchy would still remain as actively involved views.

Event dispatcher module 374 dispatches the event information to an event recognizer (e.g., event recognizer 380). In embodiments including active event recognizer determination module 373, event dispatcher module 374 delivers the event information to an event recognizer determined by active event recognizer determination module 373. In some embodiments, event dispatcher module 374 stores in an event queue the event information, which is retrieved by a respective event receiver module 382.

In some embodiments, operating system 326 includes event sorter 370. Alternatively, application 340-1 includes event sorter 370. In yet other embodiments, event sorter 370 is a stand-alone module, or a part of another module stored in memory 302, such as contact/motion module 330.

In some embodiments, application 340-1 includes a plurality of event handlers 390 and one or more application views 391, each of which includes instructions for handling touch events that occur within a respective view of the application's user interface. Each application view 391 of the application 340-1 includes one or more event recognizers 380. Typically, an application view 391 includes a plurality of event recognizers 380. In other embodiments, one or more of event recognizers 380 are part of a separate module, such as a user interface kit (not shown) or a higher level object from which application 340-1 inherits methods and other properties. In some embodiments, a respective event handler 390 includes one or more of: data updater 376, object updater 377, GUI updater 378, and/or event data 379 received from event sorter 370. Event handler 390 optionally utilizes or calls data updater 376, object updater 377 or GUI updater 378 to update the application internal state 392.

Alternatively, one or more of the application views **391** includes one or more respective event handlers **390**. Also, in some embodiments, one or more of data updater **376**, object updater **377**, and GUI updater **378** are included in an application view **391**.

A respective event recognizer **380** receives event information (e.g., event data **379**) from event sorter **370**, and identifies an event from the event information. Event recognizer **380** includes event receiver **382** and event comparator **384**. In some embodiments, event recognizer **380** also includes at least a subset of: metadata **383**, and event delivery instructions **388** (which optionally include sub-event delivery instructions).

Event receiver **382** receives event information from event sorter **370**. The event information includes information about a sub-event, for example, a touch or a touch movement. Depending on the sub-event, the event information also includes additional information, such as location of the sub-event. When the sub-event concerns motion of a touch, the event information optionally also includes speed and direction of the sub-event. In some embodiments, events include rotation of the device from one orientation to another (e.g., from a portrait orientation to a landscape orientation, or vice versa), and the event information includes corresponding information about the current orientation (also called device attitude) of the device.

Event comparator **384** compares the event information to predefined event or sub-event definitions and, based on the comparison, determines an event or sub-event, or determines or updates the state of an event or sub-event. In some embodiments, event comparator **384** includes event definitions **386**. Event definitions **386** contain definitions of events (e.g., predefined sequences of sub-events), for example, event 1 (**387-1**), event 2 (**387-2**), and others. In some embodiments, sub-events in an event **387** include, for example, touch begin, touch end, touch movement, touch cancellation, and multiple touching. In one example, the definition for event 1 (**387-1**) is a double tap on a displayed object. The double tap, for example, comprises a first touch (touch begin) on the displayed object for a predetermined phase, a first lift-off (touch end) for a predetermined phase, a second touch (touch begin) on the displayed object for a predetermined phase, and a second lift-off (touch end) for a predetermined phase. In another example, the definition for event 2 (**387-2**) is a dragging on a displayed object. The dragging, for example, comprises a touch (or contact) on the displayed object for a predetermined phase, a movement of the touch across display system **312** when it is a touch-sensitive display, and lift-off of the touch (touch end). In some embodiments, the event also includes information for one or more associated event handlers **390**.

In some embodiments, event definition **387** includes a definition of an event for a respective user-interface object. In some embodiments, event comparator **384** performs a hit test to determine which user-interface object is associated with a sub-event. For example, in an application view in which three user-interface objects are displayed on display system **312**, when a touch is detected on display system **312** when it is a touch-sensitive display, event comparator **384** performs a hit test to determine which of the three user-interface objects is associated with the touch (sub-event). If each displayed object is associated with a respective event handler **390**, the event comparator uses the result of the hit test to determine which event handler **390** should be activated. For example, event comparator **384** selects an event handler associated with the sub-event and the object triggering the hit test.

In some embodiments, the definition for a respective event **387** also includes delayed actions that delay delivery of the event information until after it has been determined whether the sequence of sub-events does or does not correspond to the event recognizer's event type.

When a respective event recognizer **380** determines that the series of sub-events do not match any of the events in event definitions **386**, the respective event recognizer **380** enters an event impossible, event failed, or event ended state, after which it disregards subsequent sub-events of the touch-based gesture. In this situation, other event recognizers, if any, that remain active for the hit view continue to track and process sub-events of an ongoing touch-based gesture.

In some embodiments, a respective event recognizer **380** includes metadata **383** with configurable properties, flags, and/or lists that indicate how the event delivery system should perform sub-event delivery to actively involved event recognizers. In some embodiments, metadata **383** includes configurable properties, flags, and/or lists that indicate how event recognizers interact, or are enabled to interact, with one another. In some embodiments, metadata **383** includes configurable properties, flags, and/or lists that indicate whether sub-events are delivered to varying levels in the view or programmatic hierarchy.

In some embodiments, a respective event recognizer **380** activates event handler **390** associated with an event when one or more particular sub-events of an event are recognized. In some embodiments, a respective event recognizer **380** delivers event information associated with the event to event handler **390**. Activating an event handler **390** is distinct from sending (and deferred sending) sub-events to a respective hit view. In some embodiments, event recognizer **380** throws a flag associated with the recognized event, and event handler **390** associated with the flag catches the flag and performs a predefined process.

In some embodiments, event delivery instructions **388** include sub-event delivery instructions that deliver event information about a sub-event without activating an event handler. Instead, the sub-event delivery instructions deliver event information to event handlers associated with the series of sub-events or to actively involved views. Event handlers associated with the series of sub-events or with actively involved views receive the event information and perform a predetermined process.

In some embodiments, data updater **376** creates and updates data used in application **340-1**. For example, data updater **376** stores a video file used by media player module **344**. In some embodiments, object updater **377** creates and updates objects used by application **340-1**. For example, object updater **376** creates a new user-interface object or updates the position of a user-interface object. GUI updater **378** updates the GUI. For example, GUI updater **378** prepares display information and sends it to graphics module **332** for display on display system **312** (e.g., primary display **102** of display portion **110**, FIG. 1A and/or dynamic function row **104**, FIGS. 1A-1B).

In some embodiments, event handler(s) **390** includes or has access to data updater **376**, object updater **377**, and GUI updater **378**. In some embodiments, data updater **376**, object updater **377**, and GUI updater **378** are included in a single module of an application **340-1** or application view **391**. In other embodiments, they are included in two or more software modules.

It shall be understood that the foregoing discussion regarding event handling of user touches on touch-sensitive displays also applies to other forms of user inputs to operate

electronic device **300** with input-devices, not all of which are initiated on touch screens. For example, mouse movement and mouse button presses, optionally coordinated with single or multiple keyboard presses or holds; contact movements such as taps, drags, scrolls, etc., on touch-pads; pen stylus inputs; movement of the device; oral instructions; detected eye movements; biometric inputs; and/or any combination thereof are optionally utilized as inputs corresponding to sub-events which define an event to be recognized.

FIG. 4 shows a block diagram of a peripheral electronic device **400**, in accordance with some embodiments. In some embodiments, peripheral electronic device **400** is a peripheral input and output device that at least partially contains a dynamic function row **104** and a physical input mechanism, such as a set of physical keys (e.g., the set of physical keys **106**, FIGS. 2A-2B) and/or a touch pad (e.g., touch pad **108**, FIGS. 2B-2C), within a same housing. Examples of peripheral electronic device **400** includes: peripheral keyboard (e.g., peripheral keyboard **206**, FIGS. 2A-2B), a peripheral touch-sensitive surface (e.g., first peripheral input mechanism **212**, FIG. 2C), or other peripheral input mechanisms (e.g., second peripheral input mechanism **222**, FIG. 2D). Peripheral electronic device **400** is communicatively coupled with computing device **202** (FIGS. 2A-2D). For example, peripheral electronic device **400** is communicatively coupled with computing device **202** via a wired connection, such as USB or PS/2, or via a wireless communication link, using a communication protocol such as Bluetooth, Wi-Fi, or the like. Peripheral electronic device **400** may rely on some of the components or procedures in electronic device **300** (FIG. 3A) or some of these components or procedures may be completed by, located in, or housed by peripheral electronic device **400** instead of electronic device **300**.

In some embodiments, peripheral electronic device **400** includes one or more of memory **402** (which optionally includes one or more computer readable storage mediums), memory controller **422**, one or more processing units (CPU(s)) **420**, peripherals interface **418**, RF circuitry **408**, audio circuitry **410**, speaker **411**, microphone **413**, input/output (I/O) subsystem **406**, other input or control devices **416**, and external port **424**. Peripheral electronic device **400** includes a touch-sensitive display system **412** (e.g., dynamic function row **104**, FIGS. 2A-2D) (sometimes also herein called a "touch-sensitive display," a "touch screen," or a "touch screen display").

Peripheral electronic device **400** optionally includes one or more intensity sensors **465** for detecting intensity of contacts on a touch-sensitive surface such as touch-sensitive display system **412** or a touch pad (e.g., touch pad **108**, FIGS. 2B-2C). Peripheral electronic device **400** optionally includes one or more tactile output generators **467** for generating tactile outputs on a touch-sensitive surface such as touch-sensitive display system **412** or a touch pad (e.g., touch pad **108**, FIGS. 2B-2C). These components optionally communicate over one or more communication buses or signal lines **403**.

Memory **402** optionally includes high-speed random access memory and optionally also includes non-volatile memory, such as one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices. Access to memory **402** by other components of peripheral electronic device **400**, such as CPU(s) **420** and peripherals interface **418**, is, optionally, controlled by memory controller **422**. Peripherals interface **418** can be used to couple CPU(s) **420** and memory **402** to I/O subsystem **406** and other circuitry. The one or more processing

units **420** run or execute various software programs and/or sets of instructions stored in memory **402** to perform various functions for peripheral electronic device **400** and to process data. In some embodiments, peripherals interface **418**, CPU(s) **420**, and memory controller **422** are, optionally, implemented on a single chip, such as chip **404**. In some other embodiments, they are, optionally, implemented on separate chips.

RF (radio frequency) circuitry **408** receives and sends RF signals, also called electromagnetic signals. RF circuitry **408** converts electrical signals to/from electromagnetic signals and communicates with communications networks and other communications devices via the electromagnetic signals. RF circuitry **408** optionally includes well-known circuitry for performing these functions, including but not limited to an antenna system, an RF transceiver, one or more amplifiers, a tuner, one or more oscillators, a digital signal processor, a CODEC chipset, a subscriber identity module (SIM) card, memory, and so forth. The wireless communication optionally uses any of a plurality of communications standards, protocols and technologies, including but not limited to near field communication (NFC), Bluetooth, Wireless Fidelity (Wi-Fi) (e.g., IEEE 802.11a, IEEE 802.11b, IEEE 802.11g, and/or IEEE 802.11n), Wi-MAX, or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document.

Optional audio circuitry **410**, speaker **411**, and microphone **413** provide an audio interface between a user and peripheral electronic device **400**. Audio circuitry **410** receives audio data from peripherals interface **418**, converts the audio data to an electrical signal, and transmits the electrical signal to speaker **411**. Speaker **411** converts the electrical signal to human-audible sound waves. Audio circuitry **410** also receives electrical signals converted by microphone **413** from sound waves. Audio circuitry **410** converts the electrical signals to audio data and transmits the audio data to peripherals interface **418** for processing. Audio data is, optionally, retrieved from and/or transmitted to memory **402** and/or RF circuitry **408** by peripherals interface **418**. In some embodiments, audio circuitry **410** also includes a headset jack. The headset jack provides an interface between audio circuitry **410** and removable audio input/output peripherals, such as output-only headphones or a headset with both output (e.g., a headphone for one or both ears) and input (e.g., a microphone).

I/O subsystem **406** couples the input/output peripherals of peripheral electronic device **400**, such as touch-sensitive display system **412** (e.g., dynamic function row **104**, FIGS. 2A-2D), to peripherals interface **418**. I/O subsystem **406** optionally includes display controller **456**, intensity sensor controller **459**, haptic feedback controller **461**, and one or more input controllers **460** for other input or control devices **416**. The one or more other input controllers **460** receive/send electrical signals from/to other input or control devices **416**. The other input or control devices **416** optionally include physical buttons (e.g., push buttons, rocker buttons, etc.), dials, slider switches, joysticks, click wheels, a set of physical keys, a touch pad, and so forth.

Touch-sensitive display system **412** (e.g., dynamic function row **104**, FIGS. 2A-2D) provides an input/output interface between peripheral electronic device **400** and a user. Touch-sensitive display (TSD) controller **456** receives and/or sends electrical signals from/to touch-sensitive display system **412**. Touch-sensitive display system **412** displays visual output to the user. The visual output optionally includes graphics, text, icons, video, and any combination thereof (collectively termed "graphics"). In some embodi-

ments, some or all of the visual output corresponds to user-interface objects/elements.

Touch-sensitive display system **412** (e.g., dynamic function row **104**, FIGS. 2A-2D) includes a touch-sensitive surface, sensor, or set of sensors that accepts input from the user based on haptic and/or tactile contact. As such, touch-sensitive display system **412** and TSD controller **456** (along with any associated modules and/or sets of instructions in memory **402**) detect contact (and any movement or breaking of the contact) on touch-sensitive display system **412** and convert the detected contact into signals used to select or control user-interface objects (e.g., one or more soft keys, icons, web pages, or images) that are displayed on touch-sensitive display system **412**. In one example embodiment, a point of contact between touch-sensitive display system **412** and the user corresponds to an area of touch-sensitive display system **412** in contact with a finger of the user.

Touch-sensitive display system **412** (e.g., dynamic function row **104**, FIGS. 2A-2D) optionally uses LCD (liquid crystal display) technology, LPD (light emitting polymer display) technology, LED (light emitting diode) technology, or OLED (organic light emitting diode) technology, although other display technologies are used in other embodiments. Touch-sensitive display system **412** and TSD controller **456** optionally detect contact and any movement or breaking thereof using any of a plurality of touch sensing technologies now known or later developed, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with touch-sensitive display system **412**. In one example embodiment, projected mutual capacitance sensing technology is used, such as that found in the iPhone®, iPodTouch®, and iPad® from Apple Inc. of Cupertino, California.

Touch-sensitive display system **412** (e.g., dynamic function row **104**, FIGS. 2A-2D) optionally has a video resolution in excess of 400 dpi (e.g., 500 dpi, 800 dpi, or greater). In some embodiments, the user makes contact with touch-sensitive display system **412** using a stylus, a finger, and so forth. In some embodiments, the user interface is designed to work primarily with finger-based contacts and gestures.

In some embodiments, in addition to touch-sensitive display system **412**, peripheral electronic device **400** optionally includes a touch pad (e.g., touch pad **108**, FIGS. 2B-2C). In some embodiments, the touch pad is a touch-sensitive area of peripheral electronic device **400** that, unlike touch-sensitive display system **412**, does not display visual output. In some embodiments, the touch pad is, optionally, a touch-sensitive surface that is separate from touch-sensitive display system **412**, or an extension of the touch-sensitive surface formed by touch-sensitive display system **412**.

Peripheral electronic device **400** also includes power system **462** for powering the various components. Power system **462** optionally includes a power management system, one or more power sources (e.g., battery, alternating current (AC), etc.), a recharging system, a power failure detection circuit, a power converter or inverter, a power status indicator (e.g., a light-emitting diode (LED)) and any other components associated with the generation, management and distribution of power in portable devices.

Peripheral electronic device **400** optionally also includes one or more contact intensity sensors **465** coupled with intensity sensor controller **459** in I/O subsystem **406**. Contact intensity sensor(s) **465** optionally includes one or more piezoresistive strain gauges, capacitive force sensors, elec-

tric force sensors, piezoelectric force sensors, optical force sensors, capacitive touch-sensitive surfaces, or other intensity sensors (e.g., sensors used to measure the force (or pressure) of a contact on a touch-sensitive surface). Contact intensity sensor(s) **465** receives contact intensity information (e.g., pressure information or a proxy for pressure information) from the environment. In some embodiments, at least one contact intensity sensor is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system **412** and/or touch pad **108**, FIGS. 2B-2C).

Peripheral electronic device **400** optionally also includes one or more tactile output generators **467** coupled with haptic feedback controller **461** in I/O subsystem **406**. Tactile output generator(s) **467** optionally includes one or more electroacoustic devices such as speakers or other audio components and/or electromechanical devices that convert energy into linear motion such as a motor, solenoid, electroactive polymer, piezoelectric actuator, electrostatic actuator, or other tactile output generating component (e.g., a component that converts electrical signals into tactile outputs on the device). Contact intensity sensor(s) **465** receives tactile feedback generation instructions from haptic feedback module **433** and generates tactile outputs that are capable of being sensed by a user of peripheral electronic device **400**. In some embodiments, at least one tactile output generator is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system **412** and/or touch pad **108**, FIGS. 2B-2C) and, optionally, generates a tactile output by moving the touch-sensitive surface vertically (e.g., in/out of a surface of peripheral electronic device **400**) or laterally (e.g., back and forth in the same plane as a surface of peripheral electronic device **400**).

In some embodiments, the software components stored in memory **402** include operating system **426**, communication module **428** (or set of instructions), contact/motion module **430** (or set of instructions), and dynamic function row module **450** (or sets of instructions). Furthermore, in some embodiments, memory **402** stores device state **457** including the display state, indicating what views or other information occupy various regions of touch-sensitive display system **412** (e.g., dynamic function row **104**, FIGS. 2A-2D).

Operating system **426** includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device control, power management, etc.) and facilitates communication between various hardware and software components.

Communication module **428** facilitates communication with other devices (e.g., computing device **202**, FIGS. 2A-2D) over one or more external ports **424** and/or RF circuitry **408** and also includes various software components for sending/receiving data via RF circuitry **408** and/or external port **424**. External port **424** (e.g., Universal Serial Bus (USB), FIREWIRE, etc.) is adapted for coupling directly to other devices or indirectly over a network (e.g., the Internet, wireless LAN, etc.).

Contact/motion module **430** optionally detects contact with touch-sensitive display system **412** and other touch sensitive devices (e.g., a touch pad or physical click wheel). Contact/motion module **430** includes various software components for performing various operations related to detection of contact, such as determining if contact has occurred (e.g., detecting a finger-down event), determining an intensity of the contact (e.g., the force or pressure of the contact or a substitute for the force or pressure of the contact), determining if there is movement of the contact and tracking the movement across the touch-sensitive surface (e.g., detecting one or more finger-dragging events), and deter-

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mining if the contact has ceased (e.g., detecting a finger-up event or a break in contact). Contact/motion module **430** receives contact data from the touch-sensitive surface. Determining movement of the point of contact, which is represented by a series of contact data, optionally includes determining speed (magnitude), velocity (magnitude and direction), and/or an acceleration (a change in magnitude and/or direction) of the point of contact. These operations are, optionally, applied to single contacts (e.g., one finger contacts) or to multiple simultaneous contacts (e.g., “multitouch”/multiple finger contacts). In some embodiments, contact/motion module **430** also detects contact on a touch pad (e.g., touch pad **108**, FIGS. 2B-2C).

In some embodiments, contact/motion module **430** uses a set of one or more intensity thresholds to determine whether an operation has been performed by a user (e.g., to determine whether a user has selected or “clicked” on an affordance). In some embodiments at least a subset of the intensity thresholds are determined in accordance with software parameters (e.g., the intensity thresholds are not determined by the activation thresholds of particular physical actuators and can be adjusted without changing the physical hardware of peripheral electronic device **400**). For example, a mouse “click” threshold of a trackpad or touch screen display can be set to any of a large range of predefined thresholds values without changing the trackpad or touch screen display hardware. Additionally, in some implementations a user of the device is provided with software settings for adjusting one or more of the set of intensity thresholds (e.g., by adjusting individual intensity thresholds and/or by adjusting a plurality of intensity thresholds at once with a system-level click “intensity” parameter).

Contact/motion module **430** optionally detects a gesture input by a user. Different gestures on the touch-sensitive surface have different contact patterns (e.g., different motions, timings, and/or intensities of detected contacts). Thus, a gesture is, optionally, detected by detecting a particular contact pattern. For example, detecting a finger tap contact includes detecting a finger-down event followed by detecting a finger-up (lift off) event at the same position (or substantially the same position) as the finger-down event (e.g., at the position of an icon). As another example, detecting a finger swipe gesture on the touch-sensitive surface includes detecting a finger-down event followed by detecting one or more finger-dragging events, and in some embodiments also followed by detecting a finger-up (lift off) event.

Haptic feedback module **433** includes various software components for generating instructions used by tactile output generator(s) **467** to produce tactile outputs at one or more locations on peripheral electronic device **400** in response to user interactions with peripheral electronic device **400**.

Dynamic function row (DFR) module **450** includes: focus obtaining module **451**, DFR determining module **452**, and DFR presenting module **453**. In some embodiments, focus obtaining module **451** is configured to obtain an indication of an active user interface element that is the current focus of the graphical user interface displayed on primary display **102** of peripheral display device **204** (FIGS. 2A-2D) from computing device **202** (FIGS. 2A-2D). In some embodiments, DFR determining module **452** is configured to determine graphics (e.g., a set of one or more affordances) based on the active user interface element that is current focus. Alternatively, in some embodiments, computing device **202** (FIGS. 2A-2D) determines the graphics (e.g., the set of one or more affordances) based on the active user interface

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element that is in focus and provides the graphics to peripheral electronic device **400** or a component thereof (e.g., DFR module **450**) for display on touch-sensitive display system **412** (e.g., dynamic function row **104**, FIGS. 2A-2D). In some embodiments, DFR presenting module **453** is configured to render the graphics determined by DFR determining module **452** (or provided by computing device **202**) on touch-sensitive display system **412** (e.g., dynamic function row **104**, FIGS. 2A-2D). DFR presenting module **453** includes various known software components for rendering and causing display of graphics on touch-sensitive display system **412**, including components for changing the visual impact (e.g., brightness, transparency, saturation, contrast or other visual property) of graphics that are displayed. In some embodiments, DFR module **450** includes other modules for: adjusting the sensitivity of dynamic function row **104**; adjusting the audible and/or haptic feedback provided by dynamic function row **104**; adjusting the settings of affordances and information displayed by dynamic function row **104** (e.g., size, brightness, font, language, and the like); adjusting the current power mode of dynamic function row **104** (e.g., normal and low-power modes); and the like.

In some embodiments, memory **402** includes event sorter **470** (e.g., in operating system **426**). In some embodiments, event sorter **470** performs the same functions as event sorter **370** (FIG. 3B) and includes a subset or superset of the modules, procedures, and instructions of event sorter **370** (FIG. 3B). As such, event sorter **470** will not be described for the sake of brevity.

It should be appreciated that peripheral electronic device **400** is only an example and that peripheral electronic device **400** optionally has more or fewer components than shown, optionally combines two or more components, or optionally has a different configuration or arrangement of the components. The various components shown in FIG. 4 are implemented in hardware, software, firmware, or a combination thereof, including one or more signal processing and/or application specific integrated circuits.

Each of the above identified modules correspond to a set of executable instructions for performing one or more functions described above and the methods described in this application (e.g., the computer-implemented methods and other information processing methods described herein). These modules (i.e., sets of instructions) need not be implemented as separate software programs, procedures or modules, and thus various subsets of these modules are, optionally, combined or otherwise re-arranged in various embodiments. In some embodiments, memory **402** optionally stores a subset of the modules and data structures identified above. Furthermore, memory **402** optionally stores additional modules and data structures not described above.

User Interfaces and Associated Processes

Attention is now directed towards embodiments of user interfaces (“UIs”) and associated processes that may be implemented by portable computing system **100** (FIG. 1A) or desktop computing system **200** (FIGS. 2A-2D). In some embodiments, primary display **102** is implemented in display portion **110** of portable computing system **100** (FIG. 1A). Alternatively, in some embodiments, primary display **102** is implemented in peripheral display device **204** (FIGS. 2A-2D). In some embodiments, dynamic function row **104** is a touch-sensitive display implemented in body portion **120** of portable computing system **100** (FIGS. 1A-1B). Alternatively, in some embodiments, dynamic function row

104 is a touch-sensitive display implemented in peripheral keyboard **206** (FIGS. 2A-2B), first input mechanism **212** (FIG. 2C), or second input mechanism **222** (FIG. 2D).

FIGS. 5A-5DDD illustrate example user interfaces for displaying application-specific affordances on a dynamically updated touch screen display in accordance with some embodiments. The user interfaces in these figures are used to illustrate the methods and/or processes described below, including the methods in FIGS. 6A-6D, 7A-7C, 8A-8B, 9A-9B, and 10A-10C. One of ordinary skill in the art will appreciate that the following user interfaces are merely examples. Moreover, one of ordinary skill in the art will appreciate that additional affordances and/or user interface elements, or that fewer affordances and/or user interface elements may be used in practice.

FIG. 5A illustrates primary display **102** displaying a status tray **502** indicating that the system (i.e., the operating system) is currently in focus, and an application (app) tray **514** with a plurality of executable/selectable application icons including: a mail application icon **506** (e.g., corresponding to e-mail client module **341**, FIG. 3A), a web browser application icon **508** (e.g., corresponding to web browsing module **345**, FIG. 3A), a media player application icon **510** (e.g., corresponding to media player module **344**, FIG. 3A), an application A icon **512** (e.g., corresponding to a game), and a photo application icon **515** (e.g., corresponding to image management module **343**, FIG. 3A). In some embodiments, status tray **502** indicates an application that is currently running in the foreground and also includes a plurality of menus (e.g., the file, edit, view, go, window, and help menus in FIG. 5A) each including a set of corresponding controls for the application. FIG. 5A also illustrates primary display **102** displaying cursor **504** at a location corresponding to application A icon **512**. In some embodiments, cursor **504** is controlled by touch pad **108** of portable computing system **100** (FIGS. 1A-1B), peripheral mouse **208** of desktop computing system **200** (FIGS. 2A and 2D), touch-pad **108** of peripheral keyboard **206** (FIG. 2B), touch-pad **108** of first input mechanism **212** (FIG. 2C), or the like.

FIG. 5A further illustrates dynamic function row **104** (e.g., a touch-sensitive display) displaying a plurality of affordances based on the current focus of primary display **102** (i.e., the operating system because no application windows are open). For example, in FIG. 5A, the system/operating system is currently in focus on primary display **102**. In FIG. 5A, dynamic function row **104** includes persistent controls implemented as physical and/or soft keys, including: escape affordance **516**, which, when activated (e.g., via a tap contact), invokes a corresponding function (e.g., exiting an application which is currently in focus on primary display **102** or pausing a game); and power control **534**, which, when activated (e.g., via a tap contact), causes display of a modal alert (e.g., modal alert **5308**, FIG. 5DDD) on dynamic function row **104** and/or primary display **102** for logging out, restarting, or powering-off portable computing system **100** or desktop computing system **200**.

In FIG. 5A, dynamic function row **104** also includes a plurality of system-level affordances, including: brightness affordance **518** for adjusting the brightness of primary display **102**, FIGS. 1A and 2A-2B; brightness affordance **520** for adjusting the brightness of the set of physical keys **106**, FIGS. 1A-1B and 2A-2B (when applicable) and/or the brightness of dynamic function row **104**; exposé affordance **522**, which, when activated (e.g., via a tap contact), causes display of preview windows for active applications on primary display **102**, FIGS. 1A and 2A-2B; search affordance **524** for performing a local search (e.g., for an elec-

tronic document) and/or an Internet search; launchpad affordance **526**, which, when activated (e.g., via a tap contact), causes display of default or user-selected widgets and tools on primary display **102**, FIGS. 1A and 2A-2B; notifications affordance **528**, which, when activated (e.g., via a tap contact), causes display of a notification center on primary display **102**, FIGS. 1A and 2A-2B including recent messages, notifications, calendar events, and/or the like; play/pause affordance **530** for initiating playback or pausing playback of media items (e.g., songs, podcasts, videos, and the like); and volume control affordance **532** for adjusting the volume of a media item being played. For example, when a tap is detected on brightness affordance **520**, dynamic function row **104** displays a brightness slider for adjusting the brightness of the set of physical keys **106** and/or the brightness of dynamic function row **104** (e.g., similar to the volume slider **5100** in FIG. 5K). In some embodiments, the plurality of system-level affordances also include a settings affordance (not shown) for accessing adjusting settings associated with the dynamic function row **104** such as symbol/icon size, touch detection sensitivity, haptic feedback, audible feedback, animations for change in focus, power modes, and the like.

FIG. 5B illustrates primary display **102** displaying a window **536** for application A (e.g., a fantasy RPG game) in response to detecting selection of application A icon **512** with cursor **504** in FIG. 5A. In FIG. 5B, application A is in a main menu mode (e.g., the fantasy RPG game is paused), and window **536** displays a main menu for application A. Window **536** for application A is in focus on primary display **102**. In FIG. 5B, status tray **502** indicates that application A is running in the foreground, and app tray **514** also indicates that application A is running in the foreground based on the shadow behind application A icon **512**. In FIG. 5B, window **536** for application A includes three selectable affordances in the upper left-hand corner for closing window **536**, maximizing the size of window **536**, and minimizing window **536** (from left-to-right, respectively).

FIG. 5B also illustrates dynamic function row **104** displaying affordance **538** in addition to the persistent controls (i.e., affordances **516** and **534**) and the plurality of system-level affordances (i.e., affordances **518**, **520**, **522**, **524**, **526**, **528**, **530**, and **532**) in response to detecting selection of application A icon **512** with cursor **504** in FIG. 5A. When activated (e.g., via a tap contact), affordance **538** causes dynamic function row **104** to display a first set of affordances and/or indicators corresponding to application A (e.g., control set A in FIG. 5C). FIG. 5B further illustrates dynamic function row **104** receiving and detecting contact **540** (e.g., a tap contact) at a location corresponding to affordance **538**.

FIG. 5C illustrates dynamic function row **104** displaying a first set of affordances and/or indicators (e.g., control set A) corresponding to application A and at least one system-level affordance (e.g., affordance **542**) in response to detecting selection of affordance **538** in FIG. 5B. In FIG. 5C, the first set of affordances and/or indicators (e.g., control set A) corresponding to application A (e.g., the fantasy RPG game) includes a health indicator **543** and a mana indicator **545** related to an in-game character/avatar controlled by the user of portable computing system **100** or desktop computing system **200** while playing application A. In FIG. 5C, the first set of affordances and/or indicators (e.g., control set A) corresponding to application A also includes control affordances **546-A**, **546-B**, and **546-C** for controlling the in-game character/avatar. When activated (e.g., via a tap contact), affordance **542** causes dynamic function row **104** to display

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the plurality of system-level affordances (e.g., affordances **518**, **520**, **522**, **524**, **526**, **528**, **530**, and **532** shown in FIG. 5A). FIG. 5C also illustrates dynamic function row **104** receiving and detecting an upward swipe gesture with contact **544** moving from a first location **548-A** to a second location **548-B**.

FIG. 5D illustrates dynamic function row **104** displaying a second set of affordances and/or indicators (e.g., control set B) corresponding to application A and the at least one system-level affordance (e.g., affordance **542**) in response to detecting the upward swipe gesture in FIG. 5C. In FIG. 5D, the second set of affordances and/or indicators (e.g., control set B) corresponding to application A includes control affordances **546-D**, **546-E**, **546-F**, **546-G**, **546-H**, **546-I**, **546-J**, and **546-K** for controlling the in-game character/ avatar controlled by the user of portable computing system **100** or desktop computing system **200** while playing application A. FIG. 5D also illustrates dynamic function row **104** receiving and detecting contact **552** (e.g., a tap contact) at a location corresponding to affordance **542**.

FIG. 5E illustrates dynamic function row **104** displaying persistent controls (i.e., affordances **516** and **534**), the plurality of system-level affordances (i.e., affordances **518**, **520**, **522**, **524**, **526**, **528**, **530**, and **532**), and affordance **538** corresponding to application A in response to detecting selection of affordance **542** in FIG. 5D. FIG. 5E also illustrates primary display **102** displaying cursor **504** at a new location corresponding to media player application icon **510**.

FIG. 5F illustrates primary display **102** displaying a window **554** for the media player application in response to detecting selection of media player application icon **510** with cursor **504** in FIG. 5E. For example, window **554** is overlaid on window **536**. In FIG. 5F, window **554** displays a plurality of albums associated with a music sub-section of a user's media library. In FIG. 5F, the music sub-section of the user's media library is in focus on primary display **102** as shown by "Music" displayed in bold and albums A-L at least partially displayed in window **554**. In FIG. 5F, status tray **502** indicates that the media player application is running in the foreground, and app tray **514** also indicates that the media player application is running in the foreground based on the shadow behind media player application icon **510**. FIG. 5F also illustrates primary display **102** displaying cursor **504** at a location corresponding to the podcasts sub-section of the user's media library.

FIG. 5F further illustrates dynamic function row **104** displaying a plurality of album affordances **558** (e.g., album affordances **558-A** to **558-G**) and the at least one system-level affordance (e.g., affordance **542**) in response to detecting selection of media player application icon **510** with cursor **504** in FIG. 5E. In FIG. 5F, the plurality of album affordances **558** correspond to a subset of the albums currently displayed in window **554**. In some embodiments, the plurality of album affordances **558** mirror the albums currently displayed in window **554**. For example, in response to selection of album affordance **558-D** (e.g., via a tap contact), portable computing device **100** or computing device **200** causes playback of album D by audio circuitry **310** (FIG. 3A) and also causes primary display **102** to display album D in the now playing region of window **554**.

FIG. 5G illustrates primary display **102** displaying a first plurality of podcasts in window **554** in response to detecting selection of the podcasts sub-section with cursor **504** in FIG. 5F. In FIG. 5G, window **554** displays a plurality of podcasts associated with a podcasts sub-section of the user's media library. In FIG. 5G, the podcasts sub-section of the user's

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media library is in focus on primary display **102** as shown by "Podcasts" displayed in bold in the menu and podcasts A-L at least partially displayed in window **554**.

FIG. 5G also illustrates dynamic function row **104** displaying a first plurality of podcast affordances **560** (e.g., podcast affordances **560-A** to **560-G**) and the at least one system-level affordance (e.g., affordance **542**) in response to detecting selection of the podcasts sub-section with cursor **504** in FIG. 5F. In FIG. 5G, the plurality of podcast affordances **560** correspond to a subset of the podcasts currently displayed in window **554**. For example, in response to selection of podcast affordance **560-D** (e.g., via a tap contact), portable computing device **100** or computing device **200** causes playback of podcast D by audio circuitry **310** (FIG. 3A) and also causes primary display **102** to display podcast D in the now playing region of window **554**. FIG. 5G further illustrates dynamic function row **104** detecting a right-to-left swipe gesture with contact **562** moving from a first location **564-A** to a second location **564-B**.

FIG. 5H illustrates primary display **102** displaying a second plurality of podcast affordances **560** (e.g., podcast affordances **560-E** to **560-P**) in window **554** in response to detecting the right-to-left swipe gesture in 5G. FIG. 5H also illustrates dynamic function row **104** displaying a second plurality of podcast affordances **560** (e.g., podcast affordances **560-E** to **560-K**) and the at least one system-level affordance (e.g., affordance **542**) in response to detecting the right-to-left swipe gesture in 5G. FIG. 5H further illustrates dynamic function row **104** receiving and detecting contact **566** (e.g., a tap contact) at a location corresponding to podcast affordance **560-J**.

FIG. 5I illustrates primary display **102** displaying playback of podcast J in window **554** in response to detecting selection of podcast affordance **560-J** in FIG. 5H. FIG. 5I also illustrates primary display **102** displaying cursor **504** at a location corresponding to mail application icon **506**.

FIG. 5I further illustrates dynamic function row **104** displaying persistent volume control **568** along with playback controls and indicators in response to detecting selection of podcast affordance **560-J** in FIG. 5H. In FIG. 5I, persistent volume control **568** indicates that podcast J is not muted and also displays equalizer feedback for podcast J. In FIG. 5I, the playback controls include a rewind control **571**, pause control **572**, and fast-forward control **573** for controlling the playback of podcast J. In FIG. 5I, the playback indicators include an image **574** corresponding to podcast J (e.g., cover art or an associated image), indicator **576** displaying the author and title of podcast J, and a time remaining indicator **578**.

FIG. 5J illustrates primary display **102** displaying a window **580** for the mail application in response to detecting selection of mail application icon **506** with cursor **504** in FIG. 5I. For example, window **580** is overlaid on windows **554** and **536**. In FIG. 5J, window **580** displays a list of a plurality of emails (e.g., emails A-F) in a user's inbox and the contents of selected email A. In some embodiments, a newest or the most urgent email is displayed at the top of the list of the plurality of emails in the user's inbox and the email at the top of the list is automatically selected. In FIG. 5J, email A is in focus on primary display **102** as email is displayed in bold within the list of the plurality of emails and email A's contents are displayed in window **580** below the list. In FIG. 5J, status tray **502** indicates that the mail application is running in the foreground, and app tray **514** also indicates that the mail application is running in the foreground based on the shadow behind mail application icon **506**.

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FIG. 5J also illustrates dynamic function row 104 displaying a plurality of affordances corresponding to email A (e.g., affordances 582, 584, 586, 588, 590, 592, 594, 596, and 598) and the at least one system-level affordance (e.g., affordance 542) in response to detecting selection of mail application icon 506 with cursor 504 in FIG. 5I. In FIG. 5J, the plurality of affordances corresponding to email A include: affordance 582, which, when activated (e.g., via a tap contact), refreshes the inbox; affordance 584, which, when activated (e.g., via a tap contact), causes primary display 102 to display a sub-window for composing a new email (e.g., shown in FIG. 5M); affordance 586, which, when activated (e.g., via a tap contact), causes primary display 102 to display a sub-window for replying to the sender of email A; affordance 588, which, when activated (e.g., via a tap contact), causes primary display 102 to display a sub-window for replying to all recipients of email A; affordance 590, which, when activated (e.g., via a tap contact), causes primary display 102 to display a sub-window for forwarding email A; affordance 592, which, when activated (e.g., via a tap contact), causes email A to be archived to a default mailbox or folder; affordance 594, which, when activated (e.g., via a tap contact), causes email A to be deleted; affordance 596, which, when activated (e.g., via a tap contact), causes dynamic function row 104 to display a set of affordances for selecting different flags, which optionally correspond to pre-existing folders, to be applied to email A (e.g., as shown in FIG. 5L); and affordance 598, which, when activated (e.g., via a tap contact), causes primary display 102 or dynamic function row 104 to display a search dialogue for searching the user's inbox. FIG. 5J further illustrates dynamic function row 104 receiving and detecting contact 599 (e.g., a long press gesture) at a location corresponding to persistent volume control 568.

FIG. 5K illustrates dynamic function row 104 displaying volume slider 5100 for adjusting the playback volume of podcast J, which was initiated in FIG. 5H, in response to detecting the long press gesture at the location corresponding to persistent volume control 568 in FIG. 5J. The circle/thumb in volume slider 5100 can be dragged by the user of portable computing system 100 or desktop computing system 200 to adjust the volume. Alternatively, in some embodiments, dynamic function row 104 displays playback controls (e.g., pause, fast forward, rewind, next track, previous track, and the like) for controlling the playback of podcast J, which was initiated in FIG. 5H, in response to detecting the long press gesture at the location corresponding to persistent volume control 568 in FIG. 5J. In FIG. 5K, volume slider 5100 is overlaid on the plurality of affordances corresponding to email A. FIG. 5K also illustrates dynamic function row 104 receiving and detecting contact 5102 (e.g., a tap contact) at a location corresponding to affordance 596.

In other embodiments, volume slider 5100 is displayed in a separate region from the plurality of affordances corresponding to email A. As such, while volume slider 5100 is activated, the plurality of affordances corresponding to email A (e.g., affordances 582, 584, 586, 588, 590, 592, 594, 596, and 598) are animatedly scrolled or shrunken to ensure enough display space is available to display volume slider 5100. In some embodiments, the circle/thumb is displayed under the user's finger in response to the long press gesture to allow the user to slide the circle/thumb without having to remove the contact from the affordance.

FIG. 5L illustrates dynamic function row 104 displaying a set of affordances 5103, 5104, 5106, 5108, and 5110 for selecting different flags, which optionally correspond to pre-existing folders, to be applied to email A in response to

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detecting selection of affordance 596 in FIG. 5K. In FIG. 5L, dynamic function row 104 also displays exit affordance 5112 for ceasing to display the set of affordances 5103, 5104, 5106, 5108, and 5110 on dynamic function row 104 and displaying the plurality of affordances corresponding to email A on dynamic function row 104 (e.g., as shown in FIG. 5J). FIG. 5L also illustrates primary display 102 displaying cursor 504 at a location corresponding to an email composition affordance in window 580.

FIG. 5M illustrates primary display 102 displaying a sub-window for composing a new email within window 580 in response to detecting selection of the email composition affordance with cursor 504 in FIG. 5L. In FIG. 5M, the sub-window for composing a new email is in focus on primary display 102 as the email composition affordance is displayed with thicker lines and also as indicated by the thick lines surrounding the sub-window for composing a new email.

FIG. 5M also illustrates dynamic function row 104 displaying a set of affordances corresponding to composing a new email in response to detecting selection of the email composition affordance with cursor 504 in FIG. 5L. In FIG. 5M, the set of affordances corresponding to composing a new email includes an affordance 5114, which, when activated (e.g., via a tap contact), causes portable computing system 100 or desktop computing system 200 to send the newly composed email; affordance 5116 for changing the text color of selected text of the new email; affordance 5118 for boldening selected text of the new email; affordance 5120 for italicizing selected text of the new email; and affordance 5122 for underlining selected text of the new email. FIG. 5M further illustrates dynamic function row 104 receiving and detecting contact 5124 (e.g., a tap contact) at a location corresponding to persistent volume control 568.

FIG. 5N illustrates primary display 102 displaying text in the body of the new email in the sub-window for composing a new email within window 580 and an indicator of the current insertion position located after the characters "pl."

FIG. 5N also illustrates dynamic function row 104 displaying persistent volume control 568 indicating that podcast J is muted in response to detecting the tap contact at the location corresponding to persistent volume control 568 in FIG. 5M. In FIG. 5N, persistent volume control 568 displays equalizer feedback for podcast J even while podcast J, whose playback was initiated in FIG. 5H, is muted (i.e., shows that podcast J is still playing but is muted). In FIG. 5N, dynamic function row 104 displays predictive words 5126, 5128, and 5130 for completing the word beginning with "pl" that is being typed in the body of the new email based on the insertion point in the sub-window within window 580. FIG. 5N further illustrates dynamic function row 104 receiving and detecting contact 5132 (e.g., a tap contact) at a location corresponding to predictive word 5126 (i.e., "planet").

FIG. 5O illustrates primary display 102 displaying the word "planet" in the body of the new email in the sub-window for composing a new email within window 580 in response to detecting the selection of predictive word 5126 (i.e., "planet") in FIG. 5N. FIG. 5O also illustrates primary display 102 displaying cursor 504 at a location corresponding to the "To:" field of the sub-window for composing a new email.

FIG. 5P illustrates primary display 102 displaying menu 5134 corresponding to the user's contact book in response to detecting selection of the "To:" field with cursor 504 in FIG. 5O. In FIG. 5P, menu 5134 includes a list of a plurality of

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contacts corresponding the “All Contacts” group of the user’s contact book (e.g., a list of pre-existing or automatically populated contacts). In FIG. 5P, the “All Contacts” group of the user’s contact book is in focus on primary display 102 as indicated by the thick lines surrounding the “All Contacts” group in menu 5134.

FIG. 5P also illustrates dynamic function row 104 displaying a first plurality of contact affordances 5136 (e.g., contact affordances 5136-A to 5136-F) corresponding to the “All Contacts” group of the user’s contact book and the at least one system-level affordance (e.g., affordance 542) in response to detecting selection of the “To:” field with cursor 504 in FIG. 5O. In FIG. 5P, dynamic function row 104 also displays exit affordance 5112, which, when activated (e.g., via a tap contact), causes primary display 102 to cease displaying menu 5134 on primary display 102 and also causes dynamic function row 104 to cease displaying the first plurality of contact affordances 5136. FIG. 5P further illustrates dynamic function row 104 detecting a right-to-left swipe gesture with contact 5138 moving from a first location 5140-A to a second location 5140-B (e.g., the user scrolls right-to-left through All Contacts).

FIG. 5Q illustrates dynamic function row 104 displaying a second plurality of contact affordances 5136 (e.g., contact affordances 5136-E to 5136-J) corresponding to the “All Contacts” group of the user’s contact book and the at least one system-level affordance (e.g., affordance 542) in response to detecting the right-to-left swipe gesture in 5P. FIG. 5Q also illustrates dynamic function row 104 detecting an upward swipe gesture with contact 5142 moving from a first location 5144-A to a second location 5144-B.

FIG. 5R illustrates primary display 102 displaying a list of a plurality of contacts corresponding the “Family” group of the user’s contact book in response to detecting the upward swipe gesture in FIG. 5Q. In FIG. 5R, the “Family” group of the user’s contact book is in focus on primary display 102 as indicated by the thick lines surrounding the “Family” group in menu 5134.

FIG. 5R also illustrates dynamic function row 104 displaying a plurality of contact affordances 5146 (e.g., contact affordances 5146-A to 5146-F) corresponding to the “Family” group of the user’s contact book and the at least one system-level affordance (e.g., affordance 542) in response to detecting upward swipe gesture in FIG. 5Q. FIG. 5R further illustrates dynamic function row 104 receiving and detecting contact 5148 (e.g., a tap contact) at a location corresponding to contact affordance 5146-D, which is associated with a contact named “James H.” within the “Family” group of the user’s contact book.

FIG. 5S illustrates primary display 102 displaying “James H.” in the “To:” field of the sub-window for composing a new email within window 580 in response to detecting selection of contact affordance 5146-D in FIG. 5R. FIG. 5S also illustrates dynamic function row 104 replacing display of the plurality of contact affordances 5146 (e.g., contact affordances 5146-A to 5146-F) corresponding to the “Family” group of the user’s contact book with the set of affordances (e.g., affordances 5114, 5116, 5118, 5120, and 5122) corresponding to composing a new email in response to detecting selection of contact affordance 5146-D in FIG. 5R. FIG. 5S further illustrates dynamic function row 104 receiving and detecting contact 5150 (e.g., a tap contact) at a location corresponding to the at least one system-level affordance 542.

FIG. 5T illustrates dynamic function row 104 displaying persistent controls (i.e., affordances 516 and 534), the plurality of system-level affordances (i.e., affordances 518, 520,

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522, 524, 526, 528, 530, and 532), and affordance 5152 corresponding to the mail application in response to detecting selection of affordance 542 in FIG. 5S. FIG. 5T also illustrates dynamic function row 104 receiving and detecting contact 5154 (e.g., a tap contact) at a location corresponding to play/pause affordance 530. For example, in response to detecting selection of play/pause affordance 530, portable computing system 100 or desktop computing system 200 pauses playback of podcast J, which was initiated in FIG. 5H and muted in FIG. 5M. Playback of podcast J may be re-initiated by a subsequent selection of play/pause affordance 530 in FIG. 5U.

FIG. 5U illustrates primary display 102 displaying cursor 504 at a location corresponding to an exit affordance for closing window 580. FIG. 5V illustrates primary display 102 displaying modal alert 5156 in response to detecting selection of the exit affordance with cursor 504 in FIG. 5U. In FIG. 5V, modal alert 5156 is in focus on primary display 102. In FIG. 5V, modal alert 5156 displayed on primary display 102 prompts the user to save the draft email prior to closing window 580 and includes a “Save” affordance, a “Don’t Save” affordance, and a “Cancel” affordance. FIG. 5V also illustrates primary display 102 displaying cursor 504 at a location corresponding to “Cancel” affordance. The display of modal alerts on the dynamic function row increasing efficiency and provides a better user experience because it removes the need for the user to move their eyes between the keyboard and the screen and also removes the need for the user to move their hands from the keyboard to another input device such as a mouse.

FIG. 5V further illustrates dynamic function row 104 displaying modal alert 5156 and the at least one system-level affordance (e.g., affordance 542) in response to detecting selection of the exit affordance with cursor 504 in FIG. 5U. In some embodiments, a modal alert is a notification corresponding to an email, SMS, or the like received by portable computing system 100 or desktop computing system 200, an alert associated with an application (e.g., as a save dialog, an exit confirmation dialog, or a send email confirmation dialog), or the like. In FIG. 5V, modal alert 5156 displayed on dynamic function row 104 prompts the user to save the draft email prior to closing window 580 and includes a “Save” affordance 5158, a “Don’t Save” affordance 5160, and a “Cancel” affordance 5162. Affordances 5158, 5160, and 5162 are merely examples, and other affordances may be used to control or respond to modal alerts.

FIG. 5W illustrates primary display 102 ceasing to display modal alert 5156 and maintaining display of the sub-window for composing a new email (as in FIG. 5U) in response to detecting selection of “Cancel” affordance with cursor 504 in FIG. 5V. FIG. 5W also illustrates dynamic function row 104 ceasing to display modal alert 5156 and displaying persistent controls (i.e., affordances 516 and 534), the plurality of system-level affordances (i.e., affordances 518, 520, 522, 524, 526, 528, 530, and 532), and affordance 5152 corresponding to the mail application (as in FIG. 5U) in response to detecting selection of “Cancel” affordance with cursor 504 in FIG. 5V.

FIG. 5X illustrates primary display 102 displaying an application selection window 5164 in response to receiving a signal corresponding to a specified physical key combination (e.g., alt+tab) from the set of physical keys 106 of portable computing system 100 (FIGS. 1A-1B) or from the set of physical keys 106 of peripheral keyboard 206 of desktop computing system 200 (FIGS. 2A-2D). In FIG. 5X, application selection window 5164 displayed on primary display 102 includes: mail application icon 506, which,

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when activated (e.g., via selection by cursor **504**) causes primary display **102** to display window **580** corresponding to the mail application in the foreground; media player application icon **510**, which, when activated (e.g., via selection by cursor **504**) causes primary display **102** to display window **554** corresponding to the media player application in the foreground; and application A icon **512**, which, when activated (e.g., via selection by cursor **504**) causes primary display **102** to display window **536** corresponding to application A in the foreground. FIG. **5X** also illustrates primary display **102** displaying cursor **504** at a location corresponding to photo application icon **515**.

FIG. **5X** further illustrates dynamic function row **104** displaying application selection window **5164** and the at least one system-level affordance (e.g., affordance **542**) in response to receiving a signal corresponding to a specified physical key combination (e.g., alt+tab) from the set of physical keys **106** of portable computing system **100** (FIGS. **1A-1B**) or from the set of physical keys **106** of peripheral keyboard **206** of desktop computing system **200** (FIGS. **2A-2D**). In FIG. **5X**, application selection window **5164** displayed on dynamic function row **104** includes: mail application icon **506**, which, when activated (e.g., via a tap contact) causes primary display **102** to display window **580** corresponding to the mail application in the foreground; media player application icon **510**, which, when activated (e.g., via a tap contact) causes primary display **102** to display window **554** corresponding to the media player application in the foreground; and application A icon **512**, which, when activated (e.g., via a tap contact) causes primary display **102** to display window **536** corresponding to application A in the foreground.

FIG. **5Y** illustrates primary display **102** displaying a window **5166** for the photo application in response to detecting selection of photo application icon **515** with cursor **504** in FIG. **5X**. For example, window **5166** is overlaid on windows **580**, **554**, and **536**. In FIG. **5Y**, window **5166** displays a plurality of photos associated with an all photos sub-section of a user's photo library. In FIG. **5Y**, the all photos sub-section of a user's photo library is in focus on primary display **102** as shown by "Photos" displayed in bold and photos A-L at least partially displayed in window **5166**. In FIG. **5Y**, status tray **502** indicates that the photo application is running in the foreground, and app tray **514** also indicates that the photo application is running in the foreground based on the shadow behind photo application icon **515**. FIG. **5Y** also illustrates primary display **102** displaying cursor **504** at a location corresponding to photo B within window **5166**.

FIG. **5Y** further illustrates dynamic function row **104** displaying a plurality of affordances corresponding to the all photos sub-section of the user's photo library (e.g., affordances **5168**, **5170**, and **5172**) and the at least one system-level affordance (e.g., affordance **542**) in response to detecting selection of photo application icon **515** with cursor **504** in FIG. **5X**. In FIG. **5Y**, dynamic function row **104** includes: search affordance **5168**, for searching the user's photo library; slideshow affordance **5170**, which, when activated (e.g., via a tap contact), initiates a slideshow of the selected photos or all photos in the all photos sub-section of the user's photo library in a full-screen mode (e.g., shown in FIG. **5AA**); and slider affordance **5172** for scrolling the photos displayed from the all photos sub-section of the user's photo library that are displayed in window **5166**.

FIG. **5Z** illustrates primary display **102** displaying selected photo B in window **5166** in response to detecting selection of photo B with cursor **504** in FIG. **5Y**. In FIG. **5Z**,

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selected photo B is in focus on primary display **102** as shown by the thick lines surrounding photo B in window **5166**. FIG. **5Z** also illustrates primary display **102** displaying cursor **504** at a location corresponding to a slideshow affordance.

FIG. **5Z** further illustrates dynamic function row **104** displaying a set of affordances corresponding to selected photo B in response to detecting selection of photo B with cursor **504** in FIG. **5Y**. In FIG. **5Z**, the set of affordance corresponding to selected photo B include: search affordance **5168** for searching the user's photo library; zoom affordance **5174** for zooming into selected photo B; like affordance **5176** for liking selected photo B; slideshow affordance **5170**, which, when activated (e.g., via a tap contact), initiates a slideshow of the selected photos or all photos in the all photos sub-section of the user's photo library in a full-screen mode (e.g., shown in FIG. **5AA**); information affordance **5178** for displaying information corresponding to selected photo B, such as size, location, time/date, and the like, on dynamic function row **104** and/or primary display **102**; editing affordance **5180**, which, when activated (e.g., via a tap contact), causes dynamic function row **104** to display tools for editing selected photo B (e.g., shown in FIG. **5DD**) and/or causes primary display **102** to display an editing interface for editing selected photo B; photo adding affordance **5182** for adding selected photo B to a photo album; sharing affordance **5184** for sharing selected photo B via one or more communication modes (e.g., social media networks, SMS, email, and the like); and deletion affordance **5186** for deleting selected photo B from the user's photo library.

FIG. **5AA** illustrates primary display **102** displaying a slideshow of photos from the all photos sub-section of the user's photo library in window **5188** in response to detecting selection of the slideshow affordance with cursor **504** in FIG. **5Z**. In FIG. **5AA**, primary display **102** displays window **5188** with photo B in full-screen mode.

FIG. **5AA** also illustrates dynamic function row **104** displaying a plurality of thumbnail images (e.g., thumbnail images **5192-Y**, **5192-Z**, **5192-A**, **5192-B**, **5192-C**, **5192-D**, and **5192-E**) corresponding to the photos in the all photos sub-section of the user's photo library in response to detecting selection of the slideshow affordance with cursor **504** in FIG. **5Z**. In FIG. **5AA**, the thick lines surrounding thumbnail image **5192-B** indicate that photo B is currently displayed by primary display **102**. In FIG. **5AA**, dynamic function row **104** also displays a pause affordance **5190**, which, when activated (e.g., via a tap contact), causes the slideshow to be paused and also causes primary display **102** to exit the full-screen mode. FIG. **5AA** further illustrates dynamic function row **104** receiving and detecting contact **5194** (e.g., a tap contact) at a location corresponding to pause affordance **5190**.

FIG. **5BB** illustrates primary display **102** displaying photo B in an expanded view within window **5166** in response to detecting selection of pause affordance **5190** in FIG. **5AA**. In FIG. **5BB**, the expanded view of photo B is in focus on primary display **102** as shown by the thick lines surrounding the expanded view of photo B in window **5166**.

FIG. **5BB** also illustrates dynamic function row **104** displaying a set of affordances corresponding to photo B in response to detecting selection of pause affordance **5190** in FIG. **5AA**. In FIG. **5BB**, the set of affordance corresponding to photo B include: zoom affordance **5174**, which, when activated (e.g., via a tap contact), causes dynamic function row **104** and/or primary display **102** to display zoom controls that enable the user of portable computing system **100** or desktop computing system **200** to zoom into or zoom out

from photo B; full-screen affordance **5194**, which, when activated (e.g., via a tap contact), causes primary display **102** to display photo B in full-screen mode; slideshow affordance **5170**, which, when activated (e.g., via a tap contact), initiates a slideshow of the selected photos or all photos in the all photos sub-section of the user's photo library in a full-screen mode; information affordance **5178** for displaying information corresponding to selected photo B, such as size, location, time/date, and the like, on dynamic function row **104** and/or primary display **102**; editing affordance **5180**, which, when activated (e.g., via a tap contact), causes dynamic function row **104** to display tools for editing selected photo B and/or causes primary display **102** to display an editing interface for editing selected photo B; photo adding affordance **5182** for adding selected photo B to a photo album; and sharing affordance **5184** for sharing selected photo B via one or more communication modes (e.g., social media networks, SMS, email, and the like). FIG. **5BB** further illustrates dynamic function row **104** receiving and detecting contact **5196** (e.g., a tap contact) at a location corresponding to full-screen affordance **5196**.

FIG. **5CC** illustrates primary display **102** displaying photo B in full-screen mode within window **5200** in response to detecting selection of full-screen affordance **5196** in FIG. **5BB**. FIG. **5CC** also illustrates dynamic function row **104** displaying minimize affordance **5198** in response to detecting selection of full-screen affordance **5196** in FIG. **5BB**. When activated (e.g., via a tap contact), minimize affordance **5198** causes primary display **102** to display photo B in the expanded view within window **5166** (as shown in FIG. **5BB**). FIG. **5CC** further illustrates dynamic function row **104** receiving and detecting contact **5201** (e.g., a tap contact) at a location corresponding to editing affordance **5180**.

FIG. **5DD** illustrates dynamic function row **104** displaying a set of editing tools **5205** in response to detecting selection of editing affordance **5180** in FIG. **5CC**.

FIG. **5DD** also illustrates dynamic function row **104** displaying editing affordance **5180** with a thickened/bold outline and all other affordances displayed in FIG. **5CC** (e.g., zoom affordance **5174**, minimize affordance **5198**, information affordance **5178**, photo adding affordance **5182**, and sharing affordance **5184**) with increased translucency in response to detecting selection of editing affordance **5180** in FIG. **5CC**.

In FIG. **5DD**, the set of editing tools **5205** includes: a rotate tool **5202**, which, when activated (e.g., via a tap contact) causes dynamic function row **104** to display controls for rotating photo B clockwise or counter-clockwise within window **5200**; an enhance tool **5204**, which, when activated (e.g., via a tap contact) causes dynamic function row **104** to display controls for enhancing photo B such as applying filters to photo B, adjusting the brightness of photo B, adjusting the saturation of photo B, and/or the like; a red-eye reduction tool **5206**, which, when activated (e.g., via a tap contact) causes dynamic function row **104** to display controls for reducing the red-eye of persons in photo B; a straighten tool **5208**, which, when activated (e.g., via a tap contact) causes dynamic function row **104** to display controls for straightening the orientation of photo B within window **5200**; a crop tool **5210**, which, when activated (e.g., via a tap contact) causes dynamic function row **104** to display controls for cropping photo B within window **5200**; and a retouching tool **5212**, which, when activated (e.g., via a tap contact) causes dynamic function row **104** to display controls for re-touching photo B such as removal and airbrush effects. FIG. **5DD** further illustrates dynamic func-

tion row **104** receiving and detecting contact **5214** (e.g., a tap contact) at a location corresponding to straighten tool **5208**.

FIG. **5EE** illustrates dynamic function row **104** displaying a set of controls **5209** for straightening the orientation of photo B within window **5200** in response to detecting selection of straighten tool **5208** in FIG. **5DD**. In FIG. **5EE**, the set of controls **5209** corresponding to the straightening tool **5208** includes a slider **5210** for adjusting the orientation of photo B within window **5200** and done affordance **5212**, which, when activated (e.g., via a tap contact), causes dynamic function row **104** to cease displaying the set of controls **5209** and to display the set of editing tools **5205** (as shown in FIG. **5DD**). FIG. **5EE** also illustrates dynamic function row **104** receiving and detecting contact **5216** (e.g., a tap contact) at a location corresponding to escape affordance **516**.

For example, the user of portable computing system **100** or desktop computing system **200** is able to adjust the orientation of photo B within window **5200** by performing a left-to-right swipe/drag gesture or a right-to-left swipe/drag gesture at a location originating on slider **5210** or within the set of controls **5209**. For example, in response to detecting an upward swipe gesture on dynamic function row **104**, dynamic function row **104** displays a set of controls corresponding to crop tool **5210**. In another example, in response to detecting a downward swipe gesture on dynamic function row **104**, dynamic function row **104** displays a set of controls corresponding to red-eye reduction tool **5206**.

FIG. **5FF** illustrates primary display **102** displaying photo B in the expanded view within window **5166** in response to detecting selection of escape affordance **516** in FIG. **5EE**. FIG. **5FF** illustrates dynamic function row **104** displaying a set of affordances corresponding to photo B (as shown in FIG. **5BB**) in response to detecting selection of escape affordance **516** in FIG. **5EE**. In FIG. **5FF**, the set of affordance corresponding to photo B include: zoom affordance **5174**, which, when activated (e.g., via a tap contact), causes dynamic function row **104** and/or primary display **102** to display zoom controls that enable the user of portable computing system **100** or desktop computing system **200** to zoom into or zoom out from photo B; full-screen affordance **5194**, which, when activated (e.g., via a tap contact), causes primary display **102** to display photo B in full-screen mode; slideshow affordance **5170**, which, when activated (e.g., via a tap contact), initiates a slideshow of the selected photos or all photos in the all photos sub-section of the user's photo library in a full-screen mode; information affordance **5178** for displaying information corresponding to selected photo B, such as size, location, time/date, and the like, on dynamic function row **104** and/or primary display **102**; editing affordance **5180**, which, when activated (e.g., via a tap contact), causes dynamic function row **104** to display tools for editing selected photo B and/or causes primary display **102** to display an editing interface for editing selected photo B; photo adding affordance **5182** for adding selected photo B to a photo album; and sharing affordance **5184** for sharing selected photo B via one or more communication modes (e.g., social media networks, SMS, email, and the like).

FIG. **5FF** also illustrates dynamic function row **104** displaying notification **5218** overlaid on affordances **5178**, **5180**, **5182**, and **5184** in response to reception of notification **5218** by portable computing system **100** or desktop computing system **200**. In FIG. **5FF**, notification **5218** corresponds to an SMS, instant message, or the like sent by Suzie S. to the user of portable computing system **100** or desktop computing system **200**, where the notification's content

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inquiries “Movies tonight?” FIG. 5FF further illustrates dynamic function row 104 detecting a left-to-right swipe gesture with contact 5220 from a first location 5222-A within notification 5128 to a second location 5222-B.

FIG. 5GG illustrates dynamic function row 104 ceasing to display notification 5218 in response to detecting the left-to-right swipe gesture in FIG. 5FF. FIG. 5GG also illustrates primary display 102 displaying cursor 504 at a location corresponding to web browser application icon 508.

FIG. 5HH illustrates primary display 102 displaying a window 5224 for the web browser application in response to detecting selection of web browser application icon 508 with cursor 504 in FIG. 5GG. For example, window 5224 is overlaid on window 5166. In FIG. 5HH, window 5224 includes controls for the web browser application including browsing controls (e.g., last web page, next web page, refresh, and add to favorites), an address bar, a search bar, a show-all bookmarks affordance (e.g., resembling an open book), a show-all open tabs affordance (e.g., a grid of six squares), and affordances for particular bookmarks A, B, and C. In FIG. 5HH, window 5224 shows a home interface for the web browser application including a plurality of affordances 5227 linking to favorited or most frequently visited websites A-H. In FIG. 5HH, window 5224 for application A is in focus on primary display 102. In FIG. 5HH, status tray 502 indicates that the web browser application is running in the foreground, and app tray 514 also indicates that the web browser application is running in the foreground based on the shadow behind the web browser application icon 508.

FIG. 5HH also illustrates dynamic function row 104 displaying affordance 5226 in addition to the persistent controls (i.e., affordances 516 and 534) and the plurality of system-level affordances (i.e., affordances 518, 520, 522, 524, 526, 528, 530, and 532) in response to detecting selection of web browser application icon 508 with cursor 504 in FIG. 5GG. When activated (e.g., via a tap contact), affordance 5226 causes dynamic function row 104 to display a set of controls for the web browser application (e.g., affordances 5230, 5232, and 5238, and address bar 5234 as shown in FIG. 5II). FIG. 5HH further illustrates dynamic function row 104 receiving and detecting contact 5228 (e.g., a tap contact) at a location corresponding to affordance 5226.

FIG. 5II illustrates dynamic function row 104 displaying a set of controls for the web browser application in response to detecting selection of affordance 5226 in FIG. 5HH. In FIG. 5II, the set of controls for the web browser application includes: affordance 5230 for displaying a web page visited before the one currently displayed by the web browser application within window 5224; affordance 5232 for displaying a web page visited after the one currently displayed by the web browser application within window 5224; affordance 5238 for adding the web page currently displayed by the web browser application to a favorites list or a bookmarks folder; and address bar 5234 for displaying the URL of the web page currently displayed by the web browser application. In FIG. 5II, address bar 5234 also includes a refresh affordance 5236 for refreshing the web page currently displayed by the web browser application. FIG. 5II also illustrates primary display 102 displaying cursor 504 at a location corresponding to affordance 5227-A, which links to website A.

FIG. 5JJ illustrates primary display 102 displaying an interface for tab A within window 5224 after detecting selection of affordance 5227-A corresponding to website A with cursor 504 in FIG. 5II. In FIG. 5JJ, the interface for tab A is in focus on primary display 102 as indicated by the thick

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lines surrounding tab A and the bold text for tab A. In FIG. 5JJ, the interface for tab A shows a checkout web page of website A (e.g., associated with the URL: www.website_A.com/checkout). The checkout web page corresponds to the user's virtual shopping cart, which includes Items A and B for purchase. FIG. 5JJ also illustrates primary display 102 displaying cursor 504 at a location corresponding to a purchase affordance within window 5224. FIG. 5II further illustrates dynamic function row 104 displaying the URL (e.g., www.website_A.com/checkout) for the checkout web page of website A in address bar 5234.

FIG. 5KK illustrates primary display 102 displaying modal alert 5240 overlaid on window 5224 in response to detecting selection of the purchase affordance with cursor 504 in FIG. 5JJ. In FIG. 5KK, modal alert 5240 displayed on primary display 102 prompts the user of portable computing system 100 or desktop computing system 200 to provide their fingerprint on dynamic function row 104 and also includes cancel affordance 5242, which, when activated (e.g., via selection by cursor 504) causes cancelation of the purchase. For example, modal alert 5240 is displayed in accordance with security settings (e.g., default or user-specified) that requires a fingerprint to validate purchases initiated by portable computing system 100 or desktop computing system 200. For example, in some embodiments, primary display 102 and/or dynamic function row 104 displays the modal alert prompting the user of portable computing system 100 or desktop computing system 200 to provide their fingerprint on dynamic function row 104 upon logging into portable computing system 100 or desktop computing system 200, when entering a password to access an application or website, when entering a password to decrypt the data stored by portable computing system 100 or desktop computing system 200, when deleting folders and/or data from portable computing system 100 or desktop computing system 200, when taking other destructive actions, and/or the like.

FIG. 5KK also illustrates dynamic function row 104 displaying modal alert 5240 in response to detecting selection of the purchase affordance with cursor 504 in FIG. 5JJ. In FIG. 5KK, modal alert 5240 displayed on dynamic function row 104 prompts the user of portable computing system 100 or desktop computing system 200 to provide their fingerprint in fingerprint region 5244 of dynamic function row 104 and also includes cancel affordance 5242, which, when activated (e.g., via a tap contact) causes cancelation of the purchase. In some embodiments, dynamic function row 104 is configured to detect a fingerprint within fingerprint region 5244 of dynamic function row 104, which also corresponds to power control 534 in FIGS. 5A-5JJ. In some embodiments, dynamic function row 104 is configured to detect a fingerprint at any location within its touch-sensitive area. FIG. 5KK further illustrates dynamic function row 104 receiving and detecting contact 5246 (e.g., a press and hold gesture) within fingerprint region 5244.

FIG. 5LL illustrates primary display 102 displaying an interface for tab A within window 5224 after detecting contact 5246 within fingerprint region 5244 in FIG. 5KK. In FIG. 5LL, the interface for tab A shows a receipt web page of website A (e.g., associated with the URL: www.website_A.com/reciept) indicating that the purchase was completed after validation of the fingerprint provided by the user of portable computing system 100 or desktop computing system 200 in FIG. 5KK.

FIG. 5LL also illustrates dynamic function row 104 displaying an interface 5248 associated with an incoming voice call from C. Cheung along with the at least one

system-level affordance (e.g., affordance **542**) in response to reception of the incoming voice call by portable computing system **100** or desktop computing system **200**. In FIG. **5LL**, interface **5248** includes a first affordance **5250** for answering the incoming call and a second affordance **5252** for declining the incoming call. FIG. **5LL** further illustrates dynamic function row **104** receiving and detecting contact **5254** (e.g., a tap contact) at a location corresponding to first affordance **5250**. For example, after detecting selection of first affordance **5250**, a communication connection (e.g., VOIP) between C. Cheung and the user of portable computing system **100** or desktop computing system **200** is established for the voice call.

FIG. **5MM** illustrates dynamic function row **104** displaying an interface **5256** associated with an ongoing voice call between C. Cheung and the user of portable computing system **100** or desktop computing system **200** along with the at least one system-level affordance (e.g., affordance **542**) after detecting selection of first affordance **5250** in FIG. **5LL**. In FIG. **5MM**, interface **5256** includes affordance **5258** for ending the voice call and an indicator of the total voice call time (e.g., 7 minutes and 29 seconds). In some embodiments, during the ongoing voice call, affordances associated with the focus of primary display **102** are not displayed on dynamic function row **104**. In some embodiments, after the voice call has lasted a predefined amount of time, interface **5256** is displayed in a compact mode and affordances associated with the focus of primary display **102** may be displayed on dynamic function row **104**. FIG. **5MM** also illustrates dynamic function row **104** receiving and detecting contact **5260** at a location corresponding to affordance **5258**.

FIG. **5NN** illustrates primary display **102** displaying an interface for tab B within window **5224**, where tabs A, B, and C are open within the web browser application. In FIG. **5NN**, the interface for tab B shows the home web page of website B (e.g., associated with the URL: www.website_B.com/home). In FIG. **5NN**, the interface for tab B is in focus on primary display **102** as indicated by the thick lines surrounding tab B and the bold text for tab B.

FIG. **5NN** also illustrates dynamic function row **104** ceasing to display interface **5256** after detecting selection of affordance **5258** in FIG. **5MM**. In FIG. **5NN**, dynamic function row **104** includes the URL for the home web page of website B in address bar **5234** (e.g., www.website_B.com/home). In FIG. **5NN**, dynamic function row **104** also includes: affordance **5262-A**, which, when activated (e.g., by a tap contact), causes primary display **102** to display an interface for tab A and also causes dynamic function row **104** to show the URL corresponding to tab B in address bar **5234**; and affordance **5262-B**, which, activated (e.g., by a tap contact), causes primary display **102** to display an interface for tab C and also causes dynamic function row **104** to show the URL corresponding to tab C in address bar **5234**.

FIG. **500** illustrates primary display **102** displaying notification **5264** overlaid on window **5264** in response to reception of notification **5264** by portable computing system **100** or desktop computing system **200**. In FIG. **500**, notification **5264** corresponds to an SMS, instant message, or the like sent by MAS to the user of portable computing system **100** or desktop computing system **200**, where the notification's content inquires "Landed yet?" FIG. **500** illustrates primary display **102** displaying the user of portable computing system **100** or desktop computing system **200** dragging notification **5264** with cursor **504** to a predefined location in the bottom right-hand corner of primary display **102**. For example, the user portable computing system **100** or desktop computing system **200** is able to cause display of

a respective menu, notification, modal alert, or the like on dynamic function row **104** in response to dragging the respective menu, notification, modal alert, or the like from its origin location on primary display **102** to a predefined location (e.g., the bottom right-hand corner or another similar location). In some embodiments, the predefined location is one of a plurality of predefined locations that operate in a similar manner, where the plurality of predefined locations are default or user-specific locations.

FIG. **5PP** illustrates primary display **102** ceasing to display notification **5264** overlaid on window **5264** in response to the user of portable computing system **100** or desktop computing system **200** dragging notification **5264** with cursor **504** to the predefined location in the bottom right-hand corner of primary display **102** in FIG. **500**. FIG. **5PP** also illustrates dynamic function row **104** displaying notification **5264** overlaid on affordances **5262-A** and **5262-B** in response to the user of portable computing system **100** or desktop computing system **200** dragging notification **5264** with cursor **504** to the predefined location in the bottom right-hand corner of primary display **102** in FIG. **500**. FIG. **5PP** further illustrates dynamic function row **104** receiving and detecting contact **5266** (e.g., a tap contact) at a location within notification **5264**.

FIG. **5QQ** illustrates dynamic function row **104** displaying response dialogue box **5268** in response to detecting contact **5266** at the location within notification **5264** in FIG. **5PP**. Alternatively, in some embodiments, an application corresponding to notification **5264** is opened in response to detecting contact **5266** at the location within notification **5264** in FIG. **5PP**. In FIG. **5QQ**, response dialogue box **5268** includes a plurality of predictive responses to the content of notification **5264** shown in FIGS. **500-5PP**. In FIG. **5QQ**, response dialogue box **5268** includes a first predictive response **5270** ("Yes."), a second predictive response **5272** ("No."), and a third predictive response **5274** ("On my way!"). FIG. **5QQ** also illustrates dynamic function row **104** receiving and detecting contact **5276** (e.g., a tap contact) at a location corresponding to the first predictive response **5270**. For example, in response to selection of the first predictive response **5270**, portable computing system **100** or desktop computing system **200** causes the first predictive response **5270** ("Yes.") to be sent to MAS via a default communication mode (e.g., SMS, instant message, or the like) or a same communication mode as the one by which notification **5264** was sent to the user of portable computing system **100** or desktop computing system **200**.

FIG. **5RR** illustrates primary display **102** displaying cursor **504** at a location corresponding to the show-all bookmarks affordance (e.g., resembling an open book) within window **5224**. FIG. **5SS** illustrates primary display **102** displaying a bookmarks sidebar within window **5224** in response to detecting selection of the show-all bookmarks affordance with cursor **504** in FIG. **5RR**. In FIG. **5SS**, the bookmarks sidebar is in focus on primary display **102** as indicated by the thick lines surrounding the bookmarks sidebar. FIG. **5SS** also illustrates primary display **102** displaying the user of portable computing system **100** or desktop computing system **200** dragging the bookmarks sidebar with cursor **504** to the predefined location in the bottom right-hand corner of primary display **102**.

FIG. **5TT** illustrates dynamic function row **104** displaying a set of bookmark affordances **5278** (e.g., bookmark affordances **5278-A** to **5278-G**) corresponding to all pre-existing bookmarks in response to the user of portable computing system **100** or desktop computing system **200** dragging the bookmarks sidebar with cursor **504** to the predefined loca-

tion in the bottom right-hand corner of primary display 102 in FIG. 5SS. For example, when a respective one of the set of bookmark affordances 5278 is activated (e.g., via a tap contact), primary display 102 displays a website corresponding to the respective one of the set of bookmark affordances 5278 in a new tab within window 5224. Continuing with this example, when the respective one of the set of bookmark affordances 5278 is activated (e.g., via a tap contact), dynamic function row 104 ceases to display the set of bookmark affordances 5278 and displays the set of controls for the web browser application and the URL for the website corresponding to the respective one of the set of bookmark affordances 5278 in address bar 5234 (e.g., as shown in FIG. 5SS). In FIG. 5TT, dynamic function row 104 also displays exit affordance 5112, which, when activated (e.g., via a tap contact), causes dynamic function row 104 to cease displaying the set of bookmark affordances 5278 and display the set of controls for the web browser application as shown in FIG. 5SS. FIG. 5TT also illustrates primary display 102 displaying cursor 504 at a location corresponding to an exit affordance for closing window 5224.

FIG. 5UU illustrates dynamic function row 104 displaying modal alert 5280 overlaid on the set of bookmark affordances 5278 in response to detecting selection of the exit affordance with cursor 504 in Figure 5TT. In FIG. 5UU, modal alert 5280 prompts the user of portable computing system 100 or desktop computing system 200 to confirm that they intend to close all open tabs within the web browser application. In FIG. 5UU, modal alert 5280 includes: exit affordance 5282, which, when activated (e.g., via a tap contact), causes primary display 102 to cease display of window 5224; and cancel affordance 5284, which, when activated (e.g., via a tap contact), dismisses modal alert 5280 and causes primary display 102 to maintain display of window 5224. FIG. 5UU also illustrates dynamic function row 104 receiving and detecting contact 5286 (e.g., a tap contact) at a location corresponding to exit affordance 5282.

FIG. 5VV illustrates primary display 102 ceasing to display window 5224 and displaying window 5166 for the photo application in response to detecting selection of exit affordance 5282 in FIG. 5UU. In FIG. 5VV, status tray 502 indicates that the system/operating system is currently in focus on primary display 102. FIG. 5VV also illustrates primary display 102 displaying cursor 504 at a location corresponding to window 5166. FIG. 5VV further illustrates dynamic function row 104 displaying the persistent controls (i.e., affordances 516 and 534) and the plurality of system-level affordances (i.e., affordances 518, 520, 522, 524, 526, 528, 530, and 532) in response to detecting selection of exit affordance 5282 in FIG. 5UU and based on the current focus (e.g., the system/operating system) of primary display 102.

In FIG. 5WW, the focus of primary display 102 is the photo application in response to detecting selection of window 5166 with cursor 504 in FIG. 5VV. More specifically, the all photos sub-section of a user's photo library is in focus on primary display 102 as shown by "Photos" displayed in bold and photos A-L at least partially displayed in window 5166. In FIG. 5WW, status tray 502 indicates that the photo application is running in the foreground, and app tray 514 also indicates that the photo application is running in the foreground based on the shadow behind photo application icon 515. FIG. 5WW also illustrates primary display 102 displaying cursor 504 at a location corresponding to the file menu within status tray 502. FIG. 5WW further illustrates dynamic function row 104 displaying a plurality of affordances corresponding to the all photos sub-section of the user's photo library (e.g., affordances 5168, 5170, and

5172) and the at least one system-level affordance (e.g., affordance 542) in response to detecting selection of window 5166 with cursor 504 in FIG. 5VV.

FIG. 5XX illustrates primary display 102 displaying a menu of file controls 5288 in response to detecting selection of the file menu with cursor 504 in FIG. 5WW. In FIG. 5XX, the menu of file controls 5288 is in focus on primary display 102. In FIG. 5XX, the menu of file controls 5288 includes a new album control, a new folder control, a new calendar control, an import control, an export control, a close window control, and a print control.

FIG. 5XX also illustrates dynamic function row 104 displaying a first plurality of affordances 5290 in response to detecting selection of the file menu with cursor 504 in FIG. 5WW. In FIG. 5XX, the first plurality of affordances 5290 correspond to the file controls shown in the menu of file controls 5288 displayed by primary display 102. For example, when activated (e.g., via a tap contact), affordance 5290-G (e.g., corresponding to a close window file control) causes primary display 102 to cease display of window 5166 and also causes dynamic function row 104 to cease display of the first plurality of affordances 5290. FIG. 5XX further illustrates dynamic function row 104 receiving and detecting an upward swipe gesture with contact 5292 moving from a first location 5294-A to a second location 5294-B.

FIG. 5YY illustrates primary display 102 displaying a menu of edit controls 5296 in response to detecting the upward swipe gesture in FIG. 5XX. For example, primary display 102 displays a menu of help controls in response to detecting a downward swipe on dynamic function row in FIG. 5XX. In FIG. 5YY, the menu of edit controls 5296 is in focus on primary display 102. In FIG. 5XX, the menu of edit controls 5296 includes an undo control, a redo control, a cut control, a copy control, a paste control, a select all control, a find control, a font control, and a special characters control.

FIG. 5YY also illustrates dynamic function row 104 displaying a second plurality of affordances 5298 in response to detecting the upward swipe gesture in FIG. 5XX. In FIG. 5YY, the second plurality of affordances 5298 correspond to the edit controls shown in the menu of edit controls 5296 displayed by primary display 102. For example, the user of portable computing system 100 or desktop computing system 200 is able to view the balance of the plurality of affordances 5298 (e.g., the special characters affordance 5289-1) on dynamic function row 104 by performing a left-to-right swipe gesture on dynamic function row 104.

FIG. 5ZZ illustrates dynamic function row 104 displaying a first plurality of affordances 5301 along with the persistent controls (e.g., affordances 516 and 534) in response to receiving a signal from the set of physical keys 106 of portable computing system 100 (FIGS. 1A-1B) or from the set of physical keys 106 of peripheral keyboard 206 of desktop computing system 200 (FIGS. 2A-2D) corresponding to a specified physical key (e.g., a function key) or a specified physical key combination for overriding the current affordances displayed by dynamic function row 104. In FIG. 5ZZ, dynamic function row 104 displays the first plurality of affordances 5301 (e.g., corresponding to F1-F12) and an exit affordance 5112 for ceasing to display the first plurality of affordances 5301 on dynamic function row 104. In FIG. 5ZZ, dynamic function row 104 also navigation aid 5302 indicating that the first plurality of affordances 5301 is the first of four sets of affordances. For example, the user of portable computing system 100 or desktop computing system 200 is able to reveal additional

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affordances within the first plurality of affordances **5301** (e.g., corresponding to F13, . . .) by performing a left-to-right or right-to-left swipe gesture on dynamic function row **104**. In some embodiments, the first plurality of affordances **5301** includes a customized symbol row set by the user of the computing system or a set of most frequently used symbols and/or special characters.

FIG. 5AAA illustrates dynamic function row **104** displaying a second plurality of affordances **5303** along with the persistent controls (e.g., affordances **516** and **534**) in response to receiving a second signal from the set of physical keys **106** of portable computing system **100** (FIGS. 1A-1B) or from the set of physical keys **106** of peripheral keyboard **206** of desktop computing system **200** (FIGS. 2A-2D) corresponding to a specified physical key (e.g., a function key) or a specified physical key combination for overriding the current affordances displayed by dynamic function row **104**. Alternatively, in some embodiments, dynamic function row **104** displays the second plurality of affordances **5303** in response to detecting an upward swipe gesture on dynamic function row **104** in FIG. 5ZZ. In FIG. 5AAA, dynamic function row **104** displays the second plurality of affordances **5301** (e.g., corresponding to ‘, 1, 2, 3, . . .) and an exit affordance **5112** for ceasing to display the second plurality of affordances **5303** on dynamic function row **104**. In FIG. 5AAA, navigation aid **5302** indicates that the second plurality of affordances **5303** is the second of four sets of affordances. For example, the user of portable computing system **100** or desktop computing system **200** is able to reveal additional affordances within the second plurality of affordances **5303** by performing a left-to-right or right-to-left swipe gesture on dynamic function row **104**.

FIG. 5BBB illustrates dynamic function row **104** displaying a third plurality of affordances **5304** along with the persistent controls (e.g., affordances **516** and **534**) in response to receiving a third signal from the set of physical keys **106** of portable computing system **100** (FIGS. 1A-1B) or from the set of physical keys **106** of peripheral keyboard **206** of desktop computing system **200** (FIGS. 2A-2D) corresponding to a specified physical key (e.g., a function key) or a specified physical key combination for overriding the current affordances displayed by dynamic function row **104**. Alternatively, in some embodiments, dynamic function row **104** displays the third plurality of affordances **5304** in response to detecting an upward swipe gesture on dynamic function row **104** in FIG. 5AAA. In FIG. 5BBB, dynamic function row **104** displays the third plurality of affordances **5304** (e.g., corresponding to ~, !, @, #, . . .) and an exit affordance **5112** for ceasing to display the third plurality of affordances **5304** on dynamic function row **104**. In FIG. 5BBB, navigation aid **5302** indicates that the third plurality of affordances **5304** is the third of four sets of affordances. For example, the user of portable computing system **100** or desktop computing system **200** is able to reveal additional affordances within the third plurality of affordances **5304** by performing a left-to-right or right-to-left swipe gesture on dynamic function row **104**.

FIG. 5CCC illustrates dynamic function row **104** displaying a fourth plurality of affordances **5305** along with the persistent controls (e.g., affordances **516** and **534**) in response to receiving a fourth signal from the set of physical keys **106** of portable computing system **100** (FIGS. 1A-1B) or from the set of physical keys **106** of peripheral keyboard **206** of desktop computing system **200** (FIGS. 2A-2D) corresponding to a specified physical key (e.g., a function key) or a specified physical key combination for overriding the current affordances displayed by dynamic function row

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104. Alternatively, in some embodiments, dynamic function row **104** displays the fourth plurality of affordances **5305** in response to detecting an upward swipe gesture on dynamic function row **104** in FIG. 5BBB. In FIG. 5CCC, dynamic function row **104** displays the fourth plurality of affordances **5305** (e.g., corresponding to [, {, }, . . .) and an exit affordance **5112** for ceasing to display the fourth plurality of affordances **5305** on dynamic function row **104**. In FIG. 5CCC, navigation aid **5302** indicates that the fourth plurality of affordances **5305** is the fourth of four sets of affordances. For example, the user of portable computing system **100** or desktop computing system **200** is able to reveal additional affordances within the fourth plurality of affordances **5305** by performing a left-to-right or right-to-left swipe gesture on dynamic function row **104**. FIG. 5CCC further illustrates dynamic function row **104** receiving and detecting contact **5306** (e.g., a tap contact) at a location corresponding to power control **534**.

FIG. 5DDD illustrates dynamic function row **104** displaying modal alert **5308** in response to detecting selection of power control **534** in FIG. 5CCC. For example, modal alert **5308** is overlaid on the fourth plurality of functions affordance **5305**. In FIG. 5DDD, modal alert **5308** prompts the user of portable computing system **100** or desktop computing system **200** to select one of a plurality of options, including: logout affordance **5310**, which, when activated (e.g., via a tap contact), causes the current user of the user of portable computing system **100** or desktop computing system **200** to be logged out from portable computing system **100** or desktop computing system **200** (i.e., computing device **202**, FIGS. 2A-2D); restart affordance **5312**, which, when activated (e.g., via a tap contact), causes restart of portable computing system **100** or desktop computing system **200** (i.e., computing device **202**, FIGS. 2A-2D); power-off affordance **5314**, which, when activated (e.g., via a tap contact), causes portable computing system **100** or desktop computing system **200** (i.e., computing device **202**, FIGS. 2A-2D) to power-off; and cancel affordance **5316**, which, when activated (e.g., via a tap contact), causes dynamic function row **104** to cease display of modal alert **5308**.

FIGS. 6A-6D are a flowchart of a method of updating a dynamic input and output device (e.g., including dynamic function row **104**, FIGS. 1A-1B and 2A-2D), in accordance with some embodiments. The method **600** is performed at a computing system including a processor, a primary display, memory, and a housing at least partially containing a physical input mechanism and a touch screen display adjacent to the physical input mechanism. Some operations in method **600** are, optionally, combined and/or the order of some operations is, optionally, changed.

In some embodiments, the computing system is portable computing system **100** (FIG. 1A) or desktop computing system **200** (FIGS. 2A-2D). In some embodiments, the primary display is primary display **102** (FIG. 1A) which is implemented in display portion **110** of portable computing system **100** (FIG. 1A). Alternatively, in some embodiments, the primary display is primary display **102** (FIGS. 2A-2D) which is implemented in peripheral display device **204** (FIGS. 2A-2D). In some embodiments, the housing is body portion **120** of portable computing system **100** (FIGS. 1A-1B), which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIGS. 1A-1B) and the input mechanism (e.g., the set of physical keys **106**, FIGS. 1A-1B). Alternatively, in some embodiments, the housing is peripheral keyboard **206** (FIGS. 2A-2B) of desktop computing system **200**, which at least partially contains

the touch screen display (e.g., dynamic function row **104**, FIGS. 2A-2B) and the input mechanism (e.g., the set of physical keys **106**, FIGS. 2A-2B). Alternatively, in some embodiments, the housing is first input mechanism **212** (FIG. 2C) of desktop computing system **200**, which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIG. 2C) and the input mechanism (e.g., touch pad **108**, FIG. 2C).

The computing system displays (**602**) a first user interface on the primary display, the first user interface comprising one or more user interface elements. FIG. 5B, for example, shows primary display **102** displaying a first user interface with a status tray **502** indicating that application A is currently in focus, and an application (app) tray **514** with a plurality of executable/selectable application icons, including: a mail application icon **506**, a web browser application icon **508**, a media player application icon **510**, an application A icon **512**, and a photo application icon **515**. The first user interface displayed on primary display **102**, in FIG. 5B, also includes a window **536** for application A (e.g., a fantasy RPG game). In FIG. 5B, for example, window **536** includes a main menu for application A with a plurality of affordances including a start new game affordance, a continue affordance, and an options affordance. FIG. 5F, for example, shows primary display **102** displaying a first user interface with window **554** for the media player application.

In some embodiments, the computing system further comprises (**604**): (i) a primary computing device comprising the primary display, the processor, the memory, and primary computing device communication circuitry; and (ii) an input device comprising the housing, the touch screen display, the physical input mechanism, and input device communication circuitry for communicating with the primary computing device communication circuitry, where the input device is distinct and separate from the primary computing device. In some embodiments, the computing system is not a laptop, such as portable computing system **100** (FIGS. 1A-1B), but instead the computing system is desktop computing system **200** (FIGS. 2A-2D) with a computing device **202**, a peripheral display device **204** (which is optionally integrated with computing device **202**), and an input device (e.g., peripheral keyboard **206**, FIGS. 2A-2B) with a plurality of physical keys (e.g., the set of physical keys **106**, FIGS. 2A-2B) adjacent to a touch screen display (e.g., dynamic function row **104**, FIGS. 2A-2B). Alternatively, in some embodiments, the input device (e.g., first input mechanism **212**, FIG. 2C) includes a touch pad or a numpad (e.g., touch pad **108**, FIG. 2C) adjacent to a touch screen display (e.g., dynamic function row **104**, FIG. 2C). In some embodiments, the input device is in communication with the primary computing device (e.g., computing device **202**, FIGS. 2A-2D) via a wired connection (e.g., USB, PS/2, etc.) or a wireless connection (e.g., Bluetooth, Wi-Fi, etc.).

In some embodiments, the physical input mechanism comprises (**606**) a plurality of physical keys. In FIGS. 1A-1B, body portion **120** of portable computing system **100** at least partially contains the touch screen display (e.g., dynamic function row **104**) adjacent to the set of physical keys **106**.

In some embodiments, the physical input mechanism comprises (**608**) a touch pad. In FIGS. 1A-1B, body portion **120** of portable computing system **100** at least partially contains the touch screen display (e.g., dynamic function row **104**) and touch pad **108** in addition to the set of physical keys **106**.

The computing system identifies (**610**) an active user interface element among the one or more user interface

elements that is in focus on the primary display. In some embodiments, the term “in focus” can refer to the active element of the user interface (e.g., a window associated with an application, a particular toolbar or menu associated with an application, or the operating system) that is currently in the foreground and actively running or is controllable by input received from a user of the computing system such as a key press, mouse click, voice command, gestural motion, or the like.

In some embodiments, the computing system or a component thereof (e.g., focus determining module **351**, FIG. 3A) identifies a user interface element of the first user interface displayed on the primary display that is in focus. In some embodiments, the user interface element that is in focus is a window that corresponds to an application, a user interface element that corresponds to the application that is within or outside of the application’s window (e.g., a field, sub-window, menu, tool, toolbar, tool set, or the like), or a portion of a desktop/operating system-related interface (e.g., a volume control, a portion of a file explorer interface or a controls/settings panel). In some embodiments, the active user interface element is highlighted on the primary display or displayed in a foreground position on the primary display to indicate that it is in focus. In some embodiments, a display characteristic of the active user interface element is changed or emphasized (e.g., colored text, bold text, thick border, and the like) to indicate that it is in focus. Alternatively and/or additionally, in some embodiments, visual and/or aural cues are provided to indicate active user interface element that is in focus (e.g., a chime is played when the focus changes, a bouncing star is displayed above the active user interface element that is in focus, or a display characteristic of the active user interface element that is in focus is different from the balance of the first user interface displayed on the primary display).

In FIG. 5B, for example, window **536** for application A is in focus on primary display **102**. In FIG. 5B, status tray **502** indicates that application A is running in the foreground, and app tray **514** also indicates that application A is running in the foreground based on the shadow behind application A icon **512**. In FIG. 5F, for example, the music sub-section of the user’s media library shown within window **554** is in focus on primary display **102** as indicated by “Music” displayed in bold and albums A-L at least partially displayed within window **554**. In contrast, in FIG. 5G, for example, the podcasts sub-section of the user’s media library shown within window **554** is in focus on primary display **102** as shown by “Podcasts” displayed in bold and podcasts A-L at least partially displayed within window **554**. In FIG. 5M, for example, the sub-window for composing a new email within window **580** is in focus on primary display **102** as indicated by the thick lines surrounding the sub-window. In FIG. 5R, for example, the “Family” group tab of the user’s contact book within menu **5134** is in focus on primary display **102** as indicated by the thick lines surrounding the “Family” group tab of menu **5134**. In FIG. 5NN, for example, the interface for tab B within window **5224** is in focus on primary display **102** as indicated by the thick lines surrounding tab B and the bold text for tab B. In FIG. 5SS, for example, the bookmarks sidebar within window **5224** is in focus on primary display **102** as indicated by the thick lines surrounding the bookmarks sidebar. In FIG. 5YY, for example, the menu of edit controls **5296** is in focus on primary display **102**.

The computing system determines (**612**) whether the active user interface element that is in focus on the primary display is associated with an application executed by the

computing system. In some embodiments, the active user interface element is associated with either an application or the operating system. In some embodiments, the computing system or a component thereof (e.g., DFR determining module 352, FIG. 3A) determines a set of affordances for display on dynamic function row 104 based on the active user interface element that is in focus on primary display 102 and also whether the active user interface element is associated with a specific application or the operating system.

In some embodiments, the application is executed (614) by the processor in the foreground of the first user interface. For example, the application is one of an email application, a word processing application, a presentation application, a photo editing application, a music application, a game application, a spreadsheet application, or the like. FIGS. 5B-5E, for example, show the first user interface displayed by primary display 102 including window 536 corresponding to application A (e.g., a fantasy RPG game) executed in the foreground by the computing system. FIGS. 5F-5I, for example, show the first user interface displayed by primary display 102 including window 554 corresponding to a media player application executed in the foreground by the computing system. FIGS. 5J-5X, for example, show the first user interface displayed by primary display 102 including window 580 corresponding to a mail application executed in the foreground by the computing system. FIGS. 5Y-5GG, for example, show the first user interface displayed by primary display 102 including window 5166 corresponding to a photos application executed in the foreground by the computing system. FIGS. 5HH-5UU, for example, show the first user interface displayed by primary display 102 including window 5224 corresponding to a web browser application executed in the foreground by the computing system.

In accordance with a determination that the active user interface element that is in focus on the primary display is associated with the application executed by the computing system, the computing system displays (616) a second user interface on the touch screen display, including: (A) a first set of one or more affordances corresponding to the application; and (B) at least one system-level affordance corresponding to at least one system-level functionality. In some embodiments, the first set of one or more affordances includes user selectable symbols/icons and/or indicators and information that may or may not be selectable. In some embodiments, the first set of one or more affordances correspond to basic controls for the application. In some embodiments, at least one system-level affordance is displayed along with the first set of one or more affordances. In one example, in FIGS. 5A-5DDD, the at least one system-level affordance includes persistent affordances 516 and 534. In another example, in FIG. 5C, the at least one system-level affordance includes affordance 542, which, when activated (e.g., with a tap contact), causes display of a plurality of system-level affordances (e.g., affordances 518, 520, 522, 524, 526, 528, 530, and 532 shown in FIG. 5A).

FIG. 5II, for example, shows dynamic function row 104 displaying a set of basic controls and indicators (e.g., affordances 5230, 5232, and 5238, and address bar 5234) for the web browser application, which is in focus on primary display 102, along with the at least one system-level affordance 542, in response to detecting selection of affordance 5226 in FIG. 5HH. In some embodiments, the first set of one or more affordances corresponds to controls associated with the active user interface element that is in focus. As such, the user of the computing system is able to select controls from the first set of one or more affordances for providing input

to, and controlling the functions of, the application without shifting his/her hands away from the set of physical keys to another input device (e.g., a mouse) when such selectable controls are displayed on the primary display. This reduction in mode switching, for example, between keyboard and mouse for the user's hands and between keyboard and display for the user's eyes, provides a more intuitive user experience and a more efficient human-machine interface. FIG. 5J, for example, shows dynamic function row 104 displaying a plurality of affordances corresponding to email A (e.g., affordances 582, 584, 586, 588, 590, 592, 594, 596, and 598), which is in focus on primary display 102, along with the at least one system-level affordance 542, in response to detecting selection of mail application icon 506 with cursor 504 in FIG. 5I.

In some embodiments, the user of the computing system is able to modify or customize the affordances included in the first set of one or more affordances. In some embodiments, prior to identifying the active user interface element that is in focus on primary display 102 and displaying the first set of one or more affordances, dynamic function row 104 displays a default interface set by the user of the computing system or set in software. For example, the default interface includes one of: a plurality of function keys (e.g., F1, F2, F3, . . . , F12), a stock ticker, scrolling sports scores, scrolling weather forecasts and information, and/or the like.

In some embodiments, after displaying the first set of one or more affordances for a first predetermined period of time (e.g., 30, 60, 90, etc. seconds), dynamic function row 104 re-displays the default interface set by the user of the computing system or set in software. In some embodiments, after displaying the first set of one or more affordances for a first predetermined period of time (e.g., 30, 60, 90, etc. seconds), dynamic function row 104 turns off until a contact is detected by dynamic function row 104 or a keyboard or touch pad associated with the computing system. In some embodiments, after displaying the first set of one or more affordances for a first predetermined period of time (e.g., 30, 60, 90, etc. seconds), dynamic function row 104 turns off until connected to a power source (e.g., when dynamic function row is implemented in battery powered peripheral keyboard 206, FIGS. 2A-2B). In some embodiments, after displaying the first set of one or more affordances and not detecting user input with regard to the first set of one or more affordances for a second predetermined period of time (e.g., 30, 60, 90, etc. seconds), dynamic function row 104 re-displays the default interface set by the user of the computing system or set in software. In some embodiments, when the operating system is in focus or the active user interface element is not associated with an application running in the foreground, dynamic function row 104 re-displays the default interface set by the user of the computing system or set in software.

In some embodiments, after displaying the first set of one or more affordances for the first predetermined period of time (e.g., 30, 60, 90, etc. seconds) and when the computing system is set in a low-power mode, dynamic function row 104 displays a limited set of affordances including, for example, the time, the battery life remaining, the Wi-Fi signal strength, and/or the like. For example, the limited set of affordances are selected by the user of the computing system or set in software, and the user of computing system is able to set the computing into the low-power mode through a system settings panel.

In some embodiments, at least one of the affordances displayed on the second user interface is (618) a multi-

function affordance. In some embodiments, a multi-function affordance is capable of performing two or more functions/operations in response to detecting different inputs performed at a location corresponding to the multi-function affordance. For example, persistent volume control **568**, in FIG. **5J**, displayed by dynamic function row **104** within the second user interface is a multi-function affordance.

In some embodiments, the computing system detects (620) a user touch input selecting the multi-function affordance. In accordance with a determination that the user touch input corresponds to a first type, the computing system performs a first function associated with the multi-function affordance. In accordance with a determination that the user touch input corresponds to a second type distinct from the first type, the computing system performs a second function associated with the multi-function affordance. For example, a first function/operation (e.g., mute a media item) is performed in response to detecting a first gesture (e.g., a tap contact) at a location corresponding to the multi-function affordance, and a second function/operation (e.g., display a volume slider or playback controls for the media item) is performed in response to detecting a second gesture (e.g., a long press gesture) at a location corresponding to the multi-function affordance. For example, the first gesture type corresponds to a touch input detected for less than a predetermined period of time (e.g., 500 ms, 1 s, etc.) with one or more contacts (e.g., a tap contact), and the second gesture type corresponds to a touch input detected for greater than or equal to the predetermined period of time (e.g., 500 ms, 1 s, etc.) with one or more contacts (e.g., a long press gesture).

FIG. **5J**, for example, shows dynamic function row **104** receiving and detecting contact **599** (e.g., a long press contact) at a location corresponding to persistent volume control **568**. Continuing with this example, FIG. **5K** shows dynamic function row **104** displaying volume slider **5100** for adjusting the playback volume of podcast J, which was initiated in FIG. **5H**, in response to detecting the long press gesture at the location corresponding to persistent volume control **568** in FIG. **5J**. Alternatively, in some embodiments, dynamic function row **104** displays playback controls (e.g., pause, fast forward, rewind, next track, previous track, and the like) for controlling the playback of podcast J, which was initiated in FIG. **5H**, in response to detecting the long press gesture at the location corresponding to persistent volume control **568** in FIG. **5J**. FIG. **5M**, for example, shows dynamic function row **104** receiving and detecting contact **5124** (e.g., a tap contact) at a location corresponding to persistent volume control **568**. Continuing with this example, FIG. **5N** shows dynamic function row **104** displaying persistent volume control **568** indicating that podcast J is muted in response to detecting the tap contact at the location corresponding to persistent volume control **568** in FIG. **5M**.

In some embodiments, the least one system-level affordance is configured (622) upon selection to cause display of a plurality of system-level affordances corresponding to system-level functionalities on the touch screen display. In some embodiments, the at least one system-level affordance enables access to a plurality of system-level controls/affordances such as volume and brightness controls, and other system-level functionalities. For example, in FIG. **5D**, dynamic function row **104** displays a second set of affordances and/or indicators (e.g., control set B) corresponding to application A and at least one system-level affordance (e.g., affordance **542**). In FIG. **5C**, dynamic function row **104** also detects contact **552** (e.g., a tap contact) at a location

corresponding to affordance **542**. Continuing with this example, FIG. **5E** shows dynamic function row **104** the displaying persistent controls (i.e., affordances **516** and **534**), the plurality of system-level affordances (i.e., affordances **518**, **520**, **522**, **524**, **526**, **528**, **530**, and **532**), and affordance **538** corresponding to application A in response to detecting selection of affordance **542** in FIG. **5D**.

In some embodiments, the at least one system-level affordance corresponds to (624) one of a power control or escape control. In some embodiments, the at least one system-level affordance includes persistent controls that are displayed on dynamic function row **104** regardless of the focus of primary display **102** (e.g., escape affordance **516** and power control **534**, FIGS. **5A-5DDD**). When activated (e.g., via a tap contact), escape affordance **516**, causes performance of a corresponding function (e.g., exiting an application which is currently in focus on primary display **102**). When activated (e.g., via a tap contact), power control **534** causes display of a modal alert (e.g., modal alert **5308**, FIG. **5DDD**) on dynamic function row **104** for logging out, restarting, or powering-off the computing system.

In some embodiments, the computing system detects (626) a user touch input selecting one of the first set of affordances, and, in response to detecting the user touch input, the computing system: displays a different set of affordances corresponding to functionalities of the application; and maintains display of the at least one system-level affordance. In some embodiments, the first set of one or more affordances corresponding to the application includes a single affordance for accessing a set of tools or functions associated with the application. FIG. **5B**, for example, shows dynamic function row **104** displaying affordance **538** corresponding to application A, which is in focus on primary display **102**, in addition to the persistent controls (i.e., affordances **516** and **534**) and the plurality of system-level affordances (i.e., affordances **518**, **520**, **522**, **524**, **526**, **528**, **530**, and **532**) in response to detecting selection of application A icon **512** with cursor **504** in FIG. **5A**. Continuing with the example, FIG. **5C** shows dynamic function row **104** displaying a first set of affordances and/or indicators (e.g., control set A) corresponding to application A in response to detecting selection of affordance **538** in FIG. **5B**.

In some embodiments, the computing system detects (628) a subsequent user touch input selecting the at least one system-level affordance, and, in response to detecting the subsequent user touch input, the computing system displays a plurality of system-level affordances corresponding to system-level functionalities and at least one application-level affordance corresponding to the application. FIG. **5D**, for example, shows dynamic function row **104** displaying a second set of affordances and/or indicators (e.g., control set B) corresponding to application A, which is in focus on primary display **102**, and the at least one system-level affordance (e.g., affordance **542**). Continuing with this example, in response to detecting selection of affordance **542** in FIG. **5D**, FIG. **5E** shows dynamic function row **104** displaying persistent controls (i.e., affordances **516** and **534**), the plurality of system-level affordances (i.e., affordances **518**, **520**, **522**, **524**, **526**, **528**, **530**, and **532**), and affordance **538** corresponding to application A.

In some embodiments, after displaying the second user interface on the touch screen display, the computing system identifies (630) a second active user interface element among the one or more user interface elements that is in focus on the primary display and determines whether the second active user interface element corresponds to a different application executed by the computing device. In

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accordance with a determination that the second active user interface element corresponds to the different application, the computing system displays a fourth user interface on the touch screen display, including: (D) a third set of one or more affordances corresponding to the different application; and (E) the at least one system-level affordance corresponding to the at least one system-level functionality. FIG. 5F, for example, shows dynamic function row **104** displaying a plurality of album affordances **558** (e.g., album affordances **558-A** to **558-G**) corresponding to the sub-section of the user's media library that is in focus on primary display **102** and the at least one system-level affordance (e.g., affordance **542**). FIG. 5F also shows primary display **102** displaying cursor **504** at a location corresponding to the podcasts sub-section of the user's media library. Continuing with the example, in response to detecting selection of the podcasts sub-section with cursor **504** in FIG. 5F, FIG. 5G shows dynamic function row **104** displaying a first plurality of podcast affordances **560** (e.g., podcast affordances **560-A** to **560-G**) corresponding to the podcasts sub-section of the user's media library that is now in focus on primary display **102** and the at least one system-level affordance (e.g., affordance **542**).

In some embodiments, the computing system provides audible cues indicating the different user interface element that is in focus. In some embodiments, primary display **102** displays a visual cue indicating the different user interface element that is in focus. For example, with respect to FIG. 5G, primary display **102** displays a star indicator (not shown) above the "podcasts" text in the left-hand column within window **554**, makes the "podcasts" text bold in the left-hand column within window **554**, flashes the "podcasts" text in the left-hand column within window **554** in a different color or the same color, or otherwise indicates that the podcasts sub-section is now in focus on primary display **102**. In some embodiments, dynamic function row **104** displays a transition animation whereby the plurality of album affordances **558** (shown in FIG. 5F) are rolled over or slid out of the way to so as to display the first plurality of podcast affordances **560** (shown in FIG. 5G).

In some embodiments, after identifying that the second active user interface element, the computing system determines (632) whether a media (e.g., audio or video) item is being played by the computing system, where the media item is not associated with the different application, and, in accordance with a determination that the media item is being played by the computing system, the computing system displays at least one persistent affordance on the touch screen display for controlling the media item (e.g., volume and/or playback controls). In some embodiments, the at least one affordance is a persistent control that enables the user of the computing system to mute/unmute the media item from the touch screen display even if the focus changes or the media is muted and/or being played in the background. FIG. 5I, for example, shows dynamic function row **104** displaying persistent volume control **568** in response to detecting selection of podcast affordance **560-J** in FIG. 5H, which initiates playback of podcast J. FIG. 5I, for example, shows dynamic function row **104** displaying persistent volume control **568** even while email A of the application is in focus on primary display **102**.

In some embodiments, the at least one persistent affordance displays (634) feedback that corresponds to the media item (e.g., an equalizer (EQ) bar, a run-time indicator, or the like) FIG. 5I, for example, shows dynamic function row **104** displaying persistent volume control **568** with an equalizer bar corresponding to playback of podcast J. Similarly, FIG.

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5N, for example, shows dynamic function row **104** displaying persistent volume control **568** with the equalizer bar even while playback of podcast J is muted. In some embodiments, dynamic function row **104** includes an affordance that is continuously updated based on a media item being played or some other real-time information such as a weather indicator, a microphone capture indicator, or a Wi-Fi signal strength indicator.

In some embodiments, the computing device detects (636) a user input corresponding to an override key, and, in response to detecting the user input, the computing system: ceases to display at least the first set of one or more affordances of the second user interface on the touch screen display; and displays a first set of default function keys. In some embodiments, the user input corresponding to the override key is detected in response to receiving a signal from the input mechanism (e.g., actuation of a specified physical key such as a function key) or from the dynamic function row (e.g., selection of a virtual key such as "escape" affordance **516**). For example, the user actuates an "fn" key to dismiss current affordances displayed by dynamic function row **104** and to display the default F1, F2, F3, . . . , F12 row. FIG. 5ZZ, for example, shows dynamic function row **104** displaying interface **5300** with affordances **5301** (e.g., corresponding to F1, F2, F3, . . . , F12) along with the persistent controls (e.g., affordances **516** and **534**) in response to receiving a signal from the set of physical keys **106** of portable computing system **100** (FIGS. 1A-1B) or from the set of physical keys **106** of peripheral keyboard **206** of desktop computing system **200** (FIGS. 2A-2D) corresponding to a specified physical key (e.g., a function key) for overriding dynamic function row **104**.

In some embodiments, in response to detecting the user input corresponding to the override key, dynamic function row **104** displays a default interface set by the user of the computing system or set in software. For example, the default interface includes one of: a plurality of function keys (e.g., F1, F2, F3, . . . , F12), a stock ticker, scrolling sports scores, scrolling weather forecasts, or the like.

In some embodiments, after displaying the first set of default function keys, the computing system detects (638) a gesture in a direction substantially parallel to a major dimension of on the touch screen display, and in response to detecting the swipe gesture, displays a second set of default function keys with at least one distinct function key (e.g., a previously undisplayed function key). For example, with respect to FIG. 5ZZ, the user of the computing system is able to reveal additional function keys (e.g., F13, F14, F15, . . .) within interface **5300** on dynamic function row **104** by performing a substantially horizontal swipe gesture on dynamic function row **104** (e.g., one of a right-to-left or left-to-right swipe gesture).

In some embodiments, in accordance with a determination that the active user interface element is not associated with the application executed by the computing system, the computing system displays (640) a third user interface on the touch screen display, including: (C) a second set of one or more affordances corresponding to operating system controls of the computing system, where the second set of one or more affordances are distinct from the first set of one or more affordances. In FIG. 5A, for example, the system/operating system is currently in focus on primary display **102**. For example, the active user interface element that is in focus on the primary display is associated with the operating system such as volume controls, system controls (e.g., brightness or volume controls), system settings, a start menu, file explorer, system search, or the like. FIG. 5A

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shows dynamic function row **104** displaying a plurality of system-level affordances (e.g., affordances **518**, **520**, **522**, **524**, **526**, **528**, **530**, and **532**) along with the persistent affordances (e.g., affordances **516** and **534**).

In some embodiments, the second set of one or more affordances is (**642**) an expanded set of operating system controls that includes (B) the at least one system-level affordance corresponding to the at least one system-level functionality. FIG. **5T**, for example, shows dynamic function row **104** displaying a plurality of system-level affordances (e.g., the expanded set of operating system controls, including affordances **518**, **520**, **522**, **524**, **526**, **528**, **530**, and **532**) along with the persistent affordances (e.g., affordances **516** and **534**) in response to detecting selection of affordance **542** in FIG. **5S**.

It should be understood that the particular order in which the operations in FIGS. **6A-6D** have been described is merely an example and is not intended to indicate that the described order is the only order in which the operations could be performed. One of ordinary skill in the art would recognize various ways to reorder the operations described herein. Additionally, it should be noted that details of other processes described herein with respect to other methods described herein (e.g., methods **700**, **800**, **900**, and **1000**) are also applicable in an analogous manner to method **600** described above with respect to FIGS. **6A-6D**.

FIGS. **7A-7C** are a flowchart of a method of updating a dynamic input and output device (e.g., including dynamic function row **104**, FIGS. **1A-1B** and **2A-2D**), in accordance with some embodiments. The method **700** is performed at a computing system including a processor, a primary display, memory, and a housing at least partially containing a physical input mechanism and a touch screen display adjacent to the physical input mechanism. Some operations in method **700** are, optionally, combined and/or the order of some operations is, optionally, changed.

In some embodiments, the computing system is portable computing system **100** (FIG. **1A**) or desktop computing system **200** (FIGS. **2A-2D**). In some embodiments, the primary display is primary display **102** (FIG. **1A**) which is implemented in display portion **110** of portable computing system **100** (FIG. **1A**). Alternatively, in some embodiments, the primary display is primary display **102** (FIGS. **2A-2D**) which is implemented in peripheral display device **204** (FIGS. **2A-2D**). In some embodiments, the housing is body portion **120** of portable computing system **100** (FIGS. **1A-1B**), which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIGS. **1A-1B**) and the input mechanism (e.g., the set of physical keys **106**, FIGS. **1A-1B**).

Alternatively, in some embodiments, the housing is peripheral keyboard **206** (FIGS. **2A-2B**) of desktop computing system **200**, which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIGS. **2A-2B**) and the input mechanism (e.g., the set of physical keys **106**, FIGS. **2A-2B**). Alternatively, in some embodiments, the housing is first input mechanism **212** (FIG. **2C**) of desktop computing system **200**, which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIG. **2C**) and the input mechanism (e.g., touch pad **108**, FIG. **2C**).

The computing system displays (**702**) a first user interface for an application executed by the computing system on the primary display. FIG. **5P**, for example, shows primary display **102** displaying a first user interface with menu **5134** corresponding to the user's contact book. In FIG. **5P**, menu **5134** includes a list of a plurality of contacts corresponding

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the "All Contacts" group of the user's contact book (e.g., a list of pre-existing or automatically populated contacts), which is in focus on primary display **102** as indicated by the thick lines surrounding the "All Contacts" group in menu **5134**. FIG. **5XX**, for example, shows primary display **102** displaying a first user interface with a menu of file controls **5288** overlaid on window **5166**, where the menu of file controls **5288** is in focus on primary display **102**.

The computing system displays (**704**) a second user interface on the touch screen display, the second user interface comprising a first set of one or more affordances corresponding to the application, where the first set of one or more affordances corresponds to a first portion of the application. In some embodiments, the first set of one or more affordances associated with a top menu or a file menu (i.e., the first portion or sub-section) of the application. FIG. **5P**, for example, shows dynamic function row **104** displaying a second user interface with a first plurality of contact affordances **5136-A** to **5136-F** (i.e., the first set of affordances) corresponding to the "All Contacts" group of the user's contact book that is in focus on primary display **102**. FIG. **5XX**, for example, shows dynamic function row **104** displaying a second user interface with a first plurality of affordances **5290** (i.e., the first set of affordances) that correspond to the menu of file controls **5288** that is in focus on primary display **102**.

The computing system detects (**706**) a swipe gesture on the touch screen display. In a first example, FIG. **5P** shows dynamic function row **104** detecting a right-to-left swipe gesture with contact **5138** moving from a first location **5140-A** to a second location **5140-B**. In another example, FIG. **5XX** shows dynamic function row **104** detecting an upward swipe gesture with contact **5292** moving from a first location **5294-A** to a second location **5294-B**.

In accordance with a determination that the swipe gesture was performed in a first direction (e.g., horizontal), the computing system displays (**708**) a second set of one or more affordances corresponding to the application on the touch screen display, where at least one affordance in the second set of one or more affordances is distinct from the first set of one or more affordances, and where the second set of one or more affordances also corresponds to the first portion of the application. In some embodiments, in response to detecting a swipe gesture in the first direction, the dynamic function row displays different a second set of affordances corresponding to the first portion of the application (e.g., a toolset or menu) with at least one different affordance (e.g., a tool or item), as compared to the first set of affordances that also correspond to the first portion of the application. For example, in response to detecting the right-to-left swipe gesture in FIG. **5P**, FIG. **5Q** shows dynamic function row **104** displaying a second plurality of contact affordances **5136-E** to **5136-J** (i.e., the second set of affordances) corresponding to the "All Contacts" group of the user's contact book.

In some embodiments, the first direction is (**710**) substantially parallel to a major dimension of the touch screen display. For example, with reference to portable computing system **100** (FIGS. **1A-1B**), the first direction is substantially perpendicular (e.g., vertical) relative to the major dimension of dynamic function row **104** and/or the set of physical keys **106**.

In some embodiments, the first direction is (**712**) substantially perpendicular to a major dimension of the touch screen display. For example, with reference to portable computing system **100** (FIGS. **1A-1B**), the first direction is substantially

parallel (e.g., horizontal) relative to the major dimension of dynamic function row **104** and/or the set of physical keys **106**.

In accordance with a determination that the swipe gesture was performed in a second direction substantially perpendicular to the first direction (e.g., vertical), the computing system displays (**714**) a third set of one or more affordances corresponding to the application on the touch screen display, where the third set of one or more affordances is distinct from the second set of one or more affordances, and where the third set of one or more affordances corresponds to a second portion of the application that is distinct from the first portion of the application. In some embodiments, in response to detecting a swipe gesture in the second direction, the dynamic function row displays different a third set of affordances corresponding to the second portion of the application (e.g., a different toolset or menu), as compared to the first portion of the application associated with the first set of affordances. For example, in response to detecting the upward swipe gesture in FIG. **5XX**, FIG. **5YY** shows dynamic function row **104** displaying a second plurality of affordances **5298** (i.e., the third set of affordances) that correspond to the menu of edit controls **5296** that is in focus on primary display **102**. In some embodiments, dynamic function row **104** displays an transition animation whereby the first plurality of affordances **5290** (shown in FIG. **5XX**) are rolled over or slid out of the way to so as to display the second plurality of podcast affordances **598** (shown in FIG. **5YY**).

As such, in one embodiment, a horizontal swipe detected on the dynamic function row **104** moves or scrubs within a group or category and a vertical swipe changes the category/filter. As a result, the user of the computing system is able to move easily navigate tools and menus without moving his/her hands away from the keyboard to a mouse. This also enables more efficient display of information and allows for a more efficient man-machine interaction.

In some embodiments, the second portion is displayed (**716**) on the primary display in a compact view within the first user interface prior to detecting the swipe gesture, and the system displays the second portion on the primary display in an expanded view within the first user interface in accordance with the determination that the swipe gesture was performed in the second direction substantially perpendicular to the first direction. For example, in FIG. **5XX**, the first portion of status tray **502** for the photos application (e.g., the menu of file controls **5288**) is displayed by primary display **102** in an expanded mode, and the second portion of status tray **502** for the photos application (e.g., the menu of edit controls **5296**) is displayed by primary display **102** in a compact mode. Continuing with this example, in response to detecting the swipe gesture performed in the second direction in FIG. **5XX** (e.g., the upward swipe gesture with contact **5292**), the second portion of status tray **502** for the photos application is displayed by primary display **102** in the expanded mode in FIG. **5YY**, and the first portion of status tray **502** for the photos application is displayed by primary display **102** in the compact mode.

In another example, in FIG. **5Q**, the first portion of menu **5134** of the mail application (e.g., the “All Contacts” group of the user’s contact book within menu **5134**) is displayed by primary display **102** in an expanded mode, and the second portion of menu **5134** of the mail application (e.g., the “Family” group of the user’s contact book within menu **5134**) is displayed by primary display **102** in a compact mode. Continuing with this example, in response to detecting the swipe gesture performed in the second direction in

FIG. **5Q** (e.g., the upward swipe gesture with contact **5142**), the second portion of menu **5134** of the mail application is displayed by primary display **102** in the expanded mode in FIG. **5R**, and the first portion of menu **5134** of the mail application is displayed by primary display **102** in the compact mode.

In some embodiments, the first portion is (**718**) one of a menu, tab, folder, tool set, or toolbar of the application, and the second portion is one of a menu, tab, folder, tool set, or toolbar of the application. In FIGS. **5P-5R**, for example, the first portion of menu **5134** of the mail application corresponds to the “All Contacts” group of the user’s contact book within menu **5134**, and the second portion of menu **5134** of the mail application corresponds to the “Family” group of the user’s contact book within menu **5134**. In FIGS. **5XX-5YY**, for example, the first portion of status tray **502** for the photos application corresponds to the menu of file controls **5288**, and the second portion of status tray **502** for the photos application corresponds to the menu of edit controls **5296**.

In some embodiments, after displaying the third set of one or more affordances on the touch screen display, the computing system (**720**): detects a user input selecting the first portion on the first user interface; and, in response to detecting the user input: ceases to display the third set of one or more affordances on the touch screen display, where the third set of one or more affordances corresponds to the second portion of the application; and displays the second set of one or more affordances, where the second set of one or more affordances corresponds to the first portion of the application. For example, with respect to FIG. **5YY**, after displaying the second plurality of affordances **5298** (i.e., the third set of affordances) on dynamic function row **104** that correspond to the menu of edit controls **5296** that is in focus on primary display **102**, primary display **102** displays cursor **504** at a location corresponding to the file menu within status tray **502** (not shown). Continuing with example, in response to detecting selection of the file menu within status tray **502** with cursor **504**, dynamic function row **104** ceases to display the second plurality of affordances **5298** (i.e., the third set of affordances) and, instead, displays the menu of file controls **5288** (i.e., the second set of affordances).

In some embodiments, the second set of one or more affordances and the third set of one or more affordances includes (**722**) at least one system-level affordance corresponding to at least one system-level functionality. For example, in FIG. **5Q**, dynamic function row **104** displays the second plurality of contact affordances **5136-E** to **5136-J** (i.e., the second set of affordances) corresponding to the “All Contacts” group of the user’s contact book along with the at least one system-level affordance (e.g., affordance **542**). In FIG. **5YY**, for example, dynamic function row **104** displays a second plurality of affordances **5298** (i.e., the third set of affordances) that correspond to the menu of edit controls **5296** that is in focus on primary display **102** along with the at least one system-level affordance (e.g., affordance **542**).

In some embodiments, the first user interface for the application executed by the computing system is displayed (**724**) on the primary display in a full-screen mode, and the first set of one or more affordances displayed on the touch screen display includes controls corresponding to the full-screen mode. FIG. **5EE**, for example, shows primary display **102** displaying photo B in full-screen mode within window **5200**, and also shows dynamic function row **104** displaying a set of controls **5209** for straightening the orientation of photo B within window **5200**. For example, in response to detecting a swipe gesture in the second direction (e.g.,

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vertical relative to the major dimension of dynamic function row **104**), dynamic function row **104** displays a set of controls corresponding to crop tool **5210** (e.g., if the swipe gesture is an upward swipe gesture) or a set of controls corresponding to red-eye reduction tool **5206** (e.g., if the swipe gesture is a downward swipe gesture). Continuing with this example, in response to detecting a swipe gesture in the first direction (e.g., horizontal relative to the major dimension of dynamic function row **104**), dynamic function row **104** adjusts the orientation of photo B within window **5200**.

In another example, with respect to FIG. **5EE**, in response to detecting a swipe gesture in the second direction (e.g., vertical relative to the major dimension of dynamic function row **104**), dynamic function row **104** displays a set of controls associated with information affordance **5178** (e.g., if the swipe gesture is an upward swipe gesture) or a set of controls associated with photo adding affordance **5182** (e.g., if the swipe gesture is a downward swipe gesture). Continuing with this example, in response to detecting a swipe gesture in the first direction (e.g., horizontal relative to the major dimension of dynamic function row **104**), dynamic function row **104** displays a set of controls corresponding to crop tool **5210** (e.g., if the swipe gesture is an right-to-left swipe gesture) or a set of controls corresponding to red-eye reduction tool **5206** (e.g., if the swipe gesture is a left-to-right swipe gesture).

It should be understood that the particular order in which the operations in FIGS. **7A-7C** have been described is merely an example and is not intended to indicate that the described order is the only order in which the operations could be performed. One of ordinary skill in the art would recognize various ways to reorder the operations described herein. Additionally, it should be noted that details of other processes described herein with respect to other methods described herein (e.g., methods **600**, **800**, **900**, and **1000**) are also applicable in an analogous manner to method **700** described above with respect to FIGS. **7A-7C**.

FIGS. **8A-8B** are a flowchart of a method of maintaining functionality of an application while in full-screen mode, in accordance with some embodiments. The method **800** is performed at a computing system including a processor, a primary display, memory, and a housing at least partially containing a physical input mechanism and a touch screen display adjacent to the physical input mechanism. Some operations in method **800** are, optionally, combined and/or the order of some operations is, optionally, changed.

In some embodiments, the computing system is portable computing system **100** (FIG. **1A**) or desktop computing system **200** (FIGS. **2A-2D**). In some embodiments, the primary display is primary display **102** (FIG. **1A**) which is implemented in display portion **110** of portable computing system **100** (FIG. **1A**). Alternatively, in some embodiments, the primary display is primary display **102** (FIGS. **2A-2D**) which is implemented in peripheral display device **204** (FIGS. **2A-2D**). In some embodiments, the housing is body portion **120** of portable computing system **100** (FIGS. **1A-1B**), which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIGS. **1A-1B**) and the input mechanism (e.g., the set of physical keys **106**, FIGS. **1A-1B**). Alternatively, in some embodiments, the housing is peripheral keyboard **206** (FIGS. **2A-2B**) of desktop computing system **200**, which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIGS. **2A-2B**) and the input mechanism (e.g., the set of physical keys **106**, FIGS. **2A-2B**). Alternatively, in some embodiments, the housing is first input mechanism **212**

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(FIG. **2C**) of desktop computing system **200**, which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIG. **2C**) and the input mechanism (e.g., touch pad **108**, FIG. **2C**).

The computing system displays (**802**), on the primary display in a normal mode, a first user interface for the application executed by the computing system, the first user interface comprising at least a first set of one or more affordances associated with an application. FIG. **5Z**, for example, shows primary display **102** displaying a first user interface with window **5166** for the photo application in a normal display mode. In FIG. **5Z**, selected photo B within window **5166** is in focus on primary display **102**, and window **5166** includes selectable affordances (i.e., the first set of affordances) for performing functions/operations with selected photo B: search, zoom, slideshow, share, and settings.

The computing system detects (**804**) a user input for displaying at least a portion of the first user interface for the application in a full-screen mode on the primary display. For example, the detected user input corresponds to a selection of a full-screen view affordance or a slideshow affordance displayed on primary display **102** or dynamic function row **104**.

In some embodiments, the user input for displaying at least the portion of the first user interface for the application in full-screen mode on the primary display is (**806**) at least one of a touch input detected on the touch screen display and a control selected within the first user interface on the primary display. In a first example, FIG. **5Z** shows primary display **102** displaying cursor **504** at a location corresponding to a slideshow affordance. In a second example, FIG. **5BB** shows dynamic function row **104** receiving and detecting contact **5196** (e.g., a tap contact) at a location corresponding to full-screen affordance **5196**.

In response to detecting the user input, the computing system (**808**): ceases to display the first set of one or more affordances associated with the application in the first user interface on the primary display; displays, on the primary display in the full-screen mode, the portion of the first user interface for the application; and automatically, without human intervention, displays, on the touch screen display, a second set of one or more affordances for controlling the application, where the second set of one or more affordances correspond to the first set of one or more affordances. Continuing with the first example above, FIG. **5AA** shows primary display **102** displaying a slideshow of photos from the all photos sub-section of the user's photo library in window **5188** in response to detecting selection of slideshow affordance with cursor **504** in FIG. **5Z**. FIG. **5AA** also shows dynamic function row **104** displaying thumbnail images **5192-Y**, **5192-Z**, **5192-A**, **5192-B**, **5192-C**, **5192-D**, and **5192-E** (i.e., the second set of affordances) in response to detecting selection of slideshow affordance with cursor **504** in FIG. **5Z**. Thumbnail images **5192-Y**, **5192-Z**, **5192-A**, **5192-B**, **5192-C**, **5192-D**, and **5192-E** correspond to the sequence of photos for the slideshow that are associated with the all photos sub-section of the user's photo library. In FIG. **5AA**, affordance **5192-B** corresponding to photo B is prominently displayed by dynamic function row **104** (e.g., with bold text and a thick border) to indicate that photo B is currently in focus on primary display **102**.

Continuing with the second example above, FIG. **5CC** shows primary display **102** displaying photo B in full-screen mode within window **5200** in response to detecting selection of full-screen affordance **5196** in FIG. **5BB**. FIG. **5DD** also shows dynamic function row **104** displaying a set of editing

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tools **5205** (i.e., the second set of affordances) in response to detecting selection of editing affordance **5180** in FIG. 5CC.

In another example, when the web browser application or a portion of the user interface that corresponds to the web browser application is in focus on primary display **102** and the computing system detects a user input to enter full-screen mode, primary display **102** displays a currently active tab in full-screen mode, and dynamic function row **104** displays thumbnail images corresponds to tabs open within the web browser application along with the at least one system-level affordance. For example, the user of the computing system is able to display a respective tab on primary display **102** in full-screen mode by selecting a thumbnail corresponding to the respective tab on dynamic function row **104**.

In some embodiments, the second set of one or more affordances is **(810)** the first set of one or more affordances. For example, the second set of affordances includes at least a portion of the first set of affordances associated with the application. In another example, the second set of affordances includes the first set of affordances associated with the application. In another example, the second set of affordances is the same as the first set of affordances associated with the application. For example, the second set of affordances includes controls associated with the application executed by the computing system such as photo editing controls, gaming controls, slideshow controls and previews, currently opened web page tabs for a web browser, etc.

In some embodiments, the second set of one or more affordances includes **(812)** controls corresponding to the full-screen mode. FIG. 5AA, for example, shows dynamic function row **104** displaying thumbnail images **5192-Y**, **5192-Z**, **5192-A**, **5192-B**, **5192-C**, **5192-D**, and **5192-E** (i.e., the second set of affordances) corresponding to the sequence of photos for the slideshow that are associated with the all photos sub-section of the user's photo library. For example, the user of the computing system is able to skip ahead to a specific photo or skip back to a specific photo by selecting one of the affordances **5192**. Furthermore, for example, the user of the computing system is able to browse ahead by performing a right-to-left swipe gesture on dynamic function row **104** or browse behind in the sequence of photos by performing a left-to-right swipe gesture on dynamic function row **104**. Additionally, in FIG. 5AA, dynamic function row **104** displays pause affordance **5190**, which, when activated (e.g., via a tap contact), causes the slideshow to be paused and also causes primary display **102** to exit the full-screen mode.

In some embodiments, the second set of one or more affordances includes **(814)** at least one system-level affordance corresponding to at least one system-level functionality. FIG. 5AA, for example, shows dynamic function row **104** displaying thumbnail images **5192** and pause affordance **5190** along with the at least one system-level affordance (e.g., affordance **542**) and the persistent controls (e.g., affordances **516** and **534**).

In some embodiments, the computing system detects **(816)** a user touch input selecting one of the second set of affordances displayed on the touch screen display, and, in response to detecting the user touch input, the computing system changes the portion of the first user interface for the application being displayed in the full-screen mode on the primary display according to the selected one of the second set of affordances. In FIG. 5EE, for example, the user of the computing system is able to adjust the orientation of photo B within window **5200** displayed by primary display **102** by

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performing a left-to-right swipe/drag gesture or a right-to-left swipe/drag gesture at a location originating on slider **5210** or within the set of controls **5209**.

In some embodiments, after displaying the portion of the first user interface for the application in the full-screen mode on the primary display, the computing system **(818)**: detects a subsequent user input for exiting the full-screen mode; and, in response to detecting the subsequent user input: displays, on the primary display in the normal mode, the first user interface for the application executed by the computing system, the first user interface comprising the first set of one or more affordances associated with the application; and maintains display of at least a subset of the second set of one or more affordances for controlling the application on the touch screen display, where the second set of one or more affordances correspond to the first set of one or more affordances. In one example, FIG. 5AA shows dynamic function row **104** receiving and detecting contact **5194** (e.g., a tap contact) at a location corresponding to pause affordance **5190**. Continuing with this example, primary display **102** exits the full-screen mode, and FIG. 5BB shows primary display **102** displaying photo B in an expanded view within window **5166** in response to detecting selection of pause affordance **5190** in FIG. 5AA. In another example, FIG. 5EE shows dynamic function row **104** receiving and detecting contact **5216** (e.g., a tap contact) at a location corresponding to escape affordance **516**. Continuing with this example, primary display **102** exits the full-screen mode, and FIG. 5FF shows primary display **102** displaying photo B in the expanded view within window **5166** in response to detecting selection of escape affordance **516** in FIG. 5EE. Continuing with this example, FIG. 5FF also shows dynamic function row **104** maintaining display of at least a subset of the second set of affordances displayed FIG. 5EE.

It should be understood that the particular order in which the operations in FIGS. 8A-8B have been described is merely an example and is not intended to indicate that the described order is the only order in which the operations could be performed. One of ordinary skill in the art would recognize various ways to reorder the operations described herein. Additionally, it should be noted that details of other processes described herein with respect to other methods described herein (e.g., methods **600**, **700**, **900**, and **1000**) are also applicable in an analogous manner to method **800** described above with respect to FIGS. 8A-8B.

FIGS. 9A-9B are a flowchart of a method of displaying notifications on a touch screen display, in accordance with some embodiments. The method **900** is performed at a computing system including a processor, a primary display, memory, and a housing at least partially containing a physical input mechanism and a touch screen display adjacent to the physical input mechanism. Some operations in method **900** are, optionally, combined and/or the order of some operations is, optionally, changed.

In some embodiments, the computing system is portable computing system **100** (FIG. 1A) or desktop computing system **200** (FIGS. 2A-2D). In some embodiments, the primary display is primary display **102** (FIG. 1A) which is implemented in display portion **110** of portable computing system **100** (FIG. 1A). Alternatively, in some embodiments, the primary display is primary display **102** (FIGS. 2A-2D) which is implemented in peripheral display device **204** (FIGS. 2A-2D). In some embodiments, the housing is body portion **120** of portable computing system **100** (FIGS. 1A-1B), which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIGS. 1A-1B) and the input mechanism (e.g., the set of physical keys **106**,

FIGS. 1A-1B). Alternatively, in some embodiments, the housing is peripheral keyboard **206** (FIGS. 2A-2B) of desktop computing system **200**, which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIGS. 2A-2B) and the input mechanism (e.g., the set of physical keys **106**, FIGS. 2A-2B). Alternatively, in some embodiments, the housing is first input mechanism **212** (FIG. 2C) of desktop computing system **200**, which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIG. 2C) and the input mechanism (e.g., touch pad **108**, FIG. 2C).

The computing system displays (**902**), on the primary display, a first user interface for an application executed by the computing system. FIG. 5TT, for example, shows primary display **102** displaying a first user interface with an interface for tab B (e.g., corresponding to www.website_B.com/home) along with a bookmarks sidebar within window **5224**. In FIG. 5TT, the bookmarks sidebar is in focus on primary display **102** as indicated by the thick lines surrounding the bookmarks sidebar.

The computing system displays (**904**), on the touch screen display, a second user interface, the second user interface comprising a set of one or more affordances corresponding to the application. Continuing with the example above, FIG. 5TT shows dynamic function row **104** displaying a second user interface with a set of bookmark affordances **5278** corresponding to all pre-existing bookmarks as a result of the bookmarks sidebar being in focus on primary display **102**.

In some embodiments, prior to detecting the notification, the computing system detects (**906**) a user input selecting a notification setting so as to display notifications on the touch screen display and to not display notifications on the primary display. In some embodiments, the user of the computing system is able to specify within a settings panel whether received notifications are to be displayed on one or more of primary display **102** and dynamic function row **104**. In some embodiments, the computing system displays received notifications on dynamic function row **104** but not on primary display **102** by default.

The computing system detects (**908**) a notification. In some embodiments, the notification is associated with the application (e.g., the web browser application in FIG. 5TT). In some embodiments, the notification is associated with the application that is currently being executed in the foreground or with a background application. For example, the notification is one of: a modal alert or real-time notification such as an alert associated with a newly received email, instant message, or SMS; a notification associated with a newly detected occurrence within an application such as a post or response within a social media application; a modal alert associated with an application executed by the computing system such as a save/exit dialogue box or other prompt; or the like.

In response to detecting the notification, the computing system concurrently displays (**910**), in the second user interface, the set of one or more affordances corresponding to the application and at least a portion of the detected notification on the touch screen display, where the detected notification is not displayed on the primary display. For example, in some embodiments, based on the notification setting discussed in operation **906**, at least a portion of the detected notification is overlaid on the second user interface displayed on dynamic function row **104**. Additionally and/or alternatively, in some embodiments, the notification is overlaid on the first user interface displayed by primary display **102**. As such, the user of the computing system is able to

view and respond to notifications without shifting his/her hands away from the set of physical keys to another input device (e.g., a mouse) when such selectable controls are displayed on the primary display. This reduction in mode switching, for example, between keyboard and mouse for the user's hands and between keyboard and display for the user's eyes, provides for a more intuitive user experience and a more efficient human-machine interface.

For example, FIG. 5V shows primary display **102** and dynamic function row **104** displaying modal alert **5156** in response to detecting selection of the exit affordance with cursor **504** in FIG. 5U. In FIG. 5V, modal alert **5156** prompts the user to save the draft email prior to closing window **580** and includes a "Save" affordance **5158**, a "Don't Save" affordance **5160**, and a "Cancel" affordance **5162**. In another example, FIG. 5X shows primary display **102** and dynamic function row **104** displaying application selection window **5164** in response to receiving a signal corresponding to a specified physical key combination (e.g., alt+tab) from the set of physical keys **106** of portable computing system **100** (FIGS. 1A-1B) or from the set of physical keys **106** of peripheral keyboard **206** of desktop computing system **200** (FIGS. 2A-2D). In FIG. 5X, application selection window **5164** enables the user of the computing system to toggle between currently running applications which may be in the background.

In yet another example, FIG. 5FF shows dynamic function row **104** displaying notification **5218** overlaid on affordances **5178**, **5180**, **5182**, and **5184** in response to reception of notification **5218** by the computing system. In FIG. 5FF, notification **5218** corresponds to an SMS, instant message, or the like sent by Suzie S. to the user of the computing system, where the notification's content inquiries "Movies tonight?" In yet another example, FIG. 5KK shows primary display **102** and dynamic function row **104** displaying modal alert **5240** in response to detecting selection of the purchase affordance with cursor **504** in FIG. 5JJ. In FIG. 5KK, modal alert **5240** displayed on dynamic function row **104** prompts the user of the computing system to provide their fingerprint in fingerprint region **5244** of dynamic function row **104** and also includes cancel affordance **5242**, which, when activated (e.g., via a tap contact) causes cancelation of the purchase.

In a yet another example, FIG. 500 shows primary display **102** displaying notification **5264** overlaid on window **5264** in response to reception of notification **5264** by the computing system. In FIG. 500, notification **5264** corresponds to an SMS, instant message, or the like sent by MAS to the user of the computing system, where the notification's content inquiries "Landed yet?" In yet another example, FIG. 5UU shows dynamic function row **104** displaying modal alert **5280** overlaid on the set of bookmark affordances **5278** in response to detecting selection of the exit affordance with cursor **504** in FIG. 5TT. In FIG. 5UU, modal alert **5280** prompts the user of the computing system to confirm that they intend to close all open tabs within the web browser application. In yet another example, FIG. 5AAA shows dynamic function row **104** displaying modal alert **5308** in response to detecting selection of power control **534** in FIG. 5CCC. In FIG. 5DDD, modal alert **5308** prompts the user of the computing system to select a logout/power-off option from one of logout affordance **5310**, restart affordance **5312**, power-off affordance **5314**, and cancel affordance **5316**.

In some embodiments, the portion of the notification displayed on the touch screen display prompts (**912**) a user of the computing system to select one of a plurality of options for responding to the detected notification. For example, modal alert **5156** displayed by primary display **102**

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and dynamic function row **104**, in FIG. **5V**, prompts the user to save the draft email prior to closing window **580** and includes a “Save” affordance **5158**, a “Don’t Save” affordance **5160**, and a “Cancel” affordance **5162**. In another example, modal alert **5280** displayed by dynamic function row **104**, in FIG. **5UU**, prompts the user of the computing system to confirm that they intend to close all open tabs within the web browser application. In yet another example, modal alert **5308** displayed by dynamic function row **104**, in FIG. **5DDD**, prompts the user of the computing system to select a logout/power-off option from one of logout affordance **5310**, restart affordance **5312**, power-off affordance **5314**, and cancel affordance **5316**.

In some embodiments, the portion of the notification displayed on the touch screen display includes (914) one or more suggested responses to the detected notification. FIG. **5QQ**, for example, shows dynamic function row **104** displaying response dialogue box **5268** in response to detecting contact **5266** at the location within notification **5264** in FIG. **5PP**. In FIG. **5QQ**, response dialogue box **5268** includes a plurality of predictive responses to the content of notification **5264** shown in FIGS. **500-5PP**. In FIG. **5QQ**, response dialogue box **5268** includes a first predictive response **5270** (“Yes.”), a second predictive response **5272** (“No.”), and a third predictive response **5274** (“On my way!”). FIG. **5QQ** also illustrates dynamic function row **104** receiving and detecting contact **5276** (e.g., a tap contact) at a location corresponding to the first predictive response **5270**. For example, in response to selection of the first predictive response **5270**, the computing system causes the first predictive response **5270** (“Yes.”) to be sent to MAS via a same communication mode (e.g., SMS, instant message, or the like) as the one by which notification **5264** was sent to the user of the computing system. In another example, in response to selection of the first predictive response **5270**, the computing system causes the first predictive response **5270** (“Yes.”) to be sent to MAS via a default communication mode (e.g., selected by the user or set in software).

In some embodiments, the notification corresponds (916) to an at least one of an incoming instant message, SMS, email, voice call, or video call. In FIG. **5FF**, for example, notification **5218** corresponds to an SMS, instant message, or the like sent by Suzie S. to the user of the computing system. In another example, in FIG. **5LL**, interface **5248** corresponds to an incoming voice call from C. Cheung, and, in FIG. **5MM**, interface **5256** correspond to an ongoing voice call between C. Cheung and the user of the computing system. In yet another example, notification **5264**, in FIGS. **500-5PP**, corresponds to an SMS, instant message, or the like sent by MAS to the user of the computing system.

In some embodiments, the notification corresponds (918) to a modal alert issued by an application being executed by the processor of the computing system in response to a user input closing the application or performing an action within the application. In a first example, FIG. **5V** shows modal alert **5156** prompting the user of the computing system to save the draft email prior to closing window **580** in response to detecting selection of the exit affordance with cursor **504** in FIG. **5U**. In a second example, FIG. **5KK** shows modal alert **5240** prompting the user of the computing system **100** to provide their fingerprint to validate the purchase in response to detecting selection of the purchase affordance with cursor **504** in FIG. **5JJ**. In a third example, FIG. **5UU** shows modal alert **5280** prompting the user of the computing system to confirm that they intend to close all open tabs within the web browser application in response to detecting selection of the exit affordance with cursor **504** in FIG. **5TT**.

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In some embodiments, the set of one or more affordances includes (920) at least one a system-level affordance corresponding to at least one system-level functionality, and the notification corresponds to a user input selecting one or more portions of the input mechanism (e.g., alt+tab or another keyboard combination) or the least one of a system-level affordance (e.g., a power control affordance). In one example, FIG. **5X** shows dynamic function row **104** displaying application selection window **5164** in response to receiving a signal corresponding to a specified physical key combination (e.g., alt+tab) from the set of physical keys **106** of portable computing system **100** (FIGS. **1A-1B**) or from the set of physical keys **106** of peripheral keyboard **206** of desktop computing system **200** (FIGS. **2A-2D**). In another example, FIG. **5DDD** shows dynamic function row **104** displaying modal alert **5308** in response to detecting selection of power control **534** in FIG. **5CCC**.

In some embodiments, the computing system detects (922) a user touch input on the touch screen display corresponding to the portion of the detected notification. In accordance with a determination that the user touch input corresponds to a first type (e.g., swipe to dismiss), the computing system ceases to display in the second user interface the portion of the detected notification on the touch screen display. In accordance with a determination that the user touch input corresponds to a second type (e.g., tap to perform an action) distinct from the first type, the computing system performs an action associated with the detected notification (e.g., open a dialogue for responding to the newly received email, SMS, or IM; save a document; exit a program; or the like). For example, FIG. **5FF** shows dynamic function row **104** detecting a left-to-right swipe gesture with contact **5220** from a first location **5222-A** within notification **5128** to a second location **5222-B**. Continuing with this example, FIG. **5GG** shows dynamic function row **104** ceasing to display notification **5218** in response to detecting the left-to-right swipe gesture in FIG. **5FF**. In another example, FIG. **5PP** shows dynamic function row **104** receiving and detecting contact **5266** (e.g., a tap contact) at a location within notification **5264**. Continuing with this example, FIG. **5QQ** shows dynamic function row **104** displaying response dialogue box **5268** in response to detecting contact **5266** at the location within notification **5264** in FIG. **5PP**.

It should be understood that the particular order in which the operations in FIGS. **9A-9B** have been described is merely an example and is not intended to indicate that the described order is the only order in which the operations could be performed. One of ordinary skill in the art would recognize various ways to reorder the operations described herein. Additionally, it should be noted that details of other processes described herein with respect to other methods described herein (e.g., methods **600**, **700**, **800**, and **1000**) are also applicable in an analogous manner to method **900** described above with respect to FIGS. **9A-9B**.

FIGS. **10A-10C** are a flowchart of a method of moving user interface portions, in accordance with some embodiments. The method **1000** is performed at a computing system including a processor, a primary display, memory, and a housing at least partially containing a physical input mechanism and a touch screen display adjacent to the physical input mechanism. Some operations in method **1000** are, optionally, combined and/or the order of some operations is, optionally, changed.

In some embodiments, the computing system is portable computing system **100** (FIG. **1A**) or desktop computing system **200** (FIGS. **2A-2D**). In some embodiments, the

primary display is primary display **102** (FIG. 1A) which is implemented in display portion **110** of portable computing system **100** (FIG. 1A). Alternatively, in some embodiments, the primary display is primary display **102** (FIGS. 2A-2D) which is implemented in peripheral display device **204** (FIGS. 2A-2D). In some embodiments, the housing is body portion **120** of portable computing system **100** (FIGS. 1A-1B), which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIGS. 1A-1B) and the input mechanism (e.g., the set of physical keys **106**, FIGS. 1A-1B). Alternatively, in some embodiments, the housing is peripheral keyboard **206** (FIGS. 2A-2B) of desktop computing system **200**, which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIGS. 2A-2B) and the input mechanism (e.g., the set of physical keys **106**, FIGS. 2A-2B). Alternatively, in some embodiments, the housing is first input mechanism **212** (FIG. 2C) of desktop computing system **200**, which at least partially contains the touch screen display (e.g., dynamic function row **104**, FIG. 2C) and the input mechanism (e.g., touch pad **108**, FIG. 2C).

The computing system displays (**1002**), on the primary display, a user interface, the user interface comprising one or more user interface elements. For example, FIG. **500** shows primary display **102** displaying an interface for tab B within window **5224** and notification **5264** overlaid on window **5224**. In FIG. **500**, the interface for tab B shows the home web page of website B (e.g., associated with the URL: www.website_B.com/home).

The computing system identifies (**1004**) an active user interface element of the one or more user interface elements that is in focus on the primary display, where the active user interface element is associated with an application executed by the computing system. In FIG. **500**, for example, the interface for tab B is in focus on primary display **102** as indicated by the thick lines surrounding tab B and the bold text for tab B.

In response to identifying the active user interface element that is in focus on the primary display, the computing system displays (**1006**), on the touch screen display, a set of one or more affordances corresponding to the application. In FIG. **500**, for example, dynamic function row **104** displays controls (i.e., the set of one or more affordances) for the web browser application, including: the home web page of website B in address bar **5234** (e.g., www.website_B.com/home), affordances **5230** and **5232** for navigating between recently viewed web pages, affordances **5238** for adding the current web page to a favorites or bookmarks list, and affordances **5262-A** and **5262-B** for accessing tabs A and C, respectively.

The computing system detects (**1008**) a user input to move a respective portion of the user interface. In some embodiments, the portion of the user interface is a menu, toolbar, tool set, notification, or the like. For example, the computing system detects a gesture that drags a menu to the bottom (or other user defined location) of primary display **102** or an instruction to move the menu to dynamic function row **104** via a right-click action or other corresponding affordance. In one example, FIG. **500** shows primary display **102** displaying the user of the computing system dragging notification **5264** with cursor **504** to a predefined location in the bottom right-hand corner of primary display **102**. In another example, FIG. **5SS** shows primary display **102** displaying the user of the computing system dragging the bookmarks sidebar with cursor **504** to the predefined location in the bottom right-hand corner of primary display **102**.

In some embodiments, the respective portion of the user interface is (**1010**) a menu corresponding to the application executed by the computing system. For example, the respective portion of the user interface is a menu or a toolbar for a word processing application. For example, the respective portion of the respective user interface being drug by cursor **504**, in FIG. **5SS**, is the bookmarks sidebar within window **5224**.

In some embodiments, the respective portion of the user interface is (**1012**) at least one of a notification or a modal alert. For example, the respective portion of the respective user interface being drug by cursor **504**, in FIG. **500**, is notification **5264**.

In response to detecting the user input, and in accordance with a determination that the user input satisfies predefined action criteria, the computing system (**1014**): ceases to display the respective portion of the user interface on the primary display; ceases to display at least a subset of the set of one or more affordances on the touch screen display; and displays, on the touch screen display, a representation of the respective portion of the user interface. In one example, FIG. **5PP** shows primary display ceasing to display notification **5264** and dynamic function row **104** displaying notification **5264** overlaid on affordances **5262-A** and **5262-B** in response to the user of the computing system dragging notification **5264** with cursor **504** to the predefined location in the bottom right-hand corner of primary display **102** in FIG. **500**. In another example, FIG. **5TT** shows dynamic function row **104** ceasing to display the controls associated with the web browser application as shown in FIG. **5SS** and displaying a set of bookmark affordances **5278** corresponding to all pre-existing bookmarks in response to the user of the computing system dragging the bookmarks sidebar with cursor **504** to the predefined location in the bottom right-hand corner of primary display **102** in FIG. **5SS**.

In some embodiments, the predefined action criteria are satisfied (**1016**) when the user input is a dragging gesture that drags the respective portion of the user interface to a predefined location of the primary display. In some embodiments, the predefined location is one of a plurality of predefined location set by the user of the computing system or set by default in software. In one example, in FIG. **5PP**, the user of the computing system drags notification **5264** to a predefined location (e.g., the bottom right-hand corner of primary display **102**) with cursor **504**. In another example, in FIG. **5SS**, the user of the computing system drags the bookmarks sidebar to a predefined location (e.g., the bottom right-hand corner of primary display **102**) with cursor **504**.

In some embodiments, the predefined action criteria are satisfied when the user input is (**1018**) a predetermined input corresponding to moving the respective portion of the user interface to the touch screen display. For example, primary display **102** displays a window for a word processing application along with a formatting toolbar overlaid on the window for the word processing application. Continuing with this example, in response to selecting a specific display option after right-clicking on the formatting toolbar or selecting the specific display option while the formatting toolbar is in focus on primary display **102**, primary display **102** ceases to display the formatting toolbar and dynamic function row displays the formatting toolbar.

In some embodiments, the representation of the respective portion of the user interface is overlaid (**1020**) on the set of one or more affordances on the touch screen display. For example, FIG. **5PP** shows dynamic function row **104** displaying notification **5264** overlaid on affordances **5262-A** and **5262-B** in response to the user of the computing system

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dragging notification **5264** with cursor **504** to the predefined location in the bottom right-hand corner of primary display **102** in FIG. **500**.

In some embodiments, the set of one or more affordances includes (**1022**) at least one system-level affordance corresponding to at least one system-level functionality, and, after displaying the representation of the respective portion of the user interface on the touch screen display, the computing system maintains display of the at least one system-level affordance on the touch screen display. In FIG. **5PP**, for example, dynamic function row **104** displays notification **5264** overlaid on affordances **5262-A** and **5262-B** along with the at least one system-level affordance (e.g., affordance **542**) and the persistent controls (e.g., escape affordance **516** and power control **534**). In FIG. **5TT**, for example, dynamic function row **104** displays the set of bookmark affordances **5278** along with the at least one system-level affordance (e.g., affordance **542**) and the persistent controls (e.g., escape affordance **516** and power control **534**).

In some embodiments, in response to detecting the user input, and in accordance with a determination that the user input does not satisfy the predefined action criteria, the computing system (**1024**): maintains display of the respective portion of the user interface on the primary display; and maintains display of the set of one or more affordances on the touch screen display. For example, with respect to FIG. **500**, if the user of the computing system drags notification **5264** to the bottom left-hand corner of primary display **102** with cursor **504**, notification **5264** will be displayed in the bottom left-hand corner of primary display **102** and dynamic function row **104** will do display notification **5264** as the bottom left-hand corner is not the predefined location (e.g., the bottom right-hand corner is the predefined location).

It should be understood that the particular order in which the operations in FIGS. **10A-10C** have been described is merely an example and is not intended to indicate that the described order is the only order in which the operations could be performed. One of ordinary skill in the art would recognize various ways to reorder the operations described herein. Additionally, it should be noted that details of other processes described herein with respect to other methods described herein (e.g., methods **600**, **700**, **800**, and **900**) are also applicable in an analogous manner to method **1000** described above with respect to FIGS. **10A-10C**.

In accordance with some embodiments, FIG. **11** shows a functional block diagram of an electronic device **1100** configured in accordance with the principles of the various described embodiments. The functional blocks of the device are, optionally, implemented by hardware, software, firmware, or a combination thereof to carry out the principles of the various described embodiments. It is understood by persons of skill in the art that the functional blocks described in FIG. **11** are, optionally, combined or separated into sub-blocks to implement the principles of the various described embodiments. Therefore, the description herein optionally supports any possible combination or separation or further definition of the functional blocks described herein. For case of discussion, the electronic device **1100** is implemented as portable computing system **100** (FIGS. **1A-1B**) whose components correspond to electronic device **300** (FIG. **3**). One of skill in the art will appreciate how the electronic device **1100** may also be implemented within desktop computing system **200** (FIGS. **2A-2D**).

As shown in FIG. **11**, the electronic device **1100** includes a primary display unit **1102** configured to display information (e.g., primary display **102**, FIGS. **1A** and **2A-2D**), a physical input unit **1104** configured to receive user inputs, a

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touch screen display (TSD) unit **1106** configured to display information (sometimes also herein called “a touch screen display” or a “touch screen”) (e.g., dynamic function row **104**, FIGS. **1A-1B** and **2A-2D**), a touch-sensitive surface unit **1108** configured to receive contacts on the touch screen display unit **1106** with one or more sensors, and a processing unit **1110** coupled to the primary display unit **1102**, the physical input unit **1104**, the touch screen display unit **1106**, and the touch-sensitive surface unit **1108**. In some embodiments, the processing unit **1110** includes: a primary display control unit **1112**, a touch screen display (TSD) control unit **1114**, a focus identifying unit **1116**, a determining unit **1118**, an input detecting unit **1120**, an input type determining unit **1122**, a performing unit **1124**, and a media playback determining unit **1126**.

The processing unit **1110** is configured to: cause display of (e.g., with the primary display control unit **1112**) a first user interface on the primary display unit **1102**, the first user interface comprising one or more user interface elements; identify (e.g., with the focus identifying unit **1116**) an active user interface element among the one or more user interface elements that is in focus on the primary display unit **1102**; and determine (e.g., with the determining unit **1118**) whether the active user interface element that is in focus on the primary display unit **1102** is associated with an application executed by the processing unit **1110**. In accordance with a determination that the active user interface element that is in focus on the primary display unit **1102** is associated with the application, processing unit **1110** is configured to cause display of (e.g., with the touch screen display control unit **1114**) a second user interface on the touch screen display unit **1106**, including: (A) a first set of one or more affordances corresponding to the application; and (B) at least one system-level affordance corresponding to at least one system-level functionality.

In some embodiments, the electronic device further includes: (i) a primary computing unit comprising the primary display unit **1102**, the processing unit **1110**, and a first communication circuitry unit; and (ii) an integrated input unit comprising the touch screen display unit **1106**, the touch-sensitive surface unit **1108**, the physical input unit **1104**, and a second communication circuitry unit for communicating with the first communication circuitry unit, where the integrated input device is distinct and separate from the primary computing unit.

In some embodiments, the physical input unit **1104** comprises a plurality of physical keys.

In some embodiments, the physical input unit **1104** comprises a touch pad.

In some embodiments, the processing unit **1110** is configured to execute the application in the foreground of the first user interface.

In some embodiments, the least one system-level affordance is configured upon selection to cause display of a plurality of system-level affordances corresponding to system-level functionalities on the touch screen display unit **1106**.

In some embodiments, the least one system-level affordance corresponds to one of a power control or escape control.

In some embodiments, at least one of the affordances displayed on the touch screen display unit **1106** within the second user interface is a multi-function affordance.

In some embodiments, the processing unit **1110** is configured to: detect (e.g., with the input detecting unit **1120**) a user touch input selecting the multi-function affordance on the touch-sensitive surface unit **1108**; in accordance with a

determination (e.g., with the input type determining unit **1122**) that the user touch input corresponds to a first type, perform (e.g., with the performing unit **1124**) a first function associated with the multi-function affordance; and, in accordance with a determination (e.g., with the input type determining unit **1122**) that the user touch input corresponds to a second type distinct from the first type, perform (e.g., with the performing unit **1124**) a second function associated with the multi-function affordance.

In some embodiments, in accordance with a determination that the active user interface element is not associated with the application, the processing unit **1110** is configured to cause display of (e.g., with the touch screen display control unit **1114**) a third user interface on the touch screen display unit **1106**, including: (C) a second set of one or more affordances corresponding to operating system controls of the electronic device **1100**, where the second set of one or more affordances are distinct from the first set of one or more affordances.

In some embodiments, the second set of one or more affordances is an expanded set of operating system controls that includes (B) the at least one system-level affordance corresponding to the at least one system-level functionality.

In some embodiments, the processing unit **1110** is configured to: detect (e.g., with the input detecting unit **1120**) a user touch input selecting one of the first set of affordances on the touch-sensitive surface unit **1108**; and, in response to detecting the user touch input: cause display of (e.g., with the touch screen display control unit **1114**) a different set of affordances corresponding to functionalities of the application on the touch screen display unit **1106**; and maintain display of (e.g., with the touch screen display control unit **1114**) the at least one system-level affordance on the touch screen display unit **1106**.

In some embodiments, the processing unit **1110** is configured to: detect (e.g., with the input detecting unit **1120**) a subsequent user touch input selecting the at least one system-level affordance on the touch-sensitive surface unit **1108**; and, in response to detecting the subsequent user touch input, cause display of (e.g., with the touch screen display control unit **1114**) a plurality of system-level affordances corresponding to system-level functionalities and at least one application-level affordance corresponding to the application on the touch screen display unit **1106**.

In some embodiments, the processing unit **1110** is configured to: identify (e.g., with the focus identifying unit **1116**) a second active user interface element that is in focus on the primary display unit **1102** after displaying the second user interface on the touch screen display unit **1106**; and determine (e.g., with the determining unit **1118**) whether the second active user interface element corresponds to a different application executed by the processing unit **1110**. In accordance with a determination that the second active user interface element corresponds to the different application, the processing unit **1110** is configured to cause display of (e.g., with the touch screen display control unit **1114**) a fourth user interface on the touch screen display unit **1106**, including: (D) a third set of one or more affordances corresponding to the different application; and (E) the at least one system-level affordance corresponding to the at least one system-level functionality.

In some embodiments, the processing unit **1110** is configured to: determine (e.g., with the media playback determining unit **1126**) whether a media item is being played by the electronic device **1100** after identifying (e.g., with the focus identifying unit **1116**) that the second active user interface element, where the media item is not associated

with the different application; and, in accordance with a determination (e.g., with the media playback determining unit **1126**) that media item is being played by the electronic device **1100**, cause display of (e.g., with the touch screen display control unit **1114**) at least one persistent affordance on the fourth user interface for controlling the media item on the touch screen display unit **1106**.

In some embodiments, the at least one persistent affordance displays feedback that corresponds to the media item.

In some embodiments, the processing unit **1110** is configured to: detect (e.g., with the input detecting unit **1120**) a user input corresponding to an override key; and, in response to detecting the user input: cease to display (e.g., with the touch screen display control unit **1114**) at least the first set of one or more affordances of the second user interface on the touch screen display unit **1106**; and cause display of (e.g., with the touch screen display control unit **1114**) a first set of default function keys on the touch screen display unit **1106**.

In some embodiments, the processing unit **1110** is configured to: detect (e.g., with the input detecting unit **1120**) a gesture on the touch-sensitive surface unit **1108** in a direction that is substantially parallel to a major axis of the touch screen display unit **1106** after displaying the first set of default function keys on the touch screen display unit **1106**; and, in response to detecting the substantially horizontal swipe gesture, cause display of (e.g., with the touch screen display control unit **1114**) a second set of default function keys with at least one distinct function key on the touch screen display unit **1106**.

The operations in the information processing methods described above are, optionally implemented by running one or more functional modules in information processing apparatus such as general purpose processors (e.g., as described above with respect to FIGS. 3A and 4) or application specific chips.

The operations described above with reference to FIGS. 6A-6D are, optionally, implemented by components depicted in FIGS. 3A-3B or FIG. 11. For example, detection operations **626** and **628** are, optionally, implemented by event sorter **370**, event recognizer **380**, and event handler **390**. Event monitor **371** in event sorter **370** detects a contact on display system **312** when implemented as a touch-sensitive display, and event dispatcher module **374** delivers the event information to application **340-1**. A respective event recognizer **380** of application **340-1** compares the event information to respective event definitions **386**, and determines whether a first contact at a first location on the touch-sensitive display corresponds to a predefined event or sub-event. When a respective predefined event or sub-event is detected, event recognizer **380** activates an event handler **390** associated with the detection of the event or sub-event. Event handler **390** optionally uses or calls data updater **376** or object updater **377** to update the application internal state **392**. In some embodiments, event handler **390** accesses a respective GUI updater **378** to update what is displayed by the application. Similarly, it would be clear to a person having ordinary skill in the art how other processes can be implemented based on the components depicted in FIGS. 3A-3B.

In accordance with some embodiments, FIG. 12 shows a functional block diagram of an electronic device **1200** configured in accordance with the principles of the various described embodiments. The functional blocks of the device are, optionally, implemented by hardware, software, firmware, or a combination thereof to carry out the principles of the various described embodiments. It is understood by

persons of skill in the art that the functional blocks described in FIG. 12 are, optionally, combined or separated into sub-blocks to implement the principles of the various described embodiments. Therefore, the description herein optionally supports any possible combination or separation or further definition of the functional blocks described herein. For case of discussion, the electronic device 1200 is implemented as portable computing system 100 (FIGS. 1A-1B) whose components correspond to electronic device 300 (FIG. 3). One of skill in the art will appreciate how the electronic device 1200 may also be implemented within desktop computing system 200 (FIGS. 2A-2D).

As shown in FIG. 12, the electronic device 1200 includes a primary display unit 1202 configured to display information (e.g., primary display 102, FIGS. 1A and 2A-2D), a physical input unit 1204 configured to receive user inputs, a touch screen display (TSD) unit 1206 configured to display information (sometimes also herein called “a touch screen display” or a “touch screen”) (e.g., dynamic function row 104, FIGS. 1A-1B and 2A-2D), a touch-sensitive surface unit 1208 configured to receive contacts on the touch screen display unit 1206 with one or more sensors, and a processing unit 1210 coupled to the primary display unit 1202, the physical input unit 1204, the touch screen display unit 1206, and the touch-sensitive surface unit 1208. In some embodiments, the processing unit 1210 includes: a primary display control unit 1212, a touch screen display (TSD) control unit 1214, an input detecting unit 1216, and an input type determining unit 1218.

The processing unit 1210 is configured to: cause display of (e.g., with the primary display control unit 1212) a first user interface for an application executed by the processing unit 1210 on the primary display unit 1202; cause display of (e.g., with the touch screen display control unit 1214) a second user interface on the touch screen display unit 1206, the second user interface comprising a first set of one or more affordances corresponding to the application, where the first set of one or more affordances corresponds to a first portion of the application; and detect (e.g., with the input detecting unit 1216) a swipe gesture on the touch-sensitive surface unit 1208. In accordance with a determination (e.g., with the input type determining unit 1218) that the swipe gesture was performed in a first direction (e.g., horizontal), the processing unit 1210 is configured to cause display of (e.g., with the touch screen display control unit 1214) a second set of one or more affordances corresponding to the application on the touch screen display unit 1206, where at least one affordance in the second set of one or more affordances is distinct from the first set of one or more affordances, and where the second set of one or more affordances also corresponds to the first portion of the application. In accordance with a determination (e.g., with the input type determining unit 1218) that the swipe gesture was performed in a second direction substantially perpendicular to the first direction (e.g., vertical), the processing unit 1210 is configured to cause display of (e.g., with the touch screen display control unit 1214) a third set of one or more affordances corresponding to the application on the touch screen display unit 1206, where the third set of one or more affordances is distinct from the second set of one or more affordances, and where the third set of one or more affordances corresponds to a second portion of the application that is distinct from the first portion of the application.

In some embodiments, the second portion is displayed on the primary display unit 1202 in a compact view within the first user interface prior to detecting the swipe gesture, and the processing unit 1210 is configured to cause display of

(e.g., with the primary display control unit 1212) the second portion on the primary display unit 1202 in an expanded view within the first user interface in accordance with the determination that the swipe gesture was performed in the second direction substantially perpendicular to the first direction.

In some embodiments, the first user interface for the application is displayed on the primary display unit 1202 in a full-screen mode, and the first set of one or more affordances displayed on the touch screen display unit 1206 includes controls corresponding to the full-screen mode.

In some embodiments, the second set of one or more affordances and the third set of one or more affordances includes at least one system-level affordance corresponding to at least one system-level functionality.

In some embodiments, after displaying the third set of one or more affordances on the touch screen display unit 1206, the processing unit 1210 is configured to: detect (e.g., with the input detecting unit 1216) a user input selecting the first portion on the first user interface; and, in response to detecting the user input: cease to display (e.g., with the touch screen display control unit 1214) the third set of one or more affordances on the touch screen display unit 1206, where the third set of one or more affordances corresponds to the second portion of the application; and cause display of (e.g., with the touch screen display control unit 1214) the second set of one or more affordances on the touch screen display unit 1206, where the second set of one or more affordances corresponds to the first portion of the application.

In some embodiments, the first direction is substantially parallel to a major dimension of the touch screen display unit 1206.

In some embodiments, the first direction is substantially perpendicular to a major dimension of the touch screen display unit 1206.

In some embodiments, the first portion is one of a menu, tab, folder, tool set, or toolbar of the application, and the second portion is one of a menu, tab, folder, tool set, or toolbar of the application.

The operations described above with reference to FIGS. 7A-7B are, optionally, implemented by components depicted in FIGS. 3A-3B or FIG. 12. For example, detection operation 710 is, optionally, implemented by event sorter 370, event recognizer 380, and event handler 190. Event monitor 371 in event sorter 370 detects a contact on display system 312 when implemented as a touch-sensitive display, and event dispatcher module 374 delivers the event information to application 340-1. A respective event recognizer 380 of application 340-1 compares the event information to respective event definitions 386, and determines whether a first contact at a first location on the touch-sensitive display corresponds to a predefined event or sub-event. When a respective predefined event or sub-event is detected, event recognizer 380 activates an event handler 390 associated with the detection of the event or sub-event. Event handler 390 optionally uses or calls data updater 376 or object updater 377 to update the application internal state 392. In some embodiments, event handler 390 accesses a respective GUI updater 378 to update what is displayed by the application. Similarly, it would be clear to a person having ordinary skill in the art how other processes can be implemented based on the components depicted in FIGS. 3A-3B.

In accordance with some embodiments, FIG. 13 shows a functional block diagram of an electronic device 1300 configured in accordance with the principles of the various described embodiments. The functional blocks of the device are, optionally, implemented by hardware, software, firm-

ware, or a combination thereof to carry out the principles of the various described embodiments. It is understood by persons of skill in the art that the functional blocks described in FIG. 13 are, optionally, combined or separated into sub-blocks to implement the principles of the various described embodiments. Therefore, the description herein optionally supports any possible combination or separation or further definition of the functional blocks described herein. For case of discussion, the electronic device 1300 is implemented as portable computing system 100 (FIGS. 1A-1B) whose components correspond to electronic device 300 (FIG. 3). One of skill in the art will appreciate how the electronic device 1300 may also be implemented within desktop computing system 200 (FIGS. 2A-2D).

As shown in FIG. 13, the electronic device 1300 includes a primary display unit 1302 configured to display information (e.g., primary display 102, FIGS. 1A and 2A-2D), a physical input unit 1304 configured to receive user inputs, a touch screen display (TSD) unit 1306 configured to display information (sometimes also herein called “a touch screen display” or a “touch screen”) (e.g., dynamic function row 104, FIGS. 1A-1B and 2A-2D), a touch-sensitive surface unit 1308 configured to receive contacts on the touch screen display unit 1306 with one or more sensors, and a processing unit 1310 coupled to the primary display unit 1302, the physical input unit 1304, the touch screen display unit 1306, and the touch-sensitive surface unit 1308. In some embodiments, the processing unit 1310 includes: a primary display control unit 1312, a touch screen display (TSD) control unit 1314, an input detecting unit 1316, and a changing unit 1318.

The processing unit 1310 is configured to: cause display of (e.g., with the primary display control unit 1312) a first user interface for the application executed by the processing unit 1310 on the primary display unit 1302 in a normal mode, the first user interface comprising a first set of one or more affordances associated with the application; and detect (e.g., with the input detecting unit 1316) a user input for displaying at least a portion of the first user interface for the application in a full-screen mode on the primary display unit 1302. In response to detecting the user input, the processing unit 1310 is configured to: cease to display (e.g., with the primary display control unit 1312) the first set of one or more affordances associated with the application in the first user interface on the primary display unit 1302; cause display of (e.g., with the primary display control unit 1312) the portion of the first user interface for the application on the primary display unit 1302 in the full-screen mode; and automatically, without human intervention, cause display of (e.g., with the touch screen display control unit 1314) a second set of one or more affordances for controlling the application on the touch screen display unit 1306, where the second set of one or more affordances correspond to the first set of one or more affordances.

In some embodiments, the second set of one or more affordances is the first set of one or more affordances.

In some embodiments, the second set of one or more affordances include controls corresponding to the full-screen mode.

In some embodiments, the processing unit 1310 is configured to detect (e.g., with the input detecting unit 1316) a user touch input selecting one of the second set of affordances on the touch-sensitive surface unit 1308, and, in response to detecting the user touch input, the processing unit 1310 is configured to change (e.g., with the changing unit 1318) the portion of the first user interface for the application being displayed in the full-screen mode on the

primary display unit 1302 according to the selected one of the second set of affordances.

In some embodiments, after displaying the portion of the first user interface for the application in the full-screen mode on the primary display unit 1302, the processing unit 1310 is configured to: detect (e.g., with the input detecting unit 1316) a subsequent user input for exiting the full-screen mode; and, in response to detecting the subsequent user input: cause display of (e.g., with the primary display control unit 1312) the first user interface for the application executed by the processing unit 1310 on the primary display unit 1302 in the normal mode, the first user interface comprising the first set of one or more affordances associated with the application; and maintain display of (e.g., with the touch screen display control unit 1314) at least a subset of the second set of one or more affordances for controlling the application on the touch screen display unit 1306, where the second set of one or more affordances correspond to the first set of one or more affordances.

In some embodiments, the user input for displaying at least the portion of the first user interface for the application in full-screen mode on the primary display unit 1302 is at least one of a touch input detected on the touch-sensitive surface unit 1308 and a control selected within the first user interface on the primary display unit 1302.

In some embodiments, the second set of one or more affordances includes at least one system-level affordance corresponding to at least one system-level functionality.

The operations described above with reference to FIGS. 8A-8B are, optionally, implemented by components depicted in FIGS. 3A-3B or FIG. 13. For example, detection operations 804 and 816 are, optionally, implemented by event sorter 370, event recognizer 380, and event handler 190. Event monitor 371 in event sorter 370 detects a contact on display system 312 when implemented as a touch-sensitive display, and event dispatcher module 374 delivers the event information to application 340-1. A respective event recognizer 380 of application 340-1 compares the event information to respective event definitions 386, and determines whether a first contact at a first location on the touch-sensitive display corresponds to a predefined event or sub-event. When a respective predefined event or sub-event is detected, event recognizer 380 activates an event handler 390 associated with the detection of the event or sub-event. Event handler 390 optionally uses or calls data updater 376 or object updater 377 to update the application internal state 392. In some embodiments, event handler 390 accesses a respective GUI updater 378 to update what is displayed by the application. Similarly, it would be clear to a person having ordinary skill in the art how other processes can be implemented based on the components depicted in FIGS. 3A-3B.

In accordance with some embodiments, FIG. 14 shows a functional block diagram of an electronic device 1400 configured in accordance with the principles of the various described embodiments. The functional blocks of the device are, optionally, implemented by hardware, software, firmware, or a combination thereof to carry out the principles of the various described embodiments. It is understood by persons of skill in the art that the functional blocks described in FIG. 14 are, optionally, combined or separated into sub-blocks to implement the principles of the various described embodiments. Therefore, the description herein optionally supports any possible combination or separation or further definition of the functional blocks described herein. For case of discussion, the electronic device 1400 is implemented as portable computing system 100 (FIGS.

1A-1B) whose components correspond to electronic device 300 (FIG. 3). One of skill in the art will appreciate how the electronic device 1400 may also be implemented within desktop computing system 200 (FIGS. 2A-2D).

As shown in FIG. 14, the electronic device 1400 includes a primary display unit 1402 configured to display information (e.g., primary display 102, FIGS. 1A and 2A-2D), a physical input unit 1404 configured to receive user inputs, a touch screen display (TSD) unit 1406 configured to display information (sometimes also herein called “a touch screen display” or a “touch screen”) (e.g., dynamic function row 104, FIGS. 1A-1B and 2A-2D), a touch-sensitive surface unit 1408 configured to receive contacts on the touch screen display unit 1406 with one or more sensors, and a processing unit 1410 coupled to the primary display unit 1402, the physical input unit 1404, the touch screen display unit 1406, and the touch-sensitive surface unit 1408. In some embodiments, the processing unit 1410 includes: a primary display control unit 1412, a touch screen display (TSD) control unit 1414, a notification unit 1416, an input detecting unit 1418, an input type determining unit 1420, and a performing unit 1422.

The processing unit 1410 is configured to: cause display of (e.g., with the primary display control unit 1412) a first user interface, on the primary display unit 1402, for an application executed by the processing unit 1410; cause display of (e.g., with the touch screen display control unit 1414) a second user interface, on the touch screen display unit 1406, the second user interface comprising a set of one or more affordances corresponding to the application; detect a notification (e.g., with the notification unit 1416); and, in response to detecting the notification, cause concurrent display of (e.g., with the touch screen display control unit 1414), in the second user interface on the touch screen display unit 1406, the set of one or more affordances corresponding to the application and at least a portion of the detected notification, where the detected notification is not displayed on the primary display unit 1402.

In some embodiments, prior to detecting the notification, the processing unit 1410 is configured to detect (e.g., with the input detecting unit 1418) a user input selecting a notification setting so as to display notifications on the touch screen display unit 1406 and to not display notifications on the primary display unit 1402.

In some embodiments, the processing unit 1410 is configured to detect (e.g., with the input detecting unit 1418) a user touch input on the touch-sensitive surface unit 1408 corresponding to the portion of the detected notification. In accordance with a determination (e.g., with the input type determining unit 1420) that the user touch input corresponds to a first type (e.g., a swipe gesture), the processing unit 1410 is configured to cease to display (e.g., with the touch screen display control unit 1414) in the second user interface the portion of the detected notification on the touch screen display unit 1406. In accordance with a determination (e.g., with the input type determining unit 1420) that the user touch input corresponds to a second type (e.g., a tap contact) distinct from the first type, the processing unit 1410 is configured to perform (e.g., with the performing unit 1422) an action associated with the detected notification.

In some embodiments, the portion of the notification displayed on the touch screen display unit 1406 prompts a user of the electronic device 1400 to select one of a plurality of options for responding to the detected notification.

In some embodiments, the portion of the notification displayed on the touch screen display unit 1406 includes one or more suggested responses to the detected notification.

In some embodiments, the notification corresponds to at least one of an incoming instant message, SMS, email, voice call, or video call.

In some embodiments, the notification corresponds to a modal alert issued by an application being executed by the processing unit 1410 in response to a user input closing the application or performing an action within the application.

In some embodiments, the set of one or more affordances includes at least one a system-level affordance corresponding to at least one system-level functionality, and the notification corresponds to a user input selecting one or more portions of the input mechanism or the least one of a system-level affordance.

The operations described above with reference to FIGS. 9A-9B are, optionally, implemented by components depicted in FIGS. 3A-3B or FIG. 14. For example, detection operation 922 is, optionally, implemented by event sorter 370, event recognizer 380, and event handler 190. Event monitor 371 in event sorter 370 detects a contact on display system 312 when implemented as a touch-sensitive display, and event dispatcher module 374 delivers the event information to application 340-1. A respective event recognizer 380 of application 340-1 compares the event information to respective event definitions 386, and determines whether a first contact at a first location on the touch-sensitive display corresponds to a predefined event or sub-event. When a respective predefined event or sub-event is detected, event recognizer 380 activates an event handler 390 associated with the detection of the event or sub-event. Event handler 390 optionally uses or calls data updater 376 or object updater 377 to update the application internal state 392. In some embodiments, event handler 390 accesses a respective GUI updater 378 to update what is displayed by the application. Similarly, it would be clear to a person having ordinary skill in the art how other processes can be implemented based on the components depicted in FIGS. 3A-3B.

In accordance with some embodiments, FIG. 15 shows a functional block diagram of an electronic device 1500 configured in accordance with the principles of the various described embodiments. The functional blocks of the device are, optionally, implemented by hardware, software, firmware, or a combination thereof to carry out the principles of the various described embodiments. It is understood by persons of skill in the art that the functional blocks described in FIG. 15 are, optionally, combined or separated into sub-blocks to implement the principles of the various described embodiments. Therefore, the description herein optionally supports any possible combination or separation or further definition of the functional blocks described herein. For ease of discussion, the electronic device 1500 is implemented as portable computing system 100 (FIGS. 1A-1B) whose components correspond to electronic device 300 (FIG. 3). One of skill in the art will appreciate how the electronic device 1500 may also be implemented within desktop computing system 200 (FIGS. 2A-2D).

As shown in FIG. 15, the electronic device 1500 includes a primary display unit 1502 configured to display information (e.g., primary display 102, FIGS. 1A and 2A-2D), a physical input unit 1504 configured to receive user inputs, a touch screen display (TSD) unit 1506 configured to display information (sometimes also herein called “a touch screen display” or a “touch screen”) (e.g., dynamic function row 104, FIGS. 1A-1B and 2A-2D), a touch-sensitive surface unit 1508 configured to receive contacts on the touch screen display unit 1506 with one or more sensors, and a processing unit 1510 coupled to the primary display unit 1502, the physical input unit 1504, the touch screen display unit 1506,

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and the touch-sensitive surface unit **1508**. In some embodiments, the processing unit **1510** includes: a primary display control unit **1512**, a touch screen display (TSD) control unit **1514**, an identifying unit **1516**, an input detecting unit **1518**, and a determining unit **1520**.

The processing unit **1510** is configured to: cause display of (e.g., with the primary display control unit **1512**) a user interface, on the primary display unit **1502**, the user interface comprising one or more user interface elements; identify (e.g., with the identifying unit **1516**) an active user interface element of the one or more user interface elements that is in focus on the primary display unit **1502**, where the active user interface element is associated with an application executed by the processing unit **1510**; and, in response to identifying the active user interface element that is in focus on the primary display unit **1502**, cause display of (e.g., with the touch screen display control unit **1514**) a set of one or more affordances corresponding to the application on the touch screen display unit **1506**. The processing unit **1510** is configured to: detect (e.g., with the input detecting unit **1518**) a user input to move a respective portion of the user interface; and, in response to detecting the user input, and in accordance with a determination (e.g., with the determining unit **1520**) that the user input satisfies predefined action criteria: cease to display (e.g., with the primary display control unit **1512**) the respective portion of the user interface on the primary display unit **1502**; cease to display (e.g., with the touch screen display control unit **1514**) at least a subset of the set of one or more affordances on the touch screen display unit **1506**; and cause display of (e.g., with the touch screen display control unit **1514**) a representation of the respective portion of the user interface on the touch screen display unit **1506**.

In some embodiments, the respective portion of the user interface is a menu corresponding to the application executed by the processing unit **1510**.

In some embodiments, the respective portion of the user interface is at least one of a notification or a modal alert.

In some embodiments, the predefined action criteria are satisfied when the user input is a dragging gesture that drags the respective portion of the user interface to a predefined location of the primary display unit **1502**.

In some embodiments, the predefined action criteria are satisfied when the user input is predetermined input corresponding to moving the respective portion of the user interface to the touch screen display unit **1506**.

In some embodiments, in response to detecting the user input, and in accordance with a determination (e.g., with the determining unit **1520**) that the user input does not satisfy the predefined action criteria, the processing unit **1510** is configured to: maintain display of (e.g., with the primary display control unit **1512**) the respective portion of the user interface on the primary display unit **1502**; and maintain display of (e.g., with the touch screen display control unit **1514**) the set of one or more affordances on the touch screen display unit **1506**.

In some embodiments, the set of one or more affordances includes at least one system-level affordance corresponding to at least one system-level functionality, and the processing unit **1510** is configured to maintain display of (e.g., with the touch screen display control unit **1514**) the at least one system-level affordance on the touch screen display unit **1506** after displaying the representation of the respective portion of the user interface on the touch screen display unit **1506**.

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In some embodiments, the representation of the respective portion of the user interface is overlaid on the set of one or more affordances on the touch screen display unit **1506**.

The operations described above with reference to FIGS. **10A-10C** are, optionally, implemented by components depicted in FIGS. **3A-3B** or FIG. **15**. For example, detection operation **1008** is, optionally, implemented by event sorter **370**, event recognizer **380**, and event handler **190**. Event monitor **371** in event sorter **370** detects a contact on display system **312** when implemented as a touch-sensitive display, and event dispatcher module **374** delivers the event information to application **340-1**. A respective event recognizer **380** of application **340-1** compares the event information to respective event definitions **386**, and determines whether a first contact at a first location on the touch-sensitive display corresponds to a predefined event or sub-event. When a respective predefined event or sub-event is detected, event recognizer **380** activates an event handler **390** associated with the detection of the event or sub-event. Event handler **390** optionally uses or calls data updater **376** or object updater **377** to update the application internal state **392**. In some embodiments, event handler **390** accesses a respective GUI updater **378** to update what is displayed by the application. Similarly, it would be clear to a person having ordinary skill in the art how other processes can be implemented based on the components depicted in FIGS. **3A-3B**.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications, to thereby enable others skilled in the art to best use the invention and various described embodiments with various modifications as are suited to the particular use contemplated.

What is claimed is:

1. A method, comprising:

at an electronic device with a first display portion, a second display portion and a hinge connecting the first display portion to the second display portion:
displaying, on the first display portion, a first application icon that corresponds to a first application;
while displaying the first application icon on the first display portion, detecting a first input corresponding to selection of the first application icon;
in response to detecting the first input corresponding to selection of the first application icon on the first display portion, displaying an application user interface for the first application on the second display portion;
while displaying the application user interface for the first application on the second display portion, detecting a second input directed to the application user interface for the first application on the second display portion; and
in response to detecting the second input directed to the application user interface for the first application on the second display portion, displaying content corresponding to the first application on the first display portion that was not displayed prior to detecting the second input directed to the application user interface for the first application on the second display portion.

2. The method of claim 1, further comprising:
in response to detecting the first input:

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concurrently displaying with the application user interface for the first application on the second display portion, a user interface for the first application on the first display portion.

3. The method of claim 2, wherein the user interface for the first application on the first display portion is different from the application user interface for the first application on the second display portion.

4. The method of claim 1, wherein the first application is a music application.

5. The method of claim 1, wherein the first application is a media viewing application.

6. The method of claim 1, wherein the first input and the second input are touch gestures.

7. The method of claim 1, wherein the second display portion includes a power button.

8. The method of claim 1, wherein the second input includes selecting content.

9. The method of claim 1, wherein the second input includes navigating through content.

10. The method of claim 1, wherein the first application icon is displayed in an arrangement of multiple application icons.

11. The method of claim 1, further comprising:
in response to detecting an event, displaying a notification on the second display portion without displaying the notification on the first display portion.

12. An electronic device, comprising:
a first display portion;
a second display portion;
a hinge connecting the first display portion to the second display portion;
one or more processors;
memory; and

one or more programs, wherein the one or more programs are stored in the memory and configured to be executed by the one or more processors, the one or more programs including instructions for:

displaying, on the first display portion, a first application icon that corresponds to a first application;

while displaying the first application icon on the first display portion, detecting a first input corresponding to selection of the first application icon;

in response to detecting the first input corresponding to selection of the first application icon on the first display portion, displaying an application user interface for the first application on the second display portion;

while displaying the application user interface for the first application on the second display portion, detecting a second input directed to the application user interface for the first application on the second display portion; and

in response to detecting the second input directed to the application user interface for the first application on the second display portion, displaying content corresponding to the first application on the first display portion that was not displayed prior to detecting the second input directed to the application user interface for the first application on the second display portion.

13. The electronic device of claim 12, the one or more programs comprising instructions that, when executed by the electronic device, cause the electronic device to:
in response to display the first input:

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concurrently displaying with the application user interface for the first application on the second display portion, a user interface for the first application on the first display portion.

14. The electronic device of claim 13, wherein the user interface for the first application on the first display portion is different from the application user interface for the first application on the second display portion.

15. The electronic device of claim 12, wherein the first application is a music application.

16. The electronic device of claim 12, wherein the first application is a media viewing application.

17. The electronic device of claim 12, wherein the first input and the second input are touch gestures.

18. The electronic device of claim 12, wherein the second display portion includes a power button.

19. The electronic device of claim 12, wherein the second input includes selecting content.

20. The electronic device of claim 12, wherein the second input includes navigating through content.

21. The electronic device of claim 12, wherein the first application icon is displayed in an arrangement of multiple application icons.

22. The electronic device of claim 12, the one or more programs comprising instructions that, when executed by the electronic device, cause the electronic device to:

in response to detecting an event, display a notification on the second display portion without displaying the notification on the first display portion.

23. A non-transitory computer readable storage medium storing one or more programs, the one or more programs comprising instructions that, when executed by an electronic device that includes a first display portion, a second display portion and a hinge connecting the first display portion to the second display portion, cause the electronic device to:

display, on the first display portion, a first application icon that corresponds to a first application;

while displaying the first application icon on the first display portion, detect a first input corresponding to selection of the first application icon;

in response to detecting the first input corresponding to selection of the first application icon on the first display portion, display an application user interface for the first application on the second display portion;

while displaying the application user interface for the first application on the second display portion, detect a second input directed to the application user interface for the first application on the second display portion; and

in response to detecting the second input directed to the application user interface for the first application on the second display portion, display content corresponding to the first application on the first display portion that was not displayed prior to detecting the second input directed to the application user interface for the first application on the second display portion.

24. The non-transitory computer readable storage medium of claim 23, the one or more programs comprising instructions that, when executed by the electronic device, cause the electronic device to:

in response to detecting the first input:

concurrently display with the application user interface for the first application on the second display portion, a user interface for the first application on the first display portion.

25. The non-transitory computer readable storage medium of claim 24, wherein the user interface for the first applica-

tion on the first display portion is different from the application user interface for the first application on the second display portion.

26. The non-transitory computer readable storage medium of claim 23, wherein the first application is a music application. 5

27. The non-transitory computer readable storage medium of claim 23, wherein the first application is a media viewing application.

28. The non-transitory computer readable storage medium of claim 23, wherein the first input and the second input are touch gestures. 10

29. The non-transitory computer readable storage medium of claim 23, wherein the second display portion includes a power button. 15

30. The non-transitory computer readable storage medium of claim 23, wherein the second input includes selecting content.

31. The non-transitory computer readable storage medium of claim 23, wherein the second input includes navigating through content. 20

32. The non-transitory computer readable storage medium of claim 23, wherein the first application icon is displayed in an arrangement of multiple application icons.

33. The non-transitory computer readable storage medium of claim 23, the one or more programs comprising instructions that, when executed by the electronic device, cause the electronic device to: 25

in response to detecting an event, display a notification on the second display portion without displaying the notification on the first display portion. 30

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